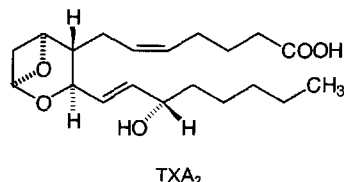


thromboxane-A synthase

5,8,11,14,17-pentaenoate, the product is TXA₁, TXA₂, or TXA₃ respectively. TXA₂ is the major thromboxane *in vivo*. It is synthesized in platelets when these are activated by other aggregatory compounds and strongly reinforces the aggregatory reaction. Its biological lifetime is a few minutes, TXA₂ being converted to the inactive metabolite thromboxane B₂ (TXB₂) by addition of water across the endoperoxide. Oxidation of the 11-hydroxy group of TXB₂ to an oxo group yields 11-dehydro-thromboxane A₂, a major long-lived metabolite in human plasma, useful for measurement as an index of thromboxane production. Beta-oxidation of TXB₂ yields 2,3-dinor-thromboxane B₂, the major human urinary metabolite. **Dideoxy-epoxymethanoprostaglandin F_{2α}** is a stable analogue that mimics TXA₂.



thromboxane-A synthase EC 5.3.99.5; *other names*: thromboxane synthase; thromboxane synthetase (*abbr.*: TXS); an enzyme that catalyses the formation from (5Z,13E)-(15S)-9α,11α-epidioxy-15-hydroxyprosta-5,13-dienoate of (5Z,13E)-(15S)-9α,11α-epoxy-15-hydroxythromba-5,13-dienoate (**thromboxane A₂**). It is a heme-thiolate protein. 1-(7-Carboxyheptyl)imidazole and benzylimidazole inhibit thromboxane-A synthase, as does the drug U-63557A, sodium furegrelate. Example from human: database code THAS_HUMAN, 533 amino acids (60.42 kDa).

thrombus (*pl.* **thrombi**) a blood clot forming within a blood vessel. Thrombus formation may occur as a normal response during physiological blood coagulation. It may also occur pathologically, triggered by **atheroma**, and is the immediate cause of ischemia in a heart attack. Thrombi may also form, especially in veins, at other times, e.g. after surgery, and if they reach the lungs cause pulmonary embolism.

Thunberg method an early method for detecting and estimating **dehydrogenases** depending on the rate and extent of decolorization of Methylene Blue in solution in the presence of dehydrogenase and substrate. The apparatus, known as a **Thunberg tube**, consists of a stoppered glass tube (sometimes with a side-arm for addition of reagents) that may be evacuated through the stopper, and then sealed by a twist of the stopper. [After Torsten Ludvig Thunberg (1873–1952), Swedish physiological chemist.]

Thx *symbol* for a residue of the α-amino acid and thyroid hormone L-thyroxine (alternative to T₄).

Thy *symbol* for a residue of the pyrimidine base thymine (alternative to T).

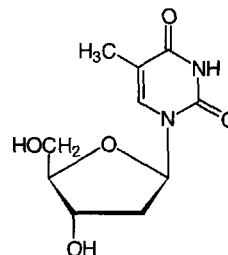
Thy-1 an allelic variable molecule, encoded on mouse chromosome 9, that is present on all T lymphocytes and also occurs in brain. It is a GPI-anchored membrane glycoprotein of the immunoglobulin superfamily, with a simple structure homologous to the **variable region** of an immunoglobulin molecule. Example (precursor) from human: database code THY1_HUMAN, 161 amino acids (17.94 kDa). *See also* **thymopoietin**.

thylakoid a saclike vesicle that bears the photosynthetic pigments in photosynthetic organisms. In prokaryotes the thylakoids are of various shapes and are attached to the plasma membrane. In eukaryotes they are flattened, membrane-bound disclike structures located in the **chloroplasts**; in the chloroplasts of higher plants the thylakoids form dense stacks called **grana**. Isolated thylakoid preparations can carry out photosynthetic electron transport and the associated phosphorylation.

thymic hormones *see* **thymopoietin**, **thymosin**.

thymidine 5'-phosph

thymidine or deoxyribosylthymine *symbol*: dT or dThd; thymine 2'-deoxyriboside; 1-(2-deoxy-β-D-ribofuranosyl)-5-methyluracil; a deoxynucleoside very widely distributed but occurring almost entirely as phosphoric esters in deoxynucleotides and deoxyribonucleic acid, DNA. *Compare* **ribosylthymidine**. Because thymidine is absent from ribonucleic acid, the incorporation of radioisotope-labelled thymidine into cells has been made the basis of a procedure frequently used for measuring the rates of DNA synthesis and cell growth.



thymidine 5'-diphosphate *symbol*: dThd5'PP or ppdT; the *recommended name* for thymidine diphosphate (*abbr.*: dTDP), 5'-diphosphothymidine, 5'-thymidylyl phosphate, thymidine 5'-(trihydrogen diphosphate). It is an intermediate in the formation of thymidine 5'-triphosphate from thymidine 5'-phosphate. Derived compounds are the rare **nucleosidediphosphosugars** thymidinediphosphorhamnose and its biosynthetic precursor thymidinediphosphoglucose.

thymidine kinase EC 2.7.1.21; *systematic name*: ATP:thymidine 5'-phosphotransferase; an enzyme that catalyses the phosphorylation by ATP of thymidine to thymidine 5'-phosphate with release of ADP. It acts on thymidine taken into a cell, or on free thymidine released within a cell, as a first step in its further metabolism. Mitochondrial and cytosolic forms are known, the latter showing much greater activity in proliferating than in resting cells. Example from herpes simplex type 2: database code KITH_HSV23, 375 amino acids (40.32 kDa).

thymidine monophosphate *abbr.*: dTMP; *an alternative name* for any **thymidine phosphate**, but in particular for thymidine 5'-phosphate, especially when its distinction from thymidine (5'-diphosphate and thymidine (5'-triphosphate requires emphasis.

thymidine phosphate *symbol*: dThdP; thymidine monophosphate (*abbr.*: dTMP); any phosphoric monoester or diester of thymidine. There are two monoesters – thymidine 3'-phosphate and **thymidine 5'-phosphate** – and one diester – thymidine 3',5'-phosphate – although thymidine 5'-phosphate is the ester commonly denoted (the locant being omitted if no ambiguity may arise). Thymidine 3'-phosphate (*symbol*: dThd3'P), also named thymidine 3'-monophosphate (*abbr.*: 3'dTMP) or 3'-thymidylic acid, is released from deoxyribonucleic acid by the action of certain endo- or exonucleases, e.g. deoxyribonuclease II (EC 3.1.22.1) or spleen exonuclease (EC 3.1.16.1). The monophosphoric diester thymidine 3',5'-phosphate (*symbol*: dThd-3',5'-P), also named thymidine 3',5'-(cyclic)phosphate or cyclic thymidine 3',5'-monophosphate (*abbr.*: 3',5'-cyclic dTMP or cyclic dTMP or cdTMP), has been isolated as an intermediate formed during the acid hydrolysis of deoxyribonucleic acid but is not known to be of significance otherwise.

thymidine 3'-phosphate *see* **thymidine phosphate**.

thymidine 3',5'-phosphate *see* **thymidine phosphate**.

thymidine 5'-phosphate or 5'-thymidylic acid or 5'-phosphothymidine or 5'-O-phosphonothymidine *symbol*: dThd5'P; *alternative recommended names* for thymidine monophosphate (*abbr.*: dTMP), thymidine 5'-(dihydrogen phosphate), thymine (mono)deoxynucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation.) It

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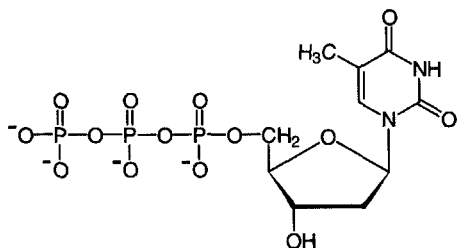
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

thymidine phosphorylase

is synthesized (very indirectly) from **uridine 5'-phosphate**, its immediate precursor being 2'-deoxyuridine 5'-phosphate – see **thymidylate synthase**. In addition, resynthesis of dTMP from free thymidine formed by degradation of DNA can be effected via a **salvage pathway**.

thymidine phosphorylase see **pyrimidine-nucleoside phosphorylase**.

thymidine 5'-triphosphate *symbol*: dThd5'PPP or pppdT; *the recommended name for* thymidine triphosphate (*abbr.*: dTTP), 5'-triphosphothymidine, 5'-thymidylyl diphosphate, thymidine 5'-(tetrahydrogen triphosphate); a substrate for DNA synthesis. It is formed from thymidine 5'-phosphate by the successive action of nucleoside-phosphate kinase (EC 2.7.4.4) and nucleoside-diphosphate kinase (EC 2.7.4.6), which transfer in turn the terminal phosphoric residues from two molecules of adenosine 5'-triphosphate.



thymidyl either of the chemical groups formed by the loss of a 3'- or a 5'-hydroxyl group from the deoxyribose moiety of thymidine.

thymidylate **1** either the monoanion or the dianion of thymidylic acid. **2** any mixture of thymidylic acid and its anions. **3** any salt or ester of thymidylic acid.

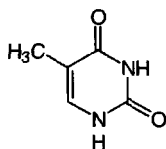
thymidylate synthase EC 2.1.1.45; *systematic name*: 5,10-methylenetetrahydrofolate:dUMP C-methyltransferase; an enzyme responsible for the formation of thymidylate, dTMP, by the *de novo* pathway. It catalyses reductive methylation of 2'-deoxy-5'-uridylyl, dUMP, by reaction with 5,10-methylenetetrahydrofolate to form dihydrofolate and dTMP. (The dihydrofolate thus formed is reduced again to tetrahydrofolate by **dihydrofolate reductase**.) Thymidylate synthase is important as a target for the antineoplastic agents **fluorouracil** and **fluorodeoxyuridine**. These drugs are metabolized to fluorodeoxyuridylyl, which is a potent irreversible inhibitor of the enzyme. Example from *Escherichia coli* (homodimer); database code TYSY_ECOLI, 264 amino acids (30.44 kDa); 3-D structure known; five motifs. See also **thymineless death**.

thymidylic acid *the trivial name for* any phosphoric monoester of thymidine. The position of the phosphoric residue on the deoxyribose moiety of a given ester may be specified by a prefixed locant – see **thymidine phosphate**. However, 5'-thymidylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-Thymidylic acid is also an alternative recommended name for **thymidine 5'-phosphate**.

thymidylyl the thymidine[mono]phospho group; the acyl group derived from **thymidylic acid**.

thymine *a former name for* **thymopoietin**.

thymine *symbol*: T or Thy; 5-methyluracil; 2,4-dihydroxy-5-methylpyrimidine; one of the two major pyrimidine bases present (as thymidine) in DNA but not found in RNA other than (as ribothymidine) in transfer RNA, where it is a minor base.



thymus nuclei

thymine dimer a **pyrimidine dimer** formed within a DNA strand from two adjacent thymine residues by **photodimerization**. It is most commonly of the cyclobutadipyrimidine type.

thymineless death the death of an animal or bacterial cell following a period of thymine (or thymidylate) deprivation. Although such a deprived cell may continue to produce protein and RNA, it can no longer synthesize DNA in the absence of thymidylate, and it eventually loses its vitality and dies, perhaps because it is also unable to repair altered or damaged DNA. The phenomenon may be demonstrated with a thymine-requiring auxotrophic bacterial mutant by omitting thymine from the suspending medium, or with any cell by adding **fluorouracil** or its metabolite fluorodeoxyuridine, with consequent inhibition of **thymidylate synthase**. A rapidly metabolizing animal cell, such as a tumour cell, is particularly susceptible to thymineless death.

thymine deoxyriboside *an alternative name for* **thymidine**.

thymine riboside *an alternative name for* **ribothymidine**.

thyminose *a former name for* 2-deoxy-D-ribose (see **deoxyribose**).

thymo+ *comb. form of, or pertaining to, the thymus*.

thymocyte (*sometimes*) any **lymphocyte** found in the thymus.

thymocyte mitogenic factor *abbr.*: TMF; *an alternative name for* **interleukin 2**.

thymocyte-stimulating factor *abbr.*: TSF; *an alternative name for* **interleukin 2**.

thymoma a tumour of thymic tissue.

thymonuclease *another name for* **deoxyribonuclease I**.

thymonucleic acid *an alternative term for* **thymus nucleic acid**.

thymopentin a synthetic pentapeptide that corresponds to residues 32–36 of **thymopoietin**.

thymopoietin *abbr.*: TP; any one of three polypeptide hormones that result from alternative splicing of the same gene, and originally isolated from bovine thymus. They have pleiotropic actions on prothymocytes, inducing expression of the differentiation antigens such as **Thy-1**, and are important in T-cell development and function. They were discovered in studies on myasthenia gravis and impair neuromuscular transmission after injection in mice, due to an effect on the nicotinic acetylcholine receptor. A synthetic pentapeptide corresponding to positions 32–36, named **thymopentin** and with the sequence Arg-Lys-Asp³⁴-Val-Tyr, has full activity. **Splenin**, a corresponding peptide in spleen, contains a similar pentapeptide, called **splenopentin**; bovine splenin has Glu in place of Asp³⁴, and human splenin has Ala in the corresponding position. These replacements affect the biological activity. Thymopoietin was formerly known as thymine.

thymosin any of several related polypeptide hormones of thymic origin that are involved in differentiation of T lymphocytes in the thymus. They include α_1 and β_4 thymosins, which are of known structure. Thymosin β_4 (human, bovine, rat) has the sequence SDKPDMAEIEKFDKSKLKKTTETQEKNP-PSKETIEQEKQAGES.

thymulin a zinc-requiring immunomodulatory thymic nonapeptide hormone known to induce intra- and extra-thymic T-cell differentiation. Its level is decreased in immunodeficiency and in autoimmune disease.

thymus (gland) a bilobed glandular organ located in the posteroventral part of the neck. In mammals the thymus is encapsulated and divided into lobules, each lobule consisting of cortex and medulla; the cortex consists mainly of lymphocytes (thymocytes). The thymus is responsible for populating the blood, lymph, and thymus-dependent areas with **T lymphocytes** during the neonatal period. The thymus is large at birth but diminishes in size with increasing age and may be difficult to identify in adults. It is important in **cell-mediated immunity**. — **thymic adj.**

thymus nucleic acid *a former name for* **deoxyribonucleic acid**. It was so termed because the thymus gland was found to be a particularly rich source of nucleic acid of a type that was thought at the time to be characteristic of animal tissues and



that was distinct from the type believed to be characteristic of plant tissues and known as **yeast nucleic acid**.

thyro+ *comb. form of*, or pertaining to, the **thyroid gland**.

thyrocalcitonin *see* **calcitonin**.

thyroglobulin a protein in the **thyroid gland**, tyrosyl residues of which are modified to form the principal thyroid hormone, **thyroxine**, which is released by proteolysis. Thyroglobulin, a dimer of molecular mass about 660 kDa, is the main constituent of the so-called colloid of the thyroid follicles. Iodination of certain tyrosyl residues results in the formation of 3,5-iodotyrosyl residues; there is then transfer of the iodinated phenolic moiety of one of these tyrosyls (in the free form this would be a 2,6-iodophenyl group) to the phenolic oxygen of another iodinated tyrosyl to form a tetraiodo compound which, on release by proteolysis, is thyroxine (T_4) (some triiodo forms are also produced). The proteolysis occurs as colloid migrates from the basal to the apical surfaces of the thyroid cell. Example (precursor) from human: database code THYG_HUMAN, 2767 amino acids (304.45 kDa). The colloid containing the modified thyroglobulin is taken into vesicles by phagocytosis to form periodic acid-Schiff positive (PAS) vesicles. Dense granules containing proteolytic enzymes then fuse with the vesicles to digest the thyroglobulin with the loss of PAS-positive material. The vesicles containing triiodothyronine and thyroxine are then extruded.

thyroidectomy the surgical removal of the thyroid gland.

thyroid gland a ductless gland, located near the larynx in most species, that secretes the hormones **thyroxine** and **triiodothyronine**. Thyroid tissue is confined to vertebrates. The gland contains closely packed follicles, or acini, each consisting of a sac that has a peripheral single layer of cuboidal follicular cells forming an epithelium on an outer basement membrane; the centre of the follicle is occupied by a colloid of which the major component is **thyroglobulin**. This is synthesized and secreted into the colloid space by the follicular cells; microvilli extend from the apical surface of the follicular cells into the colloid. The thyroid also contains parafollicular cells, or C cells, that synthesize **calcitonin**. Enlargement of the thyroid in iodine deficiency is the basis of **goitre**. The thyroid concentrates iodide, a property that can be used for thyroid imaging, and for the treatment of cancer of the thyroid.

thyroid hormone either of the compounds secreted by the **thyroid gland**, namely **thyroxine** and **triiodothyronine**.

thyroid hormone receptor a nuclear receptor for **triiodothyronine** that mediates the effects of thyroid hormones on cells. These receptors bind triiodothyronine and directly regulate gene expression. The proteins belong to a superfamily of eukaryotic transcription factors and the steroid/thyroid/retinoic acid family of nuclear receptors for hormones. The ligand-binding domain is in the C-terminal region, the zinc-finger DNA-binding domain in a central region, and a function-modulating domain in the N-terminal region. Binding of hormone induces changes in receptor conformation that control transcriptional activation and repression, and also regulates homo- or heterodimerization. In the absence of ligand, these receptors repress basal gene expression, probably through co-repressor proteins. Example, α -1 receptor from *Mus musculus*: database code THA1_MOUSE, 410 amino acids (46.79 kDa).

thyroid hormone either of the compounds secreted by the **thyroid gland**, namely **thyroxine** and **triiodothyronine**.

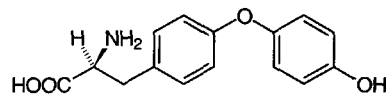
thyroid response element *abbr.*: TRE; *see* **response element**.

thyroid-stimulating hormone *abbr.*: TSH; *recommended name*: thyrotropin; *other name*: thyrotropic hormone; a hormone, secreted by the adenohypophysis, that stimulates the activity of the thyroid gland. The bovine hormone is a glycoprotein, M_r 28 300, and a heterodimer of two chains, α (M_r 13 600) and β (M_r 14 700). The α chain has an amino-acid sequence identical with that of the α chain of **lutinizing hormone** of the same species. TSH increases the uptake of iodide by the thyroid, increases the rate of conversion of diiodotyrosine to thyroxine, and increases the release of thyroid hormones from

the thyroid. The release of TSH from the adenohypophysis is influenced by **thyrotropin-releasing hormone**. Examples, α chain (precursor) from human: database code GLHA_HUMAN, 116 amino acids (13.06 kDa); β chain (precursor) from human: database code TSHB_HUMAN, 138 amino acids (15.59 kDa).

thyroliberin *recommended name for thyrotropin-releasing hormone*.

thyronine 4-(4-hydroxyphenoxy)phenylalanine; the structure from which thyroid hormones may be notionally derived by iodination. It is not naturally occurring, even in bound form, since in the synthesis of thyroid hormones the tyrosyl residues of thyroglobulin are iodinated before addition of the further phenolic group (*see* **thyroglobulin**).



thyrotoxicosis *see* **hyperthyroidism**.

thyrotroph a thyrotropic cell.

thyrotrophic having the effect of nourishing the thyroid gland, stimulating its growth and maintaining its size.

thyrotropic or (*esp. Brit.*) **thyrotrophic** having a modulating effect on the thyroid gland, especially in stimulating the synthesis of thyroid hormones. *See* **thyrotropin**.

thyrotropic hormone an alternative name for **thyroid-stimulating hormone**.

thyrotropic hormone releasing factor (*abbr.*: THRF) or **thyrotropic hormone releasing hormone** (*abbr.*: THRH) an alternative name for **thyrotropin-releasing hormone**.

thyrotropin the recommended name for **thyroid-stimulating hormone**.

thyrotropin receptor an integral membrane protein that binds thyrotropin (i.e. **thyroid-stimulating hormone**) and mediates its intracellular effects. It is a G-protein-linked receptor, with different forms produced by alternative splicing. Example from human (precursor): database code TSHR_HUMAN, 764 amino acids (86.70 kDa). Mutations are associated with hyperfunctioning thyroid adenomas. The hyperthyroidism of **Graves' disease** is thought to be due to the presence in the blood of these patients of autoantibodies that stimulate the thyroid through the thyrotropin receptor.

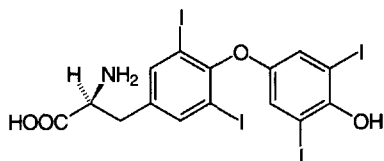
thyrotropin-releasing factor *abbr.*: TRF; an alternative name for **thyrotropin-releasing hormone**.

thyrotropin-releasing hormone *abbr.*: TRH; *recommended name*: thyroliberin; *other names*: thyrotropin-releasing factor (*abbr.*: TRF); thyrotropic hormone releasing factor (*abbr.*: THRF); thyrotropic hormone releasing hormone (*abbr.*: THRH); a hormone, released by the mammalian hypothalamus into the hypophyseal-portal circulation in response to neural and/or chemical stimuli, that regulates the secretion of **thyroid-stimulating hormone** (TSH) by the adenohypophysis. It has a wide distribution, being present not only in primitive vertebrates but also in *Amphioxus* (a provertebrate) and in nerve ganglia of snail. It is the tripeptide pyroGlu-His-Pro-NH₂, the cyclized glutamate and intact prolylamide being essential for activity. It is synthesized as part of a large prohormone that has six repeats of the TRH sequence. TRH is present in many other parts of the brain, in pancreatic islets, and in the gastrointestinal tract; it probably serves widely as a neurotransmitter or neuromodulator, being known in mammals to alter sleep patterns, increase motor activity, increase blood pressure, and affect norepinephrine turnover, among many activities. Its regulation of TSH secretion is controlled by the pituitary-thyroid axis.

thyroxine *abbr.*: Thx or T_4 ; 4-(4-hydroxy-3,5-diodophenoxy)-3,5-diodophenylalanine; 3,3',5,5'-tetraiodothyronine; the main hormone secreted by the thyroid gland, although its ac-

thyroxine-binding globulin

tivity is lower than that of **triiodothyronine** (T_3). Its synthesis from L-tyrosyl residues of **thyroglobulin** means that the L-isomer is the naturally occurring form. Thyroxine is essential for normal metabolism and physical development. It stimulates the metabolic rate, causing increased oxygen consumption and heat production in tissues. Although in humans the plasma levels of T_4 are 20- to 50-fold higher than those of T_3 , T_4 is converted to T_3 in tissues and the effects are thought to be mediated by T_3 binding to receptors in the cell nucleus (chromatin fraction), leading to mRNA and protein synthesis. The normal range in human blood is 60–150 nmol L⁻¹ (total of bound and free), 9–26 pmol L⁻¹ (free). The chemical constitution of thyroxine was deduced by the British biochemist Sir Charles R. Harington (1897–1972), and confirmed by synthesis (1927).



thyroxine-binding globulin *abbr.*: TBG; a major thyroid hormone transport protein found in mammalian serum. It is synthesized in liver, and variations in its concentration in serum result in similar variations in the level of circulating thyroid hormones. Alterations in its affinity for thyroid hormone also affect circulating thyroid hormone levels, decreased affinity being accompanied by decreased serum concentration. In the human, this variation may result from non-thyroidal illness, the presence of drugs (salicylates and phenytoin block thyroid-binding sites), or genetic variation, many variants being known, and listed in the example from human: database code THBG_HUMAN, 415 amino acids (46.32 kDa).

Ti *symbol* for titanium.

tight coupling a term applied to mitochondrial function implying a high degree of efficiency in the yield of ATP (or other functional response) in relation to electron flow. The nearer the yield approaches the optimal value, the tighter the coupling is said to be. Tight coupling is obtained in isolated mitochondria only if gentle procedures, preserving membrane integrity, are employed during isolation. *See also chemiosmotic coupling hypothesis, phosphorus:oxygen ratio.*

tight junction or **zona occludens** (*in vertebrate tissues*) a belt-like region of very close contact between the plasma membranes of adjacent cells, such that the intercellular space is completely occluded. Tight junctions occur in epithelia and brain endothelia, and are effective barriers to the passage of water and solutes. *Compare gap junction.*

tight turn *see beta turn.*

time *symbol*: *t*; one of the seven SI base physical quantities, usually indicating duration or a precise moment. The SI base unit of time is the **second**.

time constant a characteristic time taken by a system or a process to respond to a perturbation.

time course a graph plot of a parameter (that changes with time) scaled on the ordinate and time scaled on the abscissa.

time-resolved fluorescence spectrometry a method in which a sample is irradiated using a laser delivering a short (sub-nanosecond) pulse of light, after which the exponential decay of fluorescence is measured using a photomultiplier. The sensitivity of detection must be at the single photon level ('single photon counting'), and fluorescence lifetimes as short as a few nanoseconds can be determined. However, a disadvantage of the technique is that many measurements over possibly several hours may be needed to obtain reliable and sufficient data. *See also steady-state fluorescence anisotropy.*

TIMP *abbr.* for tissue inhibitor of metalloproteinases; any of a

group of proteins that inhibit metalloproteinases. An example is TIMP-1, also known as EPA (erythroid potentiating activity), a protein that mediates erythropoiesis. It forms complexes with enzymes such as collagenases, bringing about irreversible inactivation. Example (precursor) from human: database code TIM1_HUMAN, 207 amino acids (23.14 kDa). *See also matrix metalloproteinase.*

tintometer an apparatus formerly used for determining the colour of a solution by comparison with a graded colour scale.

Ti plasmid *abbr.* for tumour-inducing plasmid; any of a class of large conjugative plasmids found in the soil bacterium *Agrobacterium tumefaciens* and responsible for **crown-gall disease** of broad-leaved plants – a segment of the Ti plasmid, the **T-DNA**, is found in the genome of the tumour tissue of affected plants. With appropriate modifications the Ti plasmid can carry foreign DNA sequences of any desired kind into the genome of a susceptible plant.

Tiselius, Arne Wilhelm Kaurin (1902–71), Swedish protein chemist; Nobel Laureate in Chemistry (1948) 'for his research on electrophoresis and adsorption analysis, especially for his discoveries concerning the complex nature of the serum proteins'.

Tiselius apparatus an apparatus for performing **moving boundary electrophoresis** in which there is a U-tube with sliding joints. This enables the protein (or other) solution and the buffer to be placed in separate parts of the U-tube and sharp boundaries made between them when the parts are slid into position to complete the U-tube.

tissue any collection of cells that is organized to perform one or more specific function. *Compare organ.*

tissue culture 1 the technique or process of growing or maintaining tissue cells (**cell culture**), whole organs (**organ culture**), or parts of an organ, from an animal or plant, in artificial conditions. 2 any living material grown or maintained by such a technique.

tissue extract *an alternative name for factor III; see blood coagulation.*

tissue factor *abbr.*: TF; a transmembrane glycoprotein (eight motifs), also known as factor III, that initiates **blood coagulation** by forming a complex with factor VII or VIIA. The complex activates factors IX or X. Tissue factor plays a role in normal hemostasis by initiating cell-surface assembly and propagation of the coagulation protease cascade. Example from human (precursor): database code TF_HUMAN, 295 amino acids (33.03 kDa).

tissue grinder any device for disrupting tissue by a crushing or shearing action. The term may embrace devices ranging from a meat grinder (mincer), mechanical pestle and mortar, or mill, to the more precise **Potter-Elvehjem homogenizer** or the **Dounce homogenizer**.

tissue kallikrein *see kallikrein.*

titer *the US spelling of titre.*

titin or **connectin** a giant protein that forms a single molecule elastic filament extending from the M line to the Z line in the striated muscle sarcomere and is one of the largest polypeptides yet described. The sequence consists mainly of repeats of two types of approximately 100-amino-acid motifs, known as class I and class II that show homology to the fibronectin type-III domain and **immunoglobulin superfamily** C2 domain, respectively. There is also a domain characteristic of protein kinases near the C terminus. It is thought to play an important role in sarcomere alignment during muscle contraction. Example from human: database code HSTITIN, 4650 amino acids (521.48 kDa). *See also twitchin.*

titratable acidity 1 a measure of the acidity of a urine sample, expressed as the volume of 0.1-M NaOH required to neutralize, usually to a phenolphthalein endpoint, a 24-hour volume of urine. 2 a measure of the acidity of any solution, etc.

titrate 1 to add acid of known concentration to base of unknown concentration (or vice versa) until the point of equivalence is reached, from which can be determined the con-

titration curve

centration of the unknown. **2** more generally, to add any standard solution to an unknown until some detectable equivalence point is reached, enabling the amount of the unknown to be determined. **3** (esp. in microbiology or immunology) to determine the maximum dilution at which activity can be detected. See **titre** (def. 2). —**titratable** *adj.*; **titration** *n.*

titration curve originally, a curve relating pH to the equivalents of strong base added per equivalent of acid in the solution. Similar curves are obtained when the logarithm of ligand concentration is plotted against the degree of association (or dissociation) of a ligand to its acceptor, or when redox potential is plotted against the degree of oxidation (or reduction) of a redox system.

titration pK *symbol:* pG; a kind of **pK** that characterizes the titration curve of a multibasic acid. If this acid has *n* hydrons that titrate over the pH range of interest, then its titration curve is identical to that of an equimolar mixture of *n* (hypothetical) monobasic acids. The pK values of these acids are the titration pK values of the multibasic acid. They differ little from the **molecular pK** values if the differences between sequential molecular values exceed unity. See also **group pK**.

titre or (esp. US) **titer** **1** a value found by titration; see **titrate** (def. 1, 2). **2** (in microbiology and immunology) a measure of the concentration or activity of an active substance, e.g. an antibody, in a solution, usually expressed as the highest dilution of the solution in which the activity can be detected. By convention, if the highest dilution giving activity is 100-fold, the titre is said to be 100.

titrimetry the technique of measurement by titration, particularly using instrumentation and automation.

Tl *symbol* for thallium.

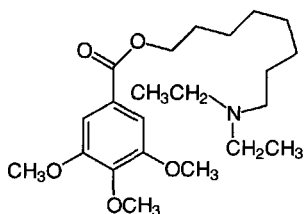
TLC *abbr.* for thin-layer chromatography.

tld a gene in *Drosophila* for **Tolloid** protein, which is required for normal dorsal development. Mutations in *tld* lead to a partial transformation of dorsal ectoderm into ventral ectoderm.

T lymphocyte or **T cell** any of a class of lymphocytes that undergo maturation and differentiation in thymus. They are responsible for immune reactions involving cell-cell interaction, i.e. **cell-mediated immunity**. Subsets of T lymphocytes include: (1) cytotoxic T-lymphocytes, directly responsible *in vivo* for the death of cells identified for elimination; they include nonspecific killer cells (**NK cells**) and specific cytotoxic T-lymphocytes primed by antigen activation; (2) **helper cells**, which collaborate with **antigen-presenting cells** in the initiation of an immune response; and (3) **suppressor cells**, which down-regulate the response of helper cells.

Tm *symbol* for thulium.

TMB-8 *abbr.* for (8-diethylamino)octyl-3,4,5-trimethoxybenzoate; a Ca^{2+} antagonist that blocks intracellular Ca^{2+} channels.



TMF *abbr.* for thymocyte mitogenic factor; i.e. **interleukin 2**.

TMP *abbr.* for **1** ribosylthymine monophosphate or ribothymidine monophosphate, the common names for ribosylthymine 5'-phosphate, ribothymidine 5'-phosphate, 5'-ribothymidyl phosphate, thymine riboside 5'-phosphate. **2** (sometimes) thymidine monophosphate. dTMP is recommended.

TMR spectroscopy *abbr.* for topical magnetic resonance spectroscopy.

TMS *abbr.* for **1** tetramethylsilane; a water-soluble reference.

substance used in nuclear magnetic resonance spectroscopy.

2 the trimethylsilyl group; the symbol $\text{Me}_3\text{Si}-$ is preferred.

TMV *abbr.* for tobacco mosaic virus, an RNA plant virus.

Tn *abbr.* for **1** troponin. **2** transposon.

TNBS *abbr.* for 2,4,6-trinitrobenzenesulfonate; a reagent that combines readily with amino groups. It may be used either to protect these groups or to introduce a chromophore. It is removable using hydrazine.

Tn-C *abbr.* for troponin C.

TNF *abbr.* for tumour necrosis factor.

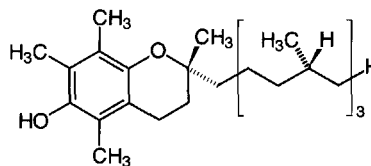
Tn-I *abbr.* for troponin I.

TNMR or **tnmr** *abbr.* for tritium nuclear magnetic resonance (spectroscopy).

TNS *abbr.* for 2-p-toluidylnaphthalene-6-sulfonate; a **fluor** used in **intrinsic fluorescence** studies of proteins.

Tn-T *abbr.* for troponin T.

tocopherol a generic term for di- and trimethyltocols; α -tocopherol is 5,7,8-trimethyltocol, and is active as **vitamin E**; β -tocopherol is 5,8-dimethyltocol; γ -tocopherol is 7,8-dimethyltocol; and δ -tocopherol is 8-methyltocol.



α -tocopherol

Todd, Alexander Robertus Todd, Lord Todd (1907–97), British organic chemist; Nobel Laureate in Chemistry (1957) 'for his work on nucleotides and nucleotide co-enzymes'.

T-odd phage any one of the **bacteriophages** T1, T3, T5, or T7 of *Escherichia coli*. These DNA phages share a number of morphological, biochemical, and other features that distinguish them as a group from **T-even phages**.

togavirus or (formerly) **arbovirus** any of a group of RNA animal viruses consisting of enveloped, icosahedral, 50–70 nm capsids with single-stranded infectious RNA (plus strand) of 4 MDa. The group contains mosquito- and tick-borne viruses causing encephalitis and yellow fever.

tolbutamide 1-butyl-3-(*p*-tolylsulfonyl)urea, *N*-[(butylamino)carbonyl]-4-methylbenzenesulfonamide; one of a number of sulfonylurea drugs, of which it is the prototype, that exert a hypoglycemic action by stimulating release of insulin from the pancreas and are used therapeutically to treat type II **diabetes mellitus**.

tolerance **1** the progressive attenuation of the response to an agent (usually a drug) whereby increasing concentrations of the agent are required to maintain the response. Underlying mechanisms confer either functional (or pharmacodynamic) tolerance, where the loss of response is due to desensitization of effector mechanisms, or metabolic (or pharmacokinetic) tolerance, whereby elimination of the agent is accelerated usually by induction of catabolic enzymes or other inactivating mechanism. **2** the ability of an organism to grow and thrive in an unfavourable environment. **3** an allowable deviation from a standard. **4** **immunological tolerance**.

tolerogen an antigen that, in given circumstances, induces **immunological tolerance** when administered to an animal. Tolerogens may, when presented in different circumstances, induce an immune response. —**tolerogenic** *adj.*

Toll a developmental protein in *Drosophila melanogaster*, the product of *toll* gene, that is required for embryonic dorsal-ventral polarity. It is a type I transmembrane signalling receptor protein with a **leucine-rich repeat** segment. Precursor: database code TOLL_DROME, 1097 amino acids (124.51 kDa).

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tolloid

tolloid a developmental protein in *Drosophila melanogaster* encoded by the *ttld* gene; it is a member of the *astatin* subfamily of metalloproteinases. Database code TLD_DROME, 1057 amino acids (120.44 kDa).

tomography a radiographic technique that can image a thin plane or section of a body. An X-ray beam is focused on an area of a body, the beam is then rotated, and successive images are stored and subsequently analysed by computer (computer-assisted tomography, or CAT scanning). **Positron emission tomography** (or PET scanning) involves the detection of X-rays emitted from radionuclides that decay by positron emission and are located within the patient's body, e.g. from ^{15}O (oxygen-15) after administration of H_2^{15}O .

ton 1 or (esp. US) **long ton** a unit of mass equal to 2240 pounds avoirdupois (1016.05 kg). 2 or (esp. US) **short ton** or **net ton** a unit of mass equal to 2000 pounds avoirdupois (907.18 kg). 3 metric ton; see **tonne**.

tone 1 (in physiology) or **tonus** the state of sustained partial tension adopted by muscles in order to maintain posture, etc. It is brought about by the contraction of only a certain proportion of the muscle fibres at any given time. 2 (in clinical medicine) the state or degree of firmness of a muscle on palpation.

tonicity 1 the osmotic pressure of a solution; often used in the sense of the ratio of the osmotic pressure of a solution to that of a given reference solution. 2 (of a muscle) the condition of possessing **tone** or tension. See also **hypertonic**, **hypotonic**, **isotonic**.

tonin EC 3.4.21.35; other names: esterase 1; S2 kallikrein; RKG-2; RSKG-5; an endopeptidase, present in several rat tissues, that can cleave angiotensinogen and angiotensin-tetradecapeptide renin substrate to produce angiotensin II directly. It also acts on the Phe-His bond of angiotensin I to form angiotensin II. It is a trypsin- and chymotrypsin-like serine protease. Example (precursor) from rat: database code KLK2_RAT, 259 amino acids (28.22 kDa).

tonne or **metric ton** symbol: t; a unit of mass equal to 10^3 kg. 1 tonne = 0.984 207 UK ton or 1.102 31 short (US) tons.

tonometer 1 an apparatus, usually a closed glass vessel, used to equilibrate a liquid, especially blood, with a particular gas of known partial pressure. When equilibrium is established the phases are separated and the gas in question determined in both phases. 2 a device for measuring vapour pressure.

tonus an alternative term for **tone** (def. 1).

tooth-lid factor a name originally given to **epidermal growth factor**.

top+ a variant form of **topo+** (before a vowel).

TOPA abbr. for 3,4,6-trioxyphenylalanine; 6-hydroxydopa. See also **TOPA-quinone**.

TOPA-quinone 6-hydroxydopa quinone; the covalently bound redox prosthetic group of bovine plasma amine oxidase and some other copper-containing oxidases.

topical relating to a particular place or part of the body; local; (of a medicament, nutrient, etc.) applied locally rather than systemically. —**topically** adv.

topical magnetic resonance spectroscopy abbr.: TMR spectroscopy; a method for obtaining high-resolution nuclear magnetic resonance spectra from a selected place in a larger specimen (e.g. a specific organ of a living body). The sensitive volume is generated by superimposing high-order magnetic field gradients onto the main magnetic field of the spectrometer in such a manner as to define a central region of uniform field and adjustable size, surrounded by rapidly changing fields. Signals may be detected from individual nuclei, and the signals can be attributed to specific metabolites, enabling measurement of the level of these metabolites *in situ*.

topo+ or (before a vowel) **top+** comb. form denoting place or region.

topochemical 1 of, or pertaining to **topochemistry**. 2 of, pertaining to, or constituting a locally confined chemical reaction. 3 describing a combined tactile and chemical sense in

some animals that is important in the perception of odours in relation to track or position.

topochemistry the characteristic chemical activity of a **topomer**.

topogenesis an alternative term for **morphogenesis**, especially in regard to biomolecular structures. —**topogenic** adj.

topogenic sequence any part of the sequence of a nascent polypeptide that is involved in getting the mature protein to its proper position in the cell. See (e.g.) **mannose 6-phosphate**, **signal peptide**.

topogenic signal any feature of the structure of a macromolecule that directs it to a particular organelle in a cell. The term comprehends **topogenic sequence**.

topography 1 the science and practice of describing the arrangement in space of shapes and surfaces, and of components in relation to one another. 2 the detailed description, or representation on a map, diagram, etc., of the surface features of a three-dimensional object, or part of one, with reference to the underlying structure. Compare **topology**. —**topographic** or **topographical** adj.; **topographically** adv.

topoinhibition the inhibition of a cellular process by an action at the cell membrane, especially through contact with, or proximity to, other cells.

topoisomer or **topological isomer** any of two or more **isomers** of a macrocyclic molecule, or system of two or more macrocyclic molecules or molecular subunits, that differ in the degree of knotting of a loop or the degree of interlocking of rings. Topoisomers of a given molecule, or system of molecules, display the same connectivity, bond orders, and configurations but can be interconverted only by breaking and reforming a covalent bond; they have different **linking numbers**. Compare **topomer**. —**topoisomeric** adj.; **topoisomerism** n.

topoisomerase see **DNA topoisomerase**.

topological bond the force equivalent to a chemical bond that may be conceived as holding together any two interlocked rings of a catenated molecular structure or two loops of a knot in a cyclic molecule. Such a bond is the property not of a pair of atoms but of the complete molecule, the strength of the union being that of the weakest chemical bond in the entire structure.

topological isomer an older name for **topoisomer**.

topological isomerism an alternative term for **topoisomerism**; see **topoisomer**.

topological winding number see **supercoil**.

topology 1 (in molecular biology) the study of deformability and deformation of structures, including macromolecules. The basic concept is founded on whether a particular figure or surface can be continuously deformed without rupture and re-making of, e.g., a chemical bond. 2 (in botany) the study of localities where plants are found. 3 (in mathematics) the branch of geometry that deals with general types of shape rather than with particular shapes and sizes; the study of relationships that remain invariant under one-to-one transformation. Compare **topography**. —**topological** or **topologic** adj.; **topologically** adv.

topomer any of two or more **conformers** of a macromolecule that have differing **topography** (def. 2). Any topomer (or its exact mirror image) is characterized by a unique set of interactions between any two residues of the complete sequence of the macromolecule. Compare **topoisomer**.

topotactic describing a chemical reaction in which the product has an ordered geometric relationship to the reactant.

top yeast see **bottom yeast**.

toroid 1 an alternative term for **torus**. 2 the solid enclosed by a torus. —**toroidal** adj.

torr symbol: torr; a non-SI unit of pressure, used especially in connection with low pressures. It is defined as equal to 1/760 of an atmosphere; i.e. 1 torr = 101 325/760 Pa = 133.322 Pa. Its use is now discouraged, in favour of the **pascal**. [After Evangelista Torricelli (1608–47), Italian physicist.]

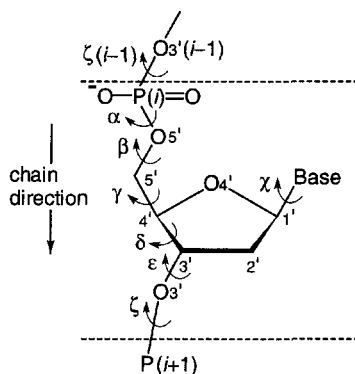
torsion 1 the twisting of something about an axis by the appli-



torsion angle

cation of equal and opposite rotational forces acting in parallel planes. **2** the twist so produced. —**torsional** *adj.*

torsion angle *symbol:* θ or ω ; consider an assembly of attached atoms represented by the generalized partial structure X–A–B–Y where neither X nor Y is collinear with A and B; then the torsion angle is the smaller angle subtended by the bonds X–A and Y–B in a plane projection obtained by viewing the assembly along the axis A–B from either direction. The angle is considered positive or negative according as the bond to the atom nearest the viewer, X or Y, requires to be rotated to the right or left, respectively, for its direction to coincide with that of the bond to the furthestmost atom, Y or X. The multiplicity of the bonding of the various atoms is not relevant. A torsion angle may similarly be defined for a partial structure X–A–Z–B–Y where Z represents one or more additional atoms and A, Z, and B are collinear. The torsion angles for the backbone of a polypeptide chain are symbolized as follows: that for rotation about the N–C α bond is denoted by ϕ ; that for rotation about the C α –C' bond is denoted by ψ ; that for rotation about the C'–N bond is denoted by ω . The torsion angles for the backbone of a polynucleotide chain are as shown below. *See also* conformation, Ramachandran plot, tight turn.



designation of torsion angles for polynucleotides

torsion balance an instrument, used for the rapid weighing of small amounts of substances, in which the sample is balanced by the torsion of a wire and the amount of torsion that has to be applied to restore balance is indicated as a scale calibrated in units of mass.

torso a gene for a tyrosine kinase (EC 2.7.1.112) receptor in *Drosophila melanogaster*. The receptor is a type I membrane glycoprotein and is involved in the determination of anterior and posterior structures in the embryo. Precursor: database code KTOR_DROME, 923 amino acids (105.04 kDa).

torulin an obsolete name for extracts of yeast that cure polyneuritis in pigeons fed on a diet of polished rice, no doubt due to the vitamin B₁ content of yeast extract.

torus (*pl. tori*) a shape generated by rotating a circle or disk about an axis that is displaced outside the diameter of the circle or disk, but in the same plane; i.e. shaped like a ring doughnut (donut). *See also* toroid. —**toric** *adj.*

Tos *symbol* for the tosyl group.

tosyl *symbol:* Tos; a trivial name for the toluene-4-sulfonyl group, CH₃–C₆H₄–SO₂–, with which amino groups may be modified by allowing them to react with *p*-toluenesulfonyl chloride. *Compare* pipsyl.

total body water *abbr.:* TBW; *see* body water.

totipotent (of a cell) having the potentiality to develop in any of the directions inherently possible, given its particular genetic constitution, and thus to form a new organism or to regenerate any part of an organism.

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toxic **1** poisonous. **2** of, relating to, or caused by a **toxin** or poison.

toxicity **1** the quality or state of being poisonous or toxic. **2** the relative degree of being poisonous or toxic; the potency of a **toxin**.

toxicogenic an alternative term for **toxigenic**.

toxicology the science dealing with the effects of poisons on living organisms. The term is sometimes restricted to the effects of synthetic poisons and the term **toxology** used when dealing with naturally occurring poisons.

toxify to poison; to render poisonous. —**toxification** *n.*

toxigenic or **toxicogenic** or **toxigenic** (especially of bacteria and fungi) producing toxin.

toxin any of various specific poisonous substances that are formed biologically. Not all such poisons are so termed, their classification as toxins being somewhat arbitrary and tending to vary with the discipline concerned; furthermore, the term is sometimes extended to include synthetic poisonous substances. Various types of toxin may be designated according to the source of the toxin, e.g. **endotoxin**, **exotoxin**, **mycotoxin**, **phycotoxin**, **phytoxin**, and **zootoxin**, or according to the specific or prime site of action of the toxin, e.g. **hepatotoxin** and **neurotoxin**.

toxigenic an alternative term for **toxigenic**.

toxology *see* **toxicology**.

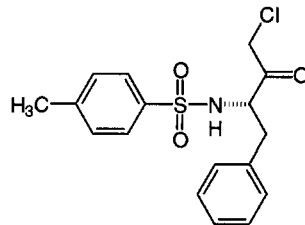
toxoid an **exotoxin** that has been modified (e.g. by formalin treatment) so that its toxicity has been lost while its antigenicity (both immunogenicity and reactivity) is retained.

toxophore a term introduced by Ehrlich to denote the specific chemical group in a toxin molecule that is responsible for its toxic effect. Unlike the **haptophore**, it is destroyed on conversion of the toxin to a **toxoid**.

TP *abbr.* for thymopoietin.

TPA *abbr.* for tissue-type plasminogen activator; *see* **plasminogen activator**.

TPCK *abbr.* for L-(1-tosylamido-2-phenyl)ethylchloromethyl ketone; a modifying reagent for histidine residues.



TPI (*formerly*) *abbr.* for **1** triphosphoinositide; use PtdInsP₂ instead (PIP₂ is also used). **2** phosphatidylinositol 4,5-bisphosphate; use PtdIns-4,5-P₂ instead. *See* **phosphoinositide**.

TPN *abbr.* for triphosphopyridine nucleotide; obsolete, use **NADP**.

TPNH *abbr.* for reduced triphosphopyridine nucleotide; obsolete, use **NADPH**.

TPP *abbr.* for thiamine diphosphate (*formerly* thiamin pyrophosphate); *see* **vitamin B₁**.

TPR *abbr.* for tetracosapeptide repeat.

tp_r-met an oncogene formed by the translocation of the *tp_r* (translocated promoter) locus of chromosome 1 and *MET* (the gene for **hepatocyte growth factor receptor**).

tracee (*in tracer kinetics*) the non-radioactive compound of which the tracer is the radioactive equivalent; coined to represent unambiguously what has been variously called **stable material**, **mother substance**, etc. *Compare* **compartment**.

trace element or **microelement** any element that is present in the body of an organism at relatively minute concentrations (often arbitrarily defined as <0.005%), and consequently is required in the organism's diet in extremely small amounts. In higher animals the most important trace elements include:



trace metal

iron, zinc, copper, manganese, iodine, cobalt, molybdenum, selenium, and chromium. The extended list includes lead, arsenic, antimony, nickel, bromine, fluorine, boron, silicon, aluminium, silver, cadmium, bismuth, mercury, vanadium, titanium, indium, barium, strontium, and lithium. The term may be extended to cover elements of primarily toxicological interest.

trace metal any trace element that is a metal.

tracer an element or compound containing atoms that can be distinguished from their normal counterparts by physical means, e.g. by virtue of their radioactivity, mass spectrum, colour, etc. Tracers are used in biochemistry for following the metabolic pathways of substances in an organism, cell, or cell-free system.

track an individual lane resolving the components of a single sample on a slab electrophoresis gel.

tracking dye a dye that is added to a sample on an electrophoresis gel to be the fastest component and thus to indicate visually how far the samples have run at any instant in time.

tragacanth **1** any of various thorny leguminous shrubs of the genus *Astragalus*. **2** a gum obtained especially from *A. gummifer* and used extensively in commerce, including pharmacology. It is a complex mixture of polysaccharides, including D-galacturonic acid, D-xylose, L-fucose, and D-galactose.

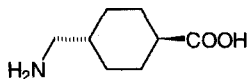
trailer sequence a sequence following the termination signal at the 3' end of messenger RNA.

trait any observable, phenotypic feature of a developing or developed individual.

TRAM or **TRAM protein** *abbr. for* translocating chain associating membrane protein; a transmembrane protein that spans the rough endoplasmic reticulum of mammalian cells and plays a role in the translocation of nascent proteins destined for export from the endoplasmic reticulum. Example from *Canis familiaris*: database code TRAM_CANFA, 373 amino acids (43.03 kDa).

TRAMP *abbr. for* tyrosine-rich acidic matrix protein; *other names*: 22 kDa extracellular matrix protein; dermatopontin; a protein of the extracellular matrix that may have various binding and adhesion functions. Example from *Bos taurus*: database code DERM_BOVIN, 183 amino acids (21.99 kDa).

tranexamic acid *trans*-4-(aminomethyl)cyclohexanecarboxylic acid; a compound that inhibits binding of plasminogen and plasmin to fibrin thereby inhibiting fibrinolysis.



trans+ or (sometimes before *s-*) **tran+** *comb. form* denoting across, beyond, on the other side, transverse; transfer, interchange. See also **trans-acting**.

trans- *abbr.*: *t-*; *prefix (in chemical nomenclature)* denoting a *trans*-isomer; see **cis-trans isomerism**.

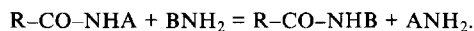
trans-acting describing a regulatory genetic element whose effects are insensitive to its position. Examples include the genes for bacterial repressors. Compare **cis-acting**.

trans-activation gene activation by means of a **trans-acting** mechanism.

transacylase an older name for **acyltransferase**.

transaldolase EC 2.2.1.2; *other names*: dihydroxyacetone transferase, glyceronetransferase; an enzyme of the pentose phosphate pathway that catalyses the transfer of a 3-carbon glycerone moiety from sedoheptulose 7-phosphate to glyceraldehyde 3-phosphate forming erythrose 4-phosphate and fructose 6-phosphate. It has a broad substrate specificity. Example from human: database code TAL1_HUMAN, 336 amino acids (37.59 kDa).

transamidation a chemical reaction in which the amino group of an amide is exchanged for another amino group, i.e.:

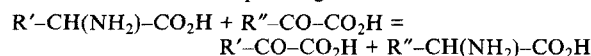


Transamidation is catalysed by such proteolytic enzymes as **trypsin**, **chymotrypsin**, **papain**, and **subtilisin**.

transamidination a chemical reaction, catalysed by a transamidinase (aminotransferase) enzyme of sub-subclass EC 2.1.4, in which the amidino moiety of arginine is reversibly transferred to an amino group to form a different guanidino compound. An example is **glycine amidinotransferase**.

transaminase an alternative name for **aminotransferase**. See also **alanine transaminase**, **aspartate transaminase**, **transamination**.

transamination a chemical reaction, catalysed by any **amino-transferase** (transaminase) enzyme, in which the α -amino group of one amino acid is transferred to the α -carbon atom of an α -oxo acid. As a result, the α -oxo acid is transformed into an amino acid, and the amino acid that donated the amino group is converted to the corresponding α -oxo acid:



Transamination plays an important role in the catabolism of amino acids, in which the amino group is eventually or directly transferred to oxaloacetate (to form aspartate) or to α -oxoglutarate (to form glutamate). If transferred to aspartate, the amino group enters the **urea cycle** at the argininosuccinate step; if transferred to glutamate, the amino group is removed by glutamate dehydrogenase, yielding ammonia, which then enters the urea cycle. Transamination is also involved in amino-acid synthesis. Pyridoxal phosphate is an important coenzyme of aminotransferases. Transamination was discovered by the Russian biochemists Alexander Evseyevich Braunstein (1902–86) and Maria Grigorievna Kritzmann (1904–71).

transbilayer signifying across or through the membrane **lipid bilayer**, as in **transbilayer distribution**, transbilayer movement.

transbilayer distribution the distribution of various components between the two leaflets of a membrane lipid bilayer, but often referring specifically to lipids. In the plasma membrane this distribution is asymmetric, particularly with regard to phospholipids: phosphatidylcholine and sphingomyelin are relatively enriched in the outer leaflet, and phosphatidylethanolamine and phosphatidylserine are more predominant in the inner leaflet (i.e. the cytoplasmic face).

transcarboxylase **1** methylmalonyl-CoA carboxyltransferase; EC 2.1.3.1; *systematic name*: (S)-2-methyl-3-oxopropanoyl-CoA:pyruvate carboxyltransferase; a biotinyl-protein, containing cobalt and zinc, that is present in prokaryotes and catalyses the transfer of a carboxyl group from methylmalonyl-CoA [(S)-2-methyl-3-oxopropanoyl-CoA] to pyruvate forming oxaloacetate and propanoyl-CoA. **2** a component of the prokaryote form of **acetyl-CoA carboxylase** that transfers the carboxyl group from N-carboxybiotin to acetyl-CoA.

transcarboxylation the transfer of a carboxyl group from one compound to another, e.g. as catalysed by a **transcarboxylase**.

transconjugant a bacterial cell that has received genetic material from another bacterium by **conjugation**. A transconjugant should be referred to as a **recombinant** only if the transferred genetic material has been inserted into a pre-existing **replicon** in the recipient. If the transferred material is perpetuated *per se* as a plasmid, then the cell is called a **plasmid transconjugant**. If such material is not a plasmid and does not become a resident replicon, it will fail to replicate; the bacterium will be called an **abortive transconjugant**.

transcortin a specific plasma α_1 -globulin, M_r 52 000, that binds **cortisol** at one binding site and serves to transport the hormone. It also binds corticosterone and certain other steroid hormones; aldosterone is bound only weakly. Transcortin is synthesized in the liver. The normal serum concentration of transcortin is 3–4 mg dL⁻¹; saturation occurs at a cortisol serum concentration of about 28 μ g dL⁻¹, other cortisol being



transcribe

less tightly bound to albumin or (about 5%) unbound. *Other names:* corticosteroid-binding globulin; corticosteroid-binding protein; cortisol-binding globulin (*abbr.:* CBG); cortisol-binding protein.

transcribe 1 to copy. 2 to use DNA as a template for the synthesis of mRNA, or, in some cases (e.g. as in retrovirus infection), to synthesize DNA from an RNA template. *See transcription.*

transcribing strand an alternative name for **noncoding strand** (of duplex DNA).

transcript the product of a **transcription** process; the **primary transcript** is the immediate product of a polymerase engaged in transcription (e.g. RNA polymerase).

transcriptase an alternative name for **DNA-dependent RNA polymerase**.

transcription the synthesis of either RNA on a template of DNA or DNA on a template of RNA. The latter is important in the replication of **retroviruses** by the enzyme **reverse transcriptase** and will not be further described here. Concerning the transcription of double-stranded DNA the older literature is confusing in terms of the use of the word transcription until the discovery of transfer RNA in 1956 when it became clear that RNA was an intermediary in the transfer of information contained in the base sequence of DNA to the polypeptide chain of protein. In the transcription of double-stranded DNA the RNA is formed on a template comprising only one strand, which we will name the transcribed strand. This is also called the antisense strand since the mRNA is termed the sense strand. Similarly the transcribed strand may also be named the noncoding strand, the coding strand having the same base sequence as the mRNA. Transcription involves many steps. (1) The region of DNA to be transcribed must be unwound approximately one turn to form an open loop to allow the transcribed strand to form a DNA-RNA double helix with the 3' end of the newly synthesized RNA. (2) The mRNA is transcribed by **RNA polymerase** utilizing nucleoside triphosphates as substrates. The enzyme binds to the **promoter**, which is a region of DNA on the nontranscribed strand that has a characteristic base sequence (e.g. **TATA box**). Promoters are mostly located upstream of the start of transcription. However, in the case of eukaryotic RNA polymerase III, the promoter may sometimes be located downstream of the start point. There are substantial differences between prokaryotic systems and eukaryotic systems for the regulation of transcription. In prokaryotes, for transcription to proceed, a complex must be formed on the double-stranded DNA helix with the promoter and an accessory factor, **sigma factor**. In eukaryotes, an extensive and complex system of **transcription factors** regulates the process. While prokaryotes appear to utilize only a single RNA polymerase, eukaryotes possess three RNA polymerases, which are responsible for transcribing the various specific types of RNA. Thus RNA polymerase I is located in the nucleolus and is responsible for the synthesis of ribosomal RNA. This polymerase accounts for by far the greatest quantity of RNA synthesized. RNA polymerase II is situated in the **nucleoplasm** and is responsible for the synthesis of heterogeneous nuclear RNA, for **primary transcript** from which messenger RNA is formed. This polymerase is thus instrumental in determining the nature of the protein profile of the cell. RNA polymerase III is also in the nucleoplasm and synthesizes transfer RNA. In so far as eukaryotes are concerned, what follows refers mainly to the activity of RNA polymerase II. (3) As the enzyme advances along the DNA template the DNA unwinds ahead of the growing RNA and rewinds behind it thereby stripping the newly synthesized RNA from the transcribed strand of the DNA. The origin of the energy requirements for this movement is not known at present. Eventually the movement is terminated by specific sequences in the transcribed DNA strand. In some cases **rho protein** plays a part in termination. (4) **Transcription factors**, which may be **cis-acting** or **trans-acting**, control transcription. In prokaryotes the control of transcrip-

tion is induced by the environment and changes are very rapid since transcription and translation are tightly coupled. In eukaryotes such changes are slower. The activity of eukaryotic transcription factors is enhanced by proteins called **enhancers**, which although lacking promoter activity may nevertheless exert their influence over distances of several thousand base pairs. They may be **upstream**, **downstream**, or even in the midst of a transcribed gene. They may also be tissue specific. In the case of eukaryotes where the gene is split by **introns** the entire gene is transcribed and the introns are subsequently excised from the primary transcript by splicing.

transcriptional repressor any **transcription factor** (usually bacterial) that down-regulates gene expression by binding to an **operator**.

transcriptional unit the segment of DNA within which the **transcription** occurs that eventually leads to RNA formation. *Compare* **cistron**, **gene**.

transcription factor *abbr.:* TF; any protein required to initiate or regulate **transcription**; the term includes **enhancers** as well as the general transcription factors. Transcription factors bind to DNA sequence elements upstream of the **start point** and thus participate in the formation of the complex that includes RNA polymerase and initiates transcription. *See, e.g.* **NF-KB**, **Oct**, **TFII**, **TFIII**.

transcytosis the transport of molecules across a cell, especially across a **polarized cell**, such as an intestinal epithelial cell, which has a basolateral and an apical surface each with its own defined composition, which thus provides a spatially orientated transport system. The molecules undergoing transcytosis are usually contained within vesicles. One example is the absorption of maternal antibodies from the gut in newborn animals. The antibodies are taken into epithelial cells by endocytosis at the apical surface, and are transcytosed through the cell before being exocytosed at the basal surface into the blood. Transcytosis also occurs in thyroid cells: following endocytosis of thyroglobulin from the thyroid follicle, the thyroglobulin is transported across the cell in vesicles with concomitant proteolysis, with release of thyroid hormone into a blood capillary at the opposite face of the cell (*see* **thyroid gland**). —**transcytotic** *adj.*; **transcytose** *vb*.

transdeamination a former term for the coupling of a transamidation reaction and the dehydrogenation of L-glutamate whereby an amino group of an amino acid is transferred to α -oxoglutarate by a transaminase enzyme and the resulting L-glutamate is dehydrogenated to yield α -oxoglutarate, ammonia, and a reduced coenzyme molecule.

trans-dominant mutation a mutation in a regulatory gene that can control the expression of a structural gene on another chromosome.

transducer any device that converts a quantity of energy from one form into another.

transducin *abbr.:* G_T or G_i ; a heterotrimeric **G-protein** of the retina, and a component in the visual transduction pathway. In the vertebrate light response, inactive heterotrimeric transducin with bound GDP ($G_{\text{ta}\beta\gamma}$ -GDP) binds the photoactivated rhodopsin, metarhodopsin II, and releases GDP. GTP then binds to the α subunit, which then separates from both the receptor and the $\beta\gamma$ heterodimer. G_{ta} then activates cyclic GMP phosphodiesterase, and the resulting fall in cyclic GMP concentration closes cyclic GMP-gated cation channels and hyperpolarizes the retinal rod cell, generating the nerve impulse. Examples from human, α_2 subunit: database code GBT2_HUMAN, 353 amino acids (40.04 kDa); β_2 subunit: database code GBB2_HUMAN, 340 amino acids (37.33 kDa); γ subunit: database code GBG1_HUMAN, 73 amino acids (8.36 kDa).

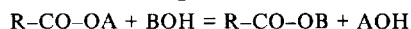
transduction 1 (*in microbiology*) the transfer of genetic information to a bacterium from a bacteriophage or between bacterial or yeast cells mediated by a phage vector. In **generalized transduction** any of the donor genes may be transduced, whereas in **specialized transduction**, i.e. by a **lysogenic bacteriophage**.



transesterification

phage, only those genes at one of the ends of the prophage can be transduced. **Abortive transduction** occurs when phage DNA is not incorporated into the bacterial genome, but can nevertheless be transmitted in the phage to one of the daughter cells. **2 (in biochemistry)** the activation of an intracellular effector pathway as a result of receptor activation by an extracellular signal. *See also* **second messenger, signal transduction.**

transesterification a chemical reaction in which the alcohol moiety of an ester is exchanged for another alcohol:



It is catalysed by such proteolytic enzymes as trypsin, chymotrypsin, papain, and subtilisin, and by esterases.

transfection originally the process of infection of competent bacterial cells by free bacteriophage or plasmid nucleic acid that results in the subsequent production of normal bacteriophage or plasmids in the infected bacterial cell. It has come to mean the process of bringing about genetic alteration of any cell or organism using recombinant DNA technology.

transferase an enzyme that catalyses the transfer of a group – e.g. the methyl group, glycosyl group, acyl group, phosphorus-containing, or other groups – from one compound (generally regarded as the donor) to another compound (generally regarded as the acceptor). Transferases, of which there are many, are EC class 2 enzymes.

transference number or **transport number** *symbol:* *t*; the fraction of the current being carried by a given type of ion when an electric current is passing through an electrolytic solution.

transfer factor a **lymphokine** extractable from leukocytes and thought to transfer the ability to elicit a **delayed hypersensitivity** reaction from one person or animal to another.

transfer gene any gene carried by a **conjugative plasmid** that is responsible for the donor phenotype.

transfer membrane a membrane used with **blotting** techniques in recombinant DNA technology or protein studies. Molecules are transferred from an electrophoretic gel to the transfer membrane either by capillary attraction or by using an electric potential (**electroelution**). Transfer membranes may be of three types: (1) nitrocellulose – mixed esters of cellulose acetate and cellulose nitrate; high cellulose nitrate improves binding of biomolecules while a low percentage of cellulose acetate improves handling properties; nucleic acids can be fixed to nitrocellulose membranes by baking; (2) nylon, which is physically more robust and reusable, and to which nucleic acids can be fixed by exposure to ultraviolet light; or (3) polyvinylidene-based membranes.

transfer potential *another name for* **group transfer potential**.

transfer reaction *another name for* **group transfer reaction**.

transferrin any of a class of single-chain (76–81 kDa), metal-binding glycoproteins, widely distributed in physiological fluids and cells of vertebrates, with the characteristic property of a stringent association of their metal-combining properties with an anion-binding requirement; iron-binding transport proteins, that bind two Fe^{3+} in association with bicarbonate (or other anion). They are monomers with internal homology. An essential component in cell culture media, transporting iron into cells through the **transferrin receptor**. The transferrin family includes proteins such as **lactoferrin**, **melanotransferrin** (with electrophoretic polymorphisms), **ovotransferrin** (*see* **conalbumin**), and **serotransferrin**. Examples: serotransferrin (*other names:* siderophilin; β -1-metal binding globulin) from human (precursors): database code TRFE_HUMAN, 698 amino acids (76.96 kDa); transferrin precursor from *Xenopus laevis*: database code TRFE_XENLA, 717 amino acids (79.53 kDa).

transferrinjection a method of gene transfer in which DNA complexed to transferrin–polycation conjugates is introduced into cells by receptor-mediated endocytosis.

transferrin receptor a transmembrane protein involved in the cellular uptake of iron, bound to transferrin, which binds to the receptor and is taken into the cell via receptor-mediated

endocytosis. Structurally it is a dimer (a soluble serum form is a monomer lacking amino acids 1–100), the extracellular domains of the two chains being linked by a disulfide bridge, and carrying the glycan moieties. The intracellular domain is phosphorylated and palmitoylated, i.e. acylated with a palmitoyl group by a process analogous to **myristoylation**. Synthesis of the receptor is regulated by binding of aconitase to mRNA when intracellular iron concentrations are low, stimulating receptor synthesis, possibly by stabilizing the mRNA. Example from human: database code TRSR_HUMAN, 760 amino acids (84.81 kDa).

transfer RNA *abbr.:* tRNA; any of a class of relatively small RNA molecules each of which mediates the insertion of an amino acid at the correct point in the sequence of a nascent polypeptide during protein synthesis. Crick predicted that the amino acids would not directly interact with the triplet nucleotide bases comprising the codons in mRNA but that interaction would involve a small nucleic acid to which the 20 different amino acids would be attached. The discovery of such an RNA molecule in the soluble cell extract required for the synthesis of protein in an *in vitro* system derived from rat liver by Hoagland was greeted as confirmation of Crick's hypothesis. Since the RNA was present in the soluble (non-particulate) extract it was called at first **soluble RNA** or **S-RNA**. With 73–93 nucleotides, tRNA ($M_r \approx 25\,000$) was rather larger than had been expected for adaptor RNA (*see* **adaptor hypothesis**). It was soon shown that the amino-acid residue was attached by ester link to the 3' terminal of tRNA at either the 2'- or 3'-OH on the ribose. Specific tRNA molecules are found for the various amino acids, but in all cases the amino acid is attached to the adenosine of a CCA sequence at the 3' end of the tRNA molecule. The amino acids are attached to tRNA by specific amino acid-tRNA ligases, which first catalyse a reaction between the amino acid and ATP to form an aminoacyl-adenylate, and then transfer the aminoacyl group to the tRNA. Each tRNA possesses an **anticodon** of three nucleotides that base pairs in an antiparallel fashion with the codons of mRNA. While there are 61 sense codons there are only about 41 different tRNAs in the cytoplasm and even fewer in mitochondria due to flexibility in the base pairing between codon and anticodon (*see* **wobble hypothesis**). When the sequence of tRNA for phenylalanine was determined by Holley it could be shown that the secondary structure contained three loops reminiscent of a clover-leaf if the maximum amount of base pairing was allowed for. This involved about half the residues. The anticodon was predicted to be on a loop lacking base pairing. This was shown to be true, it being located on the middle of the three loops. The few tertiary structures that have been determined for tRNAs have supported the clover-leaf structure. It has been predicted that **initiator transfer RNA** would have a structure that differentiated it from the other tRNAs but this has not been firmly established. tRNA is characterized by the presence of many unusual **minor bases** the function of which has not been completely established. *Other names:* acceptor RNA; amino-acid-accepting RNA; (*formerly*) soluble RNA (*abbr.:* S-RNA or sRNA). *See also* **suppressor tRNA**.

transform **1** to bring about **transformation**. **2 (in mathematics)** **a** a process for deriving one mathematical entity from another. **b** the transformed entity itself.

transformant a bacterial cell that has undergone **transformation** (*def. 1*), i.e. one that contains integrated donor genes that can be detected by plating on media selective for some or all of the donor genes.

transformation **1** or **genetic transformation** the intraspecific and interspecific (between members of different but related species) transfer of genetic information by means of 'naked' extracellular DNA in bacteria. In transformation, in contrast to **transduction**, the DNA not integrated into the recipient's genome cannot function and manifest its genetic information phenotypically. **2** characteristic and inherited changes produced in cells in culture when they have been treated with cer-



tain viruses (which may be DNA- or RNA-containing), or with chemical carcinogens, or with X-rays, or when they have been subjected to genetic modification by DNA technology. In animal cells, these changes include certain specific alterations in morphology (especially of the nucleus) and stainability, and the cells usually show a loss of **contact inhibition** and have a capacity to give rise to neoplastic growth on injection into animals. The term is usually extended to refer to corresponding changes occurring in cells *in vivo*. **3 or blast transformation or lymphocyte transformation** the morphological and other changes that occur in both B and T lymphocytes when they are cultured in the presence of an antigen to which they are primed, or that occur nonspecifically in B lymphocytes when they are exposed to a mitogen (e.g. certain lectins). The lymphocytes increase in size and develop a basophilic ribosome-rich cytoplasm, a prominent nucleolus, and a paler-staining nucleus. After about three days these cells resemble **blast cells**. **4** the alteration produced in certain specific hormone-receptor proteins, present in the extranuclear region of hormone-responsive cells, when the hormone binds to them. This alteration is accompanied by the migration of the hormone-receptor complexes to the nucleus. **5 or nuclear transformation** (of an atomic nucleus) the change of one **nucleide** to another.

transforming growth factor *abbr.*: TGF; either of two types of mitogenic **cytokine**, TGF α and TGF β . TGF α was isolated and purified on the basis of its ability to stimulate cell growth in soft agar. It is produced by macrophages, brain cells, and keratinocytes, and is mitogenic for fibroblasts, is angiogenic *in vivo*, and induces epithelial development. It is a glycoprotein and a member of the **epidermal growth factor** (EGF) family, interacting with the EGF receptor. TGF α comprises 50 amino acids with the following sequence (residues invariant in the EGF family are underlined): VVSFHNDQPSHTQFCFHGTCRFLVQEDKPA $\underline{C}$ V $\underline{C}$ HSGYV $\underline{G}$ ARCEHADLLA. Example from human (precursor): database code TGFA_HUMAN, 160 amino acids (16.99 kDa). TGF β is a multifunctional peptide that controls proliferation, differentiation, and other functions in many cell types. It is produced by fibroblasts, platelets, monocytes, chondrocytes, and osteoblasts. A glycoprotein with EGF repeats, at least five types (TGF β 1 to TGF β 5) are known; all have 112 amino acids except TGF β 5, which has 114. They have a great many biological activities and can be mitogenic or antiproliferative depending on cell type. Platelets contain large amounts, but these proteins are also found in a variety of cell types. Example, TGF β 1 precursor (human): database code TGFA_HUMAN, 1394 amino acids (152.62 kDa); alternative splicing results in different forms in platelets and fibroblasts.

transforming growth factor receptor any of the membrane proteins that bind transforming growth factor β (TGF β) and mediate its effects. Three types of receptor for TGF β have been distinguished: types I and II have high affinity for TGF β 1, with lower affinity for TGF β 2; type III has high affinity for TGF β 1, 2, and 3. Types I and II have a single transmembrane domain with a cytoplasmic protein serine/threonine domain. The following examples are all from human. Type I receptor: database code KIR4_HUMAN, 503 amino acids (55.96 kDa); type II receptor: database code TGR2_HUMAN, 567 amino acids (64.54 kDa); type III receptor (has a single transmembrane domain; little cytoplasmic structure): database code TGR3_HUMAN, 849 amino acids (93.38 kDa); residues 1–16 are the signal; 17–781 extracellular domain (heavily glycosylated, 808–849 intracellular. It may present TGF β to the signalling receptors.

transforming principle a historical name for the purified DNA that, when incorporated into a bacterial cell, brings about the **transformation** (def. 1) of that cell. The term was used, most famously, by Oswald Theodore Avery (1877–1955, US bacteriologist and immunologist) and colleagues in their work establishing that DNA, not protein, was responsible for genetic transformation. In a classic experiment, they used two

strains of *Pneumococcus*. The first was a wild strain, known as S (for smooth) strain because of the kind of colony that it forms, that is pathogenic. The other was a mutant strain, known as R (for rough) strain, that is non-pathogenic. When mice were injected with live R strain or heat-killed S strain, they remained healthy, but when injected with a combination of live R strain and heat-killed S strain they died. These workers concluded that the live R strain could be transformed into a pathogenic strain by material from the heat-killed S strain, and were able to demonstrate that the material was DNA.

transfuse to transfer blood from one animal into the circulation of another animal, or to inject liquid into the circulation of an animal to maintain the circulating fluid volume. —**transfusion** *n*.

transgene a gene that is inserted, using cDNA technology, into the germ line in a manner that ensures its function, replication, and transmission as a normal gene. *See also transgenesis, transgenic.* —**transgenic** *adj*.

transgenesis the creation in plants or animals of a stably incorporated gene or genes derived from another cell or organism which can be passed on to successive generations.

transgenic describing an organism harbouring in its germ line a gene that has been introduced using cDNA technology (*see complementary DNA*). In animals, the less transgene is introduced during transient culturing of an egg or early embryo, which is then implanted into a foster mother. Introduction of the transgene may involve any of several distinct techniques: cells may be transferred from one embryo to another (to produce a **chimera**); **pluripotent** cells can be introduced into an embryo; cells infected with a retrovirus can be introduced; or cDNA can be introduced by microinjection into the pronucleus of a fertilized egg. In plants, plasmids from *Agrobacterium tumefaciens* have been widely used to produce transgenic varieties. The **Ti plasmid** is a natural vector for genetically engineering plant cells, because it can transfer its **T-DNA** to the plant genome. A number of vectors have been constructed based on this system. Expression of the introduced genes can be controlled by suitable promoters, such as the cauliflower mosaic virus promoter.

transgenomes large complexes of DNA found inside recipient cells following transfection of the latter with calcium phosphate-precipitated DNA.

transgenosis the overall phenomenon of transfer of genetic information between unrelated organisms and the subsequent expression of such information (including gene maintenance, transcription, translation, and function), especially when donor and recipient cells are widely separated by evolution (e.g. bacterial and eukaryotic cells respectively) and when the mechanisms of gene transfer and maintenance are obscure. *Compare transgenesis.*

transglycosylation a chemical (enzymic) reaction in which one glycosyl residue is transferred from a glycoside onto a receptor molecule so as to form a new glycosidic linkage, e.g. in the biosynthesis of polysaccharides. The process therefore involves breaking simple glycosidic linkages and reforming them into different types of glycosidic linkages. **Glycosyltransferases** are in subclass EC 2.4; also included in the term are glycosyl transfers to water and to inorganic phosphate.

transient 1 temporary, transitory, short-lived. 2 a sudden, short-lived disturbance in a system. 3 a short-lived chemical species such as an excited atom or molecule or a free radical.

transient-phase kinetics an alternative name for **pre-steady-state kinetics**.

trans isomer *see cis-trans isomer.*

transition (*in molecular biology*) a particular type of mutation by single base-pair substitution that occurs by replacement of one purine by another purine, or one pyrimidine by another pyrimidine whether by chemical change or by substitution. *Compare transversion.*

transition element or transition metal any element whose atom has an incomplete d-subshell of extranuclear electrons.



transition state

or which gives rise to a cation or cations with an incomplete d-subshell. Such elements fall into groups 3 to 11 inclusive of the IUPAC **periodic table**. (Elements of group 12, which formerly were also considered as transition elements, have complete d-subshells and are now excluded.) The first series of transition elements is in period 4 of the periodic table and comprises those of proton numbers 21 to 29 inclusive, i.e. the elements scandium, titanium, vanadium, chromium, manganese, iron, cobalt, nickel, and copper; the second series is in period 5 and comprises those of proton numbers 39 to 47 inclusive, i.e. the elements yttrium, zirconium, niobium, molybdenum, technetium, ruthenium, rhodium, palladium, and silver; the third group is in period 6 and comprises those of proton numbers 57 to 79 inclusive, i.e. the **lanthanoids** together with the elements hafnium through to gold; and the fourth series is in period 7 and contains those of proton numbers 89 to 103 inclusive, i.e. the **actinoids**, and may well be found also to contain a number of elements of proton number 104 or more that yet remain to be characterized or discovered. The lanthanoids and actinoids are designated **inner transition elements** of periods 6 and 7 respectively; their atoms also have incomplete inner f-subshells of electrons. All transition elements are metals, they often have more than one valency state, and they form many coordination complexes, which frequently are coloured.

transition state or activated complex (in enzymology) the state of most positive molar **Gibbs energy** through which an assembly of atoms must pass on going from the reactants of a single-step chemical reaction to the products. This is the state that occurs immediately after binding of the reactants to the enzyme's active site; it is during this time that transient bonds between reactants and enzyme are first formed, and initial electronic interactions occur. It is followed by a state in which more stable intermediates form, prior to the eventual formation of products.

transition-state theory the theory that relates the rate of an enzymic reaction to the difference in **Gibbs energy** between the **transition state** and **ground state** ($\Delta G^\ddagger$). If x is the ground state and $x^\ddagger$ the transition state, then the rate of decomposition of x is given by:

$$-d[x]/dt = \nu[x^\ddagger] = [x](kT/h)\exp(-\Delta G^\ddagger/RT),$$

where ν is the vibrational frequency of the bond that is breaking, h is the Planck constant, k the Boltzmann constant, R the gas constant, and T the thermodynamic temperature.

transition temperature *symbol:* T_i ; (of a membrane lipid) the surface temperature at which a heated lipid bilayer undergoes transition from one physical state (e.g. liquid crystalline) to another (liquid). *See also* **pretransition temperature**, **phase transition (of lipids)**.

transit peptide any ≈ 4 kDa peptide sequence at the N or C terminus (or both) of cytoplasmically synthesized precursor proteins of certain chloroplast proteins that contains the information to ensure both the specific post-translational transport of the protein into the chloroplast and its localization within the organelle. The transit peptide is removed within the chloroplast.

transit time the time taken by a coenzyme or metabolite in travelling between two enzymes. If the two enzyme reactions were infinitely fast, the overall velocity would be determined by the reciprocal of the transit time.

transketolase EC 2.2.1.1; *systematic name:* sedoheptulose-7-phosphate:D-glyceraldehyde-3-phosphate glycolaldehyde-transferase; *other name:* glycolaldehyde transferase. An enzyme that catalyses the transfer of a glycolaldehyde ($\text{HOH}_2\text{C}-\text{CHO}-$) moiety from xylulose 5-phosphate to ribose 5-phosphate to form glyceraldehyde 3-phosphate and sedoheptulose 7-phosphate. A component of the pentose phosphate pathway, it requires thiamine diphosphate (TPP) as coenzyme; a glycolaldehyde-TPP intermediate is formed transiently during the reaction. It has broad substrate specificity.

In clinical chemistry, the measurement of transketolase activity in a red cell hemolysate, in the presence and absence of thiamin, gives an indication of the thiamin status of the individual by revealing the degree of activation of the enzyme in the absence of thiamin, compared with the activity in the presence of optimal concentrations. Example from human (homodimer): database code TKT_HUMAN, 623 amino acids (67.72 kDa).

translation 1 (in molecular biology) the process by which a particular sequence of bases in messenger RNA (mRNA) determines a sequence of amino acids in the polypeptide chain being assembled during **protein and peptide biosynthesis**. One or more specific base triplets (codons) code for each of the 20 amino acids that may be incorporated into the polypeptide chain (*see* **genetic code**). Each codon of the mRNA molecule is recognized by the corresponding anticodon on the **transfer RNA** to which the amino acid is attached. The transfer RNA binds to a site on the large subunit of the **ribosome**, which is associated with the small subunit to which the mRNA is bound. The first codon (the initiation codon) in each mRNA molecule is AUG, which codes for formylmethionine (in prokaryotes and mitochondria) or methionine (in eukaryotes). The methionine or formylmethionine, or just the formyl group, is removed before the chain is completed. Completion of translation is signalled by one of three termination codons. **Release factors** effect the release of the complete polypeptide chain and the dissociation of the two subunits of the ribosome. 2 movement or displacement laterally in space without rotation or change of orientation.

translational frictional coefficient *see* **frictional coefficient**.

translocase a system catalysing translocation of a substrate across an osmotic barrier. *See* **EF-G**, **translocator**.

translocation a vectorial transfer process, e.g. of a solute across a membrane. —**translocation adj.**

translocator or porter 1 a system catalysing a secondary translocation reaction, i.e. one not involving primary bond exchanges between different pairs of chemical groups or the donation and acceptance of electrons. 2 any molecular group that is attached to a substrate during transport of the latter across a membrane.

translocon a translocation complex that constitutes the specific site of protein translocation across the endoplasmic reticulum. It involves the **signal recognition particle receptor**. The component proteins are known as **translocon-associated proteins** (*abbr.:* TRAP). They may be involved in ensuring retention in the endoplasmic reticulum of resident proteins. TRAP proteins form a heterotetramer of α , β , γ , and δ subunits. Examples, α from *Arabidopsis thaliana*: database code SSRA_ARATH, 253 amino acids (27.5 kDa); β from dog: database code SSRB_CANFA, 183 amino acids (20.1 kDa); γ from rat: database code SSRG_RAT, 185 amino acids (21.11 kDa); δ from rat: database code SSRD_RAT, 173 amino acids (18.98 kDa).

translucent transmitting light (or other radiation) with some scattering or diffusion. Consequently, objects cannot be clearly distinguished through translucent materials. *Compare* **transparent**. —**translucence or translucency n.**

transmembrane protein any protein that traverses the membrane, especially the plasma membrane. For the different types *see* **membrane protein**.

transmethylation the transfer of a methyl or methylene group, as frequently occurs in **one-carbon metabolism**. Methyltransferases are in sub-subclass EC 2.1.1, and enzymes concerned with one-carbon metabolism are in sub-subclass EC 2.1.2. *See also* **C₁ compound**, **folate coenzymes**, **S-adenosylmethionine**.

transmission 1 the act, process, or an instance of transmitting. 2 the proportion of radiant energy that is transmitted perpendicularly through a substance or solution; *compare* **transmittance**. 3 the delivery of a nerve stimulus across a **synapse**, e.g. by a neurotransmitter.

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transmission coefficient

transmission coefficient (*in nuclear physics*) the probability that a particle striking a nucleus penetrates it.

transmission electron microscope *abbr.*: TEM; *see* electron microscope.

transmission factor *an alternative name for transmittance.*

transmittance or **transmission factor** **1** (*in chemistry*) *symbol*: T ; the ratio I/I_0 , where I_0 is the intensity of electromagnetic radiation falling on a body or substance and I the intensity after transmission through it. It is a dimensionless physical quantity. **2** (*in physics*) *symbol*: τ ; the fraction of radiant energy that, having entered an absorbing substance, reaches its far boundary. It is given by

$$\tau = \Phi_{tr}/\Phi_0 = \exp(-\kappa cl),$$

where Φ_{tr} and Φ_0 are the radiant powers of the transmitted and incident radiation, κ is the molar napierian absorption coefficient, c the speed of light in the medium, and l the path length.

transmitter-gated channel *see* ion channel.

transparent transmitting light with little scattering or diffusion so that objects beyond the transparent material can be clearly distinguished. The term can be used in appropriate circumstances to indicate materials allowing the passage of other electromagnetic, particulate, or sonic radiation without disturbance. *Compare* translucent. —**transparency** *n*.

transpeptidase **1** (*sometimes*) *an alternative term for peptidyl-transferase.* **2** (*sometimes*) the enzymic activity of a number of proteinases that catalyse, apart from hydrolysis of peptide links, the replacement of one terminal amino-acid residue in a peptide by another amino acid or other suitable molecule. **3** (*sometimes*) certain aminoacyltransferase enzymes, e.g. D-glutamyl transpeptidase, D-glutamyltransferase (EC 2.3.2.1), and glutamyl transpeptidase (i.e. **y-glutamyltransferase**, EC 2.3.2.2).

transpeptidation any of the reactions catalysed by **transpeptidases**.

transphosphoribosidase *see* adenine phosphoribosyltransferase.

transplantation antigen *an alternative term for histocompatibility antigen.*

transport the movement of a material from one place to another, especially in reference to the movement of substances around the body (e.g. in blood) or across a biological membrane, or of electrons along a series of carriers. This term implies a positive agency, in contradistinction to **passive diffusion**. *See also* active transport, antiport, facilitated diffusion, symport, transport protein.

transporter a membrane protein catalysing the passage of molecules from one face of a membrane to the other.

transport method *see* transport process.

transport number *an alternative term for transference number.*

transport piece *see* secretory piece.

transport process any process, such as sedimentation, diffusion, or electrophoresis, that is irreversible in the thermodynamic sense. The system in which such a process occurs is removed from a state of equilibrium. Such processes are of particular value in providing information about the dimensions, shapes, and masses of macromolecules.

transport protein a protein (or enzyme) that is instrumental in transporting material, often in a specific manner, across a biological membrane or within a biological fluid (e.g. blood).

transposable element *another name for transposon.*

transposase an enzyme that is responsible for the transposition of **transposable elements** or **transposons**. Transposases are involved in site-specific DNA recombination required for transposition in bacteria and other organisms. Example, Tn3 (*Escherichia coli* and *Klebsiella pneumoniae* transposon) transposase: database code TRA1_ECOLI, 1015 amino acids (114.40 kDa).

transpose **1** to alter the position of; to interchange or place in a different order. **2** (*in mathematics*) **a** to interchange the rows

and columns of a **matrix** (def. 8). **b** the matrix resulting from interchanging the rows and columns of a given such matrix.

transposition **1** the process of transposing or the state of being transposed. **2** something that has been transposed; especially a segment of a chromosome or a piece of a DNA molecule.

transposition sequence *an alternative name for translocation sequence.*

transposon or **transposable element** a specific DNA sequence that is transferred as a unit from one **replicon** to another. Transposons were first identified as a result of work on **transposable elements** and also as spontaneous insertions in bacterial operons. The simplest transposons are referred to as **insertion sequences**. These have the designation IS followed by a number related to the sequence in which they were identified, preceded by an indication of the site of insertion followed by a double colon (thus $\lambda::IS1$ indicates an IS1 element inserted into phage lambda). Each end of the transposon is characterized by repeated sequences, due to short sequences copied from the original gene called **target repeats** or **direct repeats**, and also due to short inverted repeats that are a property of the insert. More complex transposons, called **composite transposons**, may have a central portion having a variety of markers, the central portion being flanked by IS elements that are responsible for identifying the transposon for transposition. Composite transposons have the designation Tn followed by a number. In eukaryotes, transposable elements constitute much of the **repetitive sequences** in the genome, with up to a few hundred repeat copies. They are flanked by terminal inverted sequences; the terminal sequences in each strand characteristically represent the complementary bases of the inverted sequence, i.e. TCAG.....CGTA. Some transposons move physically from one site to another, while others are replicated, with one copy remaining at the original site and a duplicate inserting elsewhere. Both mechanisms involve a **transposase** enzyme; the former requires a **resolvase** enzyme in addition.

transthyretin a thyroid-binding protein occurring in the bloodstream and having an unusual structure. The molecule is a homotetramer; each monomer comprises two four-stranded beta sheets and has the shape of a prolate ellipsoid. Two of the monomers dimerize and the dimers themselves dimerize to form a structure with an internal channel. Example (precursor) from human: database code TTHY_HUMAN, 147 amino acids (15.87 kDa).

transudate any fluid, with its solutes, that has passed through a membrane or through the interstices of a system.

transude to pass or be passed through a membrane or the interstices of a system as a **transudate**. —**transudation** *n*.

transversion the particular type of mutation by single base-pair substitution that occurs by the insertion of a purine in a duplex nucleic acid, in place of a pyrimidine, or vice versa.

Compare transition.

TRAP *abbr.* for translocon-associated protein; *see* translocon.

trap hypothesis **1** a proposed mechanism for the binding of proteinases by α_2 -macroglobulin (α_2 M). It is suggested that, when α_2 M interacts with a proteinase, hydrolysis of one or more of its susceptible peptide bonds in its **bait region** occurs, whereupon the α_2 M molecule undergoes a conformational change that physically entraps the proteinase molecule, thus sterically hindering reaction between the proteinase and other large substrate molecules. **2** *see* methyltetrahydrofolate trap hypothesis.

Trasylol *a proprietary name for aprotinin.*

TRE *abbr.* for **1** TPA response element, a response element for phorbol esters. **2** thyroid response element. *See* response element.

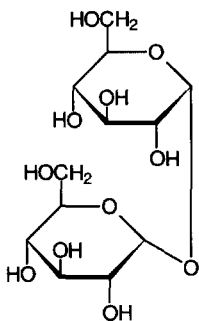
trefoil motif *an alternative name for P domain.*

trehalose α -D-glucopyranosyl- α -D-glucopyranoside; α,α -trehalose; a nonreducing disaccharide, *also known as mushroom sugar or mycose*, that acts as a reserve carbohydrate in certain fungi (especially yeasts), algae, and lichens. It is cleaved by trehalase but not by most other α -glucosidases. It



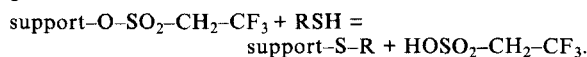
tremerogen

is a component of **cord factors**, which are a mixture of trehalose 6,6'-mycolates (see **mycolic acid**). α -D-Glucopyranosyl- β -D-glucopyranoside (α,β -trehalose) and β -D-glucopyranosyl- β -D-glucopyranoside (β,β -trehalose) occur naturally, but only very rarely.



tremerogen any of a group of oligopeptides that contain non-amino-acid components and are the sex hormones of the jelly fungi. They induce formation of a conjugation tube in compatible cell types. They may contain a farnesyl group bound to cysteine.

Tresyl *proprietary name* indicating the 2,2,2-trifluoroethane sulfonyl group; Tresyl-activated Sepharose can be used in the preparation of affinity gels, the sulfonyl groups being readily displaceable by amines, thiols, phenolic groups, or imidazole, which remain bound to the gel as the stationary component; e.g.



tretinoin a WHO-approved nonproprietary name for **retinoic acid**, especially for pharmacological use.

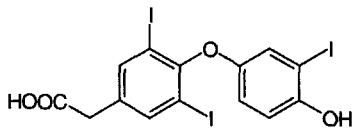
TRF *abbr. for* 1 thyrotropin-releasing factor; i.e. **thyrotropin-releasing hormone**. 2 T-cell-replacing factor; see **interleukin 5**.

TRF-III or **TRF-m** see **interleukin 1**.

TRH *abbr. for* thyrotropin-releasing hormone.

tri+ *comb. form* 1 denoting three, threefold, thrice, every third; see also **tris+** 2 (in chemical nomenclature) (distinguish from **tris+** (def. 2)) a indicating the presence in a molecular entity of three identical specified unsubstituted groups, e.g. trichloroethylene, trimethylamine. b indicating the presence in a molecular entity of three identical oxoacid residues in linear anhydride linkage; e.g. adenosine 5'-triphosphate.

triac (*informal*) *abbr. for* 3,3',5-triiodothyroacetic acid, a catabolite of **thyroxine**.



triaconta+ or (*before a vowel*) **triacont+** *comb. form* denoting thirty or thirty times.

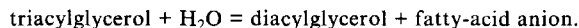
triacontanoic acid melissic acid; a 30-carbon straight-chain aliphatic acid; m.p. 93.6 °C (methyl ester 71.5 °C).

triacontapeptide a 30-residue peptide.

triacylglycerol or (*formerly*) **triglyceride** any triester of glycerol; the three fatty acids may all be the same, or differ in any permutation. Triacylglycerols are important components of plant oils, animal fats, and animal plasma lipoproteins. The degree of unsaturation of the component fatty acids determines the melting point: plant oils, which typically contain a

high percentage of (poly)unsaturated fatty acids, are liquid, while animal fats, being more highly saturated, are solid or semisolid. See also **lipoprotein** and individual plant oils and animal fats.

triacylglycerol lipase EC 3.1.1.3; *systematic name*: triacylglycerol acylhydrolase; *other names*: lipase; triglyceride lipase; tributyrase. An enzyme that catalyses the reaction:



This is a very widespread enzyme activity, with types ranging from those found in microorganisms to those such as **pancreatic lipase**, and the hormone-sensitive lipase of mammalian adipose tissue. Example from *Pseudomonas aeruginosa* (precursor): database code LIP_PSEAE, 311 amino acids (32.69 kDa). Example from *Geotrichum candidum* (yeast): database code NRL_1THG, 544 amino acids (59.62 kDa); 3-D structure known; the latter is a globulin with mainly alpha helix and turns, and contains one modified glutamine residue (pyrrolidone carboxylic acid).

triadin a major membrane glycoprotein of the sarcoplasmic reticulum triad junction of skeletal and cardiac muscle. The luminal domain has a highly conserved region and can bind both the ryanodine receptor and calsequestrin. Example from rabbit: database code A45990, 706 amino acids (79.04 kDa).

+triaeose *suffix* used in nomenclature of glycolipids to indicate a molecule containing an oligosaccharide moiety composed of three sugar residues (which may be the same or different), as in ganglioside. Distinguish from **triose**.

triazine-dye affinity chromatography (*sometimes*) a type of **dye-ligand chromatography** in which the coupled dye is a triazinyl compound. The technique is not **affinity chromatography** in the original sense.

tricaprin a *trivial name for* **tridecanoin**, the triacylglycerol containing three caproyl (decanoyl) residues. See also **tricaproin**, **tricaprylin**.

tricaproin a *trivial name for* **trihexanoin**, the triacylglycerol containing three capric (hexanoic) ester residues. See also **tricaprin**, **tricaprylin**.

tricaprylin a *trivial name for* **trioctanoin**, the triacylglycerol containing three caprylic (octanoic) ester residues. See also **tricaprin**, **tricaproin**.

tricarboxylic acid *abbr.*: TCA; any organic compound having three carboxylic-acid groups. The name often has the connotation of relationship with **citric acid**, as in the **tricarboxylic-acid cycle**.

tricarboxylic-acid cycle (*abbr.*: TCA cycle) or **citric-acid cycle** or **Krebs cycle** a nearly universal metabolic pathway in which the acetyl group of acetyl coenzyme A is effectively oxidized to two CO₂ and four pairs of electrons are transferred to coenzymes. The acetyl group combines with oxaloacetate to form citrate, which undergoes successive transformations to isocitrate, 2-oxoglutarate, succinyl-CoA, succinate, fumarate, malate, and oxaloacetate again, thus completing the cycle. In eukaryotes the tricarboxylic acid is confined to the mitochondria. See also **glyoxylate cycle**.

trichloroacetic acid *abbr.*: TCA; Cl₃CCOOH; an organic acid widely used for the precipitation of protein from tissue and other cell extracts. A final concentration of 3–9% (w/w) is normally used. TCA can be extracted into diethyl ether.

trichloroethyl alcohol 2,2,2-trichloroethanol; an active metabolite of chloral hydrate with CNS depressant actions.

trichohyalin a structural protein produced and retained in cells of the inner root sheath and medulla of the hair follicle. In the sheep, 75% of its amino acids are glutamate or glutamine, arginine, or lysine; the gene is also expressed in epithelia, hoof, and rumen. Trichohyalin associates in regular arrays with keratin intermediate filaments. Example from human: database code TRHY_HUMAN, 1898 amino acids (246.95 kDa).

tricine or **TRICINE** *trivial name for* N-tris(hydroxy-

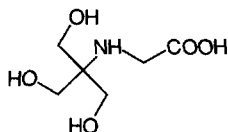
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tricos+

methyl)methyl]glycine; *N*-[2-hydroxy-1,1-bis(hydroxymethyl)-ethyl]glycine; a **Good buffer substance**; pK (20 °C) = 8.15.



tricos+ or (before a vowel) **tricos+** *comb. form* denoting twenty-three or twenty-three times.

tricosapeptide a 23-residue peptide.

trideca+ or (before a vowel) **tridec+** denoting thirteen or thirteen times.

tridentate 1 having three teeth or toothlike projections. **2** (of a **ligand**) chelating a metal ion by means of three donor atoms.

triene any alkatriene or substituted alkatriene.

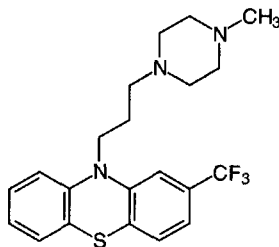
+triene(+) or (before a vowel) **+triene** (in *chemical nomenclature*) *infix or suffix* in systematic names denoting the presence in a molecular structure of an unsaturated aliphatic hydrocarbon chain containing three (conjugated or unconjugated) double bonds. The position of each double bond may be indicated by a locant for its lowest numbered carbon atom; e.g. cycloheptatriene, icoso-8,11,14-trienoic acid.

trienoic indicating a compound having a carboxylic-acid group and an aliphatic chain with three double bonds.

trienoyl indicating the acyl group of a trienoic acid.

triflavin a **disintegrin** from the snake *Trimeresurus flavoviridis*, containing an Arg-Gly-Asp (**RGD**) motif at residues 49–51, that inhibits fibrinogen interaction with platelet receptors, and thereby inhibits platelet aggregation. Database code DIST_TRIFL, 70 amino acids (7.57 kDa).

trifluoperazine 10-[3-(4-methyl-1-piperazinyl)propyl]-2-(trifluoromethyl)-10*H*-phenothiazine; an antagonist with high affinity for **D₂ dopamine receptors**. It is used in the treatment of manic and schizophrenic psychoses and as an inhibitor of **calmodulin** action. *One proprietary name*: Stelazine (dihydrochloride).



trifluoroacetic acid *abbr.*: THA; CF₃COOH; a strong organic acid. It is widely used as a catalyst in organic synthesis, and is used to cleave Cbz- and tBoc- groups from protected amino acids. The anhydride is used to prepare trifluoroacetyl derivatives for the protection of alcohols and amines, and as volatile derivatives for gas-liquid chromatography.

trigger 1 to induce (often rapidly) into activity. **2** a mechanism or agent that brings about such an action. The term is commonly used in connection with membrane receptor activation and signal transduction, as in trigger mechanism.

triglyceride a former name for **triacylglycerol**; the term is still quite commonly used as in, e.g., 'triglyceride fraction', but use is discouraged for such purposes since triglyceride is properly a generic term for compounds having three glyceryl residues (e.g. **cardiolipin**).

trigonelline nicotinic acid *N*-methylbetaine; 3-carboxy-1-methylpyridinium inner salt; a metabolite of nicotinic acid. It is excreted in the urine after the consumption of relatively large amounts of nicotinic acid.

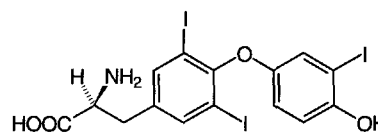
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trihydric (of a chemical compound) containing three hydroxyl groups per molecule; trihydroxy; used especially of alcohols.

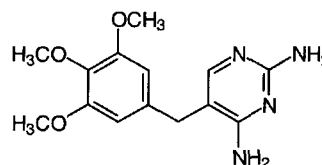
triiodothyronine *abbr.*: T₃; 3,3',5-triiodo-L-thyronine; one of the hormones secreted by the thyroid gland. Plasma levels of T₃ are much lower than those of **thyroxine** (T₄) (20- to 50-fold), but the physiological activity of T₃ is higher. The mechanism of action of thyroid hormones involves T₃ binding to a **thyroid-hormone receptor**, which is then activated to bind to **thyroid response element**, a transcription-regulatory element. Much of T₄ is converted to T₃ within cells, T₃ being the compound that binds to the thyroid-hormone receptor in target cells. *See also* **thyroglobulin**.



trilaurin a triacylglycerol in which all the fatty-acyl moieties are **lauroyl** (dodecanoyl).

trimer a molecular complex having three components or subunits. If these are identical, it is referred to as a **homotrimer**; if any one is different from any other, it is called a **heterotrimer**. —**trimeric** *adj.*

trimethoprim 2,4-diamino-5-(3,4,5-trimethoxybenzyl)pyrimidine; a specific inhibitor of formylation and a competitive inhibitor of **dihydrofolate reductase**. The trimethoprim resistance gene can be used as a selectable marker for cloning vectors in animals or plants. *See also* **sulfamethoxazole**.



trimethylaminuria *see* **fish-odour syndrome**.

trimyristin a triacylglycerol in which all the fatty-acyl moieties are **myristoyl** (tetradecanoyl).

trinactin a neutral macrocyclic antibiotic with an action similar to that of **nonactin**.

trinitroglycerin glyceryl trinitrate; trinitroglycerol; the nitrate moiety is metabolized to nitrite and, subsequently, to **nitric oxide** and nitrosothiol, which activate guanylate cyclase. The resulting increase in cyclic GMP synthesis may explain the potent vasodilator effects of this and similar organic nitrates.

trinor+ *prefix (in a trivial name of a compound)* indicating replacement of three methyl groups by hydrogen. *See also* **nor+**.

trinucleotide repeat an increase in copy number of a normally polymorphic triplet. Some genetic disorders, e.g. **fragile X syndrome**, result from this.

triol any aliphatic chain having three hydroxyl groups.

triolein a triacylglycerol in which all the fatty-acyl moieties are **oleoyl** (*cis*-9-octadecenoyl). *See also* **triolein breath-test**.

triolein breath-test a test used as an alternative to fecal fat determination to test for intestinal malabsorption of fat. When ¹⁴C-labelled triacylglycerol (10 μCi) is administered by mouth, a proportion of the ¹⁴C will be breathed out as ¹⁴CO₂, which is trapped in a solution of hyamine carbonate for scintillation counting.

triose 1 any one of three monosaccharides having a chain of three carbon atoms in the molecule, these being the chiral enantiomers **D-** and **L-glyceraldehyde**, and their achiral constitutional isomer **glycerone**. **2** trisaccharide; used as a stem name for any oligosaccharide consisting of three residues per mol-



triose phosphate

ecule of the same sugar in a single type of linkage, as in maltotriose. *Distinguish from +triose.*

triose phosphate *general term for D-glyceraldehyde 3-phosphate, glyceralone phosphate, or a mixture thereof. These two compounds are interconvertible by the enzyme triose-phosphate isomerase.*

triosephosphate dehydrogenase (NADP⁺) *see glyceraldehyde-3-phosphate dehydrogenase (NADP⁺) (phosphorylating).*

triose-phosphate isomerase EC 5.3.1.1; *systematic name:* D-glyceraldehyde-3-phosphate ketol-isomerase; *other names:* triosephosphate mutase; phosphotriose isomerase. The sugar isomerase enzyme that is important in the glycolytic pathway, and catalyses the aldose–ketose isomerization of the two **triose phosphates**, i.e. the reaction:

D-glyceraldehyde 3-phosphate = glyceralone phosphate.

Example from *Saccharomyces cerevisiae*: database code NRL_1YPIA, 247 amino acids (26.63 kDa); 3-D structure known.

trioxsalen 2,5,9-trimethyl-7H-furo[3,2-g][1]-benzopyran-7-one; 4,5',8-trimethylpsoralen (*abbr.*: TMP); a pigmentation agent (photosensitizer). *See also psoralen.*

tripalmitin a triacylglycerol in which all the fatty-acyl moieties are palmitoyl (hexadecanoyl).

tripeptide a three-residue peptide.

triphenylmethyl group *abbr.*: trityl; a group used in chemical syntheses to protect nitrogen, sulfur, and, particularly, oxygen functions as their trityl derivatives. It has the advantage that, while stable under many conditions, it is readily removed by mild acid. It can be introduced by reagents such as triphenylmethyl chloride.

triphosphate **1** the pentavalent ion, $P_3O_{10}^{5-}$, derived from triphosphoric acid, decaoxotriphosphoric(V) acid, $(HO)_2PO-O-PO(OH)-O-PO(OH)_2$ (an acid not known in the free state). **2** any mixture of the anions of triphosphoric acid. **3** any (partial or full) salt or ester of triphosphoric acid. **4** any organic compound containing three phosphoric residues linked linearly by oxygen atoms. **5** name of the univalent group $-O-PO(OH)-O-PO(OH)-O-PO(OH)_2$ (irrespective of its state of ionization). **6** (*formerly*) any organic compound containing three independent phosphoric residues attached to different parts of the molecule; now incorrect, use **trisphosphate**.

triphospho+ **1** *prefix* to a chemical name indicating the presence in ester linkage of a triphosphoric residue. **2** (*formerly*) *prefix* to a chemical name indicating the presence of three independent phosphoric residues in ester linkages; now incorrect, use **trisphospho+**.

triphosphoinositolide *abbr.*: TPI; a former name for **phosphatidylinositol 4,5-bisphosphate** (*symbol*: PtdIns-4,5- P_2). Current nomenclature requires bisphosphate, triphosphate, etc. for the substituent phosphates on the inositol moiety. Other phosphatidylinositol bisphosphates are known, e.g. PtdIns-phosphatidylinositol 3,4-bisphosphate.

triphosphopyridine nucleotide (oxidized) *abbr.*: TPN⁺; an obsolete name for **nicotinamide-adenine dinucleotide phosphate**.

triphosphopyridine nucleotide (reduced) *abbr.*: TPNH; an obsolete name for **nicotinamide-adenine dinucleotide phosphate (reduced)**.

triphosphoric *symbol*: PPP or (in the one-letter convention for nucleotides) ppp; *general name* for a residue of triphosphoric acid, $(HO)_2PO-O-PO(OH)-O-PO(OH)_2$, whether singly or multiply attached through oxygen to other groups and whether or not any residual hydroxyl groups are dissociated.

triple bolus test *see combined test of pituitary function.*

triple helix **1** the superhelically coiled arrangement of the three polypeptide chains in **tropocollagen**. **2** or (*sometimes*) **triplex** *see* **triplex DNA**.

triple membrane *see* **unit membrane**.

triple point the point on a phase diagram at which three phases may coexist.

triplet **1** (*in molecular biology*) any sequence of three purine

and/or pyrimidine bases in nucleic acids that specifies a particular amino acid. It is synonymous with **codon** in mRNA or **anticodon** in DNA or tRNA. *See also* **synonym triplet**. **2** a collection of three of a kind. **3** any one of three offspring born from a single pregnancy.

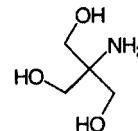
triplet code the **genetic code**, which is based on **triplets** (def. 1) of purine and pyrimidine bases.

triplex DNA a three-stranded structure formed by duplex DNA in association with an oligonucleotide, when purine or pyrimidine bases occupy the major groove of the DNA double helix adopting **Hoogsteen base pairing** with the Watson–Crick base pairs. Triplex DNA may be involved as an intermediate in **crossing over**. *See also* **tetraplex DNA**.

triploid **1** describing a cell, tissue, or organism having three times the haploid (monoploid) number of chromosomes. **2** such a cell, tissue, or organism. —**triploidy** *n*.

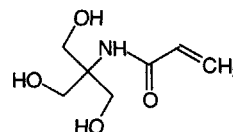
tris+ *prefix* **1** denoting three times, thrice, tripled; *see also* **tri+** (def. 1). **2** (*in chemical nomenclature*) (*distinguish from tri+* (def. 2)) **a** indicating the presence in a molecular entity of three identical specified groups each substituted in the same way; e.g. tris(2-hydroxyethyl)amine. **b** indicating the presence at separate positions in a molecular entity of three identical inorganic oxoacid residues; e.g. inositol 1,4,5-trisphosphate.

Tris or **TRIS** or **THAM** tris(hydroxymethyl)aminomethane; 2-amino-2-hydroxymethyl-1,3-propanediol; a compound widely used as a biological buffer substance in the pH range 7–9; pK_a (20 °C) = 8.3; pK_a (37 °C) = 7.82.



trisaccharide an **oligosaccharide** with three **monosaccharide** units.

Trisacryl **1** a *proprietary name* for *N*-acryloyl-2-amino-2-hydroxymethyl-1,3-propanediol; $CH_2=CH-CO-NH-C(CH_2OH)_3$; a compound used in the manufacture, by copolymerization with a cross-linking agent such as *N,N'*-diallyl L-tartardiamide (*abbr.*: DATD) or *N,N'*-methylenebisacrylamide, of a range of strongly hydrophilic artificial support materials of controlled porosity that are useful for gel-permeation chromatography. Copolymerization with a cationic or anionic derivative as a third component yields support materials of utility in ion-exchange chromatography. Such materials are thermally, chemically, and biologically stable, and resistant to high pressures. **2** *as modifier* describing a range, or any member of a range, of such chromatographic support materials; e.g. Trisacryl polymers, Trisacryl ion-exchangers, Trisacryl gels, Trisacryl GF05, DEAE-Trisacryl M.



triskelion a hexamer of **clathrin** molecules, composed of three heavy chains and three light chains. In the **electron microscope** triskelions appear to have three bent arms each linked into a central point. [From triskele, a symbolic figure consisting of three legs radiating from a common centre.]

trisomic **1** describing a cell, tissue, or organism having three copies of one or more chromosome(s) in an otherwise diploid chromosome set. In humans, trisomy 21 (i.e. presence of three



trisphosphate

copies of chromosome 21) is associated with **Down's syndrome**. 2 such a cell, tissue, or organism. —**trisomy** *n*.

trisphosphate any compound containing three independent phosphoric residues in ester linkages, at positions indicated by locants; e.g. inositol 1,4,5-trisphosphate. *Compare* **triphosphate**.

trisphospho+ *prefix* to a chemical name indicating the presence of three independent phosphoric residues in ester linkages, at positions indicated by locants. *Compare* **triphospho+**.

tristearin a triacylglycerol in which all the fatty-acyl moieties are **stearoyl** (octadecanoyl).

tritiate to combine or label with **tritium**; to replace one or more atoms of hydrogen with tritium. Tritium may be introduced into specific sites by chemical synthesis but ^3H -labelled steroids are prepared by the reduction of unsaturated precursors with $^3\text{H}_2$, the number of ^3H atoms introduced depending upon the degree of unsaturation of the precursor and the catalyst used. Biosynthesis is rarely used for ^3H -labelling since intramolecular migration of ^3H atoms is often caused by enzymes.

tritin a type I **ribosome-inactivating protein** from wheat.

tritium the radioactive isotope of hydrogen, ^3H (i.e. the nucleus contains two neutrons and one proton). It decays to produce β particles of low energy (0.018 MeV); half-life 12.26 years. *Compare* **deuterium**, **protium**.

tritium nuclear magnetic resonance spectroscopy *abbr.*: tritium NMR spectroscopy or TNMR spectroscopy; the application of **nuclear magnetic resonance spectroscopy** to determine the positions and extents of tritium substitution in substances (generally of known structure) that have been labelled with tritium (^3H). The technique depends upon an equality of the displacements of the resonance frequencies (chemical shifts) of tritium and protium (^1H) nuclei in corresponding environments relative to those occurring with the tritiated and nontritiated forms, respectively, of a reference substance. The method is quick, nondestructive, and applicable to very small amounts of material. It is thus particularly useful for determining the pattern of labelling in substances that are costly, complex in structure, and/or unstable. The technique is also used to follow the chemical and steric course of biochemical reaction sequences.

triton 1 (*in chemistry*) *symbol*: $^3\text{H}^+$; the cation derived from an atom of **tritium**; a **hydron** of nucleon number 3. 2 (*in nuclear physics*) *symbol*: *t*; a particle having a charge equal and opposite to that of an electron and having a mass of 3.0155 Da.

Triton the proprietary name for any of a series of polyoxyethylene ethers of certain alkylphenols that are surfactants and classed as nonionic detergents. Those used most commonly are based on the formula *tert*-octyl- C_6H_4 - $[\text{OCH}_2\text{CH}_2]_x\text{OH}$; e.g. in Triton X-45, $x = 5$; in Triton X-100, $x = 9$ or 10 (aggregation number 100–155, CMC 0.2–0.9 mM); in Triton X-405, $x \approx 40$; Triton CF is a benzyl-polyoxyethylene *tert*-octylphenyl ether; Triton N-101 is a polyoxyethylene nonylphenyl ether (C_9H_{19} - C_6H_4 - $[\text{OCH}_2\text{CH}_2]_x\text{OH}$), with $x = 9$ or 10.

tritosome a **lysosome** that is loaded with Triton. Tritosomes have an isopycnic density in sucrose solution of about 1.14, and can thus be separated readily from other cell constituents.

tritriaconta+ or (*before a vowel*) **tritriacont+** *comb. form* denoting thirty-three or thirty-three times.

triturate 1 to rub or grind (solid material) to a very fine powder either dry or in a liquid. 2 a finely ground powder or an even mixture of finely ground powders, especially powdered drugs.

trituration 1 the act of triturating or the condition of being triturated; *see* **triturate** (def. 1). 2 the composing of a dental amalgam.

trityl trivial name for the triphenylmethyl group.

trivial name (*in chemical nomenclature*) a name of a chemical species that has no part of it used in a systematic sense; e.g. xanthophyll, urea. It generally has historical or colloquial origins, and refers to attributes such as appearance or source. It

makes no pretension to describe molecular structure. The adjective 'trivial' has no perjorative meaning, numerous compounds of biochemical interest having approved or recommended trivial names. *Compare* **descriptive name**, **systematic name**.

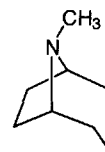
trixenic *see* **synxenic**.

trk a protooncogene encoding a transmembrane glycoprotein selectively expressed in the developing nervous system; the product is a tyrosine kinase receptor, the expression of which is restricted *in vivo* to certain neurons of neural crest origin, or, as its counterpart *trkB*, in cells of the neuroepithelium. It is required for high-affinity binding of nerve growth factor, and neurotrophins 3 and 4 and is essential for signal transduction in their action. Example of receptor from human (precursor): database code TRKA_HUMAN, 790 amino acids (86.88 kDa). It was discovered as the transforming factor of a colon carcinoma and was originally known as *oncD*. The oncogenic activity resulted from a genomic rearrangement event in which nonmuscle tropomyosin sequences were aberrantly fused onto the transmembrane and cytoplasmic domains of Trk (tropomyosin receptor kinase).

tRNA *abbr. for* transfer RNA.

Trojan horse inhibitor *see* **affinity label**.

tropane 8-methyl-8-azabicyclo[3.2.1]octane; the basic ring system in tropane alkaloids.



+trope *suffix* denoting turning.

+troph *suffix* denoting an organism with a specified mode of nutrition, e.g. **autotroph**, **chemotroph**.

trophic of, or concerning, nourishment, nutrition. *Compare* **tropic** (def. 1).

+trophic *suffix* denoting nourishment, nutrition; e.g. organotrophic, saprotrophic. *See also* **+tropic**.

+trophin *noun suffix* denoting a substance having a **trophic** function. *See also* **+trophin**.

tropho+ or (*before a vowel*) **troph+** *comb. form* denoting food or nourishment.

trophoblast the outer part of the blastocyst stage of a mammalian conceptus. Its outermost layer consists of syncytial cells concerned with penetration and implantation into the uterine endometrium; these cells secrete extracellular hydrolases that facilitate movement of the blastocyst into the previously organized endometrial structure.

+troph *noun suffix* 1 denoting food or nourishment. 2 denoting growth or size.

tropic 1 or (*sometimes, esp. Brit.*) **trophic** having a stimulating effect (especially of a hormone, etc.). 2 of or relating to a **tropism**.

+tropic or (*sometimes, esp. Brit.*) **+trophic** *suffix* denoting a stimulatory function; e.g. gonadotropin. *See also* **+trophic**, **+trophin**.

tropic acid α -phenyl- β -hydroxypropionic acid; (S)-3-hydroxy-2-phenylpropanoic acid; a compound that occurs in ester form in **atropine** and other tropane alkaloids.

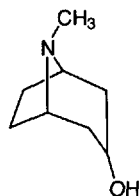
+trophin or (*sometimes, esp. Brit.*) **+trophin** *noun suffix* denoting a substance having a stimulatory function, especially a hormone produced by the adenohypophysis. *See also* **+trophin**.

tropine tropan-3 α -ol; 2-hydroxytropane; *N*-methyl-8-azabicyclo[3.2.1]octan-3-ol; **atropine** is the ester with ($\pm$)-tropic acid. Tropine is a *meso* structure. The -OH group is *trans* to the CH_3N - group. In the geometrical isomer, **pseudotropine**, this relationship is *cis*.

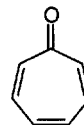
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tropine



tropone

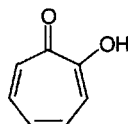
tropinone *N*-methyl-8-azabicyclo[3.2.1]octan-3-one; the oxidation product of **tropine**.

tropism movement directed towards some positive stimulus, or away from some negative stimulus.

+tropism *noun suffix* denoting a specific type of **tropism** as indicated by the prefixed word.

tropocollagen or **soluble collagen** a rod-shaped molecule, 3000 Å long and 15 Å in diameter, consisting of two α 1 and one α 2 polypeptide chains each of about 1000 amino-acid residues, of which about a third are glycine and 21% are proline plus hydroxyproline. They also contain hydroxylysine. The three polypeptide chains wind around each other to form a superhelical cable; the chains are hydrogen-bonded to each other. Tropocollagen fibres spontaneously associate with each other, with their N-terminal sequences overlapping C-terminal sequences of adjacent chains in a staggered array, to form **collagen** molecules. Compare **procollagen**. See also **registration peptide**.

tropolone 2-hydroxy-2,4,6-cycloheptatrien-1-one; a structure found in some mould metabolites and in **colchicine**.



tropomodulin a cytoskeletal tropomyosin-regulatory protein that binds to the end of erythrocyte tropomyosin and blocks its head-to-tail association along actin filaments; it does not bind to F actin. It has no homology to other proteins. Its modulation of the spectrin-actin complex in the erythrocyte membrane skeleton affects the viscoelastic properties of erythrocytes. Example from human: database code TMOD_HUMAN, 359 amino acids (40.52 kDa).

tropomyosin a protein found in smooth, skeletal, and cardiac muscle. It comprises α and β subunits and is bound to **actin** forming a component of the **tropinin** complex that regulates calcium-dependent binding of actin and myosin and, consequently, muscle contraction. Tropomyosin forms a characteristic coiled coil with a prominent seven-residue periodicity (five motifs). Different isoforms are produced by differential splicing; e.g. isoforms of α -tropomyosin differ in striated and smooth muscle. Example from human fibroblasts: database code TPM3_HUMAN, 284 amino acids (32.84 kDa).

tropomyosin kinase EC 2.7.1.132; an enzyme that catalyses the phosphorylation by ATP of tropomyosin to form *O*-phosphotropomyosin with release of ADP.

tropone 2,4,6-cycloheptatrien-1-one.

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tropinin *abbr.*: Tn; a central regulatory protein of striated muscle contraction, comprising three subunits: **tropinin T** (Tn-T), which binds to **tropomyosin**; **tropinin I** (Tn-I), which regulates actinomyosin ATPase; and **tropinin C** (Tn-C), which binds Ca^{2+} and with Ca^{2+} bound abolishes the inhibition on the binding of myosin to actin filaments produced by the other two tropinins. Examples, Tn-T (human slow skeletal muscle isoform): database code TRT1_HUMAN, 277 amino acids (32.77 kDa). Tn-I (human cardiac muscle form): database code TRIC_HUMAN, 209 amino acids (23.85 kDa). Tn-C (human slow skeletal and cardiac form; bovine and porcine sequences are the same): database code TPCC_HUMAN, 161 amino acids (18.40 kDa).

+trophy *noun suffix* denoting turning.

Trp *symbol* for a residue of the α -amino acid L-tryptophan (alternative to W).

Trp-2 tyrosinase-related protein 2; this is a product of the *tyrp-2* gene. See also **slaty**.

trp genes a family of tryptophan biosynthetic genes found in bacteria. The following examples of *trp* gene products are all from *Escherichia coli*.

(1) TrpA (database code TRPA_ECOLI, 268 amino acids (28.69 kDa)) and TrpB (database code TRPB_ECOLI, 396 amino acids (42.80 kDa)) are the α and β subunits of tryptophan synthase (EC 4.2.1.20); it catalyses the formation of L-tryptophan from L-serine and 1-(indol-3-yl) glycerol 3-phosphate; glyceraldehyde 3-phosphate is also formed. The enzyme is a tetramer ($\alpha_2\beta_2$); the subunits have half-activities (α for aldol cleavage of indoleglycerol phosphate and β for tryptophan synthesis).

(2) TrpC (database code TRPC_ECOLI, 452 amino acids (49.36 kDa)) is a monomeric bifunctional enzyme: the 'TrpF domain' is *N*-(5'-phosphoribosyl)anthranilate isomerase (EC 5.3.1.24); it catalyses the interconversion of *N*-(5'-phosphoribosyl)anthranilate and 1-(2-carboxyphenylamino)-1-deoxy-D-ribulose 5-phosphate. The 'TrpC domain' is **indole-3-glycerol-phosphate synthase** (EC 4.1.1.48; *abbr.*: IGPS); it catalyses the decarboxylation of 1-(2-carboxyphenylamino)-1-deoxy-D-ribulose 5-phosphate to form 1-(indol-3-yl) glycerol 3-phosphate, CO_2 , and H_2O .

(3) TrpE/G, **anthranilate synthase** (EC 4.1.3.27); it catalyses a reaction between chorismate and L-glutamine to form anthranilate, pyruvate, and L-glutamate. This enzyme is a tetramer with two molecules of component I – TrpE, database code TRPE_ECOLI, 520 amino acids (57.46 kDa) – and two of component II – TrpG, TRPG_ECOLI, 530 amino acids (56.68 kDa). (4) TrpR, aporepressor (*trp* is co-repressor) for the operators of the *trp* operon, *trpR* itself, and *aroH*; a homodimer of known 3-D structure; database code TRPR_ECOLI, 107 amino acids (12.21 kDa).

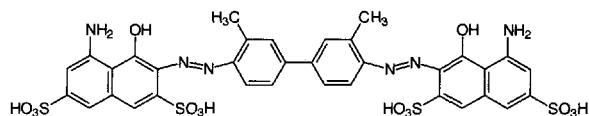
true cholinesterase an alternative name for **acetylcholinesterase**.

truncate 1 to shorten by cutting. **2** to terminate prematurely. —**truncation** *n*.

Trypan Blue a blue dye that is water-soluble and possesses sulfate and amino groups, whose charge prevents it from penetrating the plasma membrane of healthy cells. Dead cells or cells with damaged membranes take it up, making it a useful



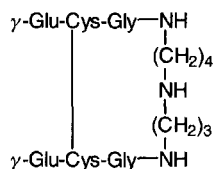
stain for determining cell viability (see **dye-exclusion test**). It does, however, stain healthy macrophages.



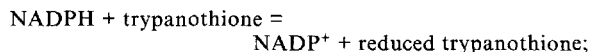
trypanosomatid a common word for any protozoan of the order **Kinetoplastida**, which includes **trypanosomes** and other genera.

trypanosome any member of the genus *Trypanosoma* (order **Kinetoplastida**), comprising parasitic flagellate protozoans. Infection causes disease (trypanosomiasis) in a range of animals. In humans there are two main forms of the disease: African trypanosomiasis (or sleeping sickness), which is caused by *T. brucei* and transmitted by tsetse flies, and South American trypanosomiasis (or Chagas' disease), which is caused by *T. cruzi* and transmitted by insects such as the assassin bug. Invasion of the nervous system, with attendant signs, is more common with the African form than with the South American form, which mainly affects the heart.

trypanothione the oxidized form of *N*¹,*N*⁶-bis(glutathionyl)spermidine; a compound purified from the trypanosomatid *Crithidia fasciculata*, a parasite of insects. Members of the **Kinetoplastida** lack **glutathione reductase**, and **glutathione** is converted to the conjugate, trypanothione. See also **trypanothione reductase**.



trypanothione reductase EC 1.6.4.8; *systematic name*: NADPH:trypanothione oxidoreductase; an enzyme found uniquely in **trypanosomatids**, and hence a potential chemotherapeutic target. It catalyses the reaction:



FAD is a cofactor. Example from *Trypanosoma cruzi* (homodimer); database code TYTR_TRCR, 492 amino acids (53.81 kDa). See also **trypanothione**.

trypsin EC 3.4.21.4; a serine endopeptidase occurring in two forms, known as α and β trypsin. These result from proteolytic cleavage, β being formed directly from the precursor, **trypsinogen**, and α and other forms by further cleavage. Trypsin is of major importance in protein digestion in the small intestine, trypsinogen being a constituent of **pancreatic juice**. Trypsin is also widely used as a proteolytic agent experimentally, e.g. in protein sequencing. It preferentially cleaves at Arg-[Xaa and Lys-[Xaa. Example (bovine): database code NRL_1NTP, 223 amino acids (23.28 kDa); 3-D structure known. See also **trypsin inhibitor**.

trypsin inhibitor any of various naturally occurring substances that inhibit the activity of **trypsin**. The most-studied example is bovine pancreatic trypsin inhibitor (see **pancreatic trypsin inhibitor**). See also **soybean trypsin inhibitor**.

trypsinize or **trypsinise** to treat something, especially intact cells or membrane preparations with trypsin. Cells are released from adhesion to tissue-culture plates by trypsinization; treatment of membranes with trypsin can be used to indicate exposure of protein domains, as subsequent analysis will reveal unattacked bonds that might be expected to be broken.

ken, indicating that they may be shielded from attack by inclusion within the membrane. — **trypsinization** or **trypsinisation** *n*.

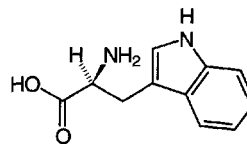
trypsinogen the pancreatic **zymogen** precursor of **trypsin**. It is secreted in **pancreatic juice** and in the duodenum is cleaved by **enteropeptidase**, which cleaves the Lys²³-Ile²⁴ (in the precursor; in the mature molecule, these are Lys⁸-Ile⁹) bond in trypsinogen, to yield β trypsin. Example, human precursor TRY1_HUMAN, 247 amino acids (26.56 kDa); residues 1–15 are the signal peptide, 16–247 trypsinogen (residues 16–23 are a propeptide).

tryptamine β -3-indolylethylamine; 3-(2-aminoethyl)indole; 2-(1*H*-indol-3-yl)ethanamine; (C₈H₆N)CH₂-CH₂-NH₂; a biogenic amine, formed by enzymic decarboxylation of tryptophan. Of widespread occurrence in mammalian tissues, it is found also in plants and is a product of microbial degradation of proteinaceous material.

tryptase EC 3.4.21.59; *other names*: mast cell tryptase; mast cell protease II; skin tryptase; lung tryptase; pituitary tryptase. An enzyme that catalyses the preferential cleavage: Arg-[Xaa, Lys-[Xaa, but with more restricted specificity than **trypsin**. It is a trypsin-like serine endopeptidase released from the secretory granules upon mast-cell activation. Example β -tryptase from human (precursor) homotetramer; database code TRYB_HUMAN, 275 amino acids (30.49 kDa). Cytotoxic T-lymphocyte tryptase, or CTL tryptase, is also known as **granzyme A**.

tryptic of, or relating to, trypsin; effected or produced by the action of trypsin; especially **tryptic digestion**, the partial hydrolysis of a protein by trypsin.

tryptophan or (formerly) **tryptophane** the trivial name for β -3-indolylalanine; α -amino- β -3-indolepropionic acid; 2-amino-3-(1*H*-indol-3-yl)propanoic acid; (C₈H₆N)CH₂-CH(NH₂)-COOH; a chiral α -amino acid. L-tryptophan (*symbol*: W or Trp), (*S*)-2-amino-3-(1*H*-indol-3-yl)propanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: UGA (only in mitochondria) or UGG. In mammals it is an essential dietary amino acid and is glycogenic. The peptide antibiotic, **tyrocidin C**, contains one residue per molecule of D-tryptophan (*symbol*: D-Trp or DTrp), (*R*)-2-amino-3-(1*H*-indol-3-yl)propanoic acid.



L-tryptophan

tryptophanal α -amino- β -3-indolepropionaldehyde; 2-amino-3-(1*H*-indol-3-yl)propanal; (C₈H₆N)CH₂-CH(NH₂)-CHO; the α -amino-aldehyde analogue of **tryptophan**.

tryptophanase 1 EC 4.1.99.1; *recommended name*: tryptophanase; *systematic name*: L-tryptophan indole-lyase (deaminating). An enzyme that catalyses a reaction between water and L-tryptophan to form indole, pyruvate, and NH₃. It is responsible for indole formation in *Escherichia coli* and other enteric bacteria and bacilli. The reaction is comparable to that of β -tyrosinase (see **tyrosinase** (def. 2)). 2 an alternative name for **tryptophan 2,3-dioxygenase**.

tryptophanate 1 tryptophan anion; the anion, (C₈H₆N)CH₂-CH(NH₂)-COO⁻, derived from **tryptophan**. 2 any salt in which tryptophan is anionic. 3 any ester of tryptophan.

tryptophan 2,3-dioxygenase EC 1.13.11.11; *systematic name*: L-tryptophan:oxygen 2,3-oxidoreductase (decyclizing); *other names*: tryptophan pyrrolase; tryptophanase; tryptophan oxygenase; tryptamine 2,3-dioxygenase. An enzyme that catalyses a reaction between dioxygen and L-tryptophan to form L-



tryptophan 5-hydroxylase

formylkynurenine. It is a heme protein in mammalian liver, and it shows broad specificity towards tryptamine and derivatives including D- and L-tryptophan, 5-hydroxytryptophan, and serotonin. Example (homotetramer) from rat: database code T230_RAT, 406 amino acids (47.80 kDa).

tryptophan 5-hydroxylase EC 1.14.16.4; *recommended name*: tryptophan 5-monooxygenase; *systematic name*: L-tryptophan, tetrahydrobiopterin:oxygen oxidoreductase (5-hydroxylating); *other name*: tryptophan oxygenase. An enzyme that catalyses a reaction between dioxygen, L-tryptophan, and tetrahydrobiopterin to form 5-hydroxy-L-tryptophan, dihydrobiopterin, and water. This is the rate-limiting enzyme in the biosynthesis of the neurotransmitter **5-hydroxytryptamine**. Example from rat: database code TR5H_RAT, 444 amino acids (51.01 kDa).

tryptophanium tryptophan cation; the cation, $(C_8H_6N)CH_2-CH(NH_3^+)-COOH$, derived from **tryptophan**.

tryptophan 2-monooxygenase EC 1.13.12.3; *systematic name*: L-tryptophan:oxygen 2-oxidoreductase (decarboxylating); *other name*: tryptophan oxygenase. An enzyme that catalyses a reaction between dioxygen and L-tryptophan to form indole-3-acetamide, CO_2 , and water. This is the first step in **auxin** biosynthesis. Example from *Pseudomonas syringae*: database code TR2M_PSESS, 557 amino acids (61.79 kDa).

tryptophan 5-monooxygenase the *recommended name* for **tryptophan 5-hydroxylase**.

tryptophano the alkylamino group, $(C_8H_6N)CH_2-CH(COOH)-NH-$, derived from **tryptophan**.

tryptophanol 3-(2-amino-3-hydroxypropyl)indole; 2-amino-3-(1H-indol-3-yl)propanol; $(C_8H_6N)CH_2-CH(NH_2)-CH_2OH$; the α -amino-alcohol analogue of **tryptophan**.

tryptophan oxygenase an *alternative name* for **1 tryptophan 2,3-dioxygenase**. **2 tryptophan 2-monooxygenase**. **3 tryptophan 5-hydroxylase**.

tryptophan pyrrolase an *alternative name* for **tryptophan 2,3-dioxygenase**.

tryptophan repressor see *trp* genes.

tryptophan synthase see *trp* genes.

tryptophan tryptophylquinone abbr.: TTQ; a quinoprotein cofactor of methylamine dehydrogenase, EC 1.4.99.3, an enzyme of methylotrophic bacteria. TTQ consists of a tryptophanyl side-chain cross-linked to the tryptophyl 6,7-dione side-chain of a modified tryptophan residue.

tryptophyl the acyl group, $(C_8H_6N)CH_2-CH(NH_2)-CO-$, derived from **tryptophan**.

TSF abbr. for thymocyte-stimulating factor, i.e. **interleukin 2**.

TSF abbr. for thyroid-stimulating hormone.

T state see **T form**.

TTP abbr. for 1 ribosylthymine triphosphate or ribothymidine triphosphate, *common names* for ribosylthymine 5'-triphosphate, ribothymidine 5'-triphosphate, 5'-ribothymidyl triphosphate, thymine riboside 5'-triphosphate. **2** (sometimes) thymidine triphosphate. dTTP is recommended.

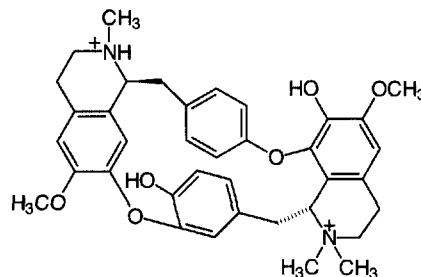
TTQ abbr. for tryptophan tryptophylquinone.

T tubule abbr. for transverse tubule; a type of tubule in a muscle fibre that arises as an invagination of the plasma membrane and penetrates the fibre, running mostly transversely among the fibrils and surrounding each of them with a ring. Each ring is shared by adjacent fibrils and the whole forms an intercommunicating system of tubules across the fibre. The function of these tubules is to pass the excitation signal from the muscle cell plasma membrane to the sarcomeres, ensuring rapid and synchronous activation. See also **sarcoplasmic reticulum**.

TTX abbr. for tetrodotoxin.

tubocurarine (+)-tubocurarine is the active principle of **curare**; it is a nicotinic cholinergic receptor competitive antagonist, and is used as a muscle relaxant. The molecule contains groups that are arranged in space in a rigid framework such that the atoms of acetylcholine that are important in its action have the correct geometry. In particular, two positively

charged nitrogens are fixed at a distance of 1.4 nm, possibly corresponding to the N^+ atoms of two acetylcholine molecules. Tubocurarine competes with acetylcholine at the neuromuscular junction and the endplate potential falls dramatically. The spacing of N^+ atoms appears crucial to the action of drugs such as **suxamethonium**.

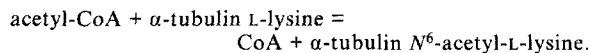


(+)-tubocurarine

tubulin a widely distributed globular subunit protein, profilaments of which form the structural elements of a **microtubule**. It is a heterodimer of two chains, α and β , each ≈ 50 kDa. The stable form of isolated tubulin is a dimer in which α - β pairing is believed to predominate; such a heterodimer is also the basic component of the microtubule. The dimer binds two molecules of GTP, one at an exchangeable site on the β chain and one at a nonexchangeable site on the α chain. The GTP is hydrolysed to GDP at the plus end, more slowly than assembly of heterodimers, to give a GTP cap at that end. GDP-tubulin is disassembled more readily than GTP-tubulin, so the rate of GTP hydrolysis could regulate tubulin assembly. Examples from human; α -1: database code TBA1_HUMAN, 451 amino acids (50.10 kDa); β -1: database code TBB1_HUMAN, 444 amino acids (49.70 kDa).

α -tubulin acetylase see **tubulin N-acetyltransferase**.

tubulin N-acetyltransferase EC 2.3.1.108; *other name*: α -tubulin acetylase; an enzyme that catalyses the reaction:



tubuliny-Tyr carboxypeptidase EC 3.4.17.17; *other names*: carboxypeptidase-tubulin; soluble carboxypeptidase; an enzyme that catalyses the cleavage of the Glu-[Tyr bond to release the C-terminal tyrosine residue from the native tyrosinated tubulin. It is inactive on carbobenzoxy-Glu-Tyr.

tuftelin or **enamelin** an acidic enamel protein of teeth with a composition similar to tuft proteins. It is found in association with the crystal component and secreted at a very early stage of enamel formation in vertebrate teeth. Example (bovine): database code ENAM_BOVIN, 389 amino acids (43.81 kDa).

tuftsin the tetrapeptide Thr-Lys-Pro-Arg, isolable from **leukokinin** (leukophilic human IgG); the sequence occurs in the Fc fragment of IgG. The tetrapeptide binds to and stimulates the bactericidal, phagocytic, and tumoricidal activities of macrophages and granulocytes, and may also stimulate the migration of polymorphonuclear leukocytes. [After Tufts University, where it was discovered.]

tumble 1 to rotate in a plane vertical to the axis of rotation; 'head over heels'. Proteins and other macromolecules tumble in solution whereas integral membrane proteins rotate on an axis normal to the plane of the membrane but cannot tumble through the membrane surface. **2** in bacterial chemotaxis, to change direction by a rapid reversal of the direction of flagellar rotation, the bacterium then 'tumbling' in response to Brownian motion.

tumor a variant spelling (esp. US) of **tumour**.

tumorigenic describing a substance or treatment that has the

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tendency to cause the formation of a **tumour**.

tumour or (*esp. US*) **tumor** a mass of proliferating cells lacking, to varying degrees, normal growth control. Tumours may be benign, i.e. proliferating very slowly, or malignant, i.e. growing aggressively, invading neighbouring tissues, and spreading to distant parts of the body by releasing cells into the bloodstream or lymph. These cells form secondary tumours (*see metastasis*). *See also adenoma, carcinoma, lymphoma, sarcoma.* —**tumorous** or **tumoral** *adj.*

tumour initiator any compound having the ability, when administered to an animal, to cause the formation of a tumour.

tumour necrosis factor *abbr.*: TNF; either of two structurally and functionally related proteins: TNF- α is produced mainly by monocytes and macrophages, whereas TNF- β is produced by lymphoid cells. The two proteins are about 30% homologous at the amino-acid level, and bind to the same cell-surface receptors; both exist as homotrimers. Both TNF- α (cachectin) and TNF- β (lymphotoxin) were originally thought of as selective antitumour agents, but are now known to have a multiplicity of actions. In binding to their receptors, present on virtually all cells examined, they activate a large array of cellular genes and also multiple signal-transduction pathways, kinases, and transcription factors. Their genes are single-copy genes, closely linked within the MHC cluster. *See tumour necrosis factor α ; tumour necrosis factor β , tumour necrosis factor receptor.*

tumour necrosis factor α (*abbr.*: TNF- α) or (*formerly*) **cachectin** a **cytokine** that is produced by macrophages, monocytes, endothelial cells, neutrophils, smooth muscle cells, activated lymphocytes, and astrocytes. It is a transmembrane glycoprotein and cytotoxin with a variety of functions, including the ability to mediate the expression of genes for growth factors, cytokines, transcription factors, and receptors. It can cause cytolysis of certain tumour cell lines, it has been implicated in the induction of cachexia (but *see cachectic factor*), it is a potent pyrogen, causing fever by direct action or by stimulation of interleukin 1 secretion, and it can stimulate cell proliferation and induce cell differentiation under certain conditions. The molecule is a homotrimer. Example: database code NRL_1TNFA, 152 amino acids (16.82 kDa); 3-D structure known (high beta content, disulfide bond). *See also tumour necrosis factor.*

tumour necrosis factor β (*abbr.*: TNF- β) or (*formerly*) **lymphotoxin** a **cytokine** produced by T lymphocytes that is cytotoxic for a wide range of tumour cells. It binds to the same ligands as **tumour necrosis factor α** . Example from human (precursor): database code TNFB_HUMAN, 205 amino acids (22.27 kDa). *See also tumour necrosis factor.*

tumour necrosis factor receptor *abbr.*: TNF receptor; either of two distinct receptors for **tumour necrosis factor** (TNF), one of M_r 55 000 and the other of 75 000. The two types have related extracellular domains, based on cDNA sequencing, and these are also related to the extracellular domains of nerve growth factor receptor and some other cell-surface molecules. These form a receptor family characterized by four domains with regularly spaced cysteine residues; each has a single transmembrane domain; the intracellular domains show no significant homologies. Either TNF receptor type alone appears sufficient for high-affinity binding and full biological activity, and binds either **tumour necrosis factor α** or **tumour necrosis factor β** with high affinity. The signal transduction mechanism remains speculative, but receptor activation leads to a number of events, including activation of protein kinase C, protein kinase A, phospholipase A_2 , and phosphatidylcholine-specific phospholipase C, and accumulation of cyclic AMP and diacylglycerol. Example, tumour necrosis factor receptor 1 precursor from *Rattus norvegicus*: database code TNFR1_RAT, 461 amino acids (50.97 kDa).

tumour promoter any compound that, when administered to an animal, sensitizes tissues to the action of a **tumour initiator**, or predisposes tissue to become tumorous in any other way.

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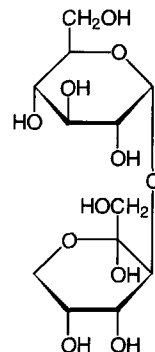
such that tumours form in greater numbers or grow more rapidly than if a tumour promoter had not been administered.

tumour-suppressor gene a normal gene one function of which is to suppress tumorigenesis. Certain cancers have been shown to be associated with mutant tumour-suppressor genes; *see (e.g.) p53, RB.*

tunnel effect the phenomenon in which an electron or a proton, because of its small mass, does not behave kinetically according to the laws of classical mechanics. These particles can sometimes penetrate an energy barrier that is greater than their kinetic energy.

turacin *see tetrapyrrole.*

turanose 3-O- α -D-glucopyranosyl-D-fructose; an early product of the acid hydrolysis of **melezitose**.



turbid (of a solution) opaque due to the presence of suspended particles. —**turbidity** *n.*

turbidimeter an instrument for measuring the degree to which turbidity hinders the passage of light through a solution.

turbidimetry the quantitative determination of the concentration of substances in suspension by measurement of the decrease in light transmission caused by the suspended particles. *Compare nephelometry.*

turnaround sequence (in DNA) a sequence of nucleotides between two **inverted repeats**. A turnaround sequence will cause a 'bubble' of noncomplementary nucleotides at one end of a **hairpin**.

turnover number an obsolete expression of enzymic activity, variously defined on different occasions as the number of substrate molecules converted to product per catalytic site per minute, or as the number of moles of substrate converted per mole of enzyme per minute or second; it has the dimension of reciprocal time. *Compare catalytic constant, katal.*

turret a multiple cuvette holder that rotates in a spectrophotometer.

Tween the *proprietary name* for a series of surfactants based on polyoxyethylene sorbitan esters. For example, Tween 20 is polyoxyethylene sorbitanmonolaurate (CMC 0.059 mM). Similarly the sorbitan moiety in other Twens is: Tween 40, sorbitanmonopalmitate; Tween 60, sorbitanmonostearate; Tween 65, sorbitantristearate; Tween 80, sorbitanmonooleate (aggregation number 58, CMC 0.012 mM); Tween 85, sorbitantri-oleate.

twist a *Drosophila* gene for a **MYC** family **helix-turn-helix** transcription factor involved in the establishment of germ layers. It is a homodimer. Database code TWST_DROME, 490 amino acids (54.34 kDa).

twist conformation 1 a median conformation of a non-planar, six-membered, saturated ring compound through which one **boat conformation** passes during conversion into another boat conformation, or through which a boat conformation passes during conversion into a **chair conformation** or vice versa. It is similar to **skew conformation** (def. 1). 2 any conformation of the five-membered ring form of a monosaccha-



ride or monosaccharide derivative when three adjacent ring atoms are coplanar and the other two lie on opposite sides of the plane. The conformational descriptor *-T* may be added to the name of such a compound. *See also* **conformation**.

twist number or **twisting number** *symbol*: *T*; the total number of base pairs in a DNA molecule divided by the number of bases per turn of the double helix. It is one of the properties that identifies **topological isomers** of DNA. *See also* **linking number**, **writhing number**.

twitchin the protein product of *unc-22*, one of about 40 genes important for muscle assembly and function in the nematode *Caenorhabditis elegans*. Twitchin is a giant myosin-associated protein (*compare* **titin**) that is autophosphorylated on threonine; repeats are homologous to **fibronectin** type III. The protein was named for a mutation that results in twitching. Database code S07571, 6048 amino acids (667.72 kDa). It has a counterpart in *Aplysia californica* and presumably other invertebrates.

two-carbon fragment **1** any group of atoms containing two carbon atoms. **2** an early term for a metabolically active acetyl radical, as in acetyl coenzyme A.

two-dimensional chromatography a chromatographic method in which separation is first effected in one direction and, after rotation of the chromatogram through 90°, then in a second direction, often with a different solvent (or a different pH). **Two-dimensional electrophoresis** works on the same principle.

two-dimensional polymer a macromolecular structure in which approximately parallel chains extending in one direction are cross-linked by another series of approximately parallel chains extending in a second direction. Bacterial cell-wall peptidoglycans are two-dimensional polymers with glycan chains extending in one direction that are cross-linked to peptide chains.

TXA *abbr.* for thromboxane A (*see* **thromboxane**).

TXB *abbr.* for thromboxane B (*see* **thromboxane**).

Tylenol *see* **acetaminophen**.

tyndallization or **tyndallisation** a method of sterilization in which the sample is heated at 80–100 °C on successive days, with a period of incubation in between. The aim is to kill cells on each occasion, but to allow spores (which are not killed by heating) to germinate to produce susceptible cells during the periods of incubation. [After John Tyndall (1820–93), British natural philosopher, alpinist, and popularizer of science.]

type III system a protein secretory system found in Gram-negative bacteria that is responsible for the secretion of proteins known as **Yop**. The system is known as type III because it was the third type of secretory system found in the membrane of Gram-negative bacteria.

typing (*in immunology*) the identification of the classes of histocompatibility antigens, blood groups, or other surface markers present in a sample of serum, or on white blood cells or red blood cells.

Tyr *symbol* for a residue of the α -amino acid L-tyrosine (alternative to **Y**).

tyramine tyrosamine; *p*-(β -aminoethyl)phenol; 2-(4-hydroxyphenyl)ethylamine; a biogenic amine with adrenergic action, formed by enzymic decarboxylation of tyrosine. Of widespread occurrence in mammalian tissues, it is also found in several plant species and in **ergot**, and is a product of microbial action in the intestine, in the putrefaction of animal tissue, and in the ripening of cheese.

Tyr(I₂) or (if the context does not imply the locants) **Tyr(3,5-I₂)** *symbol* for a residue of the α -amino acid 3,5-diiodo-L-tyrosine (*see* **diiodotyrosine**).

tyrocidine or **tyrocidin** any of several peptide antibiotics produced by *Bacillus brevis* that are cyclic decapeptides. The structure of tyrocidine A is (amino acids are L unless otherwise stated) cyclic -Leu-D-Phe-Pro-Phe-D-Phe-Asn-Gln-Tyr-Val-Orn-. Tyrocidine B is similar, but with L-Tyr in place of L-Phe, and in tyrocidine C D-Trp replaces the D-Phe attached to the

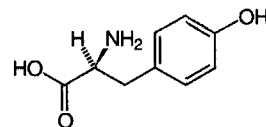
Asn. Structurally tyrocidines are similar to **gramicidin S**, and they have a similar action, increasing the permeability of bacterial membranes, promoting leakage of cytoplasmic constituents, and permitting entry into the cell of ions larger than those normally transported. *Bacillus brevis* is immune from this action. This group of compounds has little application in medicine. Biosynthesis is independent of ribosomes (*see* **protein and peptide biosynthesis**).

Tyrode's solution or **Tyrode solution** or **Ringer–Tyrode solution** a balanced salt solution modified from **Locke's solution**, initially for experiments on mammalian intestine. The solution usually contains (in g L⁻¹, approx.): NaCl (8.0); KCl (0.2); MgCl₂·6H₂O (0.1); CaCl₂ (anhydrous, 0.2); NaHCO₃ (0.1); NaH₂PO₄·H₂O (0.05); and glucose (1.0). [After Maurice Vejux Tyrode (1878–1930), US pharmacologist, who devised this solution in 1910.]

tyrosinase **1** an alternative name for **monophenol monooxygenase**. **2** β -tyrosinase EC 4.1.99.2; *recommended name*: tyrosine phenol-lyase; an enzyme that catalyses a reaction between L-tyrosine and water to form phenol, pyruvate, and NH₃. Pyridoxal-phosphate is a coenzyme. The protein is a homotetramer, similar to **tryptophanase** (def. 1). Example from *Citrobacter freundii*: database code TPL_CITFR, 456 amino acids (51.44 kDa).

tyrosinate **1** tyrosinate(1-); tyrosine monoanion. In theory, the term denotes any ion or mixture of ions formed from **tyrosine** and having a net charge of -1, although the species HO-C₆H₄-CH₂-CH(NH₂)-COO⁻ predominates in practice. **2** the systematic name for tyrosinate(2-); tyrosine dianion; ⁻O-C₆H₄-CH₂-CH(NH₂)-COO⁻. **3** any salt containing an anion of tyrosine. **4** any ester of tyrosine.

tyrosine the trivial name for β -(*p*-hydroxyphenyl)alanine; α -amino-*p*-hydroxyhydrocinnamic acid; α -amino- β -(*p*-hydroxyphenyl)propionic acid; 3-(4-hydroxyphenyl)alanine; 2-amino-3-(4-hydroxyphenyl)propanoic acid; HO-C₆H₄-CH₂-CH(NH₂)-COOH; a chiral α -amino acid. L-Tyrosine (*symbol*: **Y** or **Tyr**), (S)-2-amino-3-(4-hydroxyphenyl)propanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: UAC or UAU. In mammals it is an essential dietary amino acid, and is both glucogenic and ketogenic. One residue per molecule of D-tyrosine (*symbol*: D-Tyr or D-Tyr), (R)-2-amino-3-(4-hydroxyphenyl)propanoic acid, occurs as its *O*-methyl ether in the antibiotic cycloheptamycin.



L-tyrosine

tyrosine decarboxylase EC 4.1.1.25; an enzyme that catalyses the decarboxylation of L-tyrosine to **tyramine**. Pyridoxal-phosphate is a coenzyme. Example from parsley: database code A44405, 514 amino acids (57.38 kDa).

tyrosine hydroxylase EC 1.14.16.2; *recommended name*: tyrosine 3-monooxygenase; *systematic name*: L-tyrosine, tetrahydrobiopterin:oxygen oxidoreductase. An enzyme that catalyses a reaction between dioxygen, L-tyrosine, and tetrahydrobiopterin to form 3,4-dihydroxy-L-phenylalanine, dihydrobiopterin, and water; it requires ferric ion. It is the rate-limiting enzyme in the synthesis of **catecholamine** neurotransmitters. In humans four types are produced by alternative splicing. Example, type 3: database code TY3H_HUMAN, 528 amino acids (58.46 kDa).

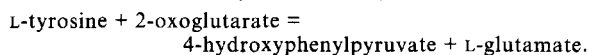
tyrosine kinase EC 2.7.1.112; *recommended name*: protein-tyrosine kinase; *other names*: tyrosylprotein kinase; protein kinase (tyrosine); hydroxyaryl-protein kinase. An enzyme that

tyrosine 3-monooxygenase

transfers the terminal phosphate of ATP to a specific tyrosine residue on its target protein; i.e. it catalyses the phosphorylation by ATP of protein-tyrosine to form protein-tyrosine phosphate with release of ADP. The enzyme molecule has five motifs. The tyrosine kinase family is of profound importance due to its involvement in the signalling processes associated with mitogenic stimulation, cell growth, and oncogenesis. The tyrosine kinases fall into two major classes, those that are cell membrane receptors and integral membrane proteins, and those that are intracellular and function downstream of the receptors. The membrane receptors have been grouped into subclasses according to their structure (all individual polypeptides have a single transmembrane domain, with the kinase activity in an intracellular domain); examples include subclass I, proteins having two cysteine-rich repeat sequences in the extracellular domain of a monomeric receptor, a group that includes the **epidermal growth factor receptor**; subclass II, disulfide-linked heterotetrameric $\alpha_2\beta_2$ structures with extracellular cysteine-rich repeat sequences similar to subclass I, an example being the **insulin receptor**; subclass III, proteins with five immunoglobulin-like repeats in the extracellular domain, and with a non-kinase insert in the kinase region, typified by the **platelet-derived growth factor receptor**; and subclass IV, proteins with three immunoglobulin-like repeats in the extracellular domain, with a non-kinase insert in the kinase region. This classification continues to be extended. The intracellular non-receptor tyrosine kinases include, among many, the products of *FGR*, *FYN*, and *src*. See also **protein kinase**.

tyrosine 3-monooxygenase the recommended name for **tyrosine hydroxylase**.

tyrosine transaminase or **tyrosine aminotransferase** (abbr.: TAT) EC 2.6.1.5; *systematic name*: L-tyrosine:2-oxoglutarate aminotransferase; an enzyme that catalyses the reaction:



It is a pyridoxal-phosphate enzyme and it catalyses the first step in tyrosine catabolism; defects lead to tyrosinemia type II

(**Richner-Ranhart syndrome**). Example from human (homodimer): database code ATTY_HUMAN, 454 amino acids (50.34 kDa).

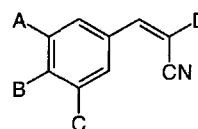
tyrosinium tyrosine cation; the cation, $\text{HO-C}_6\text{H}_4\text{-CH}_2\text{-CH(NH}_3^+)\text{-COOH}$, derived from **tyrosine**.

tyrosino the alkylamino group, $\text{HO-C}_6\text{H}_4\text{-CH}_2\text{-CH(COOH)-NH-}$, derived from **tyrosine**.

tyrosin-O⁴-yl the aryloxy group, $\text{HOOC-CH(NH}_2\text{)-CH}_2\text{-C}_6\text{H}_4\text{-O-}$, derived from **tyrosine**.

tyrosyl the acyl group, $\text{HO-C}_6\text{H}_4\text{-CH}_2\text{-CH(NH}_2\text{)-CO-}$, derived from **tyrosine**.

tyrphostin any of a series of inhibitors of protein tyrosine kinases. The structures are derived from the substituted benzylidenemalononitrile shown as the core structure; A, B, and C can be $-\text{NO}_2$, $(\text{CH}_3)_3\text{C-}$, $-\text{OH}$, or $-\text{H}$. In tyrphostin 25, [(3,4,5-trihydroxyphenyl)-methylene]-propanedinitrile, D is $-\text{CN}$. Other actions are known for some members of the series, including uncoupling of oxidative phosphorylation and inhibition of lipoygenases.

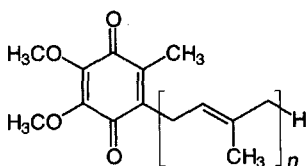


tyrphostin 25

tyvelose a trivial name for the aldohexose 3,6-dideoxy-D-mannose, 3,6-dideoxy-D-arabino-hexose, a monosaccharide isolated from various lipopolysaccharides elaborated by some Gram-negative bacteria, e.g. certain serotypes of *Salmonella typhi*. In such serotypes it contributes to the antigenic specificity of the organism's O antigens. The L-enantiomer is separately named as **ascarylose**.

Uu

u 1 symbol for unified atomic mass unit. 2 *see* **IU**.
u symbol for 1 speed (alternative to *v*, *w*, or *c*). 2 (bold italic) velocity (alternative to *v*, *w*, or *c*).
U symbol for 1 a residue of the base uracil in a nucleic-acid sequence. 2 a residue of the ribonucleoside uridine (alternative to *Urd*). 3 uniformly labelled. 4 enzyme unit (obsolete; superseded by **katal**). 5 uranium.
U symbol for 1 electric potential difference (alternative to ΔV or $\Delta\phi$). 2 internal energy.
U46619 a stable thromboxane A_2 analogue and mimetic; another name for **9,11-dideoxy-11 α ,9 α -epoxymethanoprostaglandin $F_{2\alpha}$** .
UAA a codon in mRNA for chain termination; the **ochre codon**.
UAC a codon in mRNA for L-tyrosine.
UAG a codon in mRNA for chain termination; the **amber codon**.
UAS *abbr.* for upstream activator sequence; the counterpart in yeast of an **enhancer**.
UAU a codon in mRNA for L-tyrosine.
Ubbelohde viscometer *see* **capillary viscometer**.
UBIP *abbr.* for ubiquitous immunopoietic polypeptide.
ubiquinol *abbr.*: QH₂; reduced **ubiquinone**.
ubiquinol-cytochrome c reductase EC 1.10.2.2; *systematic name*: ubiquinol:ferri-cytochrome-*c* oxidoreductase; *other name*: complex III. An enzyme complex that catalyses the oxidation of ubiquinol by two molecules of ferri-cytochrome *c* to form ubiquinone and two molecules of ferrocytochrome *c*. It contains cytochromes *b*₅₆₂, *b*₅₆₆, and *c*₁, as well as nonheme iron and Fe-S complexes.
ubiquinone *or* (formerly, but not recommended) **coenzyme Q** *abbr.*: Q; a lipid-soluble electron-transporting coenzyme that is an essential component in the **respiratory chain**. The structure is shown in the diagram; *n* has values from 6 to about 12. The side-chain length is indicated by the abbreviation Q_{*n*}. Ubiquinone is biosynthesized in mammals from tyrosine, and in bacteria and other organisms by the **shikimate pathway**. *See also* **protonmotive Q cycle**.



ubiquinone reductase *see* **NADH dehydrogenase (ubiquinone)**.

ubiquitin 1 a small, highly conserved, heat-stable protein of 76 amino-acid residues first isolated from bovine thymus but subsequently found in the cells of all tissues studied, in animals, yeast, bacteria, and higher plants. Located in both nuclei and cytoplasm, it derives its name from its widespread distribution. It undergoes an ATP-dependent reaction with proteins that results in condensation of its terminus with lysine amino groups; this process is mediated by a large multiprotein complex, the 26S **proteasome**. For example, cyclin degradation in the control of the cell cycle is triggered by ubiquitination. Thus the cell has two means of protein degradation, ubiquitination and proteolysis in lysosomes. Such modified proteins are degraded rapidly. Ubiquitin is synthesized as a poly-ubiquitin precursor with exact head-to-tail repeats, the number of repeats differing between species. In some species there is a final amino acid after the last repeat (e.g. Val in human).

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Some ubiquitin genes encode a single copy of ubiquitin fused to a ribosomal protein. Human, bovine, and porcine ubiquitins and those from long-tailed hamster, rat, mouse, rabbit, *Xenopus laevis*, and the following insects – *Drosophila melanogaster*, *Ceratitis capitata*, *Spodoptera frugiperda*, and *Manduca sexta* – all have the sequence: MQIFVKTLTGKTTIT-LEVEPSDTIENVKAKIQDKEGIPPDQQRLLIFAGKQLED-GRTLSDYNIQKESTLHLVLRRLRGG. Example from human: database code UBIQ_HUMAN, 76 amino acids (8.56 kDa); 3-D structure known; database code; NRL_1UBQ; three motifs. 2 a former name for **ubiquitous immunopoietic polypeptide**.

ubiquitin activating enzyme E1 an enzyme that activates ubiquitin by first adenylating, with ATP as co-substrate, its C-terminal glycine residue and thereafter linking this to itself via the side chain of a cysteine residue, releasing AMP. The protein is a monomer and can accommodate two ubiquitin molecules. Example from human: database code UBA1_HUMAN, 1058 amino acids (117.66 kDa).

ubiquitin-conjugating enzyme *see* **ubiquitin-protein ligase**.

ubiquitin-protein ligase EC 6.3.2.19; *other names*: ubiquitin-conjugating enzyme; ubiquitin-activating enzyme; an enzyme that couples ubiquitin to a protein lysine to form protein N-ubiquityllysine with hydrolysis of ATP to ADP and pyrophosphate. Examples from yeast: database code UBC1_YEAST, 215 amino acids (24.15 kDa); database code UBC2_YEAST, 172 amino acids (19.68 kDa); UBC3_YEAST, 295 amino acids (34.03 kDa); UBC8_YEAST, 218 amino acids (24.61 kDa). Examples from human: database code UBC2_HUMAN (also in rat and rabbit), 152 amino acids (17.29 kDa); UBC8_HUMAN, 183 amino acids (20.63 kDa).

ubiquitous immunopoietic polypeptide *abbr.*: UBIP; a polypeptide that has lymphocyte-differentiating properties and can stimulate adenylate cyclase. It is involved in maintenance of chromatin structure, regulation of gene expression, stress response, and ribosome biogenesis. It was formerly called ubiquitin. *See also* **proteasome**.

UCA a codon in mRNA for L-serine.

UCC a codon in mRNA for L-serine.

UCG a codon in mRNA for L-serine.

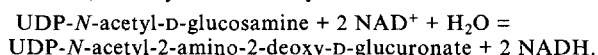
UCP *abbr.* for brown fat uncoupling protein.

UCU a codon in mRNA for L-serine.

Udo symbol for the undecanoyl group, CH₃–[CH₂]₉–CO–.

UDP *abbr.* for uridine (5'-)diphosphate.

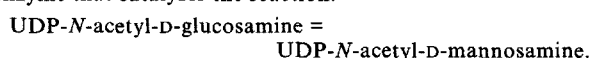
UDP-N-acetylglucosamine 6-dehydrogenase EC 1.1.1.136; an enzyme that catalyses the reaction:



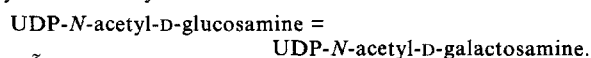
UDP-N-acetylglucosamine-dolichyl-phosphate-N-acetylglucosamine phosphotransferase EC 2.7.8.15; an enzyme that catalyses a reaction between UDP-N-acetyl-D-glucosamine and dolichyl phosphate to form N-acetyl-D-glucosaminyl-diphosphodolichol and UMP.

It catalyses the first reaction in the synthesis of oligosaccharide lipids. Example from Chinese hamster: database code UAGT_CRIGR, 408 amino acids (46.19 kDa).

UDP-N-acetylglucosamine 2-epimerase EC 5.1.3.14; an enzyme that catalyses the reaction:



UDP-N-acetylglucosamine 4-epimerase EC 5.1.3.7; an enzyme that catalyses the reaction:





UDPG

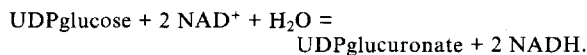
UDPG *abbr.* for uridinediphosphoglucose; UDPGlc is preferred (to avoid possible ambiguity in context with galactose).

UDPGal or **UDPGalactose** *abbr.* for uridinediphosphogalactose.

UDPGalactose 4-epimerase *an alternative name for UDPglucose 4-epimerase.*

UDPGlc (*preferred*) *abbr.* for uridinediphosphoglucose; see also **UDPG**.

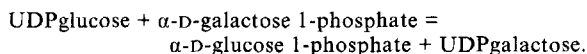
UDPGlucose 6-dehydrogenase EC 1.1.1.22; an enzyme that catalyses the reaction:



Example from *Streptococcus pyogenes*: database code UDHG_STRPY, 402 amino acids (45.43 kDa).

UDPGlucose 4-epimerase *the recommended name and systematic name for the isomerase* EC 5.1.3.2; *other names*: UDPgalactose 4-epimerase; (*formerly*) galactowaldenase; one of the enzymes that can interconvert uridinediphosphoglucose and uridinediphosphogalactose. The mechanism involves oxidation of the C-4 hydroxyl group of the hexose moiety of either uridinediphosphosugar to an oxo group by a tightly bound molecule of NAD⁺; the resultant NADH is then reoxidized by the intermediate oxo derivative with inversion of configuration of the resulting hydrogen and hydroxyl groups at C-4. *Compare* **UDPGlucose-hexose-1-phosphate uridylyltransferase**. Example from *Escherichia coli*: database code GALE_ECOLI, 338 amino acids (37.22 kDa); 3-D structure known.

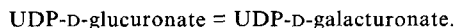
UDPGlucose-hexose-1-phosphate uridylyltransferase EC 2.7.7.12; *systematic name*: UDPglucose:α-D-galactose-1-phosphate uridylyltransferase; *other names*: **uridyl transferase**; galactose-1-phosphate uridylyltransferase; one of the enzymes that can interconvert uridinediphosphoglucose and uridinediphosphogalactose. An enzyme important in the metabolism of galactose, it catalyses the exchange reaction:



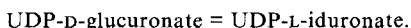
A deficiency of the enzyme is associated with one form of **galactosemia**. Example from human: HUMGALT, 380 amino acids (43.01 kDa). *Compare* **UDPGlucose 4-epimerase**.

UDPGlucose pyrophosphorylase *an alternative name for UTP-glucose-1-phosphate uridylyltransferase.*

UDPGlucuronate 4-epimerase EC 5.1.3.6; an enzyme that catalyses the reaction:



UDPGlucuronate 5'-epimerase EC 5.1.3.12; an enzyme that catalyses the reaction:



NAD⁺ is a cofactor.

UDPGlucuronosyltransferase EC 2.4.1.17; *abbr.*: UDPGT or UGT; *recommended name*: glucuronosyltransferase; an enzyme involved in the formation of **glucuronosides**. It catalyses a reaction between UDPglucuronate and an acceptor to form UDP and acceptor β-D-glucuronoside. There are many variants. The numbering follows several systems; the one followed here has UDPGT followed by a number, letter, and sometimes a number, e.g. UDPGT1F, UDPGT2B1O. Examples from human: isozyme 1a precursor, microsomal (UDPGT1a) (bilirubin-specific isozyme 1); database code UD1A_HUMAN, 533 amino acids (59.52 kDa); UDGP1b; database code UD1B_HUMAN, 530 amino acids (59.95 kDa); UDGP1c (*other name*: UGT13); database code UD1C_HUMAN, 534 amino acids.

UDPGP *abbr.* for UDPglucose pyrophosphorylase (i.e. **UTP-glucose-1-phosphate uridylyltransferase**).

UGA 1 a codon in nonmitochondrial mRNA for chain termination; the **opal codon**. 2 a codon in mitochondrial mRNA for L-tryptophan.

UGC a codon in mRNA for L-cysteine.

UGG a codon in mitochondrial mRNA, and the only codon in nonmitochondrial mRNA, for L-tryptophan.

UGU a codon in mRNA for L-cysteine.

+ulose *noun suffix* denoting a **ketose**; it may be expanded to **+ulofuranose** or **+ulopyranose** according to whether the cyclic form of the sugar is five-membered or six-membered respectively.

+ulosonic acid *noun suffix* denoting a **ketoaldonic acid**; it may be expanded to **+ulofuranosonic acid** or **+ulopyranosonic acid** according to whether the cyclic form of the sugar acid is five-membered or six-membered respectively.

ultimate carcinogen or **ultimate carcinogenic metabolite** any metabolite of a **chemical carcinogen** that reacts with cellular constituents in a specific way to initiate carcinogenesis. *See also* **proximate carcinogen**.

ultimobranchial body one of a pair of glandlike structures derived during embryogenesis from the last branchial (gill) pouches. They occur in all vertebrates except cyclostomes. In most mammals they become embedded in the respective lobes of the thyroid gland, forming **C cells**, which secrete the polypeptide hormone **calcitonin**.

ultra+ *prefix* 1 beyond in space; on the other side (of). *Compare* **infra+**, **trans+**. 2 beyond a specified (esp. in scientific usage, lower) limit, range, value, etc. *Compare* **super+**. 3 beyond the common, ordinary, natural, etc.; extreme(ly). *Compare* **hyper+**.

ultracentrifuge any centrifuge that operates at high rotational velocities, i.e. one that generates high gravitational fields. Ultracentrifuges can be operated at known speeds with small variations and without temperature fluctuations. They may be either analytical or preparative.

ultradian (of a biological activity) occurring less often than once per 24-hour period. *Compare* **circadian**, **infradian**.

ultrafilter a filter of small pore size suitable for **ultrafiltration**.

ultrafiltration a separation process whereby a solution containing a solute of molecular size significantly greater than that of the solvent molecule is removed from the solvent by the application of a hydraulic pressure, which forces only the solvent to flow through a suitable membrane (usually one having a pore size in the range 0.001–0.1 μm), thereby concentrating the solute.

ultramicrotome a device for preparing very thin sections of cells or tissues, after fixing and embedding or freezing, for inspection in the transmission electron microscope.

ultrasonic of, pertaining to, or producing **ultrasound** waves.

ultrasonication the process of exposing material to **ultrasound**. It is used for cell disruption, for denaturation of proteins, and for cleaning solid surfaces. *See also* **sonicate**.

ultrasonic disintegrator an apparatus for disintegrating cells and other material, or for cleaning solid surfaces, by exposing them to **ultrasound**. *See also* **sonicator**.

ultrasound propagated waves, of the same nature as sound but inaudible, with frequencies of >20 kHz. Frequencies of ≈25 kHz can be used for disintegrating cells and subcellular components and also in cleaning baths for the removal of deposits on a wide variety of objects. In physiotherapy ultrasound in the range of 0.5–3 MHz is used for the relief of painful muscles and has its effect by the delivery of warmth and by changing the permeability of cell membranes which allows for the removal of fluid. For diagnostic purposes ultrasound in the range of 3–12 MHz is commonly used and is a popular method of non-invasive imaging e.g. in pregnant women. There is also ultrasonic microscopy which mimics light microscopy and employs sound in the GHz range. Ultrasound is also being developed for the disintegration of tumours using sound in the MHz range.

ultrastructure the fine structure of an organ, tissue, cell, or subcellular particle, especially that beyond the resolution of a light microscope but demonstrable with an electron microscope.

ultraviolet (radiation) *abbr.*: UV or uv; electromagnetic radiation of wavelengths in the range 13.6–400 nm (frequencies

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ultraviolet-endonuclease

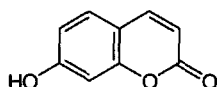
750 THz–22.1 PHz). This range overlaps the wavelengths of soft X-radiation and ends just across the short-wavelength limit of visible light. Ultraviolet radiation is often divided into **near ultraviolet** (200–400 nm) and **far ultraviolet** (or **vacuum ultraviolet**) (<200 nm); radiation in the far ultraviolet is absorbed by O₂ and requires the use of evacuated apparatus or flushing with N₂.

ultraviolet-endonuclease an enzyme involved in the repair of ultraviolet-irradiated DNA in microorganisms. *See also* **deoxyribonuclease** (pyrimidine dimer), **uvr**.

ultraviolet reactivation *see* **W reactivation**.

Ultrogel the proprietary name for a range of bead-form rigid support materials used in **gel filtration**, **affinity chromatography**, and related procedures. It consists of a lattice of polyacrylamide in a matrix of agarose gel.

umbelliferone 7-hydroxycoumarin; a common component of plant extracts, used as a fluorescent pH indicator and as a fluorescent label.



UMP *abbr. for* uridine monophosphate; i.e. uridine phosphate.

UMP synthase a loose term applied to the combination of enzymes that form uridine 5'-phosphate from orotate: these are **orotate phosphoribosyltransferase** (forming orotidylic acid) and **orotidine 5'-phosphate decarboxylase**. *See also* **uridine-5'-monophosphate synthase**.

umu bacterial genes encoding proteins that are involved in the error-prone **SOS repair** process, probably by allowing bypass synthesis across a damaged template. Protein examples from *Escherichia coli*: UmuC: database code UMUC_ECOLI, 422 amino acids (47.68 kDa); UmuD: database code UMUD_ECOLI, 139 amino acids (15.05 kDa).

un+ prefix **1** (used with adjectives, adverbs, and nouns) not; opposite of; contrary to. **2** (used with verbs) reversal of an action or state; release; removal from.

unary operator an arithmetic or logical operator (e.g. NOT) that operates on only one variable. *Compare* **binary operator**.

unc-86 a gene of *Caenorhabditis elegans* that encodes a transcription factor containing the **POU domain** and contributes to the naming of the domain. Example, Unc-86: database code UN86_CAEEL, 467 amino acids (52.31 kDa).

uncoating enzyme (sometimes) an enzyme that catalyses the removal of the protein coat from a virus.

uncoded amino acid an alternative term for **noncoded amino acid**.

uncompensated acidosis an **acidosis** in which there is a fall of the blood pH, due to the inadequacy of the physiological compensating mechanisms.

uncompensated alkalosis an **alkalosis** in which there is a rise of the blood pH, due to the inadequacy of the physiological compensating mechanisms.

uncompetitive inhibition the inhibition of the activity of an enzyme system when the inhibitor does not combine with the free enzyme, but only with one of the enzyme–substrate intermediates.

uncoupled **1** disconnected; unfastened. **2** (in *biochemistry*) describing an organism, tissue, cell, or mitochondrion in which phosphorylation is dissociated from oxidation in the **respiratory chain**. *See also* **uncoupling**.

uncoupler an alternative name for uncoupling agent (*see* **uncoupling**).

uncoupling (in *biochemistry*) the dissociation of phosphorylation from oxidation in the **respiratory chain**, so that electron transport proceeds without the esterification of phosphate and

without the need for inorganic phosphate. An **uncoupling agent** (or **uncoupler**) is a substance that causes uncoupling of phosphorylation from electron transport at one or more points in the electron-transport chain. A classical example is **dinitrophenol**. Uncoupling agents probably act by causing leakage of protons across the mitochondrial inner membrane thereby destroying the proton gradient that drives oxidative phosphorylation. *See also* **chemiosmotic coupling hypothesis**.

uncoupling protein *see* **brown fat uncoupling protein**.

undec+ or **undeca+** prefix denoting eleven.

undecaprenol or **bactoprenol** a polyunsaturated long-chain (C₅₅) primary alcohol containing 11 **prenyl** units.

undecaprenyl phosphate or **undecaprenol phosphate** a very hydrophobic phosphate ester of **undecaprenol**. It acts as a membrane-soluble carrier to which intermediates are combined in the course of the biosynthesis of bacterial cell-envelope components, e.g. peptidoglycan and lipopolysaccharide. *Other names*: bactoprenyl phosphate; carrier lipid; C₅₅ carrier lipid; C₅₅ lipid carrier.

underivatized or **underivatized** (of a substance) in the naturally occurring state.

undulin a noncollagenous glycoprotein of the interstitial extracellular matrix, associated with collagen. Isolated from skin or placenta, it is found mainly between densely packed mature collagen fibrils as bundles of uniform wavy fibres. It is a member of the fibronectin–tenascin family, and is related to collagen type XIV. It is associated with dense collagen matrices in soft tissues and is likely to be involved in the supramolecular organization of interstitial collagens. Example (sequence fragments) from human, undulin 2: database code B40970, 445 amino acids (49.06 kDa); the sequence shows homology with fibronectin type III.

ung *see* **uracil-DNA N-glycosylase**.

Uni *see* **reactancy of enzymes**.

uni+ *comb. form* consisting of, having, or relating to one.

unified atomic mass unit or (formerly) **atomic mass unit** (*abbr.*: amu or a.m.u.) *symbol*: u; a non-SI unit of mass equal to one-twelfth of the mass of one atom of carbon-12; i.e.

$$1 \text{ u} = m_{\text{a}}(^{12}\text{C})/12 \approx 1.660\,540 \times 10^{-27} \text{ kg}.$$

It is identical with **dalton**.

uniformly labelled *symbol*: U; describing a **labelled compound** in which the label is distributed in a statistically uniform, or nearly uniform, manner between all the possible positions in the molecule.

unimolecular reaction *see* **molecularity**.

uniparental disomy the inheritance of two copies of a chromosome from one parent.

uniport the process of noncoupled solute translocation or **facilitated diffusion**. The term includes reactions in which a single solute equilibrates across an osmotic barrier, normally mediated by a carrier protein known as a **uniporter**; e.g. the glucose-transport protein of mammalian erythrocytes. *Compare* **antiport**, **symport**.

unique (sequence) DNA or **single-copy DNA** any unique and specifically ordered nucleotide sequence in DNA that occurs only once in a genome. *Compare* **repetitive sequences**.

unit cell the smallest portion of a crystal that, by translation in three dimensions, is able to account for the structure of the entire crystal. The array of unit cells, arranged along parallel straight lines, makes up the **crystal lattice**.

unit membrane or **double membrane** or **triple membrane** any membrane, irrespective of cellular location, that on staining with osmium or other heavy metals appears in section in the electron microscope as a pair of parallel dark lines separated by a less dense layer, the whole being 6–10 nm thick. Unit membranes include the plasma membrane and the membranes of organelles such as mitochondria, nuclei, and endoplasmic reticulum.

unit of replication *see* **replicon**.

unit of transcription *see* **transcriptional unit**.

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univalent

univalent **1** or **monovalent** (of a chemical species) having a **valency** of one. **2** an unpaired chromosome seen during meiosis when bivalents are also present. Such a chromosome either lacks a homologue or results from **asynapsis**. —**univalency** *n*.

universal buffer mixture any solution containing several buffering systems chosen so as to provide a relatively high buffer capacity over a wide pH range.

universal donor an individual with O-type blood; such blood may be transfused into any recipient. See **ABO system**.

universal recipient an individual with AB-type blood; such individuals may be transfused with blood from any donor. See **ABO system**.

UNIX an **operating system** implemented on many current workstations and large computers.

unmediated transport or **nonmediated transport** the movement of a solute across a barrier by **diffusion** (def. 1).

unpaired electron an electron in an atom or molecule whose spin is not paired with the oppositely directed spin of another electron in the same atom or molecule. Only systems containing unpaired electrons will give an electron spin resonance signal. See **electron spin resonance spectroscopy**.

unsaponifiable lipid or **unsaponifiable material** part of a lipid sample that is not solubilized by **saponification**. It consists principally of steroids and terpenes.

unsaturated **1** (of a solution) able to dissolve more of the solute in question. **2** (of an organic compound) containing double bonds; see also **polyunsaturated**. **3** (of any chemical system, e.g. an enzyme or antibody) not fully saturated with ligand (e.g. substrate or antigen).

unsaturated-fatty-acid oxidation the degradation of unsaturated fatty acids to CO₂ and H₂O; this is achieved in large part by enzymes of the **beta-oxidation system**, but at certain points enzymes specific for unsaturated fatty acids are required. The first of these occurs when beta-oxidation reactions have removed carbons to the point at which the resulting acyl-CoA molecule has a Δ³-*cis* double bond; this is not a substrate for **enoyl-CoA hydratase**, but an **enoyl-CoA isomerase** (see **dodecenoyl-CoA Δ-isomerase**) converts it to a substrate with a Δ²-*trans* double bond. Another problem occurs when an acyl-CoA with a Δ⁴-*cis* double bond is produced, which again is not a substrate for **enoyl-CoA hydratase**, but the problem is solved by the action of **2,4-dienoyl-CoA reductase**, followed by **enoyl-CoA isomerase**; this combination yields the substrate with a Δ²-*trans* double bond.

untranscribed spacer a region of genomic DNA, lying between genes or groups of genes, that is not transcribed. See **spacer** (def. 1).

untranslated region any part of an mRNA molecule not coding for a protein; e.g. in eukaryotes the **poly(A) tail**.

unwinding protein any of the proteins concerned in DNA replication that have a higher affinity for single-stranded DNA than for double-stranded DNA and act to hold the strands apart during replication.

UPA *abbr.* for urokinase-type plasminogen activator; see **plasminogen activator**.

U5'pp1Gal *symbol* for uridinediphosphogalactose (alternative to **UrdPPGal** or **Urd-5'PP-Gal**).

U5'pp1Glc *symbol* for uridinediphosphoglucose (alternative to **UrdPPGlc** or **Urd-5'PP-Glc**).

up-promoter a mutation that increases **promoter** strength in the expression of a gene.

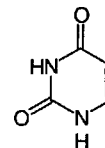
upsilon *symbol*: υ (lower case) or Υ (upper case); the twentieth letter of the Greek alphabet.

upstream in or towards positions in a DNA molecule lying in the 5' direction relative to the start of transcription of a gene.

Ur or **UR** *abbr.* for uridine (the symbol **Urd** is recommended).

Ura *symbol* for a residue of the pyrimidine base **uracil**.

uracil *symbol*: **U** or **Ura**; 2,4-dioxypyrimidine; 2,4-pyrimidinediol; one of the pyrimidine bases occurring in RNA (but not in DNA).



uracil

uracil-DNA N-glycosylase an enzyme (sub-subclass EC 3.2.2) that removes from DNA any uracil resulting from misincorporation of dUTP that has escaped the action of dUTPase (EC 3.6.1.23, dUTP pyrophosphatase). The enzyme hydrolyses the bond at C-1 of the deoxyribose to yield free uracil and DNA containing an apyrimidinic site; another enzyme, apyrimidine endonuclease, recognizes this site and cleaves the phosphodiester bond on the 5' side of the deoxyribose moiety. Uracil-DNA N-glycosylase may represent the evolutionary basis for the presence of T (not U) in DNA. Cytosine deamination leads to U residues, which would generate transition mutations on replication. In *Escherichia coli* the enzyme is the product of the *ung* gene and mutations in this can be used in certain forced site-directed mutagenesis protocols. The enzyme is a monomer, and is highly conserved in all species. Example, the product of the *ung* gene: database code UNG_ECOLI, 228 amino acids (25.53 kDa).

urate oxidase EC 1.7.3.3; *systematic name*: urate:oxygen oxidoreductase; *other name*: uricase. An enzyme that catalyses the reaction: urate + O₂ → intermediate(s) that lead(s) to **allantoin**. It is a conserved (five motifs) peroxisomal enzyme and a copper protein. Some mammals (e.g. humans, gorillas, etc.) that lack the enzyme are prone to **gout**. Example from rat: database code URIC_RAT, 302 amino acids (34.76 kDa).

Urd *symbol* for a residue of the ribonucleoside **uridine** (alternative to **U**).

UrdP *symbol* for any uridine phosphate.

Urd2'P *symbol* for uridine 2'-phosphate.

Urd-2'3'-P *symbol* for uridine 2',3'-phosphate.

Urd3'P *symbol* for uridine 3'-phosphate.

Urd-3'5'-P *symbol* for uridine 3',5'-phosphate.

Urd5'P *symbol* for uridine 5'-phosphate.

Urd5'PP *symbol* for uridine 5'-diphosphate (alternative to **ppU**).

UrdPPGal or **Urd-5'PP-Gal** *symbol* for uridinediphosphogalactose (alternatives to **U5'pp1Gal**).

UrdPPGlc or **Urd-5'PP-Glc** *symbol* for uridinediphosphoglucose (alternatives to **U5'pp1Glc**).

Urd5'PPP *symbol* for uridine 5'-triphosphate (alternative to **pppU**).

urea the water-soluble compound H₂N-CO-NH₂, produced in the liver via the **ornithine-urea cycle**. It is the main nitrogen-containing (urinary) excretion product in **ureotelic** animals. It is used as source of nonprotein nitrogen in ruminant livestock feeds.

urea cycle see **ornithine-urea cycle**.

urease EC 3.5.1.5; *systematic name*: urea amidohydrolase; a nickel-protein enzyme that catalyses the hydrolysis of urea to carbon dioxide and ammonia. A carbamoylated lysine provides an oxygen ligand to each nickel, thus explaining a requirement for CO₂ as an activator of urease apoenzyme. Example from *Proteus vulgaris*; a decamer of three subunits (probably in two αβ₂γ₂ complexes); α subunit: database code URE1_PROVU, 567 amino acids (60.86 kDa); β subunit: database code URE2_PROVU, 108 amino acids (12.14 kDa); γ subunit: database code URE3_PROVU, 100 amino acids (10.94 kDa). The 3-D structure is known for the *Klebsiella aerogenes* enzyme.

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ureido

ureido the $\text{NH}_2\text{-CO-NH-}$ group, derived from urea by loss of a hydrogen atom.

ureidosuccinase EC 3.5.1.7; an enzyme that hydrolyses *N*-carbamoyl-L-aspartate (ureidosuccinate) to L-aspartate, CO_2 , and NH_3 .

ureogenic describing an organism (or species) that possesses the full complement of **ornithine-urea cycle** enzymes, and that potentially, but not necessarily, can biosynthesize significant amounts of urea.

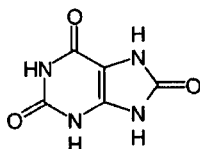
ureosmotic describing an organism (or species) that adjusts urea production or retention in a manner that assists the maintenance of osmotic equilibrium, rather than as a means of disposing of surplus nitrogen.

ureotelic describing an organism (or species) in which **urea** is the principal end product of the degradation of nitrogen-containing compounds. Mammals and marine fishes are typically ureotelic.

Urey, Harold Clayton (1893–1981), US physical chemist; Nobel Laureate in Chemistry (1934) 'for his discovery of heavy hydrogen'.

+uria *comb. form* signifying (in relation to a substance specified) presence unusually, or at a supranormal concentration, in urine. Hence also used to denote a morbid condition so marked, e.g. **phenylketonuria**.

uric acid 2,6,8-trioxypurine; purine-2,6,8-triol; the end product of purine metabolism in certain mammals (including humans) and the main nitrogenous excretory product in **uricotelic** animals. It is formed from xanthine by **xanthine oxidase**. Uric acid is very sparingly soluble in water, and crystalline deposits of monosodium urate occur in the tissues of humans affected by gout.

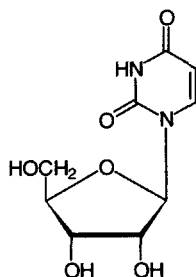


uricase an alternative name for **urate oxidase**.

uricosuric (of a drug) tending to promote urinary excretion of uric acid.

uricotelic describing an organism (or species) in which uric acid is the principal end product of the degradation of nitrogen-containing compounds. Uricotelic animals include birds, terrestrial reptiles, and insects. The low solubility of uric acid in water means that it is present as a solid in the excreta of these animals, enabling excretion with little water loss.

uridine *symbol*: U or Urd; uracil riboside; 1- β -D-ribofuranosyluracil; a ribonucleoside very widely distributed but occurring almost entirely as phosphoric esters in ribonucleotides and ribonucleic acids. *Compare pseudouridine*.

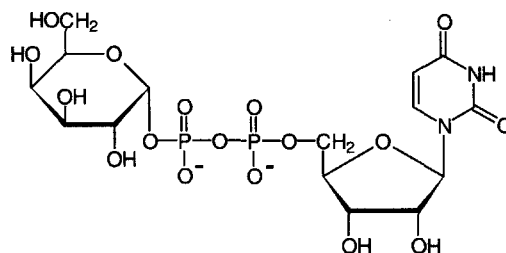


uridine 2',3'-(cyclic)phosphate *see* **uridine phosphate**.

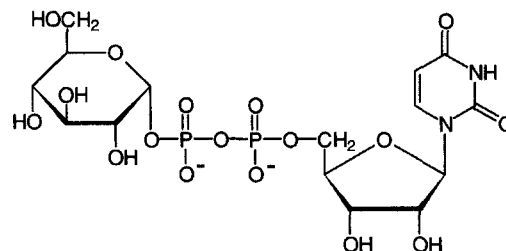
uridine 3',5'-(cyclic)phosphate or **uridine 3',5'-cyclophosphate** *see* **uridine 3',5'-phosphate**.

uridine 5'-diphosphate *symbol*: Urd5'PP or ppU; *the recommended name for* uridine diphosphate (*abbr.*: UDP); 5'-diphosphouridine; 5'-diphosphouridine; 5'-uridylyl phosphate; uridine 5'-(trihydrogen diphosphate); a universally distributed nucleotide that occurs both in the free state and as a component of a number of important **uridinediphosphosugars**. It is the intermediate in the formation of uridine 5'-triphosphate, UTP, from uridine 5'-phosphate, UMP.

uridine diphosphate galactose *symbol*: Urd5'-PP-Gal or U5'pp1Gal or UDPGal; uridine 5'-(α -D-galactopyranosyl diphosphate); *recommended name*: uridine(5')diphospho(1)- α -D-galactose; an important intermediate in the metabolism of D-galactose and in the formation of numerous derivatives. It is formed by the reaction of galactose 1-phosphate with UDP-glucose (catalysed by **uridylyl transferase**), or with UTP (catalysed by **UTP-hexose-1-phosphate uridylyltransferase**).



uridine diphosphate glucose *symbol*: Urd5'-PP-Glc or U5'pp1Glc or UDPGlc or UDPG; uridine 5'-(α -D-glucopyranosyl diphosphate); *recommended name*: uridine(5')diphospho(1)- α -D-glucose; an important intermediate in the formation of other sugars from glucose. It is formed by the reaction of glucose 1-phosphate with UTP, catalysed by **UTP-glucose-1-phosphate uridylyltransferase**.



uridinediphosphosugar *abbr.*: UDPsugar; *the recommended name for* **uridine diphosphate sugar**; any of many **nucleoside-diphosphosugars** in which the distal phosphoric residue of uridine 5'-diphosphate is in glycosidic linkage with a monosaccharide (or a derivative of a monosaccharide). UDPsugars are particularly important as coenzymes for many types of reaction involving the transfer of monosaccharide residues in the biosynthesis of oligosaccharides and polysaccharides (especially in plants) and in the formation of the oligosaccharide moieties of **glycoconjugates**. The main UDPsugars in this category are: UDP-L-arabinose, UDP-D-galactose (*see* **uridine diphosphate galactose**), UDP-D-galacturonate, UDP-D-glucose (*see* **uridine diphosphate glucose**), UDP-D-glucuronate, and UDP-D-xylose, all of which are (directly or indirectly) inter-



convertible where they occur together; UDP-L-iduronate, which is formed by epimerization of UDP-D-glucuronate; UDP-N-acetyl-D-galactosamine and UDP-N-acetyl-D-glucosamine, which are interconvertible; and UDP-N-acetylmuramate, which is synthesized from UDP-N-acetyl-D-glucosamine. In addition, certain UDPsugars, particularly UDP-D-glucose and UDP-D-glucuronate, are coenzymes for the formation in animals or plants of glycosides with a variety of aglycons such as bilirubin, phenols, plant pigments and sterols, and steroid hormones.

uridine kinase EC 2.7.1.48; *other name*: uridine monophosphokinase; an enzyme that catalyses the phosphorylation by ATP of uridine to uridine 5'-phosphate with formation of ADP. Example from yeast: database code URK1_YEAST, 501 amino acids (56.23 kDa).

uridine monophosphate *abbr.*: UMP; *an alternative name for* any uridine phosphate, but in particular for uridine 5'-phosphate, especially when its distinction from uridine (5')-diphosphate and uridine (5')-triphosphate requires emphasis.

uridine 5'-monophosphate synthase *abbr.*: UMP synthase; a multifunctional enzyme that in higher eukaryotes carries out the following two reactions: (1) orotate phosphoribosyltransferase, EC 2.4.2.10; *other names*: orotidylic acid phosphorylase, orotidine-5'-phosphate pyrophosphorylase; *systematic name*: orotidine-5'-phosphate:pyrophosphate phospho- α -D-ribosyltransferase; a reaction of the pathway for *de novo* biosynthesis of pyrimidine nucleotides. It catalyses a reaction between orotate and 5-phospho- α -D-ribose 1-diphosphate to form orotidine 5'-phosphate and pyrophosphate. (2) orotidine-5'-phosphate decarboxylase, EC 4.1.1.23; *systematic name*: orotidine-5'-phosphate carboxy-lyase; a decarboxylation reaction that forms uridine 5'-monophosphate and CO₂ from orotidine 5'-phosphate. Example from *Bos taurus*: database code PYR5_BOVIN, 480 amino acids (52.23 kDa).

uridine phosphate *symbol*: UrdP; uridine monophosphate (*abbr.*: UMP); any phosphoric monoester or diester of uridine. There are three monoesters – uridine 2'-phosphate, uridine 3'-phosphate, and **uridine 5'-phosphate** – and two diesters – uridine 2',3'-phosphate and **uridine 3',5'-phosphate** – although uridine 5'-phosphate is the ester commonly denoted (the locant being omitted if no ambiguity may arise). Uridine 2',3'-phosphate (*symbol*: Urd-2',3'-P), *also named* uridine 2',3'-(cyclic)phosphate or cyclic uridine 2',3'-monophosphate (*abbr.*: 2',3'-cyclic UMP), is formed as an intermediate during the alkaline hydrolysis of ribonucleic acid. This diester then readily hydrolyses to a mixture of the two monoesters uridine 2'-phosphate (*symbol*: Urd2'P), *also named* uridine 2'-monophosphate (*abbr.*: 2'UMP) or 2'-uridylic acid or uridylic acid a, and uridine 3'-phosphate (*symbol*: Urd3'P), *also named* uridine 3'-monophosphate (*abbr.*: 3'UMP) or 3'-uridylic acid or uridylic acid b.

uridine 2'-phosphate *see* uridine phosphate.

uridine 2',3'-phosphate *see* uridine phosphate.

uridine 3'-phosphate *see* uridine phosphate.

uridine 3',5'-phosphate *symbol*: Urd-3',5'-P; *the recommended name for* cyclic uridine 3',5'-monophosphate (*abbr.*: 3',5'-cyclic UMP or cyclic UMP or cUMP); uridine 3',5'-cyclophosphate; uridine 3',5'-(cyclic)phosphate; a monophosphoric diester of uridine found in practically all mammalian tissues. It is a cyclic nucleotide similar in structure and metabolism to **adenosine 3',5'-phosphate**, cyclic AMP. It has been reported to inhibit the growth of certain transplantable tumours.

uridine 5'-phosphate or 5'-uridylic acid or 5'-phosphouridine or 5'-O-phosphonouridine *symbol*: Urd5'P; *alternative recommended names for* uridine 5'-monophosphate (*abbr.*: 5'UMP); uridine 5'-(dihydrogen phosphate); uridine (mono)ribonucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation). It is biosynthesized from aspartate, the last stage of the reaction sequence being the decarboxylation of orotidine 5'-phosphate.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

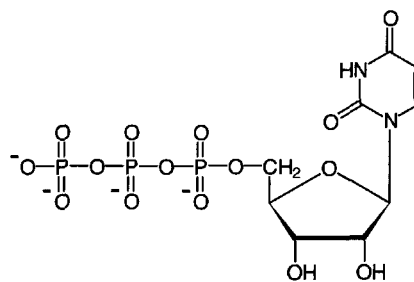
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

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orotidylic acid. In addition, resynthesis of UMP from free uridine formed by degradation of ribonucleic acid can be effected via a **salvage pathway**. UMP is a key intermediate in the formation of uridine 5'-triphosphate, UTP, and hence in the biosynthesis of other pyrimidine nucleotides.

uridine phosphorylase *see* **pyrimidine-nucleoside phosphorylase**.

uridine 5'-triphosphate *symbol*: Urd5'PPP or pppU; *the recommended name for* uridine triphosphate (*abbr.*: UTP); 5'-triphosphouridine; 5'-uridylyl diphosphate; uridine 5'-(tetrahydrogen triphosphate); an important nucleotide present in all cells. It is formed from uridine 5'-phosphate via uridine 5'-diphosphate by the successive action of nucleoside-phosphate kinase (EC 2.7.4.4) and nucleoside-diphosphate kinase (EC 2.7.4.6), which transfer in turn the terminal phosphoric residues from two molecules of adenosine 5'-triphosphate. UTP is the starting point for the biosynthesis, first of cytidine triphosphate, next of cytidine diphosphate, and then, in a succession of further reactions, of various deoxyribonucleotides of cytosine, uracil, and thymine. It is important also for the formation of uridinediphosphogalactose and uridinediphosphoglucose through the action of certain **uridylyltransferases**, and thence of a number of other uridinediphosphosugars.



uridyl 1 any chemical group formed by the loss of a 2'-, a 3'-, or a 5'-hydroxyl group from the ribose moiety of uridine. 2 a *misnomer for* **uridylyl**.

uridylylate 1 either the monoanion or the dianion of uridylic acid. 2 any mixture of uridylic acid and its anions. 3 any salt or ester of uridylic acid.

uridylic acid *the trivial name for* any phosphoric monoester of uridine. The position of the phosphoric residue on the ribose moiety of a given ester may be specified by a prefixed locant; *see* **uridine phosphate**. However, 5'-uridylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-Uridylic acid is also an alternative recommended name for **uridine 5'-phosphate**.

uridylyl transferase *another name for* **UDPglucose-hexose-1-phosphate uridylyltransferase**. This is a *misnomer*; compare **uridyl** with **uridylyl**.

uridylyl the uridine[mono]phospho group; the acyl group derived from uridylic acid. *Distinguish from* **uridyl**.

uridylylate to introduce uridylyl groups into a compound (e.g. a protein, a ribonucleic acid, or a sugar phosphate), generally through the action of a uridylyltransferase. —**uridylylated** *adj.*; **uridylylation** *n*.

uridylyl-removing enzyme *the provisional name for* a putative enzyme present in *Escherichia coli* that detaches uridylyl groups from uridylylated nitrogen regulatory protein P_{II} by hydrolysis, with formation of uridylylate. *See also* **nitrogen regulatory protein, uridylyltransferase**.

uridylyltransferase EC 2.7.7.12; *recommended name*: UDPglucose-hexose-1-phosphate uridylyltransferase; *systematic name*: UDPglucose: α -D-galactose-1-phosphate uridylyltransferase; *other names*: uridylyl removing enzyme; P_{II} uridylyl-transferase. An enzyme that catalyses the reaction:



UDPglucose + α -D-galactose 1-phosphate =
 α -D-glucose 1-phosphate + UDPgalactose.

The *Escherichia coli* enzyme comprises 890 amino acids (102.31 kDa) and modifies the **nitrogen regulatory protein P_{II}**.

urinalysis the chemical and physical analysis of urine.

urinary of, or pertaining to, urine.

urinary stone protein *see* **osteopontin**.

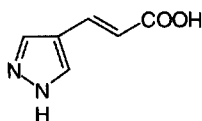
urine the aqueous excretion produced by the kidneys in higher animals. It serves as a vehicle for the elimination of certain metabolic waste products, e.g. **urea** and **uric acid**, and for the removal of excess ions and water. In mammals and fishes it is fluid, while in birds and reptiles it is semi-solid.

urinometer a **hydrometer** specifically designed and calibrated for measuring the specific gravity of (mammalian) urine.

urobilin or **urobilin IX α** a yellow coloured linear tetrapyrrole. It is formed in kidney and is an oxidation product of **urobilinogen**; the central methylene group linking pyrrole units has been converted to $-\text{CH}=\text{}$. Urobilin gives urine its characteristic yellow colour. *See also* **stercobilin**.

urobilinogen or **mesobilirubinogen** or **urobilinogen IX α** a colourless linear tetrapyrrole formed in heme catabolism. It arises by reduction of free **bilirubin** released from its glucuronide by intestinal bacterial hydrolases. The four pyrrole rings are linked by three methylene groups. *See also* **stercobilinogen**.

urocanoate 4-imidazoleacrylate; 3-(1*H*-imidazol-4-yl)propenoate: an intermediate in the degradation of L-histidine, from which it is formed through the action of **histidine ammonia-lyase** by elimination of ammonium.



urocanic acid

urodilatin a 32-amino-acid peptide, isolated from human urine that is identical in sequence to human **natriuretic peptide** (ANP) but with an N-terminal Thr-Ala-Pro-Arg tetrapeptide extension. It has actions similar to ANP.

uroflavin or **uroflavine** a former name for **riboflavin**.

urogastrone a name given (by analogy with **enterogastrone**) to a proteinaceous agent present in human urine that potently inhibits gastric secretion. It is now believed to be identical or nearly identical with human **epidermal growth factor**.

urokinase a proteolytic enzyme found in urine, blood, and kidney cells, that converts **plasminogen** to **plasmin**. *See* **plasminogen activator**.

uromodulin another name for **Tamm-Horsfall glycoprotein**.

uronate 1 the anion of uronic acid. 2 any mixture of a uronic acid and its anion. 3 any salt or ester of a uronic acid.

uronic acid any monocarboxylic acid formally derived by oxidizing to a carboxyl group the terminal hydroxymethylene group either of an **aldose** having four or more carbon atoms in the molecule, or of any glycoside derived from such an aldose. The carbon atom of the (potential) aldehydic carbonyl group is numbered C-1.

uropontin *see* **osteopontin**.

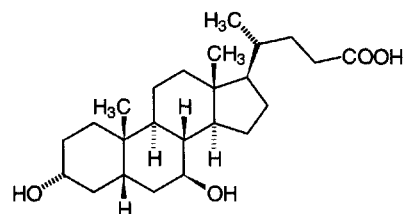
uroporphyrin-III C-methyltransferase EC 2.1.1.107; *systematic name*: S-adenosyl-L-methionine:uroporphyrin-III C-methyltransferase (*abbr.*: SUMT); *other names*: urogen-III methylase; uroporphyrin-III methylase. An enzyme that catalyses a reaction between two molecules of S-adenosyl-L-methionine and uroporphyrin III to form two molecules of S-adenosyl-L-homocysteine and sirohydrochlorin. It is a compo-

nent of **siroheme synthase**, and effects a step in the biosynthesis of siroheme and cobalamin.

uroporphyrinogen decarboxylase EC 4.1.1.37; *systematic name*: uroporphyrinogen-III carboxy-lyase. An enzyme that catalyses the decarboxylation of uroporphyrinogen-III to coproporphyrinogen with loss of four molecules of CO₂; this is a step in the pathway of heme synthesis, whereby the four carboxymethyl groups of uroporphyrinogen are converted to methyl groups. Example from yeast: database code DCUP_YEAST, 362 amino acids (41.30 kDa).

uroporphyrinogen-III synthase EC 4.2.1.75; *systematic name*: hydroxymethylbilane hydro-lyase (cyclizing); *other names*: uroporphyrinogen-III cosynthetase; uroporphyrinogen-III cosynthase. An enzyme that catalyses the conversion of hydroxymethylbilane to uroporphyrinogen-III and H₂O. Example from *Escherichia coli*: database code HEM4_ECOLI, 246 amino acids (27.77 kDa).

ursodeoxycholate 3 α ,7 β -dihydroxy-5 β -cholan-24-oate; an epimer of **chenodeoxycholate** found in the bile of bears (hence the name, from Ursidae), as a conjugate with taurine. Used therapeutically, it prevents the synthesis and absorption of cholesterol and can lead to the dissolution of gallstones.



U-snRNA *abbr. for* uridyate-rich small nuclear RNA; a family of snRNA molecules, often designated U1 to U6 snRNA, that contain 100–215 nucleotides and a 'capped' 5' end. They usually occur in combination with proteins as small nuclear ribonucleoprotein particles. Their function is unclear, but may be related to splicing heterogeneous nuclear RNA (hnRNA) to mature mRNA.

USP *abbr. for* United States Pharmacopeia.

USPHS *abbr. for* United States Public Health Service.

uterine metalloendopeptidase *see* **matrilysin**.

uterglobin or **blastokinin** a protein regulator of progesterone concentrations reaching the blastocyst and a potent inhibitor of phospholipase A₂. It is a disulfide-linked homodimer. Example (precursor) from *Oryctolagus cuniculus*: database code UTER_RABIT, 91 amino acids (9.97 kDa).

uterus (*pl.* **uteri**) a muscular part of the reproductive tract of female mammals (except Monotremata) that contains the developing embryo and fetus during pregnancy. —*uterine adj.*

UTP *abbr. for* uridine (5')-triphosphate.

UTP-glucose-1-phosphate uridylyltransferase EC 2.7.7.9; *systematic name*: UTP: α -D-glucose-1-phosphate uridylyltransferase; *other names*: UDPglucose pyrophosphorylase; glucose-1-phosphate uridylyltransferase; one of the enzymes that can form uridinediphosphoglucose. It catalyses a reaction between UTP and α -D-glucose 1-phosphate to form UDPglucose and pyrophosphate. Example from *Escherichia coli*: database code GALU_ECOLI, 302 amino-acid residues (32.90 kDa).

UTP-hexose-1-phosphate uridylyltransferase EC 2.7.7.10; *systematic name*: UTP: α -D-hexose-1-phosphate uridylyltransferase; one of the enzymes that can form uridinediphosphogalactose. It catalyses a reaction between UTP and α -D-galactose 1-phosphate to form UDPgalactose and pyrophosphate. α -D-Glucose 1-phosphate can also act (but more slowly) as an acceptor, forming uridinediphosphoglucose. Example from human: database code GAL7_HUMAN, 379

UTP-xylose-1-phosphate uridylyltransferase

amino acids (43.16 kDa). The absence of the enzyme in infancy causes a mild form of **galactosemia**.

UTP-xylose-1-phosphate uridylyltransferase EC 2.7.7.11; *other name*: xylose-1-phosphate uridylyltransferase; an enzyme that catalyses a reaction between UTP and α -D-xylose 1-phosphate to form UDPxylose and pyrophosphate.

utrophin or **dystrophin-related protein** (*abbr.*: DRP) a protein that is highly homologous to **dystrophin** and found in normal adult human muscle, primarily at the neuromuscular junction. In the absence of dystrophin in Duchenne muscular dystrophy it is also present in the sarcolemma. It is an actin-binding protein related to α -actinin. Both gene and protein sequences resemble those for dystrophin. Overexpression of utrophin in a dystrophin-deficient (*mdx*) mouse demonstrates that it can functionally replace dystrophin; it has been suggested that this could offer a therapeutic strategy for Duchenne muscular dystrophy. Example from human: database code S28381, 3433 amino acids (394.05 kDa).

UUA a codon in mRNA for L-leucine.

UUC a codon in mRNA for L-phenylalanine.

UUG a codon in mRNA for L-leucine.

UUU a codon in mRNA for L-phenylalanine.

UV or **uv** *abbr.* for ultraviolet (radiation).

uvomorulin or **epithelial cadherin** or **E-cadherin** a type I membrane protein and **cadherin** of non-neural epithelial tissues.

uvr any of four genes in *Escherichia coli* that are induced by the **SOS repair** system and whose products are involved in the repair of damage to DNA caused by UV light and DNA-reactive chemicals. Three of these genes, *uvrA*, *uvrB*, and *uvrC*, encode the three subunits of the ABC excision nuclease; this recognizes the structural distortion created by a **thymine dimer** and cleaves the damaged strand at two sites, leaving a gap that is subsequently filled by polymerase and ligase action. Because the enzyme cuts at two sites it is not a classical endonuclease, but is referred to as an **excinuclease**. The fourth gene, *uvrD*, encodes a DNA helicase with ATPase activity, similar to the *rep* (see **RepA**) gene product; it promotes the ATP-dependent unwinding of the DNA duplex during repair; example, database code UVRD_ECOLI, 720 amino acids (81.90 kDa).

UWGCG *abbr.* for University of Wisconsin Genetics Computer Group. See **GCG**.

Vv

v *symbol for* 1 specific volume. 2 volume (alternative to *V*). 3 velocity of reaction (i.e. **rate of reaction**); a subscript (in small capital letters) may be added (e.g. v_A) to denote rate of consumption, or rate of formation, of a specified reactant (e.g. *A*). 4 initial velocity of an enzymic reaction; *see also* v_0 . 5 (bold italic) velocity (alternative to *u*, *w*, or *c*).

v_0 *symbol for* initial velocity (i.e. initial rate) of an enzymic reaction. The subscript may be omitted when rates at other times are not in question.

v_i *symbol for* chemical flux through the (enzymic) step with rate constant k_i . Similarly, flux through steps with rate constants k_{-i} or k_{ij} are denoted by v_{-i} and v_{ij} respectively.

V_{max} *symbol for* the true maximum value of the **rate of reaction**. *Compare* V_{max} .

V *symbol for* 1 volt. 2 a residue of the α -amino acid L-valine (alternative to Val). 3 vanadium. 4 variable region (of an immunoglobulin chain).

V *symbol for* 1 electric potential. 2 potential energy. 3 volume (alternative to *v*). 4 maximum velocity (i.e. **limiting rate**, or maximum rate) of an enzymic reaction (alternative to V_{max}).

V_m *symbol for* molar volume.

V_m *symbol for* hold-up volume.

V_{max} *symbol for* maximum velocity (i.e. **limiting rate**) of an enzyme reaction (alternative to *V*).

Vac *symbol for* vaccenic acid.

vaccenate 1 *numerical symbol*: 18:1(11); *the trivial name for* either of the isomers (*cis*-vaccenate and *trans*-vaccenate) of octadec-11-enoate, $\text{CH}_3\text{--}[\text{CH}_2]_5\text{--CH=CH--}[\text{CH}_2]_9\text{--COO}^-$, the anions derived from *cis*-vaccenic acid, (*Z*)-octadec-11-enoic acid, and *trans*-vaccenic acid, (*E*)-octadec-11-enoic acid, a stereoisomeric pair of monounsaturated straight-chain higher fatty acids. *See also* **octadecenoic acid**. 2 any mixture of either *cis*- or *trans*-vaccenic acid and its corresponding anion. 3 any salt or ester of either isomer of vaccenic acid.

vaccenoyl *symbol*: Vac; *the trivial name for* either of the isomers (*cis*-vaccenoyl and *trans*-vaccenoyl) of octadec-11-enoyl, $\text{CH}_3\text{--}[\text{CH}_2]_5\text{--CH=CH--}[\text{CH}_2]_9\text{--CO-}$, the acyl groups derived from *cis*-vaccenic acid, (*Z*)-octadec-11-enoic acid, and *trans*-vaccenic acid, (*E*)-octadec-11-enoic acid. The *cis*-vaccenoyl group is a major fatty-acyl component of many bacterial lipids and a constituent of the oils of marine organisms. The *trans*-vaccenoyl group is present in a minor proportion of the acylglycerols in the body and milk fats of ruminants.

vaccination the process of administering a **vaccine** to produce immunity.

vaccine any preparation of immunogenic material suitable for the stimulation of active immunity in animals without inducing disease. Vaccines may be based on dead or attenuated microorganisms; altered toxins (toxoids); or viruses.

vaccinia virus a DNA virus and the type species of the genus *Orthopoxvirus* (or 'vaccinia subgroup'), which also includes the viruses responsible for cowpox, buffalopox, catpox, mousepox, and (formerly) smallpox (variola virus). Vaccinia infection usually causes only mild symptoms. Because of its serological similarity to the smallpox virus, vaccinia virus was formerly used as the basis of a live vaccine to protect humans against smallpox. Viral antigens are often on the surface of virus particles, and some of these antigens can be engineered into the coat of vaccinia. For example, engineered vaccinas with surface antigens of hepatitis B, influenza virus, and vesicular stomatitis virus (which kills cattle, horses, and pigs) have produced useful vaccines in animal tests. However, the known adverse reactions with vaccinia-virus vaccines are well documented and their incidence and severity must be carefully weighed.

vacuolar apparatus the cellular digestive system, consisting

of **lysosomes** and closely related, hydrolase-free, vacuolar structures. It is concerned with the digestion of both endogenous and exogenous materials.

vacuolar ATPase an enzyme of the synaptic vesicle membrane that pumps protons into the vesicle as it loses acetylcholine during nerve transmission; subsequently, the hydrons are released as the vesicle is recharged with acetylcholine. The enzyme (at least 10 subunits) resembles **H⁺-transporting ATP synthase**, but functions as a proton pump. Example from bovine: database code VATA_BOVIN, 617 amino acids (68.31 kDa). The catalytic subunits are highly conserved; e.g. the enzyme from carrot comprises 623 amino acids (68.76 kDa).

vacuole a closed structure, found only in eukaryotic cells, that is completely surrounded by unit membrane and contains liquid material. *Compare* **vesicle**.

vacuum (*pl.* **vacuums** or **vacua**) a space containing a gas at low pressure; i.e. a space in which there are relatively few atoms or molecules. A **perfect vacuum** would contain no atoms or molecules, but this is unobtainable owing to the vapour pressure of materials containing any vacuum. Vacuums may be classified into **low** (or **soft**) **vacuum**, $>10^{-2}$ Pa; **high** (or **hard**) **vacuum**, 10^{-2} – 10^{-7} Pa; and **ultrahigh vacuum**, $<10^{-7}$ Pa.

vacuum evaporation a technique for depositing a thin film of some solid material on a surface by evaporating the material at a high temperature in a vacuum. Atoms leaving the hot material travel directly to the cool surface to form a coating without colliding with other molecules in the gas phase; they condense on the cool surface and build up a layer of solid. *See also* **shadow casting**.

vacuum ultraviolet *see* **ultraviolet radiation**.

vagusstoff *the German name for* 'active factor'. The existence of such a factor was first demonstrated by **Loewi** (1903); it was secreted into plasma after stimulation of the vago-sympathetic trunk, and was found to inhibit cardiac function. It was later identified by **Dale** as **acetylcholine**.

Val *symbol for* 1 the α -amino acid L-valine (alternative to *V*). 2 the (*cis*- or *trans*-) vaccenoyl group.

valency or (*esp.* **US**) **valence** 1 the numerical combining power of an atom (and thus of a chemical element), ion, or chemical group, equal to the number of hydrogen atoms that the atom, ion, or group could combine with or replace. *See also* **covallence**, **electrovalence**. 2 the number of antigenic determinants with which one antibody molecule, or fragment thereof, will combine.

valerate 1 *the trivial and preferred name for* pentanoate; the anion, $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--COO}^-$, derived from valeric acid (i.e. pentanoic acid). 2 any mixture of free valeric acid and its anion. 3 any salt or ester of valeric acid.

valeryl *symbol*: Vl; *the trivial and preferred name for* pentanoyl; the acyl group, $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--CO-}$, derived from valeric acid (i.e. pentanoic acid).

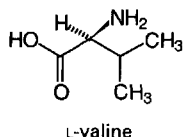
valinate 1 valine anion; the anion, $(\text{CH}_3)_2\text{CH--CH}(\text{NH}_2)\text{--COO}^-$, derived from **valine**. 2 any salt containing valine anion. 3 any ester of valine.

valine *the trivial name for* α -aminoisovaleric acid; α -amino- β -methylbutyric acid; 2-amino-3-methylbutanoic acid; $(\text{CH}_3)_2\text{CH--CH}(\text{NH}_2)\text{--COOH}$; a chiral α -amino acid. L-valine (*symbol*: V or Val), (*S*)-2-amino-3-methylbutanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: GUA, GUC, GUG, or GUU. In mammals it is an essential dietary amino acid and is glucogenic. Residues of D-valine (*symbol*: D-Val or DVal), (*R*)-2-amino-3-methylbutanoic acid, occur in a number of peptide antibiotics, e.g. several members of the actinomycin family, gramicidins A, B, C, and D, and valinomycin (the latter also contains residues of L-valine). (Illustrated overleaf).

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

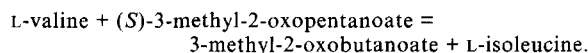
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valine-isoleucine aminotransferase *see* **valine-3-methyl-2-oxovalerate aminotransferase**.

valine-3-methyl-2-oxovalerate aminotransferase EC 2.6.1.32; *recommended name*: valine-3-methyl-2-oxovalerate transaminase; *other names*: valine-isoleucine transaminase; valine-isoleucine aminotransferase; an enzyme that catalyses the reaction:



See also **transamination**.

valinium valine cation; the cation $(\text{CH}_3)_2\text{CH}-\text{CH}(\text{NH}_3^+)-\text{COOH}$, derived from **valine**.

valino the alkylamino group, $(\text{CH}_3)_2\text{CH}-\text{CH}(\text{COOH})-\text{NH}-$, derived from **valine**.

valinomycin a cyclic 12-residue depsipeptide antibiotic that consists of three moieties, each having one molecule each of L-valine, D- α -hydroxyvaleric acid, D-valine, and L-lactic acid. The linkage from the D-valine carboxyl group is directly onto the α -carbon of L-lactic acid. Valinomycin is produced by *Streptomyces fulvissimus*. It is active especially against *Mycobacterium tuberculosis*. It can act as an ionophore, the molecule being folded to produce a hydrophobic surface with an interior that binds a univalent cation such as Rb^+ , Cs^+ , NH_4^+ , or, especially, K^+ . This property equips valinomycin to interfere with oxidative phosphorylation by rendering the mitochondrial membrane permeable to K^+ ions.

valium *see* **diazepam**.

valproic acid 2-propylpentanoic acid; $(\text{CH}_3-\text{CH}_2-\text{CH}_2)_2\text{CH}-\text{COOH}$; an antiepileptic, also used as the sodium salt, sodium valproate (*one proprietary name*: Epilim).

valyl the acyl group, $(\text{CH}_3)_2\text{CH}-\text{CH}(\text{NH}_2)-\text{CO}-$, derived from **valine**.

VAMP *abbr.* for vesicle-associated membrane protein; *another name* for **synaptobrevin**.

vancomycin a complex glycopeptide, $\text{C}_{66}\text{H}_{75}\text{Cl}_2\text{N}_9\text{O}_{24}$, from *Streptomyces orientalis*. It inhibits a specific step in the synthesis of the peptidoglycan layer in the Gram-positive bacterium *Staphylococcus aureus*. *See also* **vancosamine**.

vancosamine the aminodeoxysugar 3-amino-2,3,6-trideoxy-3-C-methyl-L-lyxo-hexose; the carbohydrate component of **vancomycin**.

van den Bergh reaction a colorimetric reaction for the determination of **bilirubin** that depends on the formation of a red azo dye by bilirubin and diazotized sulfanilic acid. A **direct van den Bergh reaction** in aqueous solution is given by bilirubin glucuronide, and an **indirect van den Bergh reaction** is given in alcoholic solution by free bilirubin. [After A. A. Hijmans van den Bergh (1869–1943), Dutch physician.]

van der Waals forces long-range forces between molecules or submolecular groups, effective from >50 nm to interatomic spacings. They are always present, albeit rather weak, but often compete with electrostatic forces. In general the van der Waals forces do not follow a simple power law; e.g. the force may be attractive at large separations but repulsive at small ones, or vice versa. The van der Waals free energy between two molecules or submolecular groups usually depends on their mutual orientation, i.e. these forces tend to align two molecules so as to minimize their free energy of interaction. [After Johannes Diderik van der Waals (1837–1923), Dutch physicist.]

van der Waals radius one half of the internuclear distance be-

tween atoms at equilibrium when the long-range attractive forces balance the short-range repulsive forces. It is larger than the atom's covalent radius and is about equal to the ionic radius of monovalent ions.

Van Dyke protein *an old name for neurophysin*.

Vane, (Sir) John Robert (1927–), British pharmacologist; Nobel Laureate in Physiology or Medicine (1982) jointly with S. K. Bergström and B. I. Samuelsson 'for their discoveries concerning prostaglandins and related biologically active substances'.

vanillylmandelate *see* **hydroxy-methoxymandelate**.

van Niel, Cornelius Bernardus (1897–1984), US microbiologist who inherited the traditions of the Delft school; he proposed the general equation for photosynthesis as



He was renowned for his summer course in microbiology at the Hopkins Marine Station, Pacific Grove, California.

Van Slyke apparatus 1 a volumetric apparatus formerly widely used for the determination of oxygen and carbon dioxide contents of blood samples. 2 an apparatus for the gasometric determination of aliphatic amino-nitrogen, e.g. in proteins, by measurement of the nitrogen gas evolved on reaction with nitrous acid. [After Donald Dexter Van Slyke (1883–1971), US biochemist.]

van't Hoff, Jacobus Henricus (1852–1911), Dutch physical chemist and stereochemist distinguished, *inter alia*, for postulating the tetrahedral disposition of the bonds to a singly bonded carbon atom, for being the first to apply the laws of thermodynamics to chemical reactions, and for showing that osmotic pressure varies directly with absolute (i.e. thermodynamic) temperature; Nobel Laureate in Chemistry (1901) 'in recognition of the extraordinary services he has rendered by the discovery of the laws of chemical dynamics and osmotic pressure in solutions'.

van't Hoff equation or **van't Hoff isochore** a relation describing the temperature variation of an equilibrium constant, K , of a reaction in terms of the change in heat content, $\Delta H'$, i.e. in terms of the heat of reaction at constant pressure. It is: $\text{dln}K/\text{d}T = \Delta H/RT^2$, where R is the gas constant and T the thermodynamic temperature.

van't Hoff law the principle that the osmotic pressure exerted by a solute is equal to the pressure the solute would exert if it were an ideal gas at the same temperature as the solution and of the same volume. Hence $\Pi V = RT$, where Π is the osmotic pressure, V is the volume of solution containing 1 mole of solute, R the gas constant, and T the thermodynamic temperature.

van't Hoff limiting law a relation giving the limiting value for the osmotic pressure exerted by an ideal solute:

$$\text{limit } (c \rightarrow 0) \Pi = RT/M_s,$$

where Π is the osmotic pressure, c the solute concentration, M_s the relative molecular mass of the solute, R the gas constant, and T the thermodynamic temperature.

variable 1 liable to change; having a range of possible values; liable to deviate from an established type. 2 something that is subject to variation; a quantity or function that can assume any of a set of specified values; a symbol for any unspecified number or quantity.

variable arm *an alternative name for* **extra arm** of tRNA.

variable number tandem repeats *abbr.*: VNTR; a multi-allelic DNA polymorphism that results from insertions or deletions of DNA between two **restriction sites**. The existence of these repeats is exploited in **genetic profiling**.

variable region *symbol*: V; a region of a chain of an **immunoglobulin** molecule having an amino-acid sequence that characteristically varies between molecules of the same immunoglobulin class. There are typically four variable regions: the N-terminal halves of the two **light chains** (V_L) and the N-terminal halves of the two **heavy chains** (V_H) of the two heavy chains. The amino-acid sequences of these variable regions de-

variance

termine the structure of the antibody-combining site. *Compare constant region.*

variance a measure of the dispersion of a population, the mean square average of the deviations from the **mean**. It is the square of the **standard deviation**, σ , of that population.

variant 1 something that deviates from the norm; used especially of abnormal proteins (e.g. hemoglobins). 2 different; showing variation; used especially of different amino-acid residues in the corresponding positions of polypeptide chains from different sources.

Varmus, Harold Eliot (1939–), US microbiologist and molecular biologist; Nobel Laureate in Physiology or Medicine (1989) jointly with M. J. Bishop 'for their discovery of the cellular origin of retroviral oncogenes'.

vascular of, or relating to vessels that conduct fluid from one part of an organism to another, e.g. the blood vessels in animals, or the xylem and phloem in plants.

vascular cell adhesion molecule *see* **VCAM**.

vascular endothelial growth factor *abbr.*: VEGF; *also known as* **vascular permeability growth factor** or **vasculotropin**; a protein that is produced by epithelial cells, macrophages, and smooth muscle cells, and is active in angiogenesis and endothelial cell growth; it induces endothelial proliferation and vascular permeability. Related to **platelet-derived growth factor**, it is a disulfide-linked homodimer. Example (precursor) from human: database code VEGF_HUMAN, 215 amino acids (25.14 kDa).

vascular endothelial growth factor receptor *abbr.*: VEGF receptor; a type I membrane protein of the immunoglobulin superfamily (seven immunoglobulin-fold domains in the extracellular part) found in the lung and elsewhere. It is a tyrosine kinase receptor related to the platelet-derived growth factor receptor, with a nonkinase insert between the kinase domains. Example (precursor of receptor 1) from human: database code VGR1_HUMAN, 1338 amino acids (150.56 kDa).

vascular permeability growth factor *see* **vascular endothelial growth factor**.

vasculature the **vascular** system in animals.

vasculotropin *see* **vascular endothelial growth factor**.

vaso+ *comb. form* denoting a blood vessel.

vasoactive describing any agent that affects the diameter of blood vessels. *See also* **vasoconstrictor**, **vasodilator**.

vasoactive intestinal contractor *see* **endothelin**.

vasoactive intestinal polypeptide *abbr.*: VIP; a basic 28-residue peptide amide, similar in sequence to **gastric inhibitory polypeptide**, **glucagon**, and **secretin**, that is widely distributed in the body, mainly in nervous tissue (especially of the gut) and appears to act as a neurotransmitter in **peptidergic** autonomic nerves. It has a wide range of biological activities, including potent hypotensive and vasodilatory effects. It has the sequence

HSDAVFTDNYTRLRKQMAVKKYLNSILN-NH₂.

vasoactive intestinal polypeptide receptor or **vasoactive intestinal peptide receptor** any of a family of proteins that bind **vasoactive intestinal polypeptide** (VIP) and mediate its intracellular effects. The receptor is a G-protein-coupled seven-transmembrane-domain protein (362 residues, human). The effector pathway involves adenylate cyclase activation; other endogenous ligands include peptide HI (PHI), GHRH, and secretin, the order of potency being VIP > PHI > GHRH > secretin.

vasoconstrictor a vasoactive agonist that causes **vasoconstriction**, i.e. a decrease in the diameter of blood vessels.

vasodilator a vasoactive agonist that causes **vasodilation**, i.e. an increase in the diameter of blood vessels.

vasopressin or **antidiuretic hormone** (*abbr.*: ADH); either of two nonapeptide hormones secreted by the neurohypophysis and structurally very similar to **vasotocin** and **oxytocin**. [**Arg**-

Arg]**vasopressin** (*abbr.*: [**Arg**⁸]**vasopressin** or **AVP**) has the sequence

Cys-Tyr-Phe-Gln-Asn-Cys-Pro-Arg-Gly-amide,

with a disulfide bridge between the half-cystine residues at positions 1 and 6. It has a pressor action by causing vasoconstriction in the peripheral arterioles and capillaries. It also has an antidiuretic action by accelerating water reabsorption from the proximal region of the distal convoluted tubules of the kidney. AVP is synthesized as part of a large precursor molecule, **vasopressin-neurophysin 2-copeptin precursor**, which includes its associated carrier protein (**neurophysin**), and is then packaged into granules in which the nonapeptide is formed by proteolytic cleavage. Subsequently it is released, on action of a specific stimulus, into the blood, in which it circulates in unbound form. Deficiency of AVP is the cause of autosomal dominant neurohypophyseal diabetes insipidus (ADNDI). The analogue, [**8**-lysine]**vasopressin** (*abbr.*: [**Lys**⁸]-**vasopressin**), in which lysine is substituted for arginine at position 8, is formed, either singly or together with arginine vasopressin, in the pig family.

vasopressin-neurophysin 2-copeptin precursor the precursor of **vasopressin**, neurophysin 2 (*see* **neurophysin**), and a third peptide, copeptin. Example from human: database code NEU2_HUMAN, 164 amino acids (17.31 kDa). Copeptin is a glycopeptide that contains 38 amino-acid residues forming the third domain of mammalian vasopressin-neurophysin precursor. It is released during intra-axonal processing of the precursor.

vasopressin receptor any of a family of membrane proteins that bind [**8**-arginine]**vasopressin** (AVP; *see* **vasopressin**) and mediate its intracellular effects; peptides related to AVP, including **oxytocin**, are also bound. Two types of vasopressin receptor have been defined: V₁ – subdivided into V_{1a} (vascular/hepatic) and V_{1b} (anterior pituitary) – and V₂ (kidney); these can be distinguished pharmacologically on the basis of selective agonists. V₂ receptors are restricted to the kidney and colocalize at the chromosomal locus for **nephrogenic diabetes insipidus**. V₁ receptors are coupled to phosphoinositide-specific phospholipase C, and V₂ receptors to adenylate cyclase, which is activated by occupied receptor. AVP is a more potent agonist for both V₁ and V₂ receptors than oxytocin. These receptors are seven-transmembrane-domain proteins characteristic of G-protein-coupled receptors. Examples, from *Rattus norvegicus*, V_{1a} receptor: database code V1AR_RAT, 395 amino acids (44.48 kDa); V_{1b} receptor: database code V1BR_RAT, 425 amino acids (47.08 kDa); V₂ receptor: database code V2R_RAT, 371 amino acids (40.75 kDa). *See also* **oxytocin receptor**.

vasostatin a 76 amino-acid peptide released by proteolytic cleavage of **chromogranin A**; it lowers vascular tension.

vasotocin a vasoactive peptide, closely related in amino-acid sequence to arginine **vasopressin** and **oxytocin**, that is secreted by the posterior pituitary gland in birds, reptiles, and some amphibians. It has the sequence

Cys-Tyr-Ile-Gln-Asn-Cys-Pro-Arg-Gly-amide,

with a disulfide bond between the two cysteines.

VASP *abbr. for* vasodilator-stimulated phosphoprotein; a substrate for cyclic AMP- and cyclic GMP-dependent protein kinases in platelets. It is associated with actin filaments and focal adhesions and is a proline-rich **profilin** ligand. Example from *Canis familiaris*: database code VASP_CANFA, 384 amino acids (40.41 kDa).

VCAM *abbr. for* vascular cell adhesion molecule; a class of **adhesion molecules** belonging to the immunoglobulin superfamily. The class is exemplified by VCAM-1 (CD106; molecular mass 90–110 kDa). This is expressed on endothelial cells, macrophages, dendritic cells, fibroblasts, and myoblasts, and is inducible within 6–10 h. by interleukin 1 β , interleukin 4,

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vector

tumour necrosis factor α , or interferon γ . It binds to VLA-4 (see **VLA**) and integrin $\alpha_4\beta_7$ (see **integrin**), and mediates adhesion of lymphocytes, monocytes, and eosinophils to activated endothelium. Example from human: database code VCA1_HUMAN, 739 amino acids (81.28 kDa). Residues 1–24 are the signal, 25–698 the extracellular domain, with seven Ig-like C_2 domains.

vector 1 or **vector quantity** a quantity specified by its direction and sense as well as by its magnitude. **2** (*in pathology*) an organism that carries an organism of a different species from one place to another, especially one that transfers a **pathogen** between hosts, either mechanically (**mechanical vector**) or by playing a specific role in the pathogen's life cycle (**biological vector**). Compare **vehicle** (def. 2). **3** (*in genetics*) see **cloning vector**.

vectorette a synthetic DNA oligonucleotide with a mismatched region. Vectorettes are ligated to genomic fragments and serve in a variety of **polymerase chain reaction** (PCR) systems in which the sequence of only one primer is available.

vectorial enzyme any enzyme, such as one fixed in a membrane, that is directional in its action, e.g. **sodium/potassium ATPase**.

vectorial processing processing of a fully formed precursor polypeptide into a mature protein when used to drive the movement of the protein from one aqueous compartment (e.g. the cytoplasm) through a membrane into a second aqueous compartment (e.g. an intramitochondrial or intrachloroplastal space). Compare **vectorial translation**.

vectorial translation the stage of **translation** in the biosynthesis of any protein destined for export from a cell, where a nascent polypeptide being produced on a **polysome** bound to the membrane of the rough endoplasmic reticulum in a eukaryote is conducted through the membrane by its N-terminal **signal peptide**. Compare **vectorial processing**.

vegetal hemisphere glycoprotein 1 *abbr.*: Vg-1; a glycoprotein, belonging to the TGF- β family (see **transforming growth factor**), that serves to facilitate the differentiation of either mesoderm or endoderm, either as a cofactor in an instructive signal or by providing a permissive environment. Example from *Xenopus*: database code DVR1_XENLA, 360 amino acids (41.72 kDa).

vegetal pole the end at which most of the yolk is located e.g. in an amphibian egg; the end opposite the **animal pole**.

vegetative describing cells or tissues that are engaged in nutrition and growth rather than sexual reproduction, and excluding dormant forms.

vegetative bacteriophage the intrabacterial genome of a **bacteriophage** when it is noninfective but is controlling the synthesis by the host of components necessary for the production of new, infective bacteriophage particles.

vegetative reproduction 1 (*in plants*) a form of **asexual reproduction** in which new individuals develop from specialized multicellular organs (e.g. bulbs, corms, tubers) that break off from the parent plant. **2** (*sometimes*) a term used to describe **budding** (def. 2) and similar forms of asexual reproduction in animals such as coelenterates and sponges.

VEGF *abbr.* for vascular endothelial growth factor.

vehicle 1 any inert medium with which a biologically active substance (e.g. a drug) is mixed or in which it is dissolved, suspended, or emulsified in order to increase its bulk and/or to facilitate its absorption after administration. **2** any agent (animate or inanimate) of transmission, especially of infection; see also **vector** (def. 2).

vein 1 (*in animals*) the blood vessels conducting blood from the tissues. **2** (*in plants*) a vascular bundle in a leaf.

velocity of reaction an alternative name for **rate of reaction**.

venom any toxic animal secretion that is produced by discrete glands and is delivered by a specific mechanism for defensive or offensive purposes.

venombin either of two serine proteinases found in snake venom. **Venombin AB** EC 3.4.21.55; *other name*: gabonase. *Ban dang doc* một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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(from the venom of the gaboon viper, *Bitis gabonica*); it cleaves Arg-|-Xaa bonds in fibrinogen to form fibrin and release fibrinopeptides A and B; it is not inhibited by antithrombin III or hirudin. **Venombin A** EC 3.4.21.74; it cleaves Arg-|-Xaa bonds in fibrinogen to form fibrin and release fibrinopeptide A, but further reaction depends on species; it is a trypsin-like enzyme; example: precursor (zymogen) from *Bothrops atrox* (fer-de-lance); database code BATX_BOTAT, 255 amino acids (28.16 kDa); six putative S-S bonds and two glycosylation sites.

venous of, or relating to, a **vein**, or to the blood in the veins.

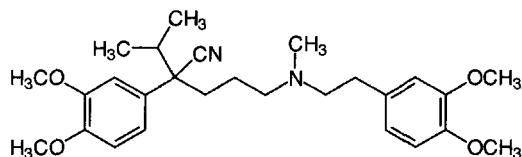
ventilation the passage of air into and out of the respiratory tract, or other confined space.

Ventolin see **sambutamol**.

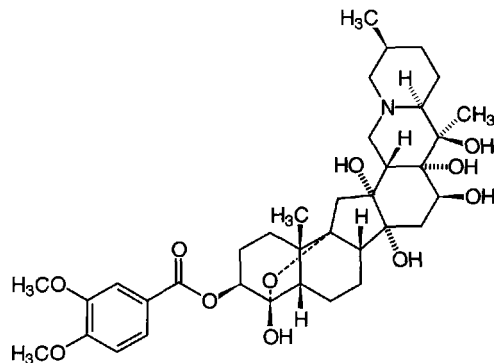
Vent polymerase or (*sometimes*) **Deep Vent polymerase** the *proprietary name* for thermostable DNA polymerases used as alternatives to **Taq DNA polymerase** in the **polymerase chain reaction**. These enzymes provide a higher fidelity than the **Taq** DNA polymerase, derived in part from their exonuclease activity. They derive from the archaeon *Thermococcus litoralis*, isolated from a submarine thermal vent, hence the enzyme's name.

ventral situated towards the belly surface of an animal.

verapamil α -3-[[2-(3,4-dimethoxyphenyl)ethyl]methylamino]propyl-3,4-dimethoxy- α -(1-methylethyl)benzeneacetonitrile; iproveratril; D365; a Ca^{2+} -blocking agent used clinically in the treatment of cardiac disorders and in the laboratory to inhibit the movement of calcium ions across biological membranes. Its action is similar to that of **nifedipine**. Nifedipine binds to the dihydropyridine binding site of vascular voltage-operated **calcium channels**, but verapamil probably acts at another related site. See also **D600**.



veratridine a toxic alkaloid, $C_{36}H_{51}NO_{11}$, found in the seeds of a plant, *Schoenocaulon officinalis*, of the lily family; it also occurs in the rhizome of *Veratrum album*. It blocks the development of the action potential involved in axonal conduction, binding to the Na^+ channels and blocking them in the open configuration.



v-erbA related protein see **COUP**

verbascose the tetrasaccharide, *O*- α -D-galactopyranosyl-(1 $\rightarrow$ 6)-*O*- α -D-galactopyranosyl-(1 $\rightarrow$ 6)-*O*- α -D-glucopyranosyl-



verdoperoxidase

(1→2)-β-D-fructofuranoside. It occurs in roots of the mullein, *Verbascum thapsus*.

verdoperoxidase see myeloperoxidase.

verification 1 establishment of the correctness of a theory, fact, etc. 2 a term used to describe the deacylation of misacylated tRNA molecules by the specific aminoacyl-tRNA synthase.

Verner-Morrison syndrome or pancreatic cholera or WDHA syndrome a disease syndrome of humans characterized by the following features, the first three of which are obligatory: tumour, severe watery diarrhoea, achlorhydria or hypochlorhydria, weight loss, and dehydration. The tumours (VIPomas) are mostly of pancreatic origin (some are neurogenic) and produce **vasoactive intestinal polypeptide** (VIP), the circulating levels of which are raised. [After John Victor Verner (1927–), US physician and Ashton Byrom Morrison (1922–), US pathologist.]

Veronal a proprietary name for **barbital**.

Versene a proprietary name for the tetrasodium salt of **edetate**.

versican or **large fibroblast proteoglycan** or **chondroitin sulfate proteoglycan core protein 2** a protein that plays a role in intercellular signalling. The N-terminal part is similar to a glial hyaluronate-binding protein; in the middle of the protein there are glucosaminoglycan attachment sites; the C-terminal part contains EGF-like repeats (see **epidermal growth factor**) and a lectin-like domain. Example (precursor) from human: database code PGCS_HUMAN, 2409 amino acids (264.76 kDa).

vertical rotor a rotor for use in an **ultracentrifuge**. The wells holding the tubes are drilled parallel to the axis of rotation and at right angles to the lines of the centrifugal force.

very-low-density lipoprotein *abbr.*: VLDL; one of the classes of lipoproteins found in blood plasma in many animals (data normally relate to humans). VLDL is also known as pre-β-lipoprotein because its electrophoretic mobility is slightly greater than that of low-density, or β-lipoprotein. VLDL particles have diameter 25–70 nm, solvent density for isolation (g mL⁻¹) <1.006, hydrated density (g mL⁻¹) 0.97. Their approximate composition (% by weight) is 5–8% unesterified cholesterol, 11–14% esterified cholesterol, 20–23% phospholipid, 44–60% triacylglycerol, 4–11% protein. Their apolipoprotein composition (% by weight total apolipoprotein) is 36.9% B-100, 49.9% C-I + C-II + C-III, 13% E-II + E-III + E-IV. They are synthesized in liver and are the precursor of low-density lipoprotein.

very-low-density-lipoprotein receptor *abbr.*: VLDL receptor; a membrane protein that binds VLDL, accumulates in clathrin-coated pits, and internalizes VLDL. It has a large extracellular domain, a single transmembrane domain, and a small intracellular domain. Example from human: database code LDVR_HUMAN, 873 amino acids (96.09 kDa); residues 1–27 are the signal, 28–797 the extracellular domain, and 820–873 the cytoplasmic domain.

vesicle a closed structure, found only in eukaryotic cells, that is completely surrounded by a unit membrane but, unlike a **vacuole**, contains material that is not (or is not known to be) in the liquid state.

Vg-1 *abbr.* for vegetal hemisphere glycoprotein 1.

VHDL *abbr.* for very-high-density lipoprotein.

viability a measure of the capacity of a cell for metabolism and division. It is often estimated experimentally by determining the number of cells that exclude normally membrane-impermeant dyes, e.g. **Trypan Blue**.

Viagra a proprietary name for sildenafil; 5-[2-ethoxy-5-(4-methylpiperazin-1-yl)phenyl]-1-methyl-3-propyl-1,6-dihydro-7H-pyrazolo[4,3-d]pyrimidin-7-one; used clinically for the oral treatment of male erectile dysfunction. It selectively and potently inhibits a specific phosphodiesterase that hydrolyses cyclic GMP; this second messenger mediates the action of nitric oxide in normal erectile function, an action which is thus potentiated.

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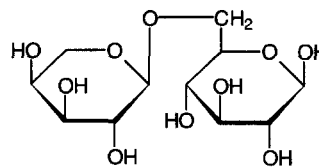
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VIC *abbr.* for vasoactive intestinal contractor; see **endothelin**.

vic- prefix (in chemical nomenclature) denoting the presence of **vicinal** (def. 2) substituents.

vicianose the disaccharide 6-O-α-L-arabinosyl-D-glucose, obtained from a glycoside in seeds of *Vicia angustifolia*.



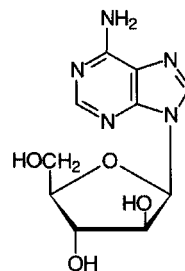
β-D-anomer

vicillinpeptidohydrolase see **legumain**.

vicinal 1 adjacent, neighbouring. 2 (in organic chemistry) describing two (usually identical) atoms or groups attached one to each of two linked carbon atoms in a molecular structure. The presence of vicinal substituents is denoted by the prefix *vic-* attached to the name of the compound. Compare **geminal**.

vicilin a major storage protein in seeds of *Pisum sativum* (pea); it comprises 124 amino acids, *M_r* 170 000. Legumin-like and vicilin-like globulins are the main storage proteins of most angiosperms and gymnosperms.

vidarabine adenine arabinoside; 9-β-D-arabinofuranosyladenine; a purine nucleoside, a metabolite of hypoxanthine arabinoside, produced by *Streptomyces antibioticus*. Its triphosphate derivative inhibits viral DNA polymerase and may be incorporated into viral DNA; it also inhibits DNA synthesis.



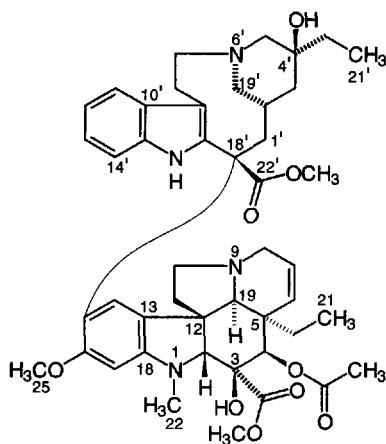
vigabatrin γ-vinyl-γ-aminobutyric acid; γ-vinyl GABA; 4-amino-5-hexenoic acid; an irreversible inhibitor of the enzyme GABA transaminase (**4-aminobutyrate transaminase**), which metabolizes the neurotransmitter γ-amino-*n*-butyric acid.

Vigneaud see **du Vigneaud**.

villin a Ca²⁺-regulated actin-binding protein occurring in the microvilli of intestinal epithelial cells and kidney proximal tubule cells. Villin consists of a large core fragment, the N-terminal portion, and a small headpiece, the C-terminal portion; the headpiece binds F-actin (see **actin**) strongly in both the presence and absence of Ca²⁺; it is similar to **gelsolin**. Example from human: database code VIL1_HUMAN, 826 amino acids (92.46 kDa).

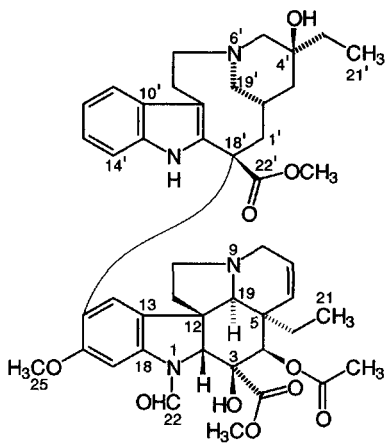
vimentin a protein found in class III **intermediate filaments** in mesenchymal and other nonepithelial cells and in the Z disk of skeletal and cardiac muscle cells. It is a phosphoprotein, phosphorylation being enhanced during cell division. Example from human: database code VIME_HUMAN, 465 amino acids (53.49 kDa).

vinblastine a structurally complex alkaloid, C₄₆H₅₈N₄O₉, obtained from *Vinca rosea*, that binds to spindle microtubules and arrests metaphase of mitosis. It is used as an antineoplastic agent.



vinblastine

vincristine an alkaloid, $C_{46}H_{56}N_4O_{10}$, obtained from *Vinca rosea* and with a structural relationship to **vinblastine**, that binds to spindle microtubules and arrests metaphase of mitosis. It is used as an antineoplastic agent.



vinculin a cytoskeletal protein associated with the cytoplasmic face of the focal adhesion plaques that anchor actin microfilaments to the plasma membrane and that attach a cell to the substratum. It associates with **talin** in binding **integrins** to the cytoskeleton. Vinculin is phosphorylated (on serines, threonines, and tyrosines) and acylated by myristic acid and/or palmitic acid. Vinculin and metavinculin are produced by alternative splicing of the same gene. Metavinculin differs from vinculin in having a 68-residue additional domain near the C terminus. Example from human: database code VINC_HUMAN, 1065 amino acids (116.46 kDa).

vindoline the major alkaloid from *Vinca rosea*; $C_{25}H_{32}N_2O_6$. It is without physiological activity but constitutes a pentacyclic moiety of **vinblastine** and **vincristine**.

vinyl ether alk-1-enyl ether; a class of compounds characterized by a bond formed between a long-chain aldehyde and one (in the case of monoalk-1-enyl ethers) or two (for dialk-1-enyl ethers) of the hydroxy groups of the glycerol moiety of a glycerolipid; only the *sn*-1 isomers occur naturally, and chains of

16 or 18 carbons are usual. Such glycerophospholipids are known as **plasmalogens**; ethanolamine glycerophospholipids characteristically contain larger amounts of plasmalogens than other classes, and heart tissue is rich in plasmalogens.

γ -vinyl GABA see **vigabatrin**.

viosamine 4-amino-4,6-dideoxy-D-glucose; a component of the lipopolysaccharide of some strains of *Escherichia coli*.

VIP abbr. for vasoactive intestinal polypeptide.

VIPoma a tumour producing **vasoactive intestinal polypeptide** (VIP). It is often associated with a watery diarrhoea and other pathological signs known as **Verner-Morrison syndrome**.

viral particle see **virion**.

viral RNA RNA that constitutes the genome of certain viruses – the RNA viruses. Examples are poliovirus, which has single-stranded RNA, and cytoplasmic polyhedrosis virus of the tussock moth, which has 10 double-stranded RNA molecules. All **retroviruses** contain RNA as the genome.

virino a name coined for a putative novel type of particulate, noncellular, submicroscopic, transmissible, pathogenic agent, as typified by the agent responsible for **scrapie**. The infective particle was believed to consist of a small nucleic-acid molecule covered with host-specified protein. See **prion**.

virion or **viral particle** the complete particle of a virus, found extracellularly and capable of surviving in crystalline form and of infecting a living cell. It comprises the nucleic-acid core and the protein **capsid**; the latter may be enclosed by an **envelope** in some viral families.

virogene see **oncogene hypothesis**.

viroid any member of a group of about 30 autonomously replicating plant pathogens. They consist of single-stranded circular RNA of about 350 nucleotides containing much self-complementarity. They have no genes, so do not express protein. It is assumed that the viroid RNA acts as its own template, subverting host cell enzymes to carry out the replication. The product consists of concatamers of daughter strands that undergo self-cleavage to form new viroid RNA genomes. It is likely that hammerhead structures (see **ribozyme**) are involved. An example is potato spindle tuber viroid (PSTV). This is similar to human hepatitis delta which is part viroid and part virus in that the mechanism of replication is like that of a viroid and part of the genome is homologous with PSTV.

virology the scientific study of viruses and the diseases caused by them. —**virological** *adj.*; **virologist** *n.*

viropexis the process by which virus particles are adsorbed onto the surface of a cell and then taken into the cytoplasm.

Virtanen, Artturi Ilmari (1895–1973), Finnish agricultural chemist; Nobel Laureate in Chemistry (1945) 'for his research and inventions in agricultural and nutrition chemistry, especially for his fodder preservation method'.

virulence the capacity of a microorganism or virus to cause disease in a host under certain conditions. —**virulent** *adj.*

virulent phage a bacteriophage that invariably lyses the bacterium it infects. Virulent phages have no alternative but to follow the **lytic cycle**.

virus a noncellular infective agent that is capable of reproducing only in an appropriate host cell. Viruses are typically smaller than bacteria, and can infect animal, plant, or bacterial cells (see **bacteriophage**); many are important agents of disease. The infective particle (**virion**) consists of a core of nucleic acid (either DNA or RNA) surrounded by a proteinaceous **capsid** and, in some cases, an outer envelope. A virus can interact with its host cell in several different ways. In a lytic interaction the cell's replicative and protein biosynthetic apparatus is directed by the viral genes to produce numerous progeny virions, which are released upon disintegration of the host cell (see **lytic cycle**). However, in a persistent infection, low levels of virus production may be sustained without destruction of the host cell. In a **latent infection**, the virus resides in a host cell without undergoing replication, although it may be provoked to enter the lytic phase by some stimulus. Oncogenic viruses, which include both DNA viruses and RNA viruses



virusoids

(retroviruses) carry **oncogenes** and are potentially capable of transforming their host cells into tumour cells. Animal viruses have been divided into seven classes according to the nature of their genome and the mechanism of synthesis of messenger RNA (see **Baltimore classification**).

virusoids see **satellite RNA**.

visceral layer see **serous membrane**.

viscometer or **viscosimeter** an apparatus for measuring the viscosity or resistance to flow of a liquid.

viscosity *symbol*: η ; the resistance to flow of a fluid due to the sum of the effects of adhesion and cohesion. Consider a liquid between two parallel plates, one of which is moving in the x direction with a velocity v . The liquid is thought of as a number of layers, each of which slides along the adjacent layer; the frictional resistance between adjacent layers generates a velocity gradient in the y direction. The deformation of the liquid produced by the velocity gradient is known as **shear** (def. 1). The frictional force, f , between the liquid layers is proportional to the area, A , of the layers and to the velocity gradient between them, such that $f = \eta A (dv/dy)$, where η is the **coefficient of viscosity** or simply the viscosity; $f/A = F$, the **shear stress**; and $dv/dy = G$, the **shear gradient** or **shear rate**. If η is independent of F or G , the fluid is a **Newtonian fluid**; if η is a function of F or G , the fluid is a **non-Newtonian fluid**.

visinin a calcium-binding protein specific to cone photoreceptors; a chicken isoform with homology to bovine **recoverin** has three Ca^{2+} -binding sites. Example from *Gallus gallus*: database code VISI_CHICK, 191 amino acids (22.39 kDa).

visual cascade the sequence of reactions occurring after the absorption of a photon by a **rhodopsin** molecule. It eventually leads to a rapid fall in cyclic GMP levels, which closes Na^+ channels in the membrane of the visual receptor and activates the nerve impulse in the optic nerve. The activated rhodopsin stimulates the G-protein **transducin** which initiates the signal-transduction cascade.

visual purple see **rhodopsin**.

vitalism a doctrine stating that that the phenomena of biology are due to a vital principle distinct from physicochemical forces, and cannot be explained by the laws of physics and chemistry alone. The accidental discovery, in 1890, by the brothers Büchner that a cell-free extract of yeast could catalyse alcoholic fermentation did much to convince any lingering adherents that such a doctrine was untenable. Also important was Wohler's synthesis of urea (a natural product) from ammonium cyanate in 1828.

vitamer any of two or more chemically and metabolically related compounds that display qualitatively the same biological activity as a vitamin. For example, retinol, retinal, retinoic acid, 3,4-didehydroretinol, and various carotenes are all vitamers of vitamin A.

vitamin an organic substance that is distributed in foodstuffs (and sometimes in intestinal bacteria) in relatively minute amounts, is distinct from the main organic components of food (i.e. protein, carbohydrate, and fat), and is needed for the normal nutrition of the organism (or species) in question. Deficiency or absence of any particular vitamin causes a corresponding specific deficiency disease. **Vitamins A, D, E, and K** are the fat-soluble vitamins; **vitamins B₁, B₂, B₆, B₁₂, and C**, plus **biotin, choline, folic acid, nicotinic acid**, and **panthothenic acid** (see **pantothenate**), are the water-soluble vitamins.

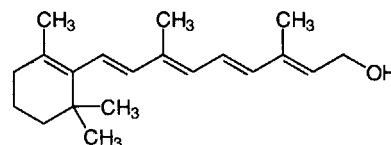
vitamin A **1** a common name for **retinol**. **2** a generic name for any retinoid exhibiting qualitatively the biological activity of retinol. Included, in addition to retinol (relative activity 100) are: **retinal** (91); **retinoic acid** (=65); **3,4-didehydroretinol** (40); **3,4-didehydroretinal**; and **3,4-didehydroretinoic acid** (the activities relative to retinol being measured by the standardized growth test on vitamin A-deficient rats). Vitamin A is fat-soluble, and is present in various lipid-containing animal preparations, especially liver and dairy products. Vitamin A activity is displayed also by various carotenes; these act as provitamins, which are converted to retinal in the small intestine in most vertebrates.

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The vitamin A of commerce was originally a preparation of, or extract from, fish-liver oils, which contain large amounts of retinol, principally as its palmitate; it is now largely derived chemically. Proper vitamin A nutrition is necessary for: (1) somatic function, including growth and differentiation, e.g. of epithelial structures and bone; (2) reproduction, including spermatogenesis, oogenesis, placental development, and embryonic growth; and (3) visual processes (see **retinal**, **3,4-didehydroretinal**). Vitamin A deficiency signs include stunted growth, thickening of various membranes by keratinization, and, in the eye, night blindness, xerophthalmia, keratomalacia, and eventually complete blindness. In moderate excess, vitamin A is highly toxic.



vitamin A₁ an alternative name (not recommended) for **retinol**.

vitamin A₂ an alternative name (not recommended) for **3,4-didehydroretinol**.

vitamin A acid or **vitamin A₁ acid** an alternative name (not recommended) for **retinoic acid**.

vitamin A₂ acid an alternative name (not recommended) for **3,4-didehydroretinoic acid**.

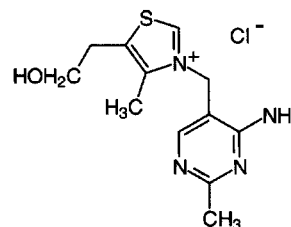
vitamin A alcohol or **vitamin A₁ alcohol** an alternative name (not recommended) for **retinol**.

vitamin A₂ alcohol an alternative name (not recommended) for **3,4-didehydroretinol**.

vitamin A aldehyde or **vitamin A₁ aldehyde** an alternative name for (not recommended) **retinal**.

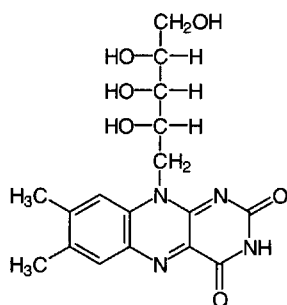
vitamin A₂ aldehyde an alternative name (not recommended) for **3,4-didehydroretinal**.

vitamin B₁ or **thiamine** or (sometimes) **thiamin** or (formerly) **aneurin(e)** 3-(4-amino-2-methylpyrimidin-5-ylmethyl)-5-(2-hydroxyethyl)-4-methylthiazolium chloride; a water-soluble vitamin, present in fresh vegetables and meats, especially liver. Deficiency in humans causes beriberi, signs of which include cardiopathy and neuropathy; in pigeons, a characteristic head-drop is an overt sign of deficiency. Its diphosphoric ester, **thiamine diphosphate** (abbr.: **TPP**), formerly known as **coccarboxylase**, is a coenzyme for oxidative decarboxylation and other enzymic reactions, including ones catalysed by pyruvate and oxo-glutarate dehydrogenases, pyruvate decarboxylase, and transketolase.



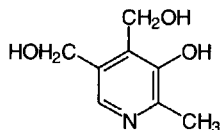
vitamin B₂ or **riboflavin** 6,7-dimethyl-9-ribitylisoalloxazine; formerly also called **lactoflavin**, **hepatoflavin**, **ovoflavin**, **uroflavin**, **vitamin G**, **vitamin H**. A water-soluble vitamin, present in most foods but especially in milk and meat products. It is a component of the **flavin nucleotide** coenzymes (e.g. **FAD**, **FMN**). Isolated from milk and eggs by Kuhn and co-

workers in 1933 and chemically synthesized by them and others in 1935. Significant deficiency seldom occurs, but can cause inflammation of the tongue and lips and sores in the corners of the mouth.



vitamin B₅ (rarely) an alternative name for pantothenic acid; see **pantothenate**.

vitamin B₆ or **pyridoxine** 2-methyl-3-hydroxy-4,5-bis(hydroxymethyl)pyridine; a water-soluble vitamin; **pyridoxal** and **pyridoxamine** also act as vitamers. It is present in many foods, including meats (largely as pyridoxal or pyridoxamine), and plant sources (largely as pyridoxine) such as cereals, lentils, nuts, and vegetables. Pyridoxal phosphate is a coenzyme for many enzyme reactions including those involving transamination, amino-acid decarboxylation, racemization, and modifications to amino-acid side chains. Deficiency is not a major problem and signs are difficult to produce.

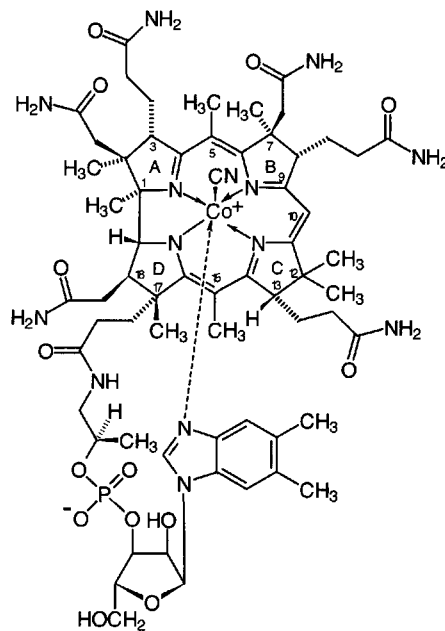


vitamin B₁₂ or **cobalamin** or **cobamide** a water-soluble vitamin, of which there are several structurally related forms. All are characterized by possession of a **corrin** nucleus containing a cobalt atom. Four of the cobalt coordinate bonds are liganded by nitrogens of the four pyrrole rings found in the corrin nucleus; the fifth position is occupied by the ribonucleotide of 5,6-dimethylbenzimidazole, which is also linked via a 1-amino-2-propanol moiety to one of the pyrrole rings; the sixth position is variously occupied by a cyanide group (cyanocobalamin, the form in which the molecule was first isolated, the cyanide group having been introduced during extraction), or a methyl group (**methylcobalamin**, one of the forms in which the molecule acts as a coenzyme), or a 5'-deoxyadenosyl group (**deoxyadenosylcobalamin**, the other active coenzyme form). Vitamin B₁₂ is present only in foods of animal origin, especially liver, fish, and eggs. The coenzyme forms are needed for two classes of enzyme: members of the first class, exemplified by **methylmalonyl-CoA mutase**, require the deoxyadenosyl form; members of the second, exemplified by 5-methyltetrahydrofolate-homocysteine S-methyltransferase, (EC 2.1.1.13) require the methyl form (see **5-methyltetrahydrofolate**). In normal individuals, deficiency rarely occurs, but is precipitated by lack of **intrinsic factor**, a protein secreted by gastric mucosa and required for efficient absorption of the vitamin. Lack of intrinsic factor leads to **pernicious anemia**, investigation of which first led to the discovery of cobalamin.

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cyanocobalamin

vitamin B₁₂ reductase see **cob(II)alamin reductase**.

vitamin C ascorbic acid; a water-soluble vitamin, and the first vitamin to be recognized. Awareness of the ability of fresh vegetables and fruit to prevent **scurvy** is attributed to the British physician James Lind (1716-94), who published *A Treatise on Scurvy* in 1757. The active compound was first isolated by A. **Szent-Györgyi** in 1928 as 'hexuronic acid', a strong reducing substance from rat adrenals and citrus fruits. This was shown four years later to be identical with the anti-scorbutic substance isolated by Zilva and associates. Vitamin C is widely distributed in vegetables and fruit, but easily destroyed by cooking. See **ascorbic acid** for structure. See also **L-gulonolactone oxidase**.

vitamin D the common and class name for any steroid that exhibits qualitatively the biological activity of calciferol (= vitamin D₃ - see below); the antirachitic vitamin. All such compounds are fat-soluble substances structurally related to either the parent hydrocarbon, **cholestane**, or its 24-methyl derivative, **ergostane**, and all are 9,10-seco-sterols, i.e. 3-hydroxysteroids in which bond breakage has occurred between C-9 and C-10 in ring B of the steroid nucleus. Separately numbered variants of vitamin D, with their recommended trivial names in parentheses, are: **vitamin D₂** or **calciferol** (**ercalciferol** or **ergocalciferol**), the variant formed in plants by the action of UV light on the endogenous provitamin **ergosterol** and produced industrially by UV-irradiation of ergosterol isolated from yeast; **vitamin D₃** (**calciferol** or **cholecalciferol**), the variant occurring naturally in particularly high concentrations in the liver oils of certain fish, especially cod, halibut, or tunny, and produced non-enzymically in the skin of animals by the action of a UV component (230-313 nm) of sunlight on the provitamin 7-dehydrocholesterol; **vitamin D₄** (22,23-dihydroercalciferol or (24S)-methylcalciferol or (formerly) 22,23-dihydroergocalciferol), the variant produced by irradiation of 22,23-dihydroergosterol; **vitamin D₅** ((24S)-ethylcalciferol); and **vitamin D₆** ((22E)-(24R)-ethyl-22,23-didehydrocalciferol). (Vitamin D₁, a designation now obsolete, was found to be a mixture of ercalciferol and lumisterol.) Of these variants, calciferol (D₃) and ercalciferol



vitamin D binding protein

ciol (D_2) are the most significant. In addition, vitamin D activity is displayed by several unnumbered synthetic derivatives or isomers of calcitriol (or of ercalcitriol) and by the 25-hydroxylated and 1 α ,25-dihydroxylated metabolites **calcidiol** and **calcitriol** (or **ercalcidiol** and **ercalcitriol**). Calcidiol (or ercalcidiol) is formed in liver from calcitriol (or ercalcitriol) and converted in kidney to calcitriol (or ercalcitriol), the physiologically active form of vitamin D_3 (or D_2). Vitamin D mediates the absorption of calcium and phosphate ions from the intestine and promotes the mineralization of bone. The vitamin D deficiency conditions of **rickets** (in children) and **osteomalacia** (in adults) may be treated by administration of the vitamin, commonly as D_2 . Overdosage of vitamin D results in disturbances of calcium metabolism and ultimately leads to decalcification of bone. See also **calcitriol**. For structures see entries for individual compounds.

vitamin D binding protein *abbr.*: DBP; a multifunctional protein found in plasma, ascitic fluid, cerebrospinal fluid, and urine, and on the surface of many cell types. In plasma it carries the vitamin D sterols; it also prevents polymerization of actin by binding its monomers. DBP associates with membrane-bound immunoglobulin on the surface of B lymphocytes, and with IgG Fc receptor on the membranes of T lymphocytes. Example (precursor) from human: database code VTDB_HUMAN, 474 amino acids (52.90 kDa).

vitamin E α -tocopherol; a fat-soluble vitamin considered to act primarily as an antioxidant. Other tocopherols (see **tocopherol** for structure) have vitamin E activity. Vitamin E is thought to prevent the formation of lipid autoxidation products, an action resulting from its ability to trap free radicals and thus to prevent the chain reaction leading to further peroxide formation. It is present in vegetable-seed oils, and also in meat, milk, and eggs. Deficiency occurs in practice only in pathological states in which absorption is affected, and can result in premature red cell death. A relative deficiency may occur when the intake of polyunsaturated fatty acids is high unless the diet is supplemented with α -tocopherol.

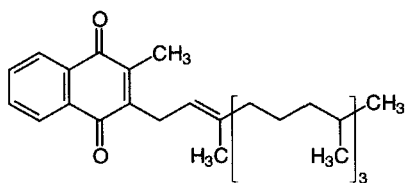
vitamin F (formerly) a collective term for **essential fatty acids**.

vitamin G an old name for riboflavin (i.e. **vitamin B₂**).

vitamin H 1 (sometimes) an alternative name for **biotin**. 2 an old name for riboflavin (i.e. **vitamin B₂**).

vitamin H₁ (sometimes) an alternative name for **p-aminobenzoic acid**.

vitamin K 2-methyl-3-phytyl-1,4-naphthoquinone; phytyl-quinone; 3-phytylmenadiene; vitamin K_1 ; a fat-soluble vitamin required for the synthesis of prothrombin and certain other **blood coagulation** factors. The **menaquinones** (vitamins K_2) are also active. Vitamin K is found in green leafy vegetables, meat, and milk. Deficiency is uncommon because it is produced by gut bacteria, but gut sterilization may trigger symptoms (decrease in plasma prothrombin) if intake is defective. Vitamin K acts as a cofactor in the post-translational formation of 4-carboxyglutamyl residues in certain proteins, e.g. **blood coagulation** factors VII, IX, X, and especially prothrombin, during which an epoxidase reaction occurs to convert vitamin K_1 to its epoxide form, which is converted back by a reductase. This reductase can be inhibited by **warfarin**, which thus reduces vitamin K_1 levels, thereby reducing prothrombin formation and so acting as an anticoagulant.



vitamin-K-epoxide reductase (warfarin-sensitive) EC 1.1.4.1; an enzyme that catalyses the formation of 2-methyl-3-phytyl-1,4-naphthoquinone, oxidized dithiothreitol, and H_2O from 3-epoxy-2,3-dihydro-2-methyl-3-phytyl-1,4-naphthoquinone and 1,4-dithiothreitol.

vitellogenin the precursor protein of **lipovitellin** and **phosphitin**, a major egg-yolk protein synthesized in the liver of chickens and *Xenopus*. It has an M_r of 135 000 and is split in the ovary into its constituent proteins. In *Xenopus*, two genes, for vitellogenin I and II, are spread over more than 21 kbp of DNA, their primary transcripts consisting of 6 kbp of **pre-mRNA** with 33 **introns**. Example from *Gallus gallus* (precursor): database code VIT2_CHICK, 1850 amino acids (204.81 kDa). A related protein is found in *Drosophila melanogaster*. Example (precursor) from *D. melanogaster*: database code VIT1_DROME, 439 amino acids (48.66 kDa).

vitronectin or **serum spreading factor** an adhesion protein, present in plasma and on the external surface of membranes of tissues, that promotes the adhesion of cells in tissue culture. The $\alpha_v\beta_3$ **integrin** binds vitronectin, fibrinogen, laminin, von Willebrand factor, fibronectin, and thrombospondin. Example from human (precursor): database code VTNC_HUMAN, 478 amino acids (54.31 kDa). Residues 1–19 are the signal, 20–478 vitronectin; residues 20–63 are **somatomedin B** (see **somatomedin**).

VI symbol for the valeryl group.

VLA a family of **integrins** all having the β_1 subunit. They comprise VLA-1, $\alpha_1\beta_1$ (β subunit also called VLA β , glycoprotein IIa or gpIIa, CD29; α subunit also called CD49a), which binds laminin (and collagen); VLA-2, $\alpha_2\beta_1$ (α subunit also called gpIIa, CD49b), which binds collagen (and laminin); VLA-3, $\alpha_3\beta_1$ (α subunit also called CD49c), which binds fibronectin, laminin, and collagen; VLA-4, $\alpha_4\beta_1$ (α subunit also called CD49d), which binds VCAM-1 and fibronectin; VLA-5 (or fibronectin receptor), $\alpha_5\beta_1$ (CD49e), which binds fibronectin; VLA-6 (or laminin receptor), $\alpha_6\beta_1$ (CD49f), which binds laminin; $\alpha_7\beta_1$, which binds laminin; $\alpha_8\beta_1$, ligand doubtful; and $\alpha_v\beta_1$ (α subunit also called vitronectin receptor subunit, CD51), which binds fibronectin.

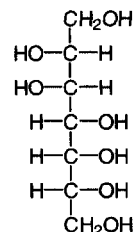
VDL *abbr.* for very-low-density lipoprotein.

VMA *abbr.* for vanillylmandelic acid; see **hydroxymethoxymandelate**.

VMS an **operating system** implemented on DEC VAX computers.

void volume 1 (in **column chromatography**) the volume of the mobile phase, i.e. total **bed volume** of the column minus the volume occupied by the support particles. 2 (in **gel chromatography**) or (sometimes) **exclusion volume** the volume of the mobile phase passing through the gel required to elute a molecule that never entered the stationary phase.

volemitol D-glycero-D-talo-heptitol; a substance widely distributed in plants, fungi, and lichens. It is used as a sweetening agent.



volkensin a ricin-like toxic glycoprotein, purified from the roots of *Adenia volkensii*, a Kenyan shrub. It consists of two subunits, A (M_r 29 000) and B (M_r 36 000), linked by disulfide bonds.



volt

volt *symbol:* V; the SI derived unit of electric potential or electromotive force. It is equal to the difference in electric potential between two points on a conductor carrying a constant current of one ampere when the power dissipated between the points is one watt. $1\text{ V} = 1\text{ W A}^{-1} = 1\text{ J C}^{-1}$. [After Alessandro Volta (1745–1827), Italian physicist.]

voltage-gated ion channel a channel in a cell membrane whose opening is governed by the membrane potential.

voltmeter an old term for **coulometer**.

voltammetry an old term for **coulometry**.

voltammetry an electroanalytical procedure for identifying and determining the concentrations of various ions in solution by studying the relation between a varied electric potential applied between two electrodes immersed in a test solution and the resultant faradaic current. It includes **anodic stripping voltammetry** and **polarography**. —**voltammetric** *adj.*; **voltammetrically** *adv.*

volume of distribution *symbol:* V_d ; the theoretical volume of fluid in which a specified dose of drug is distributed, based on measurement of the concentration of free drug (i.e. that not bound to plasma proteins) in plasma. $V_d = Q/C_p$, where Q is the amount of drug administered and C_p is the concentration in plasma.

volutin granule or **metachromatic granule** an electron-dense cytoplasmic inclusion found in various bacteria, yeasts, and other microorganisms. It exhibits **metachromasia** on staining with certain basic dyes, contains polymetaphosphate, and is believed to serve as a phosphate store. *See also* **Babes-Ernst body**.

von Euler 1 Hans von Euler, *see* **Euler** (def. 1). 2 Ulf Svante von Euler (1905–83), son of H. von Euler (*see* **Euler** (def. 1)), Swedish neurophysiologist and pharmacologist notable for his recognition of the importance of norepinephrine as the main neurotransmitter in mammalian adrenergic neurons; Nobel Laureate in Physiology or Medicine (1970) jointly with J. Axelrod and B. Katz 'for their discoveries concerning the humoral transmitters in the nerve terminals and the mechanism of their storage, release and inactivation'.

von Gierke's disease *see* **glycogen disease**.

von Hevesy *see* **Hevesy**.

von Laue *see* **Laue**.

von Willebrand factor *abbr.*: vWFa; a blood protein involved in blood clotting. It performs two important hemostatic functions: it links specific platelet membrane receptors to collagen and possibly other components of subendothelial connective tissue, so mediating platelet adhesion to sites of vascular damage, and it binds to and stabilizes factor VIII (*see* **blood coagulation**).

Deficiency results in bleeding from skin due to failure of platelet adhesion, with a prolonged bleeding time, and also results in spontaneous bleeding into joints and soft tissues due to the secondary factor VIII deficiency caused by the instability of factor VIII. The primary translation product, synthesized by vascular endothelial cells and megakaryocytes, is a protein of 2813 amino acids (309.29 kDa): database code VWF_HUMAN. Residues 1–22 are a signal peptide, 23–763 a peptide that is cleaved off and is known as **von Willebrand antigen II**, and the mature protein, residues 764–2813. Cysteine is the most abundant amino acid. Before the mature factor is secreted from cells of origin, many intermolecular disulfide bridges between vWF molecules are formed that lead to a complex that may vary in mass from dimers of about 500 kDa to species of more than 20 megadaltons. Inherited defects of vWF affect nearly 1% of the population, but most are not clinically significant. The majority of patients have a quantitative deficiency of vWF, with autosomal dominant inheritance and are known as type I. Type II variants have an abnormal vWF. In vWF deficiency (*abbr.*: vWD) type IIA intermediate and large complexes are absent. Mutations cluster near Tyr⁸⁴²-Ile⁸⁴³, a site that is sensitive to proteolysis. vWD type III is rare, and is a severe autosomal recessive disease characterized by virtual absence of vWF, in some cases known to be due to total or partial deletion of the vWF gene. Unactivated platelets bind vWF through gpIb-IX complex, deficiency of which leads to a bleeding disorder, **Bernard-Soulier syndrome**. After activation, platelets bind vWF through the **integrin** gpIIb-IIIa complex, which is defective in **Glanzmann thrombasthenia**, another bleeding disorder. [After Erik Adolph von Willebrand (1870–1949), Finnish physician.]

V8 proteinase *see* **glutamyl endopeptidase**.

vps34 or **vacuolar sorting protein 34** a protein trafficking protein that is involved in the sorting and segregation of vacuoles. It is a homologue of **1-phosphatidylinositol 3-kinase**, having that enzymic activity, which may be the mechanism by which it acts. It is in turn activated by phosphorylation by a serine/threonine protein kinase, vps15. Example from *Saccharomyces cerevisiae*: database code VP34_YEAST, 875 amino acids (100.92 kDa).

VRE *abbr.* for **vancomycin-resistant Enterococcus** (erstwhile *Streptococcus*). It is one of a number of drug-resistant pathogens (*compare* **MRSA**) found in hospital populations of bacteria. The resistance is due to an enzymic modification of the peptidoglycan precursor that is the target for vancomycin.

v-yes *see* **YES**.

Ww

w symbol for 1 mass fraction. **2** work (alternative to *W*). **3** speed (alternative to *u*, *v*, or *c*). **4** (bold italic) velocity (alternative to *v*, *u*, or *c*).

W symbol for 1 watt. **2** a residue of the α -amino acid L-tryptophan (alternative to Trp). **3** a residue of an incompletely specified base in a nucleic-acid sequence that may be adenine or may be thymine (in DNA) or uracil (in RNA). **4** a residue of the minor nucleoside **wyosine** (alternative to Y). **5** tungsten.

W symbol for 1 weight (alternative to *G* or *P*). **2** work (alternative to *w*). **3** radiant energy (alternative to *Q*). **4** writhing number.

W3 one abbr. for World Wide Web.

Waksman, Selman Abraham (1888–1973), Russian-born US soil microbiologist; Nobel Laureate in Physiology or Medicine (1952) 'for his discovery of streptomycin, the first antibiotic effective against tuberculosis'. See also **Schatz**.

Wald, George (1906–97) US physiologist; Nobel Laureate in Physiology or Medicine (1967) jointly with R. Granit and H. K. Hartline 'for their discoveries concerning the primary physiological and chemical visual processes in the eye'.

Walden inversion the inversion of the configuration at a chiral centre in a bimolecular nucleophilic substitution reaction. [After its discoverer, Paul Walden (1863–1957), Latvian chemist.]

Waldenström's macroglobulinemia a lymphoid tumour that arises from the most mature B lymphocytes and invariably produces IgM, which appear as a **paraprotein**. [After Jan Goesta Waldenström (1906–), Swedish physician.]

wall effect **1** (*in centrifugation*) the collision of sedimenting molecules with the wall of the cell. This occurs because the centrifugal field is radial, so can be avoided by using sector-shaped cells. **2** (*in chromatography*) the curving and spreading of a zone as it migrates down a chromatographic column because of inhomogeneities in solvent flow near the wall of the column.

Warburg, Otto Heinrich (1883–1970), German physiologist and biochemist; Nobel Laureate in Physiology or Medicine (1931) 'for his discovery of the nature and mode of action of the respiratory enzyme'.

Warburg apparatus a sensitive, constant-volume respirometer for measuring gas exchange of cells, homogenates, or tissue slices. It consists of a small flask, with one or more side arms for the addition of reagents, and a centre well, connected to a U-tube manometer of about 1 mm² internal cross-section, fitted with a reservoir for manometer fluid. The vessel is immersed in a constant temperature bath and shaken continually to equilibrate the gas in the liquid phase, where the reaction is taking place, with the gas in the gas phase, where the reaction is being measured. It was used extensively for metabolic studies before the wide availability of radioactive tracers, but is now virtually obsolete.

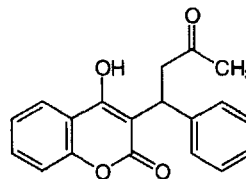
Warburg–Dickens–Horecker pathway or **Warburg–Dickens pathway** an alternative name for **hexose monophosphate pathway**.

Warburg effect **1** the relative overproduction of lactate that occurs in many tumours. **2** the inhibitory effect of high oxygen concentrations on photosynthesis.

Warburg's enzyme an alternative name for old yellow enzyme; see **yellow enzyme**.

warfarin 3-(α -phenyl- β -acetyethyl)-4-hydroxycoumarin; 1-(4'-hydroxy-3'-coumarin-3'-yl)-1-phenylbutan-3-one; an anticoagulant and rodenticide. Its action derives from its ability to inhibit the reductase that converts the epoxide form of **vitamin K** to its reduced form, which is a cofactor for the production of the 4-carboxyglutamyl residues essential for the activity of, among other proteins, the **blood coagulation** factors II (i.e. prothrombin), VII, IX, and X. [From the name of the patentees,

Wisconsin Alumni Research Foundation, + coumarin.] See also **vitamin-K-epoxide reductase (warfarin-sensitive)**.



Waring blender the proprietary name for a blender used in the preparation of tissue homogenates. It consists of a hydrodynamically designed vessel (of tetrafoil cross-section) in the bottom of which a specially shaped blade rotates at high speed. [After the manufacturers, Waring Products Corporation.]

washing soda see **soda**.

Wasserman reaction a form of **complement-fixation test** for human blood and cerebrospinal fluid, widely used in the past to detect syphilitic infection, although false positive reactions were often given by yaws, leprosy, paroxysmal hemoglobinuria, and even malarial infection. The antigen on which the test was based proved to be **cardiolipin**. [After August von Wassermann (1866–1925), German bacteriologist.]

water balance (of an organism) the **water intake** (as a liquid, the water in food, and water of oxidation) less the **water output** (water in urine, feces, or other excreta, water expended in saturating respired air, water lost through the integument).

waterfall sequence an alternative name for **cascade sequence**.

water regain (of an ion-exchange resin) for cationic resins, the weight of water taken up by 1 g dry resin in the hydrogen form; for anionic resins, the weight of water taken up by 1 g dry resin in the chloride form.

water-soluble B an old name for a fraction rich in water-soluble vitamins prepared from egg yolk.

Watson, James Dewey (1928–), US biologist; Nobel Laureate in Physiology or Medicine (1962) jointly with F. H. C. Crick and M. H. F. Wilkins 'for their discoveries concerning the molecular structure of nuclear [sic] acids and its significance for information transfer in living material'. See also **Franklin**.

Watson–Crick model of DNA a model that envisages one form of **DNA** (deoxyribonucleic acid) as consisting essentially of two antiparallel helical polynucleotide chains coiled around the same axis to form a double helix. The deoxyribose-phosphate backbones lie on the outside of the helix and the purine and pyrimidine bases lie approximately at right angles to the axis on the inside of the helix. The diameter of the helix is 2.0 nm and there is a residue on each chain every 0.34 nm in the *z* direction. The angle between each residue on the same chain is 36°, so that the structure repeats after 10 residues (3.4 nm) on each chain. The two chains are held together by hydrogen bonds between pairs of bases, each member of the pair belonging to a different polynucleotide chain. Adenine is always paired with thymine and guanine with cytosine. The two chains are therefore complementary. Watson and Crick's paper, published in *Nature* in 1953, also contained a prescient statement: 'It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.' The structure of DNA proposed by Watson and Crick has stood the test of time. It is difficult to exaggerate the importance of the short paper. To quote from L. Stryer's *Biochemistry* (fourth edi-

**watt**

tion), 'This brilliant accomplishment ranks as one of the most significant in the history of biology because it led the way to an understanding of gene function in molecular terms.' See also **double helix**. [After J. D. Watson and F. H. C. Crick.]

watt *symbol*: W; the SI derived unit of power or radiant flux; it is the power dissipated when one joule is expended in one second; i.e. $1 \text{ W} = 1 \text{ J s}^{-1}$. [After James Watt (1736–1819), British engineer.]

wave function *symbol*: ψ or Ψ or Φ or ϕ ; the amplitude of the wave associated with a particle, e.g. an electron, as obtained from the Schrödinger wave equation. The square of the value of ψ at any point is proportional to the probability of finding the particle at that point.

wavelength *symbol*: λ ; the distance between successive points of equal phase of a wave. Wavelength is equal to the velocity of the wave motion divided by its frequency. For electromagnetic radiation, $\lambda = c_0/\nu$, where c_0 is the velocity of light in *vacuo* and ν the frequency of the radiation.

wavenumber or **repetency** for an electromagnetic radiation (e.g. light), the reciprocal of the **wavelength**; i.e. the number of waves in unit distance. *Compare* **kayser**.

wax **1** any lipid fraction from living organisms or from crude petroleum that is a plastic substance, hard when cold, easily moulded when warm, and insoluble in water. **2** any fatty-acid ester of a long-chain monohydroxy alcohol.

wax synthase *see* **long-chain-alcohol O-fatty-acyltransferase**.

Wb *symbol* for **weber**.

WBC or **wbc** or **w.b.c.** *abbr.* for white blood cell(s); *see* **leukocyte**.

WDHA syndrome *abbr.* for watery diarrhoea, hypokalemia, plus achlorhydria syndrome; i.e. **Verner-Morrison syndrome**.

weak interactions noncovalent interactions between molecules and parts of molecules, including charge-charge interactions, those involving permanent dipoles, van der Waals forces, and hydrogen bonds. They are particularly important in protein and nucleic-acid structure.

weber *symbol*: Wb; the SI derived unit of magnetic flux. It is equal to the flux that, linking a circuit of one turn, produces in it an emf of one volt as it is reduced to zero at a uniform rate in one second; i.e. $1 \text{ Wb} = 1 \text{ V s}$. [After Wilhelm Eduard Weber (1804–91), German physicist.]

wee1 a gene in *Schizosaccharomyces pombe* encoding a protein kinase (EC 2.7.1.112) that can phosphorylate tyrosine and serine/threonine residues. In yeast it phosphorylates a single tyrosine of *cdc2*; in the human it phosphorylates both a tyrosine and a threonine of the equivalent protein. It has conserved catalytic domains of the other (serine/threonine) type of protein kinase. The protein acts as a negative regulator for entry to mitosis; it closely resembles the corresponding human protein kinase. Example from *S. pombe*: database code WEE1_SCHPO, 877 amino acids (96.26 kDa). *See also* **cdc gene**.

weight average molecular mass *old name* for mass average molar mass; *see* **average molar mass**.

Weigle reactivation *see* **W reactivation**.

Wernicke-Korsakoff syndrome a combination of **Wernicke's disease** (gliosis and capillary proliferation leading to ocular manifestations and polyneuropathy) and **Korsakoff's psychosis** (difficulty in recording and retaining new impressions). Wernicke's disease is due to vitamin B₁ deficiency, and manifests particularly in patients with a genetic abnormality in transketolase resulting in its reduced affinity for thiamine diphosphate. In such patients, even mild B₁ deficiency, as often occurs in alcoholics, may produce symptoms. [After Karl Wernicke (1848–1905), German alienist, and Sergei Sergeevich Korsakoff (1854–1900), Russian neurologist.]

Western blotting a technique for **blotting** proteins onto nitrocellulose, nylon, or other **transfer membrane** after they have been resolved by gel electrophoresis. The proteins can then be detected by one of several methods, including autoradiography (if labelled), or through binding to fluorescently labelled antibodies.

belled, ¹²⁵I-labelled, or enzyme-linked (*see* **ELISA**) antibody, lectin, or other specific binding agent. The name derives by extension through the cardinal points from **Southern blotting**, the first such blotting technique, with a capital initial by similar extension.

Whitaker plot a plot that, ideally, gives a linear correlation between the logarithm of the relative molecular mass of a macromolecule and the ratio of the **elution volume**, V_e , to the **void volume**, V_o , when the macromolecules in solution are passed through a suitable **gel filtration** column; i.e. $\log M_r = V/V_o$. [After John Robert Whitaker (1929–), US biochemist.]

white adipose tissue or **white fat** a highly specialized tissue developed particularly in mammals and birds to store fat to supply energy to the whole animal. The cells contain a single large droplet of fat that, in the nourished state, fills most of the cell. The nucleus is somewhat flattened and is in the narrow band of cytoplasm around the cell's periphery. Adipose tissue is found in many parts of the body but tends to be concentrated under the skin (subcutaneous) and around internal organs (heart and kidney). The stored fat consists principally of triacylglycerols. *Compare* **brown adipose tissue**.

white blood cell *see* **leukocyte**.

white fat *see* **white adipose tissue**.

white muscle pale, skeletal, usually involuntary muscle that is relatively deficient in myoglobin and cytochromes. *Compare* **red muscle**.

Wieland, Heinrich Otto (1877–1957), German chemist; Nobel Laureate in Chemistry (1927) 'for his investigations of the constitution of the bile acids and related substances'.

wild-type the **phenotype** that is characteristic of most of the members of a species occurring naturally and contrasting with the phenotype of a **mutant**. *Compare* **auxotroph**.

Wilkins, Maurice Hugh Frederick (1916–), New Zealand-born British biophysicist; Nobel Laureate in Physiology or Medicine (1962) jointly with F. H. C. Crick and J. D. Watson 'for their discoveries concerning the molecular structure of nuclear [sic] acids and its significance for information transfer in living material'. *See also* **Franklin**.

Wilkinson, (Sir) Geoffrey (1921–96), British inorganic chemist noted for his extensive studies of transition-metal complexes with organic compounds and for the discovery of rhodium complexes now used as catalysts in industrial hydrogenation reactions; Nobel Laureate in Chemistry (1973), the prize being shared with E. O. Fischer 'for their pioneering work, performed independently, on the chemistry of the organometallic, so called sandwich compounds'.

Willstätter, Richard Martin (1872–1942), German organic chemist; Nobel Laureate in Chemistry (1915) 'for his researches on plant pigments, especially chlorophyll'.

Wilm's tumour or **nephroblastoma** a malignancy in human infants (incidence 1 in 10 000) associated with protein **WT33**. [After Marx Wilms (1867–1918), German pathologist.]

Wilson, Charles Thomson Rees (1869–1959), British physicist; Nobel Laureate in Physics (1927) 'for his method of making the paths of electrically charged particles visible by condensation of vapour' [prize shared with A. H. Compton].

Wilson's disease or **hepatolenticular degeneration** a rare autosomal recessively inherited disease characterized by degenerative changes in the brain, particularly in the basal ganglia, and cirrhosis of the liver. There are excessive deposits of copper in the liver, brain, cornea, and kidney; the serum copper is usually low and there is a **ceruloplasmin** deficiency. Patients may be effectively treated with British antilewisite (**dimercaprol**) or other chelating agents. [After (Samuel Alexander) Kinnier Wilson (1878–1937), US-born British neurologist.]

Wilzbach method a method for the random labelling of a compound with tritium by exposing it to isotopically pure tritium gas in a sealed container for several weeks. The exchange is promoted by the radiation.

Windaus

Windaus, Adolf Otto Reinhold (1876–1959), German chemist; Nobel Laureate in Chemistry (1928) 'for the services rendered through his research into the constitution of the sterols and their connection with the vitamins'.

winding angle (of a DNA double helix) the angle of rotation between successive base pairs.

winding number the number of double-helical turns in a closed-circular DNA measured relative to a surface on which the DNA helix lies. *See also* **supercoil**.

windowless counter a counter for radioactive (or X-ray) ionization in which the sample (or source) is not separated from the detector by a membrane or 'window'.

wingless a gene in *Drosophila melanogaster* that codes for a protein in the **WNT family**, probably a growth factor that acts on neighbouring cells to regulate the **engrailed** gene. Database code WNTG_DROME, 468 amino acids (51.93 kDa). A gene for a homologous protein is present in the brachiopod *Terebratulina retusa*.

WNT family a protein family that includes several proteins (e.g. WNT-2 from human and mouse) that are probably signalling molecules in tissue development. They are encoded by **WNT** genes. The **WNT-1** gene is closely homologous to **wingless**. Example, WNT-2 (precursor) from human: database code WNT2_HUMAN, 360 amino acids (40.37 kDa).

wobble base the third base in the anticodon of **transfer RNA** (5' end). It can bind to one of several possible bases at the 3' end of the codon in mRNA. *See also* **wobble hypothesis**.

wobble hypothesis a hypothesis that in **translation** during protein synthesis a less strict specificity in the base pairing of the 5' base of the anticodon of **transfer RNA** (tRNA) allows it to make alternative hydrogen-bonding with the third base of the codon of messenger RNA (mRNA) beyond the usual G–C and A–U pairings. Specifically, anticodon U may recognize A or G and anticodon G may recognize C or U; then A no longer has a unique meaning at this position, since the U that recognizes it also recognizes G; the same applies to the codon U, which can also be recognized by G. In practice, although the hypothesis has been fully supported by experiment, UGG and AUG are the only examples of the first type of wobble, and no examples of the second type are known. The presence of bases other than U, A, G, or C at the anticodon 5' position also contributes to more general pairing at this position than would be the case if G–C, A–U specificity were strictly observed. Thus, bases such as **wyosine**, **queuosine**, and **inosine** may be present (I always substitutes for A) with less strict pairing constraints. The degeneracy at the third codon position is consistent with the fact that about 40 different tRNA molecules interact with the 61 possible sense codons. Not all of the different tRNA structures have been determined, so it is not possible to state with certainty the structure of all the anticodons, although possible structures can be predicted. In the case of tRNA^{Ala} the structure of the two tRNAs is known.

Wolfgram proteolipid protein one of the three principal classes of central nervous system **myelin protein**, extractable by chloroform:methanol (2:1, v/v) and consisting of three main fractions of 235, 54, and 62 kDa. *Compare* **Folch–Lees protein**. [After F. Wolfgram, who first discovered it.]

Wood, Harland Goff (1907–91), US biochemist distinguished particularly for work on fixation of CO₂; the reaction between pyruvate and CO₂ became known eponymously as the **Wood–Werkman reaction**. For 46 years he worked in the Department of Biochemistry at Case Western Reserve University. Described by Hanson as an 'American original' born and raised in the American heartland.

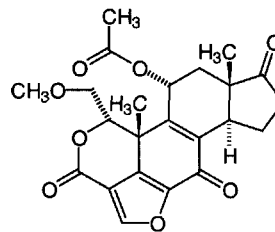
Woodward, Robert Burns (1917–79), US organic chemist; Nobel Laureate in Chemistry (1965) 'for his outstanding achievements in the art of organic synthesis'.

work *symbol*: *W* or *w*; the transfer of energy expressed as the product of a force, *F*, and the distance, *s*, through which its

point of application moves in the direction of the force; i.e. $W = \int F ds$. The SI unit of work is the joule. The manifestation of work in productive form usually involves transduction by a mechanical device of the action of a force through a distance. A relationship between the work done on (or by) a system of constant mass, the heat it gains (or loses), and the change in the internal energy of the system is provided by the first law of thermodynamics (*see* **thermodynamics**).

World Wide Web *abbr.*: WWW; or W3; an **Internet** facility for viewing and recovering archival material including an extensive range of biochemical and molecular biological data. Examples are included in **Appendix B** and **Appendix C**. *See also* **Appendix D**.

wortmannin or **MS-54** or **KY 12420** a potent inhibitor of smooth muscle myosin light chain kinase and protein kinase C produced by the fungal strain *Talaromyces wortmanni* KY12420. It is also a potent selective, cell-permeable and irreversible inhibitor of **1-phosphatidylinositol 3-kinase**.



W reactivation or **Weigle reactivation** or (formerly) **UV reactivation** the phenomenon whereby the survival of UV-irradiated bacteriophage is higher in a UV-irradiated host than in an unirradiated host. [After Jean Jacques Weigle (1901–68), who discovered it and described it in 1953.]

writhing number *symbol*: *W*; an index of the superhelical coiling of DNA, termed **writhe**. The writhing number and the **twist number** between them determine the **linking number** (*L*). The writhing number does not have a precise quantitative definition, but represents the degree of supercoiling. A decrease in *L* involves some decrease in supercoiling, and an increase in *L* an increase in supercoiling. *See also* **supercoil**.

wt or **wt.** *abbr.* for weight.

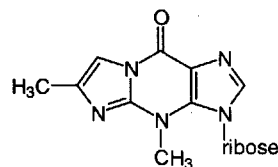
WT33 a protein that is related to **early growth response proteins** and is associated with **Wilm's tumour**; it is probably a tumour suppressor. Other mutations are associated with Denys–Drash syndrome. It has alternative splice sites, a zinc finger, and four motifs. Example from human: database code WT1_HUMAN, 449 amino acids (49.13 kDa).

w/v or **wt/vol.** *abbr.* for weight per unit volume (of a solution or mixture).

w/w or **wt/wt** *abbr.* for weight per unit weight (of a solution or mixture).

WWW *abbr.* for **World Wide Web**.

wyosine *symbol*: *W* or *Y*; any of a number of modified forms of guanosine present in the anticodon loop of some species of **transfer RNA**. All are based on the structure *N*²-(2-methyl-etheno)guanosine, and carry varying side chains on the five-carbon imidazo ring. The structure of one such compound is shown. *See also* **minor nucleoside**.



Xx

x symbol for 1 amount-of-substance fraction (i.e. mole fraction) or number fraction for condensed phases. Compare **y**. 2 a Cartesian space coordinate, complementary to *y* and *z*.

X 1 symbol for a residue of an unknown, unspecified, or (specified) unusual α -amino acid (alternative to Xaa). b a residue of the ribonucleoside **xanthosine** (alternative to Xao). 2 abbr. for X Windows.

X symbol for (electric) reactance.

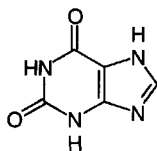
X-537A an alternative name for **lasaloacid**.

Xaa symbol for a residue of an unknown, unspecified, or (specified) unusual α -amino acid (alternative to X).

Xan symbol for a residue of the purine base **xanthine**.

xanthan gum a capsular complex heteropolysaccharide formed by strains of the pseudomonad bacterium *Xanthomonas campestris* and believed to promote adhesion of the organism to its plant host. It is widely used in the cosmetic, food, and other industries as a crystallization inhibitor, emulsifier, gelling agent, etc.

xanthine symbol: Xan; 2,6-dihydroxypurine; 2,6-dioxypurine; 3,7-dihydro-1*H*-purine-2,6-dione; a purine formed in the metabolic breakdown of guanine but not present in nucleic acids. Compare **hypoxanthine**. See also **allopurinol**.



xanthine oxidase EC 1.1.3.22; the recommended name for hypoxanthine oxidase; systematic name: xanthine:oxygen oxidoreductase; a flavoprotein (FAD) enzyme containing iron and molybdenum, abundant in milk and liver, that catalyses the oxidation of **xanthine** to urate and superoxide (or hydrogen peroxide) and also the oxidation of hypoxanthine to xanthine. It was formerly known as **Shardinger enzyme**. The enzyme can exist in two forms. One, known as xanthine dehydrogenase, has **NAD** as coenzyme, and urate and NADH as products. The other form, xanthine oxidase, reacts with dioxygen and has urate and H₂O₂ as products. The dehydrogenase is converted reversibly to the oxidase by oxidation of sulfhydryl groups or irreversibly by proteolysis. Example from *Rattus norvegicus*: database code XDH_RAT, 1330 amino acids (146.11 kDa). Example from human (homodimer): database code XDH_HUMAN, 1338 amino acids (147.76 kDa).

xanthinuria the excretion of excessive amounts of xanthine in the urine. It is sometimes due to a hereditary disease in which there is a gross deficiency of **xanthine oxidase** activity in the tissues.

xantho+ or (before a vowel) **xanth+** comb. form denoting yellow.

xanthophyll any **carotenoid** in which one or more oxygenated functions are present in the molecule.

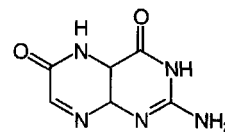
xanthoproteic test a qualitative test for protein in which a yellow colour (or precipitate from solution) is formed on addition of conc. nitric acid; the yellow colour turns orange when made alkaline.

xanthopterin a yellow pterin pigment of insect wings, e.g. of the common sulfur butterfly. Compare **leukopterin**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

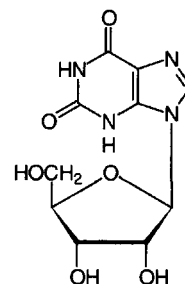
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



xanthopterin

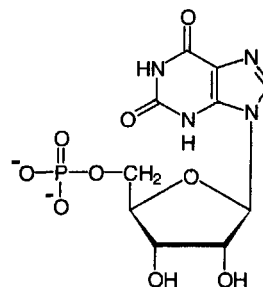
xanthosine symbol: X or Xao; xanthine riboside; 9- β -D-ribofuranosylxanthine; a nucleoside formed by the deamination of guanosine.



xanthosine monophosphate abbr.: XMP; an alternative name for any xanthosine phosphate, but in particular for xanthosine 5'-phosphate, especially when its distinction from xanthosine (5'-)diphosphate and xanthosine (5'-)triphosphate requires emphasis.

xanthosine phosphate xanthosine monophosphate (abbr.: XMP); any phosphoric monoester or diester of xanthosine. Of the three possible monoesters – the 2'-phosphate, the 3'-phosphate, and the 5'-phosphate – and the two possible diesters – the 2',3'-phosphate and the 3',5'-phosphate – only **xanthosine 5'-phosphate** is known to occur naturally (the locant being omitted if no ambiguity may arise).

xanthosine 5'-phosphate or 5'-xanthylic acid or 5'-phosphoxanthosine or 5'-O-phosphonoxanthosine symbol: Xao5'*P*; alternative recommended names for xanthosine monophosphate (abbr.: XMP); xanthosine 5'-(dihydrogen phosphate); xanthine (mono)ribonucleotide. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation.) It is an intermediate in the biosynthesis of guanosine 5'-phosphate, 5'-guanylic acid, GMP. XMP is formed from inosine 5'-phosphate, 5'-inosinic acid, by an NAD⁺-dependent dehydrogenase on the pathway for synthesis of guanine nucleotides; a glutamine-requiring amidotransferase reaction (catalysed by **GMP synthase**) then converts XMP to GMP.



xanthylic acid

xanthylic acid the trivial name for any phosphoric monoester of xanthosine. The position of the phosphoric residue on the ribose moiety of a given ester may be specified by a prefixed locant (see **xanthosine phosphate**). However, 5'-xanthylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-Xanthylic acid is also an alternative recommended name for **xanthosine 5'-phosphate**.

Xao symbol for a residue of the ribonucleoside **xanthosine** (alternative to X).

Xao5'P symbol for xanthosine 5'-phosphate.

Xao5'PP symbol for xanthosine 5'-diphosphate (alternative to ppX).

Xao5'PPP symbol for xanthosine 5'-triphosphate (alternative to pppX).

X cell see **X-Y-Z cell theory**.

X chromosome a sex chromosome. It is often found paired in the **homogametic** sex, which in many species is the female, and single in the **heterogametic** sex, in many species the male. The X chromosome carries a large number of genes which control many aspects of development and function. A mutant gene which is so carried is called X linked. Since males possess only one X chromosome an abnormal gene that it carries cannot be paired with a normal allele; the male is then said to be hemizygous for that particular mutant gene. The female in whom both members of a pair of X-linked genes are identical is said to be homozygous for that gene; if the members of a gene pair are dissimilar the woman is called a heterozygote or carrier for the abnormal allele. One X chromosome is inactivated during early embryonic development so that individuals do not have a double dose of genes on the X chromosome. Since the descendants of each cell retain the same inactivated X chromosome, a proportion of cells have the paternal X chromosome in an inactive state while in the other cells the maternal X chromosome is active. Hence the female who is heterozygous for an abnormal gene will have two types of cells, one with the normal and one with the mutant gene. She is said to be a mosaic. Compare **Y chromosome**. See also **sex determination**.

X-chromosome inactivation inactivation in females of one of the two X chromosomes during early embryonic development; this phenomenon occurs randomly in all cells of the embryo. Thus, the cells in a female are mosaic in respect to which of the X chromosomes is functional – that from the father or the mother. See also **X chromosome**.

XDP abbr. for xanthosine (5'-)diphosphate.

Xe symbol for xenon.

xenin a 25-residue peptide isolated from human gastric mucosa. The six C-terminal amino acids are identical to those of the octapeptide, **xenopsin**, present in the skin of amphibians. It stimulates exocrine pancreatic secretion, rising in plasma after a meal. Example from human: database code XENI_HUMAN, 25 amino acids (2.97 kDa).

xenite a subsidiary organism in a **synxenic** culture.

xeno+ or (before a vowel) **xen+** comb. form indicating something strange or foreign.

xenobiochemistry the biochemistry of organic compounds foreign to the organism.

xenobiotic 1 any substance that does not occur naturally but interacts with the metabolism of any organism. 2 of, or relating to, substances that are foreign to living systems.

xenobiotic-response element abbr.: XRE; a DNA-regulatory sequence that binds transcription factors responsible for activation of gene expression for enzymes that detoxify a number of xenobiotics, including halogenated and other aryl hydrocarbons. See also **Ah**.

xenogeneic or **xenogenic** originating in a different species, especially of a tissue transplant. See also **xenograft**.

xenograft a tissue graft between individuals of different species.

Xenopus laevis the South African clawed toad, the oocytes and embryos of which have been extensively exploited in molecular genetics and vertebrate embryology. The large oocytes

were used for pioneering studies of the microinjection of nucleic acids directly into the nucleus, a technique now extended to other species.

xenotope an antigenic **epitope** that is foreign to a particular animal.

xenotype a class or group of related **xenotopes**; i.e. an antigenic **epitope** that is foreign to a particular animal.

xeroderma pigmentosum an autosomal recessive hereditary skin disease of humans in which homozygotes show a marked tendency to develop skin cancers following exposure to sunlight. It is due to a reduced ability to excise UV-induced cyclobutane **pyrimidine dimers** from DNA.

xerophthalmia a condition in which there is progressive drying and wrinkling of the conjunctiva of the eye. One cause is **vitamin A** deficiency.

X-gal abbr. for 5-bromo-4-chloro-3-indolyl- β -D-galactoside, a colourless substrate that is hydrolysed by β -galactosidase to a blue product. It is used with **cloning vectors** containing part of the *lacZ* gene to select recombinants in molecular cloning: the peptide produced by the *lacZ* fragment complements the defective β -galactosidase enzyme in host cells. Insertion of foreign DNA into the vector inactivates the *lacZ* gene, so host cells containing recombinant plasmids will grow as white colonies on medium containing X-gal, whereas host cells containing 'empty' vectors will have a functional β -galactosidase and thus produce blue colonies.

xi symbol: ξ (lower case) or Ξ (upper case); the fourteenth letter of the Greek alphabet. For uses see **Appendix A**.

Xiphophorus a genus of fish, that includes the South American minnow, some species of which have been much used for the study of inherited characteristics involved in carcinogenesis. Crossing of certain species, e.g. backcrossing *X. maculatus* (platyfish) with *X. helleri* (swordtail: recurrent parent) leads to spontaneous development of melanoma.

Xis protein or **excisionase** the product of the *xis* gene in lambda phage: database code VXIS_LAMBD, 72 amino acids (8.60 kDa). It plays an essential role in the excision step of release of phage DNA from **prophage**. See also **integrase**.

XMP abbr. for xanthosine monophosphate; i.e. xanthosine (5'-)phosphate.

xotch a transmembrane protein in *Xenopus* that is similar in sequence, and presumably in role, to the product of **notch**. It is uniformly expressed in early embryos. It contains 36 EGF-like domains and three notch-type repeats. Example from *Xenopus laevis*: database code NOTC_XENLA, 2524 amino acids (275.12 kDa).

X-Pro dipeptidase EC 3.4.13.9; other names: proline dipeptidase; imidodipeptidase; prolidase; peptidase D; γ -peptidase; an enzyme that catalyses the hydrolysis of Xaa-l-Pro dipeptides; it also acts on aminoacyl-hydroxyproline analogues, but has no action on Pro-l-Pro. It requires manganese. Example from human: database code PEPD_HUMAN, 492 amino acids (54.19 kDa).

XPS abbr. for X-ray photoelectron spectroscopy; see **photoelectron spectroscopy**.

X-ray crystallography the study of the geometric forms of crystals by **X-ray diffraction**. Myoglobin was the first protein for which the structure was solved by X-ray crystallography.

X-ray diffraction a method that uses the diffraction pattern obtained by passing X-rays through crystals, or other regular molecular arrays, to measure interatomic distances and to determine the three-dimensional arrangement of atoms (or molecules) in the structure studied. For the pattern to be sharp, the wavelength of the radiation used must be shorter than the regular spacing between the elements of the structure. X-rays typically have wavelengths of only a few tenths of a nanometre, making them suitable for biological samples. X-ray diffraction is used extensively in the determination of structures of proteins and nucleic acids and many other natural products (an early landmark was, together with chemical evidence, the determination of the structure of penicillin). In order to re-



X-ray photoelectron spectroscopy

solve the phase differences, much use has been made of **isomorphous replacement** with heavy metal atoms in these structural studies.

X-ray photoelectron spectroscopy *abbr.*: XPS; *see* **photoelectron spectroscopy**.

X-rays or (sometimes, esp. US) **x-rays** or (formerly) **röntgen rays** electromagnetic radiation emitted by atoms during extranuclear loss of energy of incident radiation (e.g. high-energy electrons, gamma radiation) or by atoms of certain radionuclides during transformation by **electron capture**. X-rays have wavelengths in the range 1 pm–10 nm (frequencies 3 PHz–3 EHz). This range lies above that of gamma radiation and overlaps the wavelengths of the far ultraviolet. Characteristic X-rays have specific wavelengths of a particular nuclide or target element. *See also* **bremstrahlung**, **synchrotron**.

XRE *abbr.* for xenobiotic-response element.

XTP *abbr.* for xanthosine (5'-)triphosphate.

Xul *symbol* for a residue (or sometimes a molecule) of the ketopentose xylulose.

X Windows a graphical interface, often simply called X, for UNIX and VMS computers. Many molecular graphical and other programs (including **AcEDB**) use X.

xyl+ a variant (sometimes before a vowel) of **xyl+**.

Xyl *symbol* for a residue (or sometimes a molecule) of the aldopentose xylose.

xylan any of a group of homopolysaccharides of **xylose**, consisting mainly of β -1,4-linked xylopyranose units with various substituent groups, that occur in the cell walls of higher plants, and in some marine algae. Xylans form the major glycans of the **hemicellulose** component of the fibrous parts of plants.

xylaric acid the **aldaric acid** derived from **xylose**.

xylE a gene in *Escherichia coli* that encodes a xylose transport protein. A 12-transmembrane-domain protein, it functions as an H⁺-xylose symporter. The gene is induced by xylose. Protein product: database code XYLE_ECOLI, 491 amino acids (53.55 kDa). *See also* **sugar transporter**.

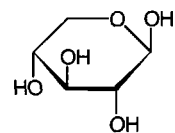
xylem the vascular tissue that conducts water and mineral salts from the roots to the aerial parts of a plant and provides it with mechanical support.

xyl+ or (sometimes before a vowel) **xyl+** *comb. form* indicating the trivial name for the aldopentose xylose.

xylo- *prefix (in chemical nomenclature)* indicating a particular configuration of a set of three contiguous >CHOH groups, as in the acyclic form of D- or L-xylose. *See* **monosaccharide**.

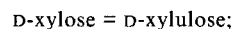
xylose *symbol*: Xyl; *xylo*-pentose; a constituent of plant polysaccharides (*see* **xylan**); the naturally occurring enantiomer is

always D-xylose. The metabolism of xylose in the human is restricted, and most is excreted unchanged in the urine. In clinical chemistry this forms the rationale for the **xylose absorption test**, the most widely used test of carbohydrate absorption.



β -D-pyranose form

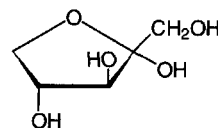
xylose isomerase EC 5.3.1.5; *systematic name*: D-xylose ketol-isomerase; an enzyme that catalyses the reaction:



magnesium ions are a cofactor. Example from *Streptomyces olivochromogenes*: database code NRL_3XIA, 377 amino acids (41.90 kDa); 3-D structure known; it is a **beta-barrel** protein.

xylose-1-phosphate uridylyltransferase *see* **UTP-xylose-1-phosphate uridylyltransferase**.

xylulose *symbol*: Xul; *the trivial name* for the ketopentose *threo*-2-pentulose; D-xylulose is the naturally occurring enantiomer.



α -D-xylulose

D-xylulose 5-phosphate the 5-phosphate ester of D-xylulose and a component of the **pentose phosphate pathway**. It is formed from D-ribulose 5-phosphate by ribulose-phosphate 3-epimerase (EC 5.1.3.1); D-xylulose 5-phosphate then reacts with D-ribose 5-phosphate, catalysed by **transketolase**, to give glyceraldehyde 3-phosphate and sedoheptulose 7-phosphate.

X-Y-Z-cell theory a theory describing immunocyte maturation in which the first committed immunocyte, the **X cell**, is transformed by antigen into a memory cell, the **Y cell**, in the primary response, and is further stimulated in the secondary response to become an antibody producer, the **Z cell**.

Yy

y symbol for yocto+ (SI prefix denoting 10^{-24} times).

Y symbol for 1 amount-of-substance fraction (i.e. mole fraction) or number fraction for gases. Compare **x**. **2** a Cartesian space coordinate, complementary to **x** and **z**.

Y symbol for 1 yotta+ (SI prefix denoting 10^{24} times). **2** a residue of the α -amino acid L-tyrosine (alternative to Tyr). **3** a residue of an incompletely specified base in a nucleic-acid sequence that may be either cytosine or thymine. **4** a residue of an unspecified pyrimidine nucleoside (alternative to Pyd). **5** a residue of the minor nucleoside **wyosine** (alternative to W). **6** yield. **7** yttrium.

Y symbol for admittance (reciprocal of impedance).

YAC abbr. for yeast artificial chromosome.

Yalow, Rosalyn Sussman (1921–), US physicist; Nobel Laureate in Physiology or Medicine (1977) 'for the development of radioimmunoassays of peptide hormones' [prize shared with R. Guillemin and A. V. Schally].

Yb symbol for ytterbium.

Y cell see **X-Y-Z-cell**.

Y chromosome a **sex chromosome** found only in the **heterogametic** sex and usually different in size from the **X chromosome**. In many animals it carries the testis-determining factor that triggers male embryonic development.

yeast any of a group of unicellular fungi that reproduce asexually – by budding or fission – and sexually – by the production of ascospores. Yeast cells may occur singly or in short chains, and some species produce a mycelium. The term 'yeast' is often used to mean members of the genus *Saccharomyces*, e.g. *S. cerevisiae*, which is an example of a budding yeast, or *Schizosaccharomyces pombe*, which is an example of a fission yeast.

yeast artificial chromosome abbr.: YAC; a specialized cloning vector that contains a **centromere**, an autonomously replicating sequence (ARS), a pair of **telomeres**, selectable marker genes, and the fragment of DNA to be cloned. Usually, genomic DNA is digested to produce fragments containing the genes, which are separated by **pulsed field gel electrophoresis**; the large fragments are then ligated into YACs. They are capable of propagation in yeasts, where they function as 'artificial chromosomes', being efficiently replicated. YACs are essential in large mapping projects, such as the Human Genome Project (see **HUGO**), as they are stable and can carry vast DNA inserts of 100 to 1000 kbp.

yeast cloning any of a number of methods of cloning using yeast as the host; cloning in yeasts and other microbial eukaryotes has a number of advantages. Expression of eukaryote genes in these organisms is subject to eukaryotic regulatory systems, and the products undergo post-translational modification in contrast to when they are expressed in prokaryotes. Thus, yeast clones are often used for the expression of glycoproteins.

yeast eluate factor a former term for **vitamin B₆**.

yeast filtrate factor a former term for pantothenic acid. See **pantothenate**.

yeast nucleic acid a former term for **ribonucleic acid**.

yeast two-hybrid system a system that exploits two hybrid proteins, one containing a protein, **X**, fused to the GAL4 DNA-binding domain, and the other containing a different protein, **Y**, fused to the GAL4 activation domain. Plasmids encoding each of the hybrid proteins are introduced together into yeast, which leads to the expression of both hybrids. The system is used to test whether protein **Y** binds to protein **X**, for example in investigating the role of newly discovered proteins. If the proteins do bind, the resulting complex will be

bound to the upstream activation sequence for the yeast **GAL** genes by the GAL4 DNA-binding domain, and the presence of the GAL4 activation domain will lead to transcriptional activation of the **reporter gene**, i.e. β -galactosidase. The system has the advantage over immunoprecipitation that if **Y** binds to **X**, the gene encoding it is already available as a clone. The hybrid containing **X** is referred to as the **bait**.

yellow enzyme **1** old **yellow enzyme**, EC 1.6.99.1; NADPH dehydrogenase; a flavoprotein (FMN in yeast, FAD in plants). **2** new **yellow enzyme** an alternative name for D-amino-acid oxidase; see **amino-acid oxidase**.

yellow marrow see **bone marrow**.

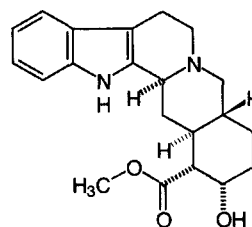
YES v-yes is the oncogene of the Yamaguchi sarcoma virus, and has a cellular counterpart, **c-yes**, the product of which is a member of the **src** family of non-receptor tyrosine kinases, having a myristoylation site, and SH2, SH3 and tyrosine kinase domains. Example, p61 from *Gallus gallus*; database code YES_CHICK, 541 amino acids (60.79 kDa).

yield symbol: **Y**; a ratio expressing the efficiency of a mass conversion process. The yield coefficient is defined as the amount of cell mass or product formed related to the consumed substrate or to the intracellular ATP production.

yin-yang hypothesis (of biological control) a concept of biological regulation based on the opposing actions of certain cyclic nucleotides. It arises from the observation that hormones or other biologically active substances that promote the cellular accumulation of cyclic GMP produce cellular responses that are antagonistic to ones occurring when the concentration of cyclic AMP is increased in the same tissues or cells. Hence cyclic GMP and cyclic AMP may have opposing or antagonistic regulatory influences in certain biological systems, somewhat analogous to the oriental concept of yin and yang symbolizing a dualism between opposing natural forces.

yocto+ symbol: **y**; SI prefix denoting 10^{-24} times.

yohimbine a complex indole alkaloid, $C_{21}H_{26}N_2O_3$, with α_2 -adrenoceptor antagonist activity. It is produced by *Corynanthe johimbe* and *Rauwolfia serpentina*.



Yop abbr. for *Yersinia* outer protein; any of a number of proteins, known as virulence proteins, originally found on the surface of Gram-negative bacteria of the genus *Yersinia*; they are now known to be characteristic of proteins pathogenic to plants, that are secreted by bacteria of genera as diverse as *Pseudomonas*, *Xanthomonas*, and *Erwinia*. They are also produced by animal pathogens such as *Shigella flexneri*, *Salmonella typhimurium*, and *Escherichia coli*. They are secreted by the **type III system**. Unlike other secreted proteins they lack a signal sequence.

yotta+ symbol: **Y**; SI prefix denoting 10^{24} times.

Yphantis method an alternative name for **meniscus depletion method** (**sedimentation equilibrium**).

Zz

z symbol for zepto+ (SI prefix denoting 10^{-21} times).

z symbol for 1 charge number of an ion. **2** a Cartesian space coordinate, complementary to *x* and *y*.

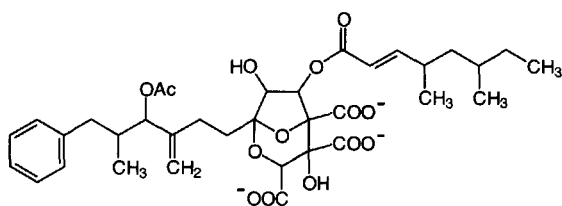
Z symbol for 1 zetta+ (SI prefix denoting 10^{21} times). **2** a residue of either of the α -amino acids L-glutamic acid or L-glutamine when the state of amidation is uncertain (alternative to Glx). It may also be used to represent a residue of any other substance, such as 4-carboxy-L-glutamic acid (as alternative to Gla) or 5-oxo-L-proline (as alternative to Glp), that yields glutamic acid on acid hydrolysis of peptides. **3** the **benzyloxycarbonyl** group (alternative to Cbz).

Z symbol for 1 proton number. **2** impedance.

(Z)- prefix (in chemical nomenclature) denoting a geometric isomer in which the highest priority substituent groups, determined according to the Cahn-Ingold-Prelog rules (see **sequence rule**), are located on the same side of a double bond. See also **E/Z** convention.

Zantac see **ranitidine**.

zaragozic acid any of a family of closely related fungal metabolites that are potent inhibitors of squalene synthase.



zaragozic acid A

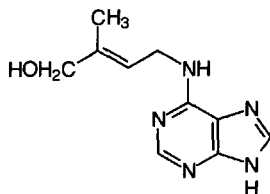
Z-average relative molecular mass see **average relative molecular mass**.

Z cell see **X-Y-Z cell theory**.

Z disk or **Z line** a platelike region of a muscle **sarcomere** to which the plus ends of actin filaments are attached. It is seen as a dark transverse line in micrographs.

Z-DNA abbr. for **Z** form of DNA.

zeatin *trans*-zeatin; (*E*)-2-methyl-4-(1*H*-purin-6-ylamino)-2-buten-1-ol; a plant hormone of the **cytokinin** class. Its derivative, *trans*-zeatin riboside (*N*⁶-(*trans*-4-hydroxy-3-methyl-*cis*-2-buten-1-yl)adenosine) is also a cytokinin. The *cis* form is a rare component in some RNA structures.



trans-zeatin-producing protein an isopentenyl transferase (dimethylallyl transferase) of sub-subclass EC 2.5.1.-. It introduces the isopentenyl moiety (2-methylbut-2-en-1-ol group) into zeatin. Example from *Burkholderia solanacearum* (*Pseudomonas solanacearum*): database code IPT_BURSO, 238 amino acids (26.41 kDa).

zebrafish *Brachydanio rerio*; a small, highly fertile species of characinoid teleost fish with a shiny blue body and four longitudinal yellowish stripes along its sides. It is much used in

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studies on the molecular biology of differentiation and development in metazoan animals.

Zeeman effect the splitting of a single line in a spectrum, indicative of the degeneration of the excited state of a particular chromophore, into two or more components of slightly different wavelengths brought about by the application of an external magnetic field. [After Pieter Zeeman (1865–1943), Dutch physicist.]

zein the principal storage protein of maize (corn) seeds. It is relatively deficient in the essential amino acids lysine and tryptophan, and tryptophan deficiency can occur in human populations dependent on maize as the major protein source. The zein genes are clustered. Example, an α zein from maize: database code ZEAW_MAIZE, 261 amino acids (28.43 kDa)

zeolite any of a family of porous, naturally occurring (or processed) alkali metal- or alkaline earth-aluminium silicates that show ion-exchange properties. They may be used for water softening or as molecular sieves. —**zeolitic** *adj.*

zepto+ symbol: *z*; SI prefix denoting 10^{-21} times.

zero-order kinetics kinetics shown by a **zero-order reaction**.

zero-order reaction a chemical reaction whose velocity is independent of the concentration of one or more of the reactants. See **order of reaction**.

zeroth law of thermodynamics see **thermodynamics, laws of**.

zervamicin see **peptaibophol**.

zeta symbol: ζ (lower case) or **Z** (upper case); the sixth letter of the Greek alphabet. For uses see **Appendix A**.

zeta potential symbol: ζ ; the potential at the surface of shear of a charged particle; for macromolecular ions it is the potential at the surface of the **hydrodynamic particles** formed by such ions.

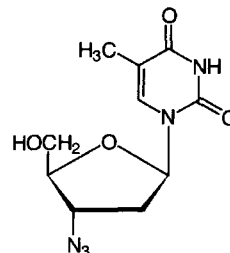
zeta protein an alternative name for cellular **fibronectin**.

zetta+ symbol: **Z**; SI prefix denoting 10^{21} times.

zeugmatography a technique in which the addition of carefully controlled inhomogeneous magnetic fields enables measurements of **nuclear magnetic resonance** (NMR) to be made at selected regions of heterogeneous samples and hence the formation of two-dimensional maps of the distribution of almost any property measurable by NMR.

Z form of DNA abbr. **Z-DNA** or **DNA-Z**; a form of DNA consisting of a left-handed double helix in which the pyrimidine residues are always in the *anti* orientation and the purine residues in the *syn*. So-called because of the zig-zag appearance of its conformation. See also **A form** (def. 1), **B form**, **C form**.

zidovudine 3'-azido-3'-deoxythymidine (abbr.: **AZT**); an analogue of thymidine, the phosphorylated form of which is an inhibitor of **reverse transcriptase** in **retroviruses**; it also terminates DNA synthesis. Zidovudine undergoes phosphorylation in human T cells to a nucleoside 5'-triphosphate, which competes with thymidine triphosphate (TTP) and serves as a chain-terminating inhibitor of **HIV** reverse transcriptase. It is used clinically to treat patients with HIV infection and AIDS. *Proprietary name:* Retrovir.





Zimm-Crothers viscometer

Zimm-Crothers viscometer a modified **Couette viscometer** in which the inner cylinder is a self-centring float containing a steel pellet. The inner cylinder is caused to rotate by an external rotating magnetic field, the speed of rotation being dependent, *inter alia*, on the viscosity of the fluid. The instrument only measures the viscosity of a sample relative to a standard.

Zimmermann reaction the reaction of *m*-nitrobenzene in strongly alkaline solution with the methylene group at position 16 in 17-ketosteroids to give a purple colour with an absorption maximum at 520 nm. It may be used to estimate such steroids.

Zimm plot a graphical double extrapolation used to analyse most macromolecular light-scattering data. Measurements are made of Kc/R_θ at various concentrations as a function of the scattering angle, 2θ , where K is an optical constant, c is the concentration, and R_θ is the Rayleigh ratio. A series of plots of Kc/R_θ against $(\sin^2 \theta + K'c)$ are made, where K' is an arbitrary constant chosen to provide a convenient spread. The data are then extrapolated separately to $c = 0$ and $\theta = 0$. The intersection of the two extrapolated curves yields the relative molecular mass.

zinc symbol Zn; a metal of group 12 of the IUPAC **periodic table**; relative atomic mass 65.38; atomic number 30. The principal ore is zincblende, ZnS. Its oxidation state is Zn(II). Zinc oxide is used as an antiseptic. Naturally occurring nuclides include (relative abundance follows each) ^{64}Zn (48%), ^{66}Zn (28%), ^{67}Zn (18%). Radionuclides include **zinc-62**, **zinc-65**, and **zinc-72**. Dietary deficiency in the rat is manifested by retarded growth, alopecia, and lesions in the skin, esophagus, and cornea. Deficiency is rare in normal individuals, but can manifest itself in alcoholics and in patients with renal disease or malabsorption. Zn^{2+} is an essential cofactor for many enzymes. Recent work suggests zinc is bactericidal.

zinc-62 the artificial radioactive nuclide ^{62}Zn ; it emits a positron (β^+ particle) (0.66 MeV), and γ -radiation of four energies (0.041, 0.25, 0.51, and 0.60 MeV). It has a half-life of 9.26 h.

zinc-65 the artificial radioactive nuclide ^{65}Zn ; it emits a positron (β^+ particle) (0.325 MeV) and γ -radiation (1.12 MeV). It has a half-life of 244.1 days.

zinc-72 the artificial radioactive nuclide ^{72}Zn ; it emits an electron (β^- particle) (0.3 MeV) and γ -radiation of three energies (0.016, 0.145, and 0.192 MeV). It has a half-life of 1.938 days.

zinc finger a polypeptide motif present in many DNA-binding proteins and having the consensus sequence



The motif takes its name from its zinc-binding site, which forms a finger-like loop in the peptide, also described as the **Cys₂/His₂ finger**, the Cys and His residues coordinating a zinc ion. Zinc fingers are most notably present in **transcription factors** (e.g. TFIIIA and SP1), which have several repeats of the motif; they are required for binding of the transcription factor to DNA. *See also* **Appendix E**.

ZIP *abbr.* for the leucine zipper domain. *See also* **bZIP**.

Z line another name for **Z disk**.

Zn *symbol* for zinc.

Zollinger-Ellison syndrome a disease syndrome caused by excessive secretion of **gastrin**, either from (type 1) **G-cell** hyperplasia in the antrum of the stomach, or from (type 2) a benign or malignant pancreatic islet tumour (**gastrinoma**). The manifestations include multiple peptic ulcers, or peptic ulcers in unusual sites, with a marked tendency to bleeding, often associated with hyperparathyroidism. [After Robert M. Zollinger (1903–) and Edwin H. Ellison (1918–70).]

zona adherens *see* **desmosome**.

zona fasciculata *see* **adrenal gland**.

zona glomerulosa *see* **adrenal gland**.

zonal centrifugation *an alternative name for* **density-gradient centrifugation**.

zonal centrifuge a centrifuge designed to allow large-scale and continuous fractionation by **density-gradient centrifugation**.

zonal rotor any centrifuge rotor of high capacity used preparatively in a **zonal centrifuge**.

zona occludens *an alternative name for* **tight junction**.

zona pellucida sperm-binding protein *abbr.*: ZP; any of several sulfated transmembrane glycoproteins (ZP1, ZP2, ZP3) that together form the zona pellucida of the oocyte. ZP3 functions as a sperm receptor. Example ZP3a from *Sus scrofa*: database code ZPB_PIG, 536 amino acids (59.33 kDa).

zona reticularis *see* **adrenal gland**.

zone electrophoresis a method of **electrophoresis** in which the protein (or other) solution is placed at the starting position as a thin band or spot in an inert supporting medium (paper, starch gel, polyacrylamide gel, etc.), containing buffer solution. An electric potential is then applied to the supporting medium, causing the proteins (or other substances) to migrate to give distinct bands or zones. These may be located *in situ* by staining, light absorption, etc., or by analysis after elution of discrete pieces of the supporting medium.

zone precipitation a technique in which protein is precipitated as a zone in a gel-filtration column by eluting with a gradient of a protein-precipitating agent. *See* **gel-permeation chromatography**.

zone spreading the broadening of a zone in electrophoresis or chromatography due to eddy currents or other disturbances in the supporting medium.

zoo blot a method for detecting conservation of DNA sequence during evolution. A segment of DNA being investigated is used as a probe to hybridize against a series of DNA samples from various species, and hence establish whether the DNA sequence has been conserved during evolution.

zoonosis (*pl.* **zoonoses**) any disease that can be transmitted to humans from other animals in natural circumstances.

zooplankton *see* **plankton**.

zootoxin any poisonous substance formed by animal cells.

Zovirax a *proprietary name* for **acyclovir**.

Zr *symbol* for zirconium.

Z scheme a term sometimes applied to two-system photosynthetic light reactions, utilizing both **photosystems** I and II. It is descriptive of the mode of photosynthesis used by higher plants and algae, and by cyanobacteria. The way in which it is normally plotted, with redox potential (on the ordinate) against time, shows there is a sudden jump on excitation by a photon of a reaction centre, followed by a slow decline through electron transport components, and gives a (horizontal) Z-like appearance, hence the name.

zuotin a nuclear Z-DNA-binding protein of *Saccharomyces cerevisiae*. Example from yeast: database code ZUO1_YEAST, 433 amino acids (49.02 kDa).

zyeastferment *an obsolete term* for an enzyme-coenzyme complex of which **glucose-6-phosphate dehydrogenase** is the apoenzyme.

Zwittergent the *proprietary name* for any of a group of zwitterionic detergents containing a **betaine** and a sulfonate group; such molecules retain their amphoteric (zwitterionic) properties over a very wide range of pH. The formula is *n*-alkyl *N,N*-dimethyl-3-ammonio-1-propanesulfonate. In Zwittergent 3-08, alkyl is octyl (CMC 330 mM); 3-10, decyl (aggregation number 41, CMC 25–40 mM); 3-12, dodecyl (aggregation number 55, CMC 2–4 mM); 3-14, tetradecyl (aggregation number 83, CMC 0.1–0.4 mM); 3-16, hexadecyl (aggregation number 155, CMC 0.01–0.06 mM).

zwitterion *an alternative name for* **dipolar ion**.

zwitterionic buffer a **buffer solution** whose acid or base component is a zwitterion (i.e. **dipolar ion**). The term is used especially of solutions of **Good buffer substances** and newer equivalents.

zygote any fertilized ovum before it has undergone cleavage. —*zygotic adj.*

zygotene the second phase of prophase I in **meiosis**.

zymase the name originally given to the heat-labile, nondif-

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zymogen

fusible fraction of a crude extract of brewers' yeast that, with the addition of the heat-stable, diffusible fraction (i.e. **co-zymase**), would enable the alcoholic fermentation of glucose to occur in a cell-free system. It was later shown to consist of a mixture of enzymes, including those of the **glycolytic pathway**.

zymogen or **proenzyme** or (formerly) **proferment** the inactive precursor of an enzyme, often convertible to the enzyme by partial proteolysis. The term is applied especially to catalytically inactive forms of pancreatic enzymes such as trypsinogen, chymotrypsinogen, proelastase, and procarboxypeptidase; these are cleaved to remove a peptide to produce the active enzyme after their secretion from granules (**zymogen granules**) in the acinar cells of the pancreas.

zymogen granule a membrane-bound, cytoplasmic **secretory granule** visible by light microscopy. Zymogen granules are formed in the Golgi apparatus of enzyme-secreting cells and serve to store a **zymogen**. The term is used especially of a secretory granule containing the proenzyme of a digestive enzyme.

zymogram 1 any record of a **zone electrophoresis** separation in which enzymes in a sample have been separated and their posi-

tions, and relative amounts, revealed by an **activity stain**. 2 any table showing the results of carbohydrate-fermentation tests carried out in the process of identifying a microorganism.

zymohexase an obsolete name for **fructose biphosphate aldolase**.

zymology the science dealing with fermentations. —**zymological** *adj.*; **zymologist** *n.*

zymosan a crude preparation of yeast cell walls, consisting chiefly of polysaccharide, that activates the alternative pathway of the complement system in the presence of properdin.

zymosterol cholesta-8,24-dien-3 β -ol; a sterol present in yeasts e.g. *Candida albicans*, *Saccharomyces cerevisiae*.

zymurgy the branch of chemistry dealing with the applications of fermentation processes, especially to brewing and wine-making.

zyxin a protein component of cell substratum and cell-cell adherens junctions. It is a component of adhesion plaques and the termini of stress fibres near the point of association with the cytoplasmic face of the adhesive membrane and interacts with α -actinin. Example from *Gallus gallus*: database code ZYX_CHICK, 542 amino acids (58.53 kDa).



Appendix A

The Greek alphabet and Greek characters used as symbols

The Greek alphabet

Name	Capital letter	Lower-case letter	English transliteration
alpha	A	α	a
beta	B	β	b
gamma	Γ	γ	g
delta	Δ	δ	d
epsilon	E	ϵ	e
zeta	Z	ζ	z
eta	H	η	$\bar{e}$ or e
theta	Θ	θ	th
iota	I	ι	i
kappa	K	κ	k
lambda	Λ	λ	l
mu	M	μ	m
nu	N	ν	n
xi	Ξ	ξ	x
omicron	O	$\omicron$	o
pi	Π	π	p
rho	P	ρ	r
sigma	Σ	σ or ς (at end of word)	s
tau	T	τ	t
upsilon	Y	υ	u
phi	Φ	ϕ	ph
chi	X	χ	kh
psi	Ψ	ψ	ps
omega	Ω	ω	$\bar{o}$ or o

Single-character symbols

α (or α) 1 a heavy chain of immunoglobulin A. 2 maximal agonist effect. 3 alpha particle.

α (or α) 1 angle of optical rotation (α_{obs} is used for observed rotation and $[\alpha]$ for specific optical rotation or specific op-

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tical rotatory power; a_m is used for molar optical rotatory power). 2 absorptance. 3 (linear) napierian absorption coefficient. 4 Bunsen coefficient. 5 partition coefficient. 6 plane angle. 7 (in organic chemical nomenclature) a locant for a substituent on the first atom from the atom carrying the principal function in an acyclic structure or in a side chain attached to a cyclic structure. 8 an anomeric configuration (of a sugar molecule or residue; see **anomer**). 9 winding number (of a helical duplex DNA molecule).

β (or β) beta particle (β^- is used for an electron and β^+ for a positron).

β (or β) 1 buffer value. 2 pressure coefficient. 3 plane angle. 4 (in organic chemical nomenclature) a locant for a substituent on the second atom from the atom carrying the principal function in an acyclic structure or in a side chain attached to a cyclic structure. 5 an anomeric configuration (of a sugar molecule or residue; see **anomer**). 6 number of secondary turns (in an unconstrained helical duplex DNA molecule).

γ (or γ) a heavy chain of immunoglobulin G. 2 a photon of gamma radiation. 3 (*obsolete*) microgram (μg now used).

γ (or γ) activity coefficient (γ_c is used for activity coefficient when expressed on a concentration basis, γ_m when expressed on a molality basis, γ_x when expressed on a mole fraction basis, and $\gamma_{\pm}$ is used for mean ionic activity coefficient). 2 electrical conductivity (alternative to κ or σ). 3 mass concentration or mass density (alternative to ρ). 4 plane angle. 5 surface tension or interfacial tension (alternative to σ). 6 gyromagnetic ratio. 7 (in organic chemical nomenclature) a locant for a substituent on the third atom from the atom carrying the principal function in an acyclic structure or in a side chain attached to a cyclic structure.

δ (or δ) 1 a heavy chain of immunoglobulin D. 2 (in mathematics) an infinitesimal change or difference.

δ (or δ) 1 chemical shift. 2 number fraction. 3 (in organic chemical nomenclature) a locant for a substituent on the fourth atom from the atom carrying the principal function in an acyclic structure or in a side chain attached to a cyclic structure.

Δ (or Δ) (in mathematics) a finite change or difference.

Appendix A

- Δ** (or Δ) (in older organic chemical nomenclature) a locant for a double bond (a right superscript number designates the lower-numbered carbon atom; see **fatty-acid nomenclature**); no longer recommended for denoting unsaturation in names of individual compounds although it may still be used in generic terms (e.g. Δ^5 -steroid) and in symbols (e.g. Δ_2 Ach).
- ϵ** (or ϵ) a heavy chain of immunoglobulin E.
- ϵ** (or ϵ) 1 molar (decadic) absorption coefficient. 2 permittivity (ϵ_r is used for relative permittivity and ϵ_0 for permittivity of vacuum). 3 (in organic chemical nomenclature) a locant for a substituent on the fifth atom from the atom carrying the principal function in an acyclic structure or in a side chain attached to a cyclic structure. 4 (in chemical nomenclature) hapticity of a ligand (less common alternative to η ; see **hapto**).
- ϵ_a** degree of activation (of an enzymic reaction).
- ϵ_i** degree of inhibition (of an enzymic reaction).
- ϵ_v** elasticity coefficient (of a specific enzymic step in a metabolic system).
- ζ** (or ζ) electrokinetic potential or zeta potential.
- η** (or η) 1 viscosity (alternative to μ). 2 (in chemical nomenclature) hapticity (of a ligand; more common alternative to ϵ ; see **hapto**).
- θ** (or θ) 1 Bragg angle (see **Bragg's law**). 2 (in surface chemistry) contact angle. 3 plane angle. 4 torsion angle (in a linear sequence of four attached atoms; alternative to ω). 5 Celsius temperature (alternative to t).
- Θ_m** melting temperature (alternative to T_m or t_m).
- [Θ]** molar ellipticity.
- κ** (or κ) one type of light chain of any class of immunoglobulin (the other being λ).
- κ** (or κ) 1 electrical conductivity (alternative to γ or σ). 2 electrolytic conductivity or (formerly) specific conductance. 3 molar napierian absorption coefficient. 4 magnetic susceptibility (alternative to χ).
- λ** (or λ) 1 one type of light chain of any class of immunoglobulin (the other being κ). 2 (obsolete) microlitre (μ l now used).
- Λ** (or Λ) 1 (in thermodynamics) absolute activity (see **activity**, def. 4). 2 disintegration constant or decay constant. 3 wavelength. 4 thermal conductivity (alternative to k).
- Λ** (or Λ) molar conductivity (of an electrolyte).
- μ** (or μ) a heavy chain of immunoglobulin M. 2 micro+ (SI prefix denoting 10^{-6} times). 3 (obsolete) micron (now called micrometre; symbol: μ m).
- μ** (or μ) 1 (in chemical nomenclature) a bridging ligand. 2 viscosity (alternative to η). 3 chemical potential (μ° or $\mu^\ominus$ is used for standard chemical potential and $\bar{\mu}$ is used for electrochemical potential). 4 electric mobility (alternative to u). 5 (in magnetism) permeability (μ_r is used for relative permeability). 6 magnetic dipole moment (bold italic) (alternative to ***m***). 7 electric dipole moment (bold italic) (alternative to ***p***).
- μ_0** permeability of vacuum (a fundamental physical constant).
- μ_B** Bohr magneton (a fundamental physical constant).
- μ_e** electron magnetic moment (a fundamental physical constant).
- μ_N** nuclear magneton (a fundamental physical constant).
- μ_p** proton magnetic moment (a fundamental physical constant).
- ν** (or ν) 1 stoichiometric number (of a chemical reactant). 2 frequency (alternative to f for electromagnetic radiation). 3 kinematic viscosity. 4 endocyclic torsion angle (in a sugar residue).
- ξ** 1 (in organic chemical nomenclature) unknown configuration at a chiral centre. 2 extent of reaction.
- $\dot{\xi}$** rate of conversion.
- π** (or π) a transcendental number that is the ratio of the circumference of a circle to its diameter (3.141 592 653 59...).
- π** (or π) 1 pros. 2 (or π^s) surface pressure.
- Π** (or Π) 1 osmotic pressure (Π_D may be used to represent colloid osmotic pressure). 2 binding potential (of a ligand).
- ρ** (or ρ) 1 density or mass density (ρ_A or ρ_S is used for surface density). 2 mass concentration or mass density (alternative to γ). 3 radiant energy density (alternative to w). 4 charge density. 5 resistivity.
- σ** (or σ) 1 surface tension or interfacial tension (alternative to γ). 2 electrical conductivity (alternative to γ or κ). 3 wavenumber in a medium. 4 surface charge density. 5 (in surface chemistry) area per molecule. 6 (in nuclear magnetic resonance spectroscopy) shielding constant. 7 superhelix density (of a helical duplex DNA molecule).
- Σ** (or Σ) (in mathematics) a summation sign.
- τ** (or τ) 1 tele. 2 bond angle. 3 relaxation time or time constant (alternative to T). 4 transmittance or transmission factor (alternative to T). 5 mean life. 6 number of superhelical turns (in a helical duplex DNA molecule).
- ϕ** (or ϕ) 1 osmotic coefficient (ϕ_m is used for osmotic coefficient when expressed on a molality basis and ϕ_x when expressed on a mole fraction basis). 2 electric potential (alternative to V). 3 fluidity. 4 plane angle. 5 phase difference. 6 volume fraction. 7 Dalziel coefficient. 8 torsion angle (from the anomeric carbon atom to the oxygen atom of a glycosidic bond). 9 quantum yield (alternative to Φ). 10 winding angle (of intercalated helical DNA).
- Φ** (or Φ) 1 heat flow rate. 2 radiant power or radiant (energy) flux (alternative to P). 3 magnetic flux. 4 potential energy (alternative to E_p or V). 5 quantum yield (alternative to ϕ).
- χ** (or χ) 1 electronegativity. 2 magnetic susceptibility (alternative to κ ; χ_m is sometimes used, but it should be reserved for molar magnetic susceptibility). 3 surface electric potential. 4 exocyclic torsion angle (in a sugar residue).
- ψ** (or ψ) 1 replacement of the $-\text{CO}-\text{NH}-$ group in a peptide by another (indicated) grouping. 2 torsion angle (from the oxygen atom of a glycosidic bond to the glycosylated molecule). 3 wavefunction (alternative to Ψ).
- Ψ** (or Ψ) 1 a residue of the ribonucleoside pseudouridine (alternative to Ψrd). 2 electric flux. 3 wave function (alternative to ψ).
- ω** (or ω) 1 angular frequency. 2 angular velocity. 3 solid angle. 4 torsion angle (in a linear assembly of four attached atoms; alternative to θ). 5 (in organic chemical nomenclature) a locant for the terminal atom or group in an acyclic organic compound.
- Ω** (or Ω) ohm.
- Ω** (or Ω) solid angle.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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Multi-character biochemical symbols commencing with a Greek character

αLnn	the linolenoyl <i>or</i> (9,12,15)-linolenoyl <i>or</i> (formerly) α -linolenoyl (i.e. (<i>all-Z</i>)-octadeca-9,12,15-trienoyl) group (alternative to Lnn).	Δ_2Ach	the (<i>all-Z</i>)-eicosa-8,11-dienoyl group.
βAad	a residue of the β -amino acid L- β -aminoadipic acid (i.e. L-3-aminohexanedioic acid).	Δ_3Ach	the (<i>all-Z</i>)-eicosa-5,8,11-trienoyl group.
βAla	a residue of the β -amino acid β -alanine (i.e. 3-amino-propanoic acid).	Δ_4Ach	the arachidonoyl (i.e. (<i>all-Z</i>)-eicosa-5,8,11,14-tetra-enoyl) group.
γLnn	the (6,9,12)-linolenoyl <i>or</i> (formerly) γ -linolenoyl (i.e. (<i>all-Z</i>)-octadeca-6,9,12-trienoyl) group.	ΔPam	the palmitoleoyl (i.e. (<i>Z</i>)-hexadec-9-enoyl) group.
		εAcp	<i>see</i> εAhx .
		εAhx	<i>or</i> (formerly) εAcp a residue of the ε -amino acid 6-aminohexanoic acid (formerly known as ε -aminocaproic acid).
		Ψrd	a residue of the ribonucleoside pseudouridine (alternative to Ψ).



Appendix B

Nomenclature publications

Introduction

Brief details are provided below of a selection from the considerable number of publications available on nomenclature and symbolism that are of possible interest to biochemists, biological chemists, clinical chemists, and molecular biologists. The majority of these publications are in the form of approved recommendations from the International Union of Biochemistry (IUB) or its successor the International Union of Biochemistry and Molecular Biology (IUBMB) through their Nomenclature Committees (respectively NC-IUB or NC-IUBMB), from the International Union of Pure and Applied Chemistry (IUPAC), or from IUPAC and IUBMB together through their Joint Commission on Biochemical Nomenclature (JCBN); on occasions the recommendations have been issued by IUPAC in conjunction with IUBMB and the International Union of Pure and Applied Biophysics (IUPAB) or by IUPAC jointly with the International Federation of Clinical Chemistry (IFCC). The full titles of the relevant nomenclature bodies to which the listed publications are attributable have generally been omitted for brevity.

This Appendix has been updated and expanded for the revised edition of the Dictionary. Now incorporated are details of significant nomenclature documents published and made available up to the end of August 1999 (or in certain instances known to be in preparation at that date). Furthermore, the interval following the compilation of the original version of this appendix has seen a considerable increase in the number of specific biochemical and chemical nomenclature recommendations whose full texts have been converted to versions readable on the Internet via the World Wide Web (WWW), and so their current uniform resource locators (URLs) have been added where appropriate. All such recommendations are also currently accessible (directly or indirectly) through links from either of the following URLs:

www.chem.qmw.ac.uk/iubmb/

(IUBMB Biochemical Nomenclature home page)

www.chem.qmw.ac.uk/iupac/

(IUPAC Chemical Nomenclature home page)

It will be appreciated that, in order for information as up-to-date as possible to be given in this Appendix, details have been included of a number of nomenclature documents that became available too recently for their contents to be taken into consideration during the compilation of this Dictionary.

Manuals and Compendia

These are volumes covering major subject areas, the majority being assemblages either of previously published nomenclature documents on specific topics or of items extracted from such documents. Some duplication of content between them has inevitably occurred. They are commonly known by the colour of their covers.

- 1 [The Blue Book] IUPAC (1979). *Nomenclature of organic chemistry — sections A, B, C, D, E, F and H — 1979 edn* (ed. J. Rigaudy and S. P. Klesney). Pergamon Press, Oxford.

A collection of several previously published separate sets of rules and recommendations, reprinted with correction of material errors. The section headings are listed in item 34. *See also* item 6.

URL for section H (isotopically modified compounds):

<http://www.chem.qmw.ac.uk/iupac/sectionH/>

- 2 [The Red Book] IUPAC (1990). *Nomenclature of inorganic chemistry — recommendations 1990* (ed. G. J. Leigh). Blackwell Scientific Publications, Oxford.

Recommendations conceived as being Part I of a third edition of the Red Book and mainly concerned with the fundamental areas of inorganic chemistry.



A series of chapters includes: the rules for naming simple molecular entities; oxoacids and their derivatives; coordination compounds; and simpler boron compounds. The chapter headings are listed in item 34. Further volumes, which will deal with more specialized aspects of the subject, are expected to be published in about 2001.

- 3 [The Purple Book] IUPAC (1991). *Compendium of macromolecular nomenclature* (ed. W. V. Metanomski). Blackwell Scientific Publications, Oxford.

A collection of reprints of previously published sets of recommendations providing definitions of terms relating to polymers and rules for naming them based on structure or source. The chapter headings are listed in item 34.

- 4 [The White Book] NC-IUBMB (1992). *Biochemical nomenclature and related documents — a compendium, 2nd edn* (ed. C. Liébecq). Portland Press, London.

An assemblage of a considerable number of previously published sets of recommendations issued by various nomenclature bodies of IUB, IUPAC, or IUB and IUPAC jointly, together with excerpts from the newsletters of the biochemical nomenclature committees. Most of the nomenclature documents contained in the first (1978) edition of the compendium are either reprinted (with updates where appropriate) or replaced by revised versions, and to these is added a number of new sets of recommendations published between 1978 and 1989. Also included are a list of titles of additional biochemical nomenclature documents (including those printed in the first edition of the compendium but not in this one) and an extensive list of titles of other nomenclature and reference publications of possible interest to biochemists. The main contents are listed in item 34.

The full texts of the individual recommendations in this compendium are available as World Wide Web versions or (if not superseded) are in course of being converted into this form; the titles and URLs of those currently accessible are included in a list of WWW versions of current biochemical nomenclature recommendations given at:

<http://www.chem.qmw.ac.uk/iubmb/nomenclature/>

- 5 [The Green Book] IUPAC (1993). *Quantities, units and symbols in physical chemistry, 2nd edn* (ed. I. M. Mills, T. Cvitaš, K. H. Homann, N. Kallay, and K. Kuchitsu). Blackwell Scientific Publications, Oxford.

The contents are drawn from the resolutions or recommendations of a number of international bodies concerned with the definition and use of physical quantities and units. Included are: the definitions and mandatory symbols for the base and derived units forming the International System (SI), with the names and symbols for SI prefixes signifying decimal multiples and submultiples of SI units; factors for

conversion of units from a variety of other systems to the corresponding SI units; sets of tables containing the internationally recommended names and symbols for the base and derived physical quantities most likely to be used by chemists and biochemists, with brief definitions and the corresponding coherent SI units; recommended mathematical symbols; a table of fundamental physical constants; and tables of the properties of chemical elements, nuclides, and particles.

- 6 [The Blue Book II] IUPAC (1993). *A guide to IUPAC nomenclature of organic compounds — recommendations 1993* (ed. R. Panico, W. H. Powell, and J.-C. Richer). Blackwell Science, Oxford.

For use in conjunction with item 1; it outlines the main principles of organic nomenclature as described in that volume, and presents important changes (published and hitherto unpublished) agreed upon since.

- 7 [The Silver Book] IUPAC and IFCC (1995). *Compendium of terminology and nomenclature of properties in clinical laboratory sciences (recommendations 1995)* (ed. J. C. Rigg, S. S. Brown, R. Dybkær, and H. Olesen). Blackwell Science, Oxford.

A guide towards a more uniform way of reporting information in clinical sciences, particularly quantitative information in clinical chemistry; it was compiled by harmonizing, updating where appropriate, and consolidating a number of previously published documents and recommendations not readily accessible. Derived physical or chemical quantities are analysed into their dimensions, with the aim of facilitating the interpretation of many of the older names of quantities used in this sphere and the recalculation of data expressed in older units.

- 8 [The Gold Book] IUPAC (1997). *Compendium of chemical terminology — IUPAC recommendations, 2nd edn* (ed. A. D. McNaught and A. Wilkinson). Blackwell Science, Oxford.

An alphabetical listing of nearly 7000 terms covering the whole range of chemistry. The terms have been selected from a considerable number of internationally approved nomenclature documents and compilations published up to the end of 1995, including all IUPAC recommendations, together with some particularly significant material from 1996. Brief but authoritative definitions are given for each term, together with references to the source documents.

- 9 [The Orange Book] IUPAC (1998). *Compendium of analytical nomenclature, 3rd edn — definitive rules 1997* (ed. J. Inczédy, T. Lengyel, and A. M. Ure). Blackwell Science, Oxford.

A compilation of previously published documents. The chapter headings are listed in item 34.



Appendix B

Enzyme list

- 10 NC-IUBMB (1992). *Enzyme nomenclature – recommendations 1992* (ed. E. C. Webb). Academic Press, San Diego.

A nomenclature and classification of enzymes according to the chemical reactions they catalyse, with systematic names, code numbers, and recommended short names assigned to each one. This, the sixth complete edition of the recommendations, lists 3196 individual enzymes (an increase of 29% on the number in the fifth (1984) edition). Included is a second printing, with corrections, of the contents of item 14 [a further correction is given at: *European Journal of Biochemistry*, 1993, **213**, 3, footnote 1].

Supplements giving corrections and additions to the enzyme list are published periodically, the following having appeared in print by the end of August 1999: Supplement (1994) *European Journal of Biochemistry*, **223**, 1–5; Supplement 2 (1995) *ibidem*, **232**, 1–6; Supplement 3 (1996) *ibidem*, **237**, 1–5; Supplement 4 (1997) *ibidem*, **250**, 1–6; Supplement 5 (1999) *ibidem*, **264**, 610–650. Supplement 6 (1999) is expected to appear in print at a later date. Except for Supplement 5, these supplements all relate to enzyme subclass EC 3.4 (peptidases).

World Wide Web versions of Supplement 5 and of a consolidated list containing the full amended text of subclass EC 3.4 are available.

URL: <http://www.chem.qmw.ac.uk/iubmb/enzyme/>

Conversion of the remaining text of *Enzyme nomenclature* to a consolidated WWW version is expected to be completed by the end of 2000; changes in format for this version will include expansion of the comments section for each enzyme, more complete and up-to-date references, and citation of references at the end of each entry.

Documents on specific topics

A selection of nomenclature documents on specific topics, most of them recently published, is given below. Information from some of them has been incorporated into certain of the compendia mentioned above.

All the recommendations by IUPAC and JCBN have been published initially in *Pure and Applied Chemistry*, and all the NC-IUB or NC-IUBMB recommendations have been first published in *European Journal of Biochemistry* (where the JCBN recommendations have usually also been published). Although many of these recommendations have been reprinted in other research journals, only the initial publication is given for each of the documents itemized below; all appearances of the JCBN, NC-IUB, or NC-IUBMB documents, including any in translation, are enumerated in the publications list cited in item 33.

- 11 Baron, D. N., Moss, D. W., Walker, P. G., and Wilkinson, J. H. (1971). Abbreviations for names of enzymes of diagnostic importance. *Journal of Clinical Pathology*, **24**, 656–657.

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A list of two- or three-letter abbreviations (strictly contractions) of the names of enzymes frequently assayed in clinical biochemistry laboratories. The abbreviations were devised to standardize their use in clinical reports and to facilitate processing of assay data by computer.

- 12 NC-IUB (1979). Units of enzyme activity – recommendations 1978. *European Journal of Biochemistry*, **97**, 319–320. Corrections: *ibidem*, 1980, **104**, 1–4.

- 13 Eicosanoid Nomenclature Committee (1989). Eicosanoid nomenclature. *Prostaglandins*, **38**, 125–133.

Proposals regarding a systematic leukotriene nomenclature and suggestions on the nomenclature of enzymes involved in eicosanoid metabolism. The spelling 'eicosanoid' is favoured in preference to 'icosanoid'. The ad hoc committee putting forward these proposals and suggestions was constituted by the IUB as an advisory panel but the recommendations have not been adopted by NC-IUB or NC-IUBMB.

- 14 NC-IUB (1991). Nomenclature of electron-transfer proteins — recommendations 1989 (ed. G. Palmer and J. Reedijk). *European Journal of Biochemistry*, **200**, 599–611. Corrections: *ibidem*, 1993, **213**, 2–3. See also item 10.

- 15 IUPAC (1994). Glossary of terms used in physical organic chemistry, 2nd edn (IUPAC recommendations 1994) (ed. P. Müller). *Pure and Applied Chemistry*, **66**, 1077–1184.

URL: <http://www.chem.qmw.ac.uk/iupac/gtpoc/>

- 16 JCBN (1994). Recommendations for nomenclature and tables in biochemical thermodynamics (recommendations 1994) (ed. R. A. Alberty). *Pure and Applied Chemistry*, **66**, 1641–1666.

- 17 IUPAC (1994). Glossary of bioanalytical nomenclature — part I: general terminology, body fluids, enzymology, immunology (IUPAC recommendations 1994) (ed. C. A. Burtis and T. A. Geary). *Pure and Applied Chemistry*, **66**, 2587–2604.

Definitions taken from IUPAC, IUB, and IFCC nomenclature documents and other sources to form a compilation useful for practitioners of clinical chemistry.

- 18 NC-IUBMB (1994). A nomenclature of junctions and branchpoints in nucleic acids — recommendations 1994 (ed. D. M. J. Lilley, R. M. Clegg, S. Diekmann, N. C. Seeman, E. von Kitzing, and P. Hagerman). *European Journal of Biochemistry*, **230**, 1–2.

- 19 Stewart, A. (ed.) (1995) Genetic nomenclature guide — including information on genomic databases. *Trends in Genetics*, **11**(3), suppl. A summary of the most important rules and guidelines for the genetic



- nomenclature of 15 favoured research organisms used by geneticists and developmental biologists.
- 20 IUPAC (1995). Glossary of class names of organic compounds and reactive intermediates based on structure (IUPAC recommendations 1995) (ed. G. P. Moss, P. A. S. Smith, and D. Tavernier). *Pure and Applied Chemistry*, **67**, 1307–1375.
A glossary of terms used to denote classes of (mostly organic) compounds, substituent groups, and reactive intermediates, in contrast to individual compounds. The terms have been selected both from IUPAC or IUB/IUBMB nomenclature publications and from the chemical literature.
URL: <http://www.chem.qmw.ac.uk/iupac/class/>
 - 21 IUPAC and IFCC (1996). Glossary of terms in quantities and units in clinical chemistry (IUPAC–IFCC recommendations 1996) (ed. H. P. Lehmann, X. Fuentes-Arderiu, and L. F. Bertello). *Pure and Applied Chemistry*, **68**, 957–1000.
A glossary containing terms considered of particular interest to the practising clinical chemist. Definitions taken from a variety of published sources have been provided in a consistent terminology.
 - 22 JCBN (1996). Nomenclature of carbohydrates (recommendations 1996) (ed. A. D. McNaught). *Pure and Applied Chemistry*, **68**, 1919–2008.
An expansion of and replacement for the document entitled ‘Tentative rules for carbohydrate nomenclature – part 1, 1969’ (reprinted in item 4 at pp. 127–148). It also replaces in essence several other sets of recommendations (also reprinted in item 4) covering various specialized areas of carbohydrate terminology.
URL: <http://www.chem.qmw.ac.uk/iupac/2carb/>
 - 23 IUPAC (1996). Basic terminology of stereochemistry (IUPAC recommendations 1996) (ed. G. P. Moss). *Pure and Applied Chemistry*, **68**, 2193–2222.
A glossary of the more important, and most widely used, stereochemical terms. It extends the list of those defined in Section E (stereochemistry) of item 1 (reprinted in item 4 at pp. 1–18), and includes some terms from inorganic, macromolecular, and physical organic chemistry, as well as some misleading terms (with guidance on correct usage or acceptable alternatives). Mention is made of many of the symbols used in stereochemical nomenclature.
URL: <http://www.chem.qmw.ac.uk/iupac/stereo/>
 - 24 IUPAC (1996). Glossary of terms used in photochemistry (IUPAC recommendations 1996) (ed. J. W. Verhoeven). *Pure and Applied Chemistry*, **68**, 2223–2286.
A revised and enhanced version of a glossary first published in 1988.
 - 25 Clark, B. F. C. and 13 others (1996) Prokaryotic and eukaryotic translation factors — ad hoc nomenclature subcommittee report. *Biochemie*, **78**, 1119–1122.
Proposals for updating and extending to prokaryotic factors the document entitled ‘Nomenclature of initiation, elongation and termination factors for translation in eukaryotes — recommendations 1988’ (reprinted in item 4 at pp. 90–92). NC–IUBMB has not formally adopted the report (although published under the name of IUBMB), but it has recommended (in item 30) that the proposals should be followed.
 - 26 JCBN (1997). Nomenclature of glycolipids (recommendations 1997) (ed. M. A. Chester). *Pure and Applied Chemistry*, **69**, 2475–2487.
A replacement for the section on glycolipids of the document entitled ‘The nomenclature of lipids — recommendations 1976’ (reprinted in item 4 at pp. 180–190).
URL: <http://www.chem.qmw.ac.uk/iupac/misc/glylp.html>
 - 27 IUPAC, IUBMB, and IUPAB (1998). Recommendations for the presentation of NMR structures of proteins and nucleic acids (recommendations 1998) (ed. J. L. Markley, A. Bax, Y. Arata, C. W. Hilbers, R. Kaptein, B. D. Sykes, P. E. Wright, and K. Wüthrich). *Pure and Applied Chemistry*, **70**, 117–142. Correction: *ibidem*, 1998, **70**, Errata pp. (or part 6, i).
 - 28 IUPAC (1998). Glossary of terms used in medical chemistry (IUPAC recommendations 1998) (ed. C. G. Wermuth, C. R. Ganellin, P. Lindberg, and L. A. Mitscher). *Pure and Applied Chemistry*, **70**, 1129–1143.
A glossary compiled from a number of sources with the aim of establishing international definition standards in a subject where rapid changes are occurring. Concise definitions in a consistent terminology are provided for about 100 commonly used terms.
URL: <http://www.chem.qmw.ac.uk/iupac/medchem/>

Newsletters

The biochemical nomenclature committees (JCBN and NC–IUB or NC–IUBMB) have together published newsletters in various journals of biochemistry from time to time since 1980. These are designed to inform scientists about the work of the committees, to call attention to features of recommendations that are recently published or are in press, to report on current nomenclature problems under study, and to make recommendations not meriting publication as separate documents.

A schedule of all the published newsletters is included in the publication list available via the World Wide Web and noted at item 33; this schedule enumerates all the appearances in print of each of the newsletters (or of excerpts therefrom) and states their partial contents, with links to the websites of any directly accessible relevant recommendations. The two most recent newsletters are listed below; WWW versions of



Appendix B

their full texts are available. Excerpts from previous newsletters are included in item 4.

- 29 JCBN and NC-IUBMB (1997). Newsletter 1996. *European Journal of Biochemistry*, **247**, 733–739; and elsewhere.

Topics covered include: vitamin A, retinoids, and retinoate analogues; absorbance and attenuation; glycosaminoglycans and proteoglycans; biochemical equations, catalytic antibodies; nomenclature of proteins and of receptors; and terminology in immunology. *See also* items 35 and 36.

URL: <http://www.chem.qmw.ac.uk/iupac/newsletter/1996/>

- 30 JCBN and NC-IUBMB (1999). Newsletter 1999. *European Journal of Biochemistry*, **264**, 607–609 and elsewhere.

Topics covered include: symbols for selenocysteine; IUBMB–Nicholson metabolic maps; nomenclature recommendations on the World Wide Web; databases for ribosomal proteins and for gene families; and the recommendation regarding translation factors noted under item 25.

URL: <http://www.chem.qmw.ac.uk/iupac/newsletter/1999/>

Listings

- 31 Lists of separately published current recommendations on specific topics of biochemical interest issued by JCBN, NC-IUB, NC-IUBMB, or certain other international committees are included in the information or instructions provided to authors by a number of journals of biochemistry (e.g. *Archives of Biochemistry and Biophysics*, *Biochemical Journal*, *European Journal of Biochemistry*, *Journal of Biological Chemistry*) and presented from time to time in the first volume or part of their printed versions issued for a calendar year or posted on their websites. *See also* item 33.

- 32 A list of all JCBN, NC-IUB, and NC-IUBMB recommendations currently accessible through the World Wide Web, with their URLs and links to WWW versions of their full texts, is provided at the Biochemical Nomenclature Committees home page, together with general information concerning the committees. Included may be newly approved recommendations in advance of the printed versions.

URL: <http://www.chem.qmw.ac.uk/iubmb/nomenclature/>
or <http://www.chem.qmw.ac.uk/iupac/jcbn/>

Similar information is presented in item 30

- 33 A complete list of all JCBN, NC-IUB, and NC-IUBMB publications, enumerating all their appearances in print and including any in translation, is available via the World Wide Web, with links to the websites of those documents directly accessible.

URL: <http://www.chem.qmw.ac.uk/iubmb/bibliog/jcbn.html>

- 34 A list of current IUPAC publications on nomenclature and symbolism is included in the instructions to authors provided in the first part issued for a calendar year of certain of the journals published by the Royal Society of Chemistry. The 1999 list includes the main contents of item 4, the section or chapter headings for items 1, 2, 3, and 9, and a subject-by-subject catalogue, complete to mid-1998, of separately published current recommendations on specific chemical topics.

URL: <http://www.rsc.org/is/journals/authrefs/jappdx.htm>

- 35 A list of approved documents on the nomenclature of immunology is included in item 29 at p. 737.
- 36 Lists of other nomenclature and reference publications of interest to biochemists are given in item 4 at pp. 341–347 and in item 29 at p. 738.



Appendix C

Organizations that are helpful to biochemists and molecular biologists

Much of the information in this Appendix was collated from sources on the World Wide Web (WWW) in December 1996. The information presented here can be checked or updated by reference either to the ICSU home page:

<http://www.icsu.org>

or the site maintained at the University of Waterloo Electronic Library Scholarly Societies Project:

<http://www.lib.uwaterloo.ca/society/>

International unions

The International Council For Science (ICSU) is a non-governmental organization, founded in 1931 to bring together natural scientists in international scientific endeavour. It comprises 94 multidisciplinary National Scientific Members (scientific research councils or science academies) and 23 international, single-discipline Scientific Unions to provide a wide spectrum of scientific expertise enabling members to address major international, interdisciplinary issues which none could handle alone. ICSU also has 30 Scientific Associates.

The International Council for Science

51 Boulevard de Montmorency
F-75016 Paris
France

Telephone: +33 1 45 25 03 29
Telefax: +33 1 42 88 94 31
E-mail: secretariat@icsu.org
WWW URL: <http://www.icsu.org>

The unions concerned with biochemistry and molecular biology which are members of ICSU are:

International Union of Biochemistry and Molecular Biology (IUBMB)

The IUBMB unites biochemists and molecular biologists in 63 countries which are members of the Union. Each country

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Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentd.com.

has either an Adhering Body or Associate Adhering Body in the Union which is represented as a biochemical society, as a national research council or as an academy of sciences.

You may reach the IUBMB and learn more about the activities by contacting your Adhering Body or the General Secretary:

Dr Frank Vella
Department of Biochemistry
University of Saskatchewan
18 Leyden Crescent
Saskatoon
Saskatchewan
Canada S7J 2S4

Telephone: +1 306 374 1304
Telefax: +1 306 955 1314
E-mail: f.vella@sk.sympatico.ca
WWW URL: <http://www.iubmb.unibe.ch>

International Union of Pure and Applied Biophysics (IUPAB)

The International Union for Pure and Applied Biophysics is a member of the International Council of Scientific Unions family. Affiliated to it are the national adhering bodies of 45 countries. Its function is to support research and teaching in biophysics. The Secretary General, to whom enquiries should be addressed, is:

Professor A. C. T. North
Department of Biochemistry and Molecular Biology
The University of Leeds
Leeds LS2 9JT
United Kingdom

Telephone: +44 113 233 3023
FAX: +44 113 233 3167
E-mail: a.c.t.north@leeds.ac.uk

WWW URL: <http://bmbsg11.leeds.ac.uk/iupab/>



Appendix C

International Union of Pure and Applied Chemistry (IUPAC)

Founded in 1919, the International Union of Pure and Applied Chemistry (IUPAC) is a voluntary, non-governmental, non-profit association of National Adhering Organizations representing the chemists, from both industry and academia, of 40 countries. In addition, IUPAC has links with 14 Observer Countries, 30 regional and international Associated Organizations, which are concerned with chemistry and its application, and approximately 160 Company Associates from 24 countries.

IUPAC Secretariat
PO Box 13757
Research Triangle Park
NC 27709-3757
USA

It is geographically located at 104 T. W. Alexander Drive, Building 19, Research Triangle Park, North Carolina

Telephone: +1 919 485 8700
Telefax: +1 919 485 8706
E-mail: secretariat@iupac.org
WWW URL: <http://www.iupac.org>

International Union of Biological Sciences (IUBS)

The International Union of Biological Sciences (IUBS) was created in 1919, and held its first General Assembly that same year. Its objectives are: to promote the study of biological sciences, to facilitate and coordinate research and other scientific activities that require international cooperation, to ensure the discussion and dissemination of international conferences, and to assist in the publication of their reports. The Executive Committee of IUBS can be contacted at:

IUBS
51 Boulevard de Montmorency
F-75016 Paris
France
Telephone: +33 1 45 25 00 09
Telefax: +33 1 45 25 20 29
E-mail: iubs@paris7.jussieu.fr
WWW URL: <http://www.icsu.org/membership/SUM/iubs.html>

International Union of Crystallography (IUCr)

The IUCr is a scientific union adhering to ICSU. Its objectives are to promote international cooperation in crystallography and to contribute to all aspects of crystallography, to promote international publication of crystallographic research, to facilitate standardization of methods, units, nomenclatures and symbols, and to form a focus for the relations of crystallography to other sciences.

The Executive Secretary of the IUCr may be contacted at:

International Union of Crystallography
2 Abbey Square
Chester CH1 2HU
United Kingdom
Telephone: +44 1244 345431

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Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

Telefax: +44 1244 344843
E-mail: execsec@iucr.org
WWW URL: <http://www.iucr.ac.uk/>

International Brain Research Organization (IBRO)

The International Brain Research Organization (IBRO) was founded in 1960 in response to the growing demand from scientists in many countries and different disciplines for the creation of a central organization for the better mobilization and utilization of the world's scientific resources for research on the brain. The general objectives and goals of IBRO are: to foster throughout the world fundamental scientific research contributing to an understanding of the brain, normal and abnormal; to promote international collaboration and worldwide education in brain research; and to work through and with existing organizations to help in reaching their own particular objectives and to solicit their aid in the objectives of IBRO. The Executive Committee can be contacted at:

IBRO
51 Boulevard de Montmorency
F-75016 Paris
France
Telephone: +33 1 46 47 92 92,
Telefax: +33 1 45 20 60 06
E-mail: ibro@pratique.fr
WWW URL: <http://www.icsu.org/membership/SUM/IBRO.html>

International Union of Immunological Societies (IUIS)

The International Union of Immunological Societies (IUIS) was established in May 1969. Its objectives are to organize international cooperation in immunology and to promote communication between the various branches of immunology and allied subjects encourage within each scientifically independent territory cooperation between the societies that represent the interests of immunology and contribute to the advancement of immunology in all its aspects. The IUIS can be contacted c/o:

Keith James (Secretary General of IUIS)
Department of Surgery
Medical School
University of Edinburgh
Teviot Place
Edinburgh EH8 9AG
United Kingdom
Telephone: +44 131 650 3557
Telefax: +44 131 667 6190
E-mail: kjames@srv2.med.ed.ac.uk
WWW URL: <http://www.icsu.org/membership/SUM/iuis.html>

International Union of Physiological Sciences (IUPS)

The objectives of IUPS are to encourage the advancement of the physiological sciences, to facilitate the dissemination of knowledge in the field of physiological sciences, to foster and encourage research in the field of physiological sciences, to promote the International Congresses of Physiological Sciences, to promote such other meetings as may be useful for



the advancement of the physiological sciences, and to promote such other measures as will contribute to the development of physiological sciences in developing countries. The IUPS Secretary General can be reached at:

LGN, Bâtiment CERVI
Hôpital de la Pitié-Salpêtrière
83 Boulevard de l'Hôpital
F-75013 Paris
France
Telephone: +33 1 42 17 75 37
Telefax: +33 1 42 17 75 75
E-mail: suorsoni@infobiogen.fr
WWW URL: <http://www.nas.edu/iups/>

International Union of Microbiological Sciences (IUMS)
The International Union of Microbiological Societies (IUMS) can be contacted c/o:

Dr Marc H. van Regenmortel (Secretary General of IUMS)
IBMC
15 rue Descartes
F-67000 Strasbourg
France
Telephone: +33 3 8841 7022
Telefax: +33 3 8861 0680
WWW URL: <http://www.icsu.org/membership/SUM/IUMS.html>

International Union of Nutritional Sciences (IUNS)
The objects of the International Union of Nutritional Sciences are to promote international cooperation in the scientific study of nutrition and its application, to encourage research and the exchange of scientific information in the nutritional sciences, by the holding of congresses and conferences, by publication, and by other suitable means, to establish such commissions, committees, and other bodies as may be required in the pursuit of the above, to provide a means of communication with other organizations, and to encourage participation in the activities of the International Council of Scientific Unions, of which the Union is a member, and to develop activity regarded as helpful and appropriate in achieving the objectives of the Union. The IUNS can be contacted c/o:

O. M. Galal, Executive Secretary, IUNS
INUS Secretariat
c/o UCLA School of Public Health
International Health Program
10833 Le Conte Avenue
Box 951772
Los Angeles
CA 90095-1772
United States
Telephone: +1 310 20 69639/68444
Telefax: +1 310 794 1805
E-mail: ogalal@ucla.edu
WWW URL: <http://www.icsu.org/membership/SUM/iuns.html>

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Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

International Union of Pharmacology (IUPHAR)

Founded in 1959 as a section of the International Union of Physiological Sciences, 'The International Union of Pharmacology', IUPHAR, has been independent since 1966. The Union is a voluntary, non-governmental, non-profit association of national organizations presently representing the pharmacologists of 47 member countries and three regional associations. IUPHAR is a member of the International Council of Scientific Unions and participates in the work of its scientific committees. It receives international recognition particularly by the United Nations Educational and Scientific Organization. IUPHAR can be contacted c/o:

Paul M. Vanhoutte, Secretary General, IUPHAR
Vice-President Research
Direction de la Recherche et du Développement
Institut de Recherches Internationales Servier (IRIS)
6 Place des Pléides
92415 Courbevoie Cedex FRANCE
WWW URL: <http://www.icsu.org/membership/SUM/iuphar.html>

Federations of biochemists and molecular biologists

In addition to the Unions there are four regional organizations for biochemists.

The Federation of European Biochemical Societies (FEBS) which now has 32 constituent societies.
(For address see **The Biochemical Society** below.)
WWW URL: <http://www.febs.unibe.ch/>

The Pan American Association of Biochemists and Molecular Biologists (PABMB).
(For address see **American Society for Biochemistry and Molecular Biology** below.)

Federation of Asian and Oceanian Biochemists and Molecular Biologists (FAOBMB) which now has 18 biochemistry and molecular biology societies associated.
(For address see **The Japanese Biochemical Society** below.)

The Federation of African Societies of Biochemistry and Molecular Biology (FASBMB)
The Chairman
Department of Biochemistry
University of Nairobi
PO Box 30197
Nairobi
Kenya

International Federation of Clinical Chemists (IFCC) which is loosely linked to IUPAC.

Human Genome Organisation (HUGO)
One Park Square West
London NW1 4LJ
United Kingdom

Appendix C

Other Regional Organizations

The European Molecular Biology Organization

Postfach 1022.40
Meyerhofstrasse 1
69117 Heidelberg
Germany

European Federation of Biotechnology

For information contact one of the following:

Society of Chemical Industry
14 Belgrave Square
London SW1X 8PS
United Kingdom

DECHEMA
Theodor-Heuss-Allee 25
D-6000
Frankfurt am Main 97
Germany

Société de Chimie Industrielle
28 rue St-Dominique
F-75007 Paris
France

National Societies

Some of the larger national Biochemical Societies may be noted as follows;

American Society for Biochemistry and Molecular Biology

9650 Rockville Pike
Bethesda
MD 20814
USA

E-mail: kmirabal@faseb.org

The Biochemical Society

59 Portland Place
London W1N 3AJ
United Kingdom

E-mail: genadmin@biochemistry.org.

The Japanese Biochemical Society

Ishikawa Building-3f
25-16, Hongo 5-Chome
Bunkyo-ku
Tokyo
Japan



Appendix D

The Internet

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Department of Biochemistry and Molecular Biology, University of Leeds, Leeds LS2 9JT, UK

Contents

709	Introduction	
709	Glossary	Commonly used Internet terms
711	Getting started	
711	URLs of use in biochemistry	

Introduction

This appendix is in three sections: first we present a glossary of terms commonly used in discussions of the Internet; secondly we consider a worked example of how to find a biochemical fact using the Internet's search facilities; finally there is a list of biochemically interesting and/or useful sites. The reader is strongly advised to become familiar with the Internet and to note that it is expanding very rapidly: we do not guarantee that all the sites will remain available for the currency of this Dictionary. Additionally more sites will undoubtedly go on line and the reader should rely on search engines rather than the specific sites that are mentioned here.

Glossary

Some of this glossary is based on N. A. Maughan and J. H. Parish (1995). The Information Superhighway and Biochemistry. *Biochemical Education* **23**, 184–189, © 1995 International Union of Biochemistry and Molecular Biology. Parts of the text are reproduced here with permission.

anonymous FTP FTP access to a site where files, etc., can be accessed. The login name for such a site is 'anonymous' (or 'ftp') and the required password is the user's e-mail address.

applet a little application; this term is used to describe a module written, usually, in Java.

Archie a database listing filenames of files in FTP servers throughout the world. It can be used to locate a particular public domain file if the exact filename is known.

archive *see* **software archive**.

archive format one of a variety of file types in which collections of computer files are put together for storage and retrieval. One of the most common is TAR (tape archive, so called because it was originally designed for this); other examples include BOO and ZOO files (for PCs); certain other archive formats are **compressed files**.

ASCII *abbr. for* American Standard Code for Information Interchange; a standard set of characters readable by all computers.

bionet names of newsgroups of interest to biochemists start with the text 'bionet'; examples include the newsgroups bionet.plants and bionet.molbio.yeast. *Distinguish from* **BioNet**.

BioNet an educational software consortium whose URL is <http://www.leeds.ac.uk/bionet.html>. *Distinguish from* **bionet**.

bookmark a URL stored in a browser file, usually to allow quick access to a frequently used WWW site without the need to type in the URL.

browser a piece of local software that allows access to WWW pages; an example is **Netscape**.

cgi *abbr. for* Common Gateway Interface: a programming language similar to C used for producing programs, etc., in HTML.

compressed file a computer file (or a collection of these in **archive format**) that has been compressed by an appropriate algorithm, such as that of Lempel-Zif. In UNIX, such files are indicated by the extensions .Z, .gz, or .sh. For PCs they are usually indicated by the extension .zip. Certain compressed files may be programs that compress themselves. Others require a decompression utility such as uncompress, gunzip (UNIX), and PKUNZIP and WINZIP (PCs). A good **software archive** will include notes (typically in a file called README or 00README) on decompression utilities that are required and copies of any decompression software required for compressed files in their own archive.

DNS *abbr. for* Domain Name Server: a computer, transparent to the average end-user, that parses IP addresses and establishes the necessary connection between the appropriate nodes.

domain part of an **IP address**; in the example given in that entry, 'leeds.ac.uk' is a domain.

e-mail *abbr. for* electronic mail.

e-mail address an address for e-mail of the form <name>@<address> where the <address> is an **IP address** in its named format, e.g. howard@mbus1.leeds.ac.uk or j.h.parish@leeds.ac.uk. Another example is president@white-house.gov. Note that e-mail addresses and IP addresses are case insensitive.

Appendix D

ethernet low-level protocols and associated hardware (cabling) for computer communications.

ethernet address *an alternative name for machine address.*

FTP *abbr. for file transfer protocol:* similar to **Telnet** in that it requires a user name and password but is only capable of transferring files. FTP has limited help facilities (obtained by typing in ?): it uses UNIX conventions for changing directories etc. The most important consideration is to tell the FTP program whether the file transfer is to be binary or ASCII (i.e. text).

Gopher a menu-based alternative to using **WWW** for finding out what sites contain files you might wish to receive.

home page **1** the first page loaded by a **WWW** browser such as Netscape when it starts up. **2** the main **WWW** site for a particular person or group, often a contents list with **links** to connected sites.

http *abbr. for HyperText transfer protocol;* also used as a suffix for computer files that are written in HTML etc.

HTML *abbr. for HyperText Mark-up Language:* an interpreted computer language used for producing HyperText.

HyperText a form of marked text used for certain computer applications.

Internet a large global network of computers that arose from a United States military network (ARPANET) that started in 1969. ARPANET was built to be reliable in that if one part of the network was damaged by enemy action, messages between computers would automatically be re-routed. The in-built reliability between sites makes this type of network popular with all users: any computer on the Internet has the capacity to communicate with any other computer on the Internet. The methods and protocols for communicating on the Internet were finalized in 1985 and programs that use the Internet continue to be developed.

IP *abbr. for Internet protocol (see protocol).*

IP address a unique address for an Internet computer. It is a software address: although actually a binary number it is usually represented in 'dotted decimal' notation, e.g. 129.11.110.24. Associated with this (for the vast majority of sites) is a 'named' version of this, in the case cited above it is `bmbus1.leeds.ac.uk`. The conventions for such named addresses are fairly straightforward. The last two letters (`uk` in that case) indicate the country: other examples include `ca` (Canada), `in` (India), `de` (Germany), and `si` (Slovenia). The exception is the USA: as IP is American, the default is the USA. The `.ac` in the case above means 'academic'; in the UK this field can also be `.gov` (government), `.co` (or, rarely, `.com`; commercial), or `.org` (organization). A similar convention is used in many countries, e.g. in the USA, `.gov` (government), `.com` (commercial), `.edu` (education), `.mil` (military). Some countries (e.g. Germany) do not usually have this field at all.

Java an object-oriented computer language designed for producing **WWW** pages etc. At the time of writing this glossary, Java is not universally used for this purpose as it has to be interpreted in software and is relatively slow. However, the importance of Java is set to increase as specialized processors designed for running Java programs become available and common. Unlike many computer neologisms, Java is not an acronym but rather a reference to Java coffee, which is popular with the original authors of the language.

link a connection between one **WWW** site and another; the presence of a link is indicated by underlined and/or coloured text; clicking on this text automatically instructs the browser to open the relevant **WWW** page.

machine address or **ethernet address** a binary number unique to a computer (more exactly its network hardware) that is used by DNS, etc. It should be transparent to the average end-user. The number is usually represented as colon-delimited hexadecimal numbers, e.g. the hardware associated with `bmbus1.leeds.ac.uk` (see **IP address**) has the machine address 00:00:f8:20:5f:3a.

Netscape a popular (and highly effective) **browser** available for a variety of personal computers, workstations, and graphics terminals.

newsgroup an electronic bulletin board on the Internet, where anyone can post a message. These messages can be read by anyone else with access to the Internet and readers can post replies to messages. The distinction between electronic mail and newsgroups is that the former is a method of sending private messages to individuals.

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Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

whereas the latter is a means of posting messages in a public arena. Newsgroups therefore provide a flexible and convenient method for exchanging ideas and opinions and getting answers to technical queries. Newsgroups are arranged by subject and the number of newsgroups currently on the Internet is in the thousands. There are several newsgroups devoted to various aspects of biochemistry and therefore some care must be taken to post queries to the most appropriate newsgroup.

news reader a program to read and reply to **newsgroup** messages. Most multi-user systems will have a news reader installed on them. Several news readers are available for PCs and Macintosh machines.

node a point in a computer network where different communication lines connect; it may be a computer or a simpler electronic device, depending on the size and complexity of the network.

protocol a set of rules governing the way that data are exchanged between computers, for example, determining the format of messages.

search engines Internet sites that have facilities for searching for text. Depending on the way in which they work they are variously referred to as Web crawlers, robots, etc. Three good examples are:

<http://www.yahoo.com>

<http://www.altavista.com>

<http://www.search.com>

software archive a site usually accessed by anonymous FTP, from which public-domain software can be downloaded. Some of the software is binary, i.e. is designed to be loaded directly into a local computer; some is code written in a high-level language (such as C or C++) and needs to be compiled locally. Typically software for PCs (Windows or Windows95) and Macintosh computers is of the former type and more generic code and software for UNIX systems is of the latter.

SRS *abbr. for Sequence Retrieval System:* a system for molecular biologists that contains links to the major sequence and 3-D structure databases etc. Examples are listed under 'Molecular biology research tools' below.

system administrator a person you should get in touch with if you are setting up an internet node for the first time.

Telnet a method of logging into a computer on the Internet from another Internet computer (which could be in the same room or across the world). Telnet is typically used to access remote computers which have facilities on them not available locally. For example, a researcher may want to query a library catalogue or access a database containing information on protein sequences.

URL *abbr. for uniform resource locator:* a method for identifying **WWW** sites, which is 'uniform', i.e. independent of any specific protocol. URLs have a characteristic convention of starting `<type>://address`. FTP and Gopher sites start 'ftp://' and 'gopher://' respectively. **WWW** sites start 'http://'. Here is a fairly typical URL:

<http://www.leeds.ac.uk/>

Note that the `www` field is optional and is not found at all sites. The URL can be extended by adding a level down and/or an actual page reference as, following that example,

<http://www.leeds.ac.uk/bmb/bmb.html>

In that example 'bmb' is the name of a Department (Biochemistry and Molecular Biology) and 'bmb.html' is an example of a **home page**.

uniform resource locator *see URL.*

Usenet collectively, the **newsgroups** on the Internet.

WAIS *abbr. for wide area information system:* a system for the retrieval of documents etc. that supports a directory of servers and is useful for certain free text searches.

W3 *abbr. for World Wide Web (see WWW).*

World Wide Web *see WWW.*

WWW or **W3** *abbr. for World Wide Web;* a network of pages on the Internet. The pages combine formatted text and graphics. The characteristic feature of the **WWW** is the way in which the pages are cross-referenced using **links**. The pages are distributed: they can be on local machines or in another country.



Getting started

The list of URLs given below may seem somewhat daunting. A useful first step in getting acquainted with the Internet is to carry out a MEDLINE search. To do this, follow the steps below.

1. Using your browser, type in the URL, <http://www3.ncbi.nlm.nih.gov/Entrez/> (NOTE: the capital E of Entrez is important). This will take you to a very valuable US government site that offers a range of information.
2. In the list of databases, under 'Search WWW Entrez', click on Search the molecular biology subset of MEDLINE. A screen appears in which there is a white box which invites you to 'Enter Term'. Place your cursor in this box, and type in a term for which you would like references. For example you could type in 'citrate synthase'. Then click on the grey box that says 'Accept'. If you do this, a screen appears that contains a grey box telling you that (at the time of writing) there are 620 documents. In fact it says 'Retrieve 620 documents'.
3. Below it there is a smaller white box that allows you to limit the number of documents to be passed to you. The default on this is 50 documents – you can alter it as you wish, but remember that transfer of a large number of documents may take a very long time.
4. When you have decided how many of the documents you wish to transfer, click on 'Retrieve 620 documents' (the number in the grey box remains the same, but only 50, or such number as you have typed in the white box, will be transferred).
5. A list of citations, starting with the most recent, appears on the screen. Below each citation, a list of options will be found. One of the most useful of these is Abstract format. If you click on this, an abstract of the document (available in most, but not all cases) will appear.

URLs of use in biochemistry

Users are advised that pages are subject to change without notice. If you are unable to find a particular page try altering the search criteria.

Search engines (general)

<http://www.yahoo.com/>

A comprehensive search engine that is easy to use but requires some guessing as to what categories should be searched. Other search engines can be accessed via Yahoo.

<http://www.altavista.com/>

A comprehensive search engine that is a good starting point if you do not know what subcategory to start with but which produces large numbers of URLs if the key word(s) are not specified carefully. Its Advanced Search option allows the use of Boolean operators such as AND and OR and the user can dictate the date of the items searched and the order in which the search results are displayed.

<http://http.demon.net/external/cusi.html>

Similar to Yahoo, worth using in Europe in the afternoon

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Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

when North American Internet activity can slow down transatlantic access to Yahoo.

<http://www.lycos.com/>

A popular if idiosyncratic browser for the WWW; not always suitable for scientific enquiries.

<http://www.search.com/>

A set of links to AltaVista and Magellan (a comparable site); it also provides direct search facilities of its own.

<http://netscape.com/home/internet-search.html>

The URL that is accessed if Netscape users choose 'Net Search' from their Netscape window. It allows searching and provides access to AltaVista, Magellan, Yahoo, Lycos, etc.

<http://www.wais.com/>

A starting point for WAIS servers.

Molecular biology and bioinformatics search engines

A very useful starting point is Pedro's list: it is available in copied (mirrored) sites in several continents as follows:

USA: http://www.public.iastate.edu/~pedro/research_tools.html

Europe: http://www.biophys.uni-duesseldorf.de/bionet/research_tools.html

http://www.fmi.ch/biology/research_tools.html

Asia: http://www.peri.co.jp/Pedro/research_tools.html

Pedro's list provides links to databases, software and analysis tools and is incorporated into a search engine at <http://www.ebi.ac.uk/htbin/bworld.pl>

Some other useful general starting points:

<http://www.molbiol.ox.ac.uk>

<http://info.er.usgs.gov/network/science/biology/index.html>

<http://www.bioinf.leeds.ac.uk/bioinformatics.html>

<http://www.hgmp.mrc.ac.uk/Bioinformatics>

<http://www.bio.cam.ac.uk>

<http://www.abc.hu/biosites.html>

<http://golgi.harvard.edu/biopages.html>

http://www.ebi.ac.uk/general/other_bioservers.html

http://rldb.rz_berlin.mpg.de/links/biolinks.html

Maps and models

Several sites that are excellent for teaching (including self-education):

<http://www.genome.ad.jp/>

(this is also a very extensive site for genome links)

<http://www.imb-jena.de/IMAGE.html>
(for 3-D structures)

<http://www.biochemgen.ucsd.edu/>
(human metabolic disorders)

<http://www2.shef.ac.uk/chemistry/web-elements/web-elements-home.html>

Appendix D

(periodic table of chemical elements)

<http://cst.lanl.gov/julie/imagemap/periodic.periodic.html>
(rather more colourful and detailed version)

Genome projects

Entries with the prefix 'bionet.' are news groups.

Arabidopsis

[bionet.genome.arabidopsis](http://bionet.genome.arabidopsis.weeds.hgh.harvard.edu/)
<http://weeds.hgh.harvard.edu/>

Caenorhabditis elegans

bionet.software.acedb
bionet.software.gcg
<http://moulon.inra.fr/acedb/acedb.html>

Drosophila melanogaster

bionet.drosophila
gopher://bio.net/11/DROS/
Flybase (many mirror sites available)
<http://morgan.harvard.edu/>

Escherichia coli genome project

<http://www.genetics.wisc.edu/>
<http://www.ai.sri.com/ecocyc/ecocyc.html>
links metabolic maps to genes etc.

Human Genome Project: several starting points include

<http://shgc.stanford.edu/>
<http://mendel.hgp.med.umich.edu/>

Mouse

<http://glengoyne.hgu.mrc.ac.uk/>
<http://mickey.utm.edu/>

Xenopus

gopher://gopher.bio.net/11/URODELES

Yeast

bionet.molbiol.yeast
<http://genome-www.stanford.edu/>

Zebrafish

bionet.organisms.zebrafish
<http://zebra.scarolina.edu/>

Databases

Ribosomal database project for taxonomy based on rRNA

<http://rdb.life.uiuc.edu/>

Protein and nucleic-acid sequence/structure databases

<http://www.ncbi.nlm.nih.gov/> (GenBank etc.)
<http://www.nih.gov/molbio/>
<http://www.dl.ac.uk/SEQNET/home.html>
<http://www.boklabs.wisc.edu/seq.html>
(mainly viral sequences etc.)
<http://pdb.pdb.bnl.gov/>
(Brookhaven structural database)

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Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

REBASE (restriction enzymes)

<http://www.gdb.org/Dan/rebase/rebase.html>

SCOP (a structural classification of proteins)

<http://www.bio.cam.ac.uk/scop/>

Molecular biology research tools

Software to run remotely

For access to software involved in protein and nucleic-acid sequence analysis including structure prediction and relationship of 'your' sequence(s) to known sequences, the number of sites is vast. The best starting point is an SRS server: choose one from the following:

Australia	http://wehiz.wehi.edu.au/srs/srsc
Belgium	http://ben.vub.ac.be/srs/srsc
Eire	http://acer.gen.tcd.ie/srs/srsc
Finland	http://cypress.csc.fi:8001/srs/srsc
France	http://www.infobiogen.fr:80/srs/srsc
Germany	http://www.embl-heidelberg.de/srs/srsc
Hungary	http://hubi.abc.hu/srs/srsc
Israel	http://dapsas.weizmann.ac.il/srs/srsc
Japan	http://srs.dna.affrc.go.jp/srs/srsc
Netherlands	http://www-srs.caos.kun.nl/srs/srsc
Norway	http://bioslave.uio.no:8001/srs/srsc
Sweden	http://www.d.bmc.uu.se:80/srs/srsc
UK	http://www.ebi.ac.uk/srs/srsc http://www.sanger.ac.uk/srs/srsc http://srs.hgmp.mrc.ac.uk/srs/srsc http://www.seqnet.dl.ac.uk/srs/srsc
USA	http://www.ch.embnet.org/srs/srsc http://www.at.embnet.org/srs/srsc http://iubio.bio.indiana.edu:81/srs/srsc http://mcabi-34.med.nyu.edu/srs/srsc

A good general-purpose European site (with links) for molecular biology applications is the Java Molecular Biology WorkBench

<http://www.embl-heidelberg.de/~toldo/JaMBW.html>

Three useful starting points for commonly used services and links are as follows:

1. mowse: given fragmentation patterns with protein-cleavage reagents and peptide mass fingerprints, mowse will guess the protein(s) that correspond to the data.
<http://www.hgmp.mrc.ac.uk/Registered/Webapp/mowse/>
2. The Leeds Bioinformatics Project: a convenient collection of alignment and other software that all runs from the same site.
<http://bmbsgill.leeds.ac.uk/>



3. BLAST (Basic Local Alignment Search Tool): a fast method for finding locally optimized sequence alignment; it is popular for identifying given sequences in a database. The most recent implementation of BLAST is Ψ -BLAST and the service is at

<http://www.ncbi.nlm.nih.gov/cgi-bin/BLAST/nph-psiblast>

Sources of public-domain software

A convenient and easily used starting point:

[gopher://gopher.molbiol.ox.ac.uk:70/1](http://gopher.molbiol.ox.ac.uk:70/1)

Convenient software archive for molecular biology (and some other topics)

<ftp://ftp.EMBL-heidelberg.de>

<ftp://ftp.ebi.ac.uk>

Large general sites and databases

<ftp://ncbi.nih.gov>

A very large archive of molecular biology and mathematics programs

<ftp://unix.hensa.ac.uk>

and

<ftp://micros.hensa.ac.uk>

UK national public-domain software archive (not specifically molecular biology)

<ftp://ftp.gdb.org/>

GDB and OMIM FTP repository (human genome and genetic traits)

<ftp://golgi.harvard.edu/pub/>

GDE (Genetic Data Environment) (Harvard)

<ftp://ftp.bchs.uh.edu/pub/gene-server/>

Gene-server: software and PIR (Houston)

<ftp://turbo.bio.net/pub/>

Intelligenetics

<ftp://ftp.gdbnet.ad.jp/pub/>

Japan Information Center of Science and Technology; biological databases and software mirror for the Pacific area

<ftp://ncbi.nlm.nih.gov/repository/>

NCBI Repository FTP archive

[gopher://fly.bio.indiana.edu:70/11/IUBio-](http://fly.bio.indiana.edu:70/11/IUBio-)

Software%2bData

IUBio-Software+Data (Indiana)

<ftp://ftp.tigr.org/estdb/>

expressed sequence tag database (TIGR)

<ftp://encyclo.jax.org/pub/>

Encyclopedia of the mouse genome (Jackson Laboratory)

<ftp://pdb.pdb.bnl.gov/>

PDB 3-D coordinates of proteins (Brookhaven)

<ftp://vent.neb.com/pub/>

REBASE restriction enzyme database (NEB)

[gopher://weeds.mgh.harvard.edu:70/11/](http://weeds.mgh.harvard.edu:70/11/)

arabidopsis/aatdb

AAtDB FTP archive

[gopher://s-crim1.daresbury.ac.uk:70/11/](http://s-crim1.daresbury.ac.uk:70/11/)

EMBNET%20Bioinformatics%20Resource%20UK/
Data%20%26%20Software/databases

Assorted databases from SERC, Daresbury UK

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Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

[gopher://rdpgopher.life.uiuc.edu:70/1/](http://rdpgopher.life.uiuc.edu:70/1/)

Ribosomal Database Project (UIUC)

[gopher://ndb.rutgers.edu:70/11/etc/](http://ndb.rutgers.edu:70/11/etc/)

ndb_link_files

Nucleic Acid Database (NDB) archive

Molecular graphics

<ftp://nemo.life.uiuc.edu/pub/>

3-D protein display and sequence analysis for the PC (UIUC)

<ftp://brise.imag.fr/pub/mac/programs/>

FoldIt (light) molecular graphics (Macintosh)

<ftp://orion.oac.uci.edu/protein/>

Kinimages, Mage, Pkin: 3-D protein display for the Macintosh and PC (UIC)

<ftp://iqm.unicamp.br/pub/chemistry/>

molecular graphics, dynamics and chemistry (Brazil)

<ftp://ion.rice.edu/pub/>

MolViewer: molecule-viewing program for NeXT (Rice)

<ftp://ftp.dcs.ed.ac.uk/home/ftp/pub/rasmol/>

RasMol: molecular graphics visualization tool for UNIX and PC (UK)

<ftp://sprite.Berkeley.edu/tkrasmol.tar.Z>

RasMol: extended interface (Berkeley)

<ftp://cuhhca.hhmi.columbia.edu/pub/programs/>

PDBview: view 3-D protein structures (Columbia University)

<ftp://stanzi.bchem.washington.edu/pub/>

raster3d/

Raster3D: molecular graphics package

<ftp://ftp.msc.edu/pub/xmol/>

XMol: display and analysis of molecular model data (MSC)

Sequence and structure alignment and comparison

<ftp://ibcp.fr/pub/>

ANTHEPROT: protein sequence and 3-D structure analysis (France)

<ftp://ftp.cs.unc.edu/pub/bioscan/>

BioSCAN: biopolymer similarity analysis (UNC)

<ftp://cmgm.stanford.edu/pub/IGdb/>

BLAST and FASTA scripts (Stanford)

<ftp://ftp.ncifcrf.gov/pub/delila/>

DELILA: sequence analysis system (NCI/FCRDC)

<ftp://virginia.edu/pub/fasta/>

FASTA database search program (Virginia)

<ftp://groucho.cs.psu.edu/>

LAV, LAD, Band, & TNB: sequence alignment and analysis (PSU)

<ftp://gnomic.stanford.edu/pub/>

SAPS: statistical analysis of protein sequences (Stanford)

<ftp://ftp.cc.umanitoba.ca/psgendb/>

XYLEM & FSAP: molecular database and analysis packages (Manitoba)

[gopher://sio.ucsd.edu:70/11/](http://sio.ucsd.edu:70/11/)

SIO-Departmental%20Gophers/MBRD/haygood
spreadsheet macros for colouring sequence alignments

Appendix D

Experimental molecular biology utilities

ftp://ccp4.dl.ac.uk/pub/ccp4/
CCP4: programs for protein crystallography (UK)
ftp://ftp.cs.yale.edu/pub/double_digester/
double digester for restriction enzymes (Yale)
ftp://peipa.essex.ac.uk/ipa/src/process/
LaboImage: image processing and analysis (including 1-D
gel analysis)
ftp://zippy.nimh.nih.gov/pub/nih-image/
NIH Image: image processing and analysis

Molecular modelling and prediction

ftp://ftp.scripps.edu/pub/
FLEX 3D (The Scripps Institute)
ftp://ftp.tigr.org/pub/
Gene Modeller, ADE (TIGR)
ftp://ftp.biochem.ucl.ac.uk/pub/MEMSAT/
MEMSAT: membrane protein structure and topology
ftp://mbcrr.harvard.edu/MBCRR-Package/
MBCRR programs (Harvard)
ftp://crocus.medicine.rochester.edu/jones/
MOBOSOL: MOPAC modified for AMSOL and NBO
(Rochester)
ftp://sheep0.chem.uic.edu/dist/
MOIL: molecular dynamics (UIC)
ftp://ouchem.chem.oakland.edu/pub/
MOPAC60, AMPAC2.1, and MOPAC50ESP (Oakland)
ftp://ns.niehs.nih.gov/
MULTI: molecular modelling (NIEHS–NIH)
ftp://expasy.hcuge.ch/pub/ProMod/
ProMod: 3-D modelling software (Geneva)
ftp://bsmchal.biochem.ucl.ac.uk/pub/hbplus/
home.html
HBPLUS: hydrogen bond calculator
ftp://nrcbsa.bio.nrc.ca/pub/
RNA secondary structure software (NRC, Canada)
ftp://ftp.itc.univie.ac.at/pub/RNA/
ViennaRNA: fold and compare RNA secondary structures
(Vienna)

Taxonomy, genomes, and genetics

ftp://spider.ento.csiro.au/delta/
DELTA: description language for taxonomy
(Australia)
ftp://toolik.stanford.edu/pub/hgmc/
Human Genome Mapping Center (Stanford)
ftp://evolution.genetics.washington.edu/pub/
PHYLP: phylogeny package (Washington)
ftp://genome.wi.edu/pub/
Primer, Mapmaker, Mouse SSLP maps (Whitehead)
ftp://zoo.toronto.edu/pub
Random: cladistic randomization routines (Toronto)
ftp://relay.cs.utoronto.ca/pub/molbio/ramha1/
RAMHA: random mutagenesis simulation (Toronto)
ftp://genome.stanford.edu/pub/
Yeast Genome Project (Stanford)
gopher://megasun.BCH.UMontreal.CA:70/11/CMB/
Phylogeny/Treecon
Treecon: phylogeny, evolutionary trees
gopher://huh.harvard.edu:70/11/software
TAXACOM systematic biology (Harvard)
gopher://megasun.BCH.UMontreal.CA:70/11/CMB/
Phylogeny
Phylogenetic analysis programs, help, etc. (megagopher)
gopher://larry.pathology.washington.edu:70/1
chromosome idiograms, human and mouse
gopher://corona.med.utah.edu:70/11/pub
linkage and pedigree software and STS probes (Utah)

Miscellaneous

ftp://ftp.vt.edu/pub/biotechnology/
National Biological Impact Assessment Program (Ag.
Biotech)
gopher://gopher.gdb.org:70/11/
softsearch/.soft-arch-merlot/sas-macros
SAS macros for the Analysis of Clinical Trial Survey and
Energy Survey Data
gopher://dean.med.uth.tmc.edu:70/11/
Biomedical_Shareware/
Biomedical_Shareware (Texas Medical School)

Appendix E

Exploring the language of bioinformatics

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<http://www.biochem.ucl.ac.uk/bsm/dbbrowser>

Contents

715	Introduction	
716	Sequence analysis	Family, superfamily, clan; homology; motifs; patterns; rules; blocks; fingerprints; profiles
718	Structure analysis	Domains, modules
719	Applications of sequence analysis methods: examples	Functional sites; domains and modules; protein families
721	Primary databases	
722	Composite databases	
722	Secondary databases	
722	Information networks	
723	References	

Introduction

Ten years from the dawn of the field of bioinformatics, we are beginning to harvest the copious fruits of the various genome projects. In spite of early flood warnings, the resulting torrent of sequence information has all but broken our databanks. The endeavour to cope with, and to rationalize, these vast quantities of data has required cooperation on a global scale. Today, there are hundreds of bio-databanks around the world, many of which are now linked together over the Internet via the World Wide Web (WWW). The WWW is now the vehicle of choice for disseminating biological information, and the resources discussed in the following pages can be pursued further using the Web addresses given in Table 3.

The first step, then, in facilitating access to biological information is to assemble it into central, shareable resources, i.e. databases. There are many different database types, de-

pending on the nature of information being stored and the manner of its storage. For example, two of the themes that dominate bioinformatics research are the acquisition and storage of protein primary and tertiary structures. But the level of complexity involved in archiving a three-dimensional structure is clearly very much higher than that required to store linear amino-acid sequences. Separate disciplines have therefore evolved to manage, or curate, these different database types and to analyse the information they contain. The computational investigation of primary structures (which will henceforth be referred to as sequences) and of tertiary structures (henceforth referred to simply as structures) is known as protein sequence and protein structure analysis respectively (sequence analysis in this context is thus not to be confused with sequence determination).

In the course of this discussion, we will encounter primary, secondary and tertiary structures. If you are interested in learning more about the book, please contact: HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
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Appendix E

secondary, composite, and other, specialist databases. Such resources store different levels of information in completely different formats, leading to a variety of communication problems. However, the advent of user-friendly Web browsers has addressed many of these difficulties, and emerging network technologies are providing exciting new tools with which to brave the information revolution.

The need to harness the sea of biological information is still acute. There are currently more than half a million sequences in public and private databases around the world. This is to be contrasted with the number of known protein structures, now around 5000. This represents a vast information deficit. But, locked up in protein sequences is a huge amount of structural, functional, and evolutionary information. The goal, then, is to try to unearth this information and convert it into biochemical knowledge.

The ultimate challenge, that of predicting protein structure from the linear amino-acid sequence, is unlikely to be realized for decades to come. We do not yet understand enough about the rules that govern protein folding, nor do we have sufficient structures in our databases to make prediction from sequence feasible in the short term. The state of the art, in terms of both sequence and structure analysis, is thus currently at the level of pattern recognition. For protein structures, fold-recognition algorithms (which rely on building reference libraries of protein-fold templates) are only about 50–60% reliable. Such approaches will improve as the database of known structures expands. For protein sequence analysis, the reliability of family/function recognition (which relies on building compendia of protein sequence descriptors) is much higher, but varies considerably with the type of analysis method employed. The difficulty is in being able to distinguish between the mathematical significance of a result returned by a database search algorithm and its biological significance.

The principle of sequence analysis hinges on the observation that sequences belonging to families have accumulated mutations, presumably within an unknown archetype, and that divergent drift is characteristic of the family. Such homologous relationships can be identified with ease where levels of sequence identity remain high, but when these drop to below 50% it becomes increasingly difficult to establish relationships reliably. Analyses can be pursued with decreasing certainty towards the 'Twilight Zone' (the 0–25% identity range), where results are no longer statistically significant. Within the Twilight Zone, alignments may appear plausible to the eye, but cannot be supported by statistical methods. Although sequences may diverge to such an extent that little or no similarity remains at the sequence level, their structures are more likely to be preserved. Such relationships can no longer be detected by sequence-based methods, but fall into the realm of fold recognition.

The following pages aim to provide a guide to the terms used (and abused) in the fields of protein sequence and structure analysis. A list of WWW addresses is given to allow readers to obtain the most up-to-date information about the various databases and resources discussed.

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Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynghuyen@huynghuyentld.com.

Sequence analysis

Identifying conserved motifs within sequences can provide insights into protein structure and function, and allow us to establish means of classification. The goal is to rationalize the data in the primary sources, by assigning sequences to families. The essence of those families is distilled into potent sequence descriptors that are housed in secondary databases, which then facilitate the diagnosis of future query sequences used to search the database. It is important to establish secondary databases because the primary sources are growing at such a rate that it is becoming increasingly difficult to resolve distant relationships from background noise.

The underlying tool of sequence analysis methods is multiple sequence alignment. Application of those methods involves the derivation of some kind of representation, or abstraction, of conserved alignment elements, which may be diagnostic of structure or function. A bewildering array of terms is used to describe the different types of data representation employed by the various approaches: e.g., motifs, segments, patterns, regular expressions, rules, profiles, signatures, fingerprints, features, domains, and blocks. Several of these have fairly well-defined meanings, but difficulties arise when they are used interchangeably. Some basic definitions of these, and other, concepts are outlined below.

Family, superfamily, clan

It should be relatively straightforward to understand the terms family and superfamily, which usually imply shared functional characteristics and an obvious sequence relationship. Nevertheless, there is still debate regarding when and how properly to use them. The G-protein-coupled receptors (GPCRs) provide a useful illustrative example. The adrenoceptors, muscarinic receptors, opioid receptors, opsins, etc., are distinct protein **families** with characteristic pharmacological and signalling profiles: for example, they may be separated on the basis of the ligands they bind, their tissue distribution, the second-messenger systems to which they couple, or their amino-acid sequences. In most cases, the families may be divided into receptor subtypes, depending on their selectivity for particular agonists and antagonists, which specificities are reflected in slight changes in their primary structures. All of these receptor families share the same structural framework, which is thought to consist of seven transmembrane (7TM) helices. They thus belong to a **superfamily** (which for historical reasons has been termed rhodopsin-like) that is characterized by a unique 7TM signature. This can be readily identified by sequence analysis approaches.

The rhodopsin-like GPCR superfamily, then, can be identified by its characteristic transmembrane sequence signature, and its individual family members resolved from it on the basis of characteristic changes, primarily in the loop regions that lie between the transmembrane helices. At least five other GPCR families are now known, all of which appear to contain 7TM domains, but share no discernible sequence similarity with the rhodopsin-like superfamily. The term **clan** has been coined to describe such a group of families, for which there are indications of an evolutionary relationship, but between which there is no statistically significant similarity of sequence.

It is important to define these terms with care, because of the underlying evolutionary implications. The shared sequence similarities within families and superfamilies suggest divergence from a single common ancestor. By contrast, the relationship between clan members is unclear: it may be the result of divergence from a remote ancestor, or it may have arisen through independent convergence to a common function and architecture.

Homology

Protein sequences are said to be homologous if they are related by divergence from a common ancestor. This very simple definition is confounded and abused in the literature. Homology is not a measure of similarity – it is an absolute statement that sequences have a divergent rather than a convergent relationship. Thus phrases that quantify homology (e.g., ‘the sequences show 50% homology’, or ‘sequences are 70% homologous’, or ‘have a high degree of homology’) are, strictly, meaningless and should be avoided.

Motifs

When building a multiple sequence alignment, as more divergent sequences are added, insertions are often required to bring equivalent parts of adjacent sequences into the correct juxtaposition. At the end of such a process, islands of conservation may be observed appearing within a sea of mutational changes, as shown in Figure 1. These conserved regions, which are often quite short (typically 10–20 amino acids in length), tend to denote the structural or functional core of the protein. Their conserved nature allows them to be used to diagnose family membership via a range of sequence analysis techniques. They are most frequently termed **motifs**. Synonymous terms are **block**, **segment**, and **feature**.

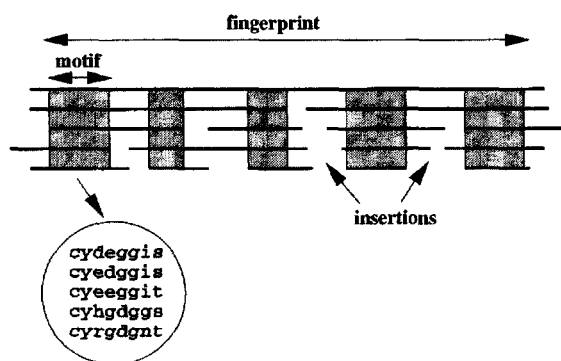


Figure 1. Conserved motifs in a multiple sequence alignment, representing the core structural and key functional sites of the aligned protein family. The broken horizontal parallel bars denote the sequence alignment; the shaded boxes show the locations of the motifs, groups of motifs forming characteristic fingerprints; the excerpt indicates typical residue frequency information contained within a motif, which provides the basis for database searching.

Patterns

For database searching, the sequence data within an individual motif can be reduced to a single **consensus** expression. That expression discards all but the most significant residue information, concentrating on the most highly conserved positions.

Alignment	Expression
GAVDFIALCDRYF	
GPIDFVCFCEERY	G-X-[IV]-[DE]-F-[IVL]-X-[PG]-C-[DE]-R-[FY]2
GRVEFLNRCDRYY	

Figure 2. Generation of a regular expression from a multiple sequence alignment. The single-letter amino acid code is used: X denotes any residue; square brackets group together the allowed residues at a given position; curly brackets group together the disallowed residues at a position.

tions. The expression shown in Figure 2 tells us that positions 1, 5, 9, and 11 are completely conserved; positions 3, 4, 10, 12, and 13 allow one of two possible residues (whatever is specified within square brackets is allowed at that position); positions 2 and 7 can be anything (X); and position 8 can be anything except proline or glycine (whatever is specified within curly brackets is *not* allowed at that position). Such expressions are used to trawl sequence databases for *identical* matches; their use is relatively limited because they tolerate no similarity. In other words, a sequence that is 99% similar is regarded as a mismatch – thus, for example, a sequence consistent with the expression in Figure 2, but with leucine in place of the allowed isoleucine or valine at position 3, would not be matched, in spite of showing only a single, conservative substitution. The result of searching a database with such an expression is therefore either an exact match, or no match at all; the outcome is either ‘on’ or ‘off’, and may hence be referred to as a binary event. Expressions of this type are most frequently termed **patterns**. Synonymous terms are **regular expression** and **signature**.

Rules

Regular expression patterns are used to greatest diagnostic effect when, within a given sequence alignment, a highly conserved motif (typically 10–20 residues in length) is found to characterize that specific protein family (this might be a core piece of the protein architecture, or it might carry critical functional residues). Often, however, it is possible to identify much shorter, generic patterns within sequence alignments. These tend to be diagnostic of specific functional sites: e.g., sugar attachment sites, phosphorylation, hydroxylation, myristoylation, sulfation sites, etc. These patterns, which are often only three or four residues long, do not provide good discrimination, but can give a guide as to whether a certain type of site *might* exist in a sequence. They are most frequently termed **rules**. Synonymous terms are **pattern** and **regular expression**.

Fingerprints

Patterns have very limited diagnostic power because they require target sequences to match the expression exactly. Thus, as we have seen, a sequence that fits a pattern at all but one position will be rejected as a mismatch. To address this problem, it is possible to build more powerful discriminators.

Within a sequence alignment, it is usual to find not one, but several motifs that characterize the aligned family (see Figure 1). Diagnostically, it makes sense to use all the conserved regions to build a family signature, so that in a database search there is a higher chance of identifying a distant relative, whether or not all parts of the signature are matched.

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Appendix E

In an approach of this sort, the motifs are treated independently. The sequence information they contain is converted into scoring tables that are populated only by the observed residue frequencies at each motif position. The scoring system is thus considered to be unweighted, in the sense that no additional scores (e.g., from mutation or substitution matrices) are used to enhance discriminating power. However, diagnostic performance is enhanced by iterative database scanning. The motifs therefore grow, and become more mature, with each database pass, as more sequences are matched and added to the scoring tables. Full potency is gained from the mutual context provided by motif neighbours, which allows sequence identification even when some parts of the signature are absent. Thus, for example, a sequence that matches only four of seven motifs may still be diagnosed as a true match if the motifs are matched in the correct order in the sequence, and the distances between them are consistent with that expected of true neighbouring motifs. Groups of aligned, ungapped, unweighted segments of this type are most frequently termed **fingerprints**; they are also called **signatures**.

Blocks

As mentioned above, the constituent motifs of a fingerprint are unweighted (i.e. the scoring system uses only observed residue frequencies), which sometimes leads to relatively poor diagnostic performance. However, it is possible to build more powerful representations of motifs by applying different weighting schemes. In one approach, sequence segments within a motif are clustered to reduce multiple contributions to residue frequencies from groups of closely related sequences. Each cluster is then treated as a single segment, each of which is assigned a score that gives a measure of its relatedness; the most distant segment is given a weight of 100. Most protein families will be defined by several such clustered, weighted motifs. For a given query sequence, as with a fingerprint, the more motifs matched, the greater the confidence that the sequence belongs to that family. Aligned, ungapped, weighted segments of this type are most frequently termed **blocks** (synonymous terms are **motif**, **segment**, and **feature**).

Profiles

By contrast with the use of patterns, blocks, and fingerprints, an alternative approach is to distil the sequence information within *complete* alignments into scoring tables. Such tables define which residues are allowed at given positions, which positions are highly conserved and which degenerate, and which positions, or regions, can tolerate insertions. The scoring system is intricate, and may include evolutionary weights and results from structural studies, as well as data implicit in the alignment. In addition, variable penalties are specified to weight against insertions and deletions occurring in core secondary structure elements. Such tables are used to search databases for similar matches, their inherent complexity rendering them very potent discriminators. They are most frequently termed **profiles**. **Weight matrix** and **position-specific scoring table** are synonymous terms.

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Structure analysis

The terms 'domain' and 'module' are not to be confused with the jargon of protein sequence analysis. These terms have specific definitions in the context of protein structure (although they are often used imprecisely and cause confusion). It is beyond the scope of this discussion to consider structure analysis in detail, but in order to draw the distinction from the terms used in sequence analysis, the concepts of domains and modules are outlined below.

Domains

Domains are compact, local, semi-independent folding units that need not be formed from contiguous regions of an amino-acid sequence: they may be discrete entities, joined only by a flexible linking region of polypeptide chain; they may have extensive interfaces, sharing many close contacts, and they may exchange chains with domain neighbours. In the simplest genetic terms, domains may be considered to have arisen via gene fusion and gene duplication events. Multifunctional enzymes, for example, are most readily understood in terms of gene fusion: the dehydrogenases are believed to have arisen from the fusion of a gene coding for a dinucleotide-binding domain with a series of genes, each coding for different catalytic domains. By contrast, the structural domains of chymotrypsin (anti-parallel beta-barrels with identical topology), are thought to have arisen by gene duplication, the interface between the domains providing the catalytic crevice.

In light of the complexity of protein architecture, these examples are somewhat simplistic. The rationalization and classification of protein folds is an arcane discipline, which has led to the development of a variety of domain and fold databases. Currently, amongst the most well-known of these are the scop (Structure Classification Of Proteins) resource, the CATH protein structure classification database, and the ProDom protein domain database. All of these databases are accessible via the WWW, and readers are referred to their Web addresses (see Table 3) for further information.

Modules

Modules are autonomous folding units, and as such can be thought of as a subset of protein domains. They are contiguous in sequence and are frequently used as building blocks to confer a variety of complex functions on the parent protein. In genetic terms, the spread of modules cannot be explained by simple gene duplication and fusion events, but is largely thought to be the result of genetic shuffling mechanisms. It is useful to make a distinction between extracellular modules (e.g. the fibronectin 3, immunoglobulin, and EGF modules) and intracellular modules (e.g. Src homology 2 and 3 (*see SH domain*), and pleckstrin homology): their cellular location is seldom interchangeable, and their methods of propagation differ. It is argued that extracellular modules are encoded by exons that are flanked by introns, and that these exons can be shuffled, while being protected by their intron buffers. This model does not hold for bacteria, which have no introns, suggesting that other methods of propagation exist. There is also no observed correlation with intron borders for intracellular modules, suggesting either a different shuffling mechanism, or that intracellular modules arose much earlier. For



further information, please refer to the WWW address given in Table 3.

Applications of sequence analysis methods: examples

The examples below indicate how functional sites, domains, modules, and families may be encoded by different sequence analysis approaches. (Different methods encode sequence alignment information in different ways – e.g. frequency tables, weight matrices, profiles, etc.) In each case, a regular expression is given (as indicated by the accession number and code retrieved from PROSITE 13.0), and corresponding references to blocks, fingerprints, and profiles are provided where such information exists. The full current lists of patterns, fingerprints, and profiles are available from the WWW addresses given in Table 3.

Functional sites

Functional sites usually contain only two or three key residues, which can be encoded by regular expression rules. Rules are too short to be diagnostically reliable, and can only give a guide as to whether a particular attachment site, glycosylation site, splice site, etc., might be present. A 'hit' with a rule should therefore be supported by biochemical evidence.

N-glycosylation site (PS00001 ASN_GLYCOSYLATION)⁸

$N-\{P\}-[ST]-\{P\}$

Possible *N*-glycosylation sites may be located with the rule shown above. Proline residues are disallowed at positions 2 and 4, as their presence at these points may hinder correct folding of the sequence and inhibit glycosylation. There is no true family, per se, defined by this functional site; consequently, equivalent blocks, profiles and fingerprints have not been derived.

Protein kinase C phosphorylation site (PS00005 PKC_PHOSPHO_SITE)⁸

$[ST]-X-[RK]$

Possible protein kinase C phosphorylation sites may be located with the rule shown above. The enzyme exhibits a preference for phosphorylation of serine or threonine residues found in close proximity to C-terminal basic residues. Additional N- or C-terminal basic residues enhance phosphorylation kinetics. As with the glycosylation site, there is no true family per se, and equivalent blocks, profiles, and fingerprints have not been derived.

Protein splicing signature (PS00881 PROTEIN_SPLICING)^{8,12,13}

$[LIVA]-[LIVMY]-[VAT]-H-N-[STC]$

Possible **inteins** may be located with the pattern shown above. Inteins, or protein introns, are parts of protein sequences that are post-translationally excised, their flanking regions (**exteins**) being spliced together to yield an additional protein product. This process is believed to be self-catalysed, apparently initiating at the C-terminal splice junction, where a conserved asparagine residue mediates the nucleophilic attack of the peptide bond between it and its neighbouring residue.

Maximum number of observed repeats: 2. This pattern, which encodes residues close to the asparagine involved in the

nucleophilic attack, is a poor discriminator for inteins, finding three times more false positive sequences than true positives. More powerful descriptors are found in the BLOCKS and PRINTS databases, with accession numbers BL00881 and PR00379 respectively.

Domains and modules

Protein domains and modules may be encoded by patterns, blocks, profiles or fingerprints. With the growing numbers of sequences available in databases, deriving effective discriminators for modules has become problematic. Module sequences are often highly divergent and tend not to contain conserved functional residues (module function may vary depending on biological context). Structurally important residues may be hard to detect at the sequence level, and even the number of cysteines, which may stabilize folds by forming disulfide bridges, may vary in a protein family. For this reason, some modules can only be effectively encoded by profiles.

Apple module (PS00495 APPLE)^{8,12,13}

$C-X(3)-[LIVMFY]-X(5)-[LIVMFY]-X(3)-[DENQ]-[LIVMFY]-X(10)-C-X(3)-C-T-X(4)-C-X-[LIVMFY]-F-X-[FY]-X(13,14)-C-X-[LIVMFY]-[RK]-X-[ST]-X(14,15)-S-G-X-[ST]-[LIVMFY]-X(2)-C$

The apple module, shown schematically in Figure 3, is a 90-residue domain stabilized by three disulfide bonds. Multiple tandem repeats of the domain are found in the N termini of the related plasma serine proteases plasma kallikrein and coagulation factor XI (see **blood coagulation**).

The pattern spans the complete apple module. Maximum number of observed module repeats: 4. Alternative descriptors: BLOCKS accession number BL00495; PRINTS accession number PR00005.

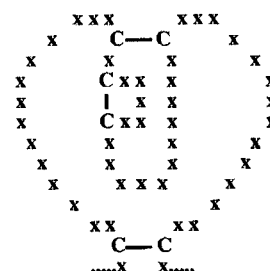


Figure 3. Schematic representation of the apple domain showing the locations of the disulfide linkages. The single-letter amino acid code is used: X denotes any residue.

Cyclic nucleotide-binding domain (PS00888 CNMP_BINDING_1)^{8,12,13}

$[LIVM]-[VIC]-X(2)-G-[DENQTA]-X-[GA]-X(2)-[LIVMFY](4)-X(2)-G$

Proteins that bind cyclic nucleotides (cAMP or cGMP) share a structural domain of around 120 residues. The domain is constructed from three alpha-helices and a distinctive eight-stranded, anti-parallel beta-barrel. Six invariant residues characterize the domain, three of which are glycines thought to be

Appendix E

essential for maintenance of the structural integrity of the barrel.

This pattern spans beta-strands 2 and 3, and contains the first two conserved glycines. Maximum number of observed domain repeats: 2. Alternative descriptors: BLOCKS accession number BL00888; PRINTS accession number PR00103.

DEAD box (PS00039 DEAD_ATP_HELICASE)^{8,13}

[LIVMF](2)-D-E-A-D-[RKEN]-X-[LIVM-FYGSTN]

A number of eukaryotic and prokaryotic proteins involved in ATP-dependent nucleic-acid unwinding have been characterized on the basis of structural similarity. All share a number of conserved motifs, one of which (the so-called 'DEAD box') represents a special version of the B motif of ATP-binding proteins.

This pattern spans the DEAD-box motif. Alternative descriptor: BLOCKS accession number BL00039.

DAG/PE-binding domain (PS00479 DAG_PE_BINDING_DOMAIN)^{8,12,13}

H-X-[LIVMFYW]-X(10,11)-C-X(2)-C-X(3)-[LIVMF]-X(2,7)-[SAGC]-[LIVMFYW]-X-C-X(2)-C-X(4)-H-X(2)-C-X(6,7)-C

The N-terminal region of protein kinase C binds phorbol esters (PEs) and diacylglycerol (DAG) in a phospholipid-dependent manner. DAG is an important second messenger; PEs, which are analogues of DAG, are potent tumour promoters that cause several physiological changes when administered to cells and tissues. The DAG/PE-binding domain is characterized by a 50-residue cysteine-rich region, which binds two zinc ions. The metal-ion ligands are thought to be provided by six conserved cysteines and two histidines.

This pattern spans the complete DAG/PE domain. Maximum number of observed domain repeats: 2. Alternative descriptors: BLOCKS accession number BL00479; PRINTS accession number PR00008.

Kringle module (PS00021 KRINGLE)^{8,12,13}

[FY]-C-R-N-P-[DNR]

Kringle modules are autonomous structural units found throughout the blood clotting and fibrinolytic proteins. They are believed to play roles in binding mediators (e.g., membranes, other proteins or phospholipids) and in regulating proteolytic activity. Kringles (so-called because cartoon representations of their primary structures resemble Danish pastries of this name) are characterized by a triple loop, three-disulfide-bridge structure, whose conformation is defined by a number of hydrogen bonds and small pieces of antiparallel beta-sheet.

The pattern encodes a well-conserved hexapeptide towards the centre of the kringle domain. Maximum number of observed module repeats: 38. Alternative descriptors: BLOCKS accession number BL00021; PRINTS accession number PR00018.

P-type 'trefoil' module (PS00025 P_TREFOIL)^{8,13}

R-X(2)-C-X-[FYPST]-X(3,4)-[ST]-X(3)-C-X(4)-C-[FYWH]

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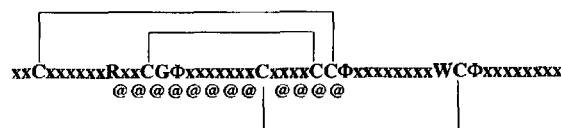


Figure 4. Schematic representation of disulfide linkages that form the P-domain. The single-letter amino acid code is used: x denotes any residue; Φ indicates the presence of large hydrophobic residues; @ marks the location of the pattern.

The P-type 'trefoil' module (or P-domain) is a 45-residue cysteine-rich region, the six cysteines of which form disulfide bridges in the order 1-5, 2-4, 3-6, as illustrated schematically in Figure 4. The domain has been found in a variety of extracellular eukaryotic proteins.

The pattern encodes the central four cysteine residues. Maximum number of observed module repeats: 6. Alternative descriptor: BLOCKS accession number BL00025.

WW module (PS01159 WW_DOMAIN_1)^{8,12,13}

W-X(9,11)-[VFY]-[FYW]-X(6,7)-[GSTNE]-[GSTQCR]-[FYW]-X(2)-P

WW modules are conserved 35-40 residue regions found in a number of disparate proteins. The domain is characterized by two conserved tryptophan residues and a conserved proline (hence its alternative name, WWP). Among the proteins known to contain WW domains is **dystrophin**, the product encoded by the gene responsible for Duchenne muscular dystrophy. The domain appears to bind proteins that contain characteristic proline motifs ([AP]-P-P-[AP]-Y), and, to an extent, resembles SH3 domains. It is often associated with other domains that typify proteins involved in signal transduction.

This pattern spans the full WW domain. Maximum number of observed module repeats: 4. Alternative descriptors: BLOCKS accession number BL01159; PRINTS accession number PR00403; PROSITE profile accession number PS50020.

Zinc finger (C₂H₂ type) domain (PS00028

ZINC_FINGER_C2H2)^{8,12,13}

C-X(2,4)-C-X(3)-[LIVMFYWC]-X(8)-H-X(3,5)-H

Zinc fingers are small, self-folding units in which zinc is a crucial structural component. The finger contains a repeated 28-residue sequence, including two conserved cysteine and two conserved histidine residues in a C-X(2)-C-X(12)-H-X(3)-H type motif. The 12 residues separating the second cysteine and the first histidine are largely polar and basic, implicating this

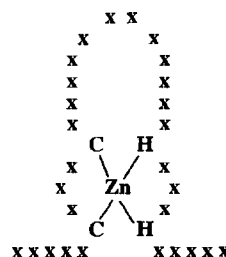


Figure 5. Schematic representation of zinc coordination by the C₂H₂-type zinc finger. The single-letter amino acid code is used: x denotes any residue.



region in particular in nucleic-acid binding. A single atom of zinc is bound in a tetrahedral array to yield a finger-like projection (shown schematically in Figure 5), which interacts with nucleotides in the major groove of the nucleic acid – the zinc binds to the conserved cysteine and histidine residues. Fingers have been found to bind to about five base pairs of nucleic acid containing short runs of guanine residues. They are found in all transcription factors, and also in proteins such as RNA and DNA polymerases. The ability to bind to both RNA and DNA suggests versatility not demonstrated by the helix-turn-helix motif: the zinc finger may thus represent the original nucleic-acid binding protein.

The pattern encodes the cysteine and histidine ligands. Maximum number of observed finger repeats: 37. Alternative descriptors: BLOCKS accession number BL00028; PRINTS accession number PR00048.

Protein families

Protein families may be effectively encoded by patterns, blocks, profiles, or fingerprints. Some typical examples are given below.

Short-chain alcohol dehydrogenases (PS00061 ADH_SHORT)^{8,12,13}

[LIVSPADK]-X(12)-Y-[PSTAGNCV]-[STAGNQ-CIVM]-[STAGC]-K-[PC]-[SAGFR]-[LIVMTAGD]-X(2)-[LIVMFYW]-X(3)-[LIVMFWGAPTH]-[GSAC-QRHM]

Primary sequence comparisons indicate an evolutionary relationship between many dehydrogenases. Most dehydrogenases possess at least two domains, the first binding the coenzyme (often NAD), and the second binding the substrate. This latter domain determines the substrate specificity and contains amino acids involved in catalysis. Little sequence similarity has been found in the coenzyme-binding domain, although there is a large degree of structural similarity. It is thus postulated that the dehydrogenase structure has arisen through gene fusion of a common ancestral coenzyme nucleotide sequence with various substrate-specific domains. A number of short conserved regions characterize the alcohol dehydrogenases, one of which includes two perfectly conserved residues: a tyrosine and a lysine. In glucose dehydrogenase, the tyrosine has been shown to be important for catalytic activity and/or for subunit binding.

The pattern encodes the functionally important tyrosine and lysine residues. Maximum number of observed repeats: 2. Alternative descriptors: BLOCKS accession number BL00061; PRINTS accession number PR00080.

Lipocalin signature (PS00213 LIPOCALIN)^{8,12,13}

[DENG]-X-[DENQGSTARK]-X(0,2)-[DENQARK]-[LIVFY]-[CP]-G-[C]-W-[FYWLRH]-X-[LIVMTA]

The lipocalins are primarily a family of extracellular ligand-binding proteins that display high specificity for small hydrophobic molecules. Their functions include transport of nutrients, control of cell regulation, pheromone transport, cryptic coloration, and the enzymatic synthesis of prostaglandins. The structure shows a novel, eight-stranded, antiparallel beta-barrel fold. Sequence similarity is low, and appears to be restricted to conserved disulfides and three mo-

tifs, which form a juxtaposed cluster that may act as a common cell-surface receptor site. The antiparallel beta-barrel fold is similarly exploited by the fatty-acid-binding proteins, which also bind small hydrophobic molecules. Similarity at the sequence level is less obvious, being confined to a single short N-terminal motif.

This pattern encodes the N-terminal motif that characterizes both the lipocalins and fatty-acid-binding proteins. More powerful descriptors: BLOCKS accession number BL00213; PRINTS accession number PR00179.

G-protein-coupled receptors (PS00237 G_PROTEIN_RECEPTOR)^{8,12,13}

[GSTALIVMFYWC]-[GSTANCPDE]-[EDPKRH]-X(2)-[LIVMNQGA]-X(2)-[LIVMFT]-[GSTANC]-[LIVMFYWSTAC]-[DENH]-R-[FYWCSTH]-X(2)-[LIVM]

G-protein-coupled receptors (GPCRs) constitute a vast protein family encompassing a wide range of functions (including various autocrine, paracrine, and endocrine processes). They show considerable diversity at the sequence level, on the basis of which they can be separated into distinct families. These families belong to a clan, within which there are indications of an evolutionary relationship, but between members of which there is no statistically significant similarity in sequence. The currently known clan members include the rhodopsin-like GPCRs, the secretin-like GPCRs, the cAMP receptors, the fungal pheromone mating factor receptors, and the metabotropic glutamate receptor family.

The rhodopsin-like GPCRs themselves represent a widespread protein family that includes hormone, neurotransmitter, and light receptors, all of which transduce extracellular signals through interaction with guanine nucleotide-binding (G) proteins. Although their activating ligands vary widely in structure and character, the amino-acid sequences of the receptors are very similar and are believed to adopt a common structural framework comprising seven transmembrane helices.

This pattern encodes part of the third transmembrane domain, which contains a completely conserved arginine residue that is implicated in binding the G-protein. More powerful descriptors: BLOCKS accession number BL00237; PRINTS accession number PR00237.

Primary databases

Nucleic acid and protein sequences are stored in primary databases. The principal primary resources are (for proteins) PIR, SWISS-PROT, mips, PSD-KYOTO, PRF, and NRL-3D and (for nucleic acids) EMBL, GenBank, and DDBJ. There are, of course, many others.

The proliferation of primary databases causes problems: do they all have the same format? Which is the most accurate? Which is the most up-to-date? Which is the most comprehensive? Which should we use?

SWISS-PROT is often regarded as the 'best', at least in terms of the quality of its annotations. But its sequence data are taken from PIR, compared with which it is neither as comprehensive, nor as up-to-date. NRL-3D is derived from the Brookhaven protein structure databank, so contains only se-

Appendix E

quences for which three-dimensional coordinates have been made available – it is therefore only small, and highly redundant, yet in view of the link it provides to structure, it is nevertheless a valuable resource. In the past, choosing which database to use was largely a matter of which was available locally. Nowadays, the problem is more likely to be a question of finding the right WWW page to allow you to perform the relevant search. In practice, to ensure the most comprehensive search, it is best to search several sources.

Cooperative projects

Maintaining databases is a laborious, time-consuming process. To relieve some of the inherent burdens, a number of groups have cooperated to share data: for protein sequences, PIR, mips, and PSD-KYOTO join forces as PIR-International; for nucleic-acid sequences, EMBL, GenBank, and DDBJ pool their resources.

Composite databases

Cooperative projects help with data distribution, but do not address the problem of multiple database formats. One approach to tackling proliferating primary databases is to compile a composite resource. Here, data from a variety of different sources are merged, rendering searches extremely efficient, the more so if redundancies between contributing sources are eliminated. But the situation is complicated by the fact that there are now several composites available (see Table 1), each of which uses different data sources, has a different format, and uses different redundancy criteria.

Table 1. Examples of composite protein sequence databases

Database	Data sources
nrdb	PDB, PIR, SWISS-PROT, GenPept, GenPept updates
Entrez	PDB, PIR, SWISS-PROT, PRF
mipsX	PIR, mips, patchX, PSD-KYTO
OWL	SWISS-PROT, PIR, GenBank, NRL-3D

Of the available composite databases, the nrdb, Entrez, and mipsX resources are in fact non-identical. Only OWL is truly non-redundant, in the sense that strict redundancy criteria are applied to the amalgamation of the primary sources (i.e., both identical and trivially different sequences are eliminated). OWL is therefore relatively small and hence efficient in database searching. But it is not as up-to-date as, for example, nrdb. So again, the user is faced with the choice of which is best to use. In practice, the ease of use of the Web makes it possible to search them all.

Secondary databases

Secondary databases contain information derived from the primary sources. There are numerous secondary, or 'value-added', databases available, all of which have different formats, use different analysis methods, and house different types of information, and most of which use different data sources (although SWISS-PROT is emerging as something like a standard) – see Table 2.

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As with the primary sources, the proliferation of secondary databases causes problems: Which is the most accurate? Which is the most up-to-date? Which is the most comprehensive? Which should we use?

PROSITE is the most widely used resource. It houses a comprehensive array of patterns, rules and profiles, encoding a variety of functional sites, domains and families. In most cases, it provides good annotations and links to other biological databases, yet diagnostically it is probably the least reliable. The profile library is still small, but profiles provide much better discrimination than can patterns. Similarly, the BLOCKS database is a more powerful resource than PROSITE, but it provides no additional annotation, and, because it is derived from families identified in PROSITE, it offers no further family coverage. The PRINTS fingerprint database does provide family signatures not seen in PROSITE and offers more powerful discrimination, yet it is not as comprehensive. The situation is similar for its BLOCKS-format counterpart (strictly a tertiary database, as it is derived from secondary data), which provides even more potent discriminators.

Table 2. Examples of secondary databases

Database	Data source	Information stored
PROSITE	SWISS-PROT	patterns, rules, profiles
Profiles	SWISS-PROT	weight matrices
BLOCKS	SWISS-PROT, PROSITE	clustered, weighted motifs
PRINTS	OWL	unweighted motifs
BLOCKS-format-PRINTS	PRINTS	clustered, weighted motifs
ProDom	SWISS-PROT	domain alignments
SBASE	SWISS-PROT	annotated domain sequences
Pfam*	SWISS-PROT	annotated alignments
PIMA	Entrez	marked-up alignments

*Pfam is based on probabilistic Hidden Markov Model.

The ProDom, SBASE, Pfam, and PIMA databases effectively contain marked-up domain sequences and alignments, derived from largely automated processes, rather than the descriptors housed in these other resources. They provide no supporting documentation and, diagnostically, are often unreliable. Nevertheless, in a comprehensive search protocol, it is wise to include as many secondary databases as possible, as no single resource can yet guarantee a diagnosis.

Information networks

Progress in bioinformatics would not have been possible without the international computer network, the Internet. The 'universe of network-accessible information' has been dubbed the World Wide Web, an 'embodiment of human knowledge'. Participating groups make information available on servers, all of which have unique addresses, referred to as Uniform Resource Locators, or URLs. Armed with a URL and a suitable WWW browser (e.g., Netscape, Mosaic, or Lynx), users may visit any site in the world and use the tools and data made available by those institutions or organizations. The URLs of some of the resources discussed above are given in Table 3.

**Table 3.** World Wide Web addresses of various bioinformatics resources

Database/resource	Reference	URL
SWISS-PROT	1	http://expasy.hcuge.ch/sprot/sprot-top.html
PIR	2	http://www.gdb.org/Dan/proteins/pir.html
mips	2	http://www.mips.biochem.mpg.de/
PRF		http://www.genome.ad.jp/htbin/bfind_prf
NRL-3D	3	http://www.gdb.org/Dan/proteins/nrl3d.html
GenBank	4	http://www.ncbi.nlm.nih.gov/
EMBL	5	http://www.ebi.ac.uk/ebi_docs/embl_db/ebi/topembl.html
OWL	6	http://www.biochem.ucl.ac.uk/bsm/dbbrowser/OWL/OWL.html
nrdb	4	http://www.ncbi.nlm.nih.gov/BLAST/blast_databases.html
Entrez	7	http://www3.ncbi.nlm.nih.gov/Entrez/
PROSITE	8	http://expasy.hcuge.ch/sprot/prosite.html
Profiles	9	http://ulrec3.unil.ch/profile/profile.html
ProDom	10	http://www.sanger.ac.uk/~esr/prodom.html
SBASE	11	http://base.icgeb.trieste.it/sbase/
PRINTS	12	http://www.biochem.ucl.ac.uk/bsm/dbbrowser/PRINTS/PRINTS.html
BLOCKS	13	http://www.blocks.fhcr.org/
PIMA	14	http://dot.imgen.bcm.tmc.edu:9331/seq-search/Help/beauty.html
Pfam		http://www.sanger.ac.uk/Pfam/
PDB	15	http://www.genome.ad.jp/htbin/bfind_pdb
SCOP	16	http://scop.mrc-lmb.cam.ac.uk/scop/
CATH	17	http://www.biochem.ucl.ac.uk/bsm/cath/
PROSITE.lis		http://expasy.hcuge.ch/cgi-bin/prosite-list.pl
PRINTS.lis		http://www.biochem.ucl.ac.uk/bsm/dbbrowser/PRINTS/printscontents.html
PROFILES.lis		http://ulrec3.unil.ch/cgi-bin/PFSCAN_form_parser2
Protfam		http://www.mips.biochem.mpg.de/public/protfam/fbrowser.html
Pfam		http://www.sanger.ac.uk/Pfam/browse_curr.html
Modules		http://swan.embl-heidelberg.de:8080/Modules/

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Appendix F

Restriction enzymes and methylases

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The restriction enzyme database, REBASE, is a collection of information about restriction enzymes and methylases (1). Recently, homing endonucleases (2) have also been included in REBASE. Among the 2825 restriction enzymes now known, there are 18 fully characterized Type I specificities, 210 different Type II specificities and 4 different Type III specificities. Tables 1–3 contain a listing of all prototype restriction enzymes (Types I, II, and III), together with their commercially available isoschizomers and neoschizomers. Table 4 contains a listing of the 23 homing endonucleases for which recognition sequences are available. REBASE currently stores 3959 published references, which include abstracts, and 464 unpublished observations.

REBASE is updated daily. Each month, a set of REBASE data files are released publicly and distributed to the scientific community, at no charge, via e-mail. They can also be retrieved by anonymous ftp ([www.neb.com, in pub/rebase](http://www.neb.com/pub/rebase)), WAIS (www.neb.com, port 210) or through the World Wide Web (<http://www.neb.com/rebase>). These data files are ASCII text files, many of which are designed specifically for use with a variety of software packages such as GCG, IGSuite, GENEPRO, Staden, DNA Strider, Pro-Cite, PC/Gene, SEQAIDII, GENETYX, DNASIS, CAD Gene, and Reference Manager. Other data files include a complete set of references, including abstracts, to papers on restriction

enzymes and methylases; a list of all commercial suppliers of restriction enzymes and methylases, complete with contact information and a listing of enzymes they currently sell. New data files are constantly being added and each release of REBASE includes a monthly newsletter indicating that the files at the WWW and ftp sites have been updated, and listing new enzymes, newly available formats, enzyme name changes, etc. To join the mailing list or for more information, contact either author (e-mail: macelis@neb.com or roberts@neb.com; telephone (508) 927-3382; fax (508) 921-1527).

In forming Tables 1–3, all endonucleases cleaving DNA at a specific sequence have been considered to be restriction enzymes, although in most cases there is no direct genetic evidence for the presence of a restriction-modification system. The endonucleases are named in accordance with the proposal of Smith and Nathans (3).

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3. Smith, H. O. and Nathans, D. (1973). *Journal of Molecular Biology*, **81**, 419–423.



Appendix F

Table 1. Type II enzymes

Prototype	Isoschizomers	Recognition sequence ¹	Me site ²	Commercial source ³
<i>AatII</i>		GACGT↑C		ADEFKLMNOPRS
<i>AccI</i>		GT↑MKAC	5(6)	ABDEGJKLMNOPQRS
<i>AceIII</i>		CAGCTC(7/11)		
<i>Acil</i>		CCGC(–3/–1)		N
<i>AcII</i>		AA↑CGTT		I
	<i>Psp1406I</i>	AA↑CGTT		DFMN
<i>Acyl</i>		GR↑CGYC		JM
	<i>BbII</i>	GR↑CGYC		K
	<i>BstACI</i>	GR↑CGYC		I
	<i>BsaHI</i>	GR↑CGYC		NS
	<i>HinII</i>	GR↑CGYC		EFO
	<i>Hsp92I</i>	GR↑CGYC		R
	<i>Msp17I</i>	GR↑CGYC		D
<i>AflII</i>		C↑TTAAG		ABJKNOS
	<i>BfrI</i>	C↑TTAAG		MO
	<i>BspTI</i>	C↑TTAAG		DF
	<i>Bst98I</i>	C↑TTAAG		R
	<i>MspCI</i>	C↑TTAAG		C
	<i>Vha464I</i>	C↑TTAAG		I
<i>AflIII</i>		A↑CRYGT		ABMN
<i>AgeI</i>		A↑CCGGT	?(5)	JNR
	<i>AsiAI</i>	A↑CCGGT		I
	<i>PinAI</i>	A↑CCGGT		BM
<i>AhaIII</i>		TTT↑AAA		
	<i>DraI</i>	TTT↑AAA		ABDEFGHIJKLMNOPQRS
<i>AluI</i>		AG↑CT	3(5)	ABCDEFHGHIJKLMNOPQRST
<i>AlwNI</i>		CAGNNN↑CTG		ABNS
<i>ApaI</i>		GGGCC↑C	4(5)	ABEGJKLMNOPQRS
	<i>Bsp120I</i>	G↑GGCCC		DFN
	<i>PspOMI</i>	G↑GGCCC		I
<i>ApaBI</i>		GCANNNNN↑TGC		
<i>ApaLI</i>		G↑TGCAC	?(5)	ADEKNS
	<i>Alw44I</i>	G↑TGCAC		FJMOR
	<i>VneI</i>	G↑TGCAC		I
<i>ApoI</i>		R↑AATTY	?(6)	N
	<i>Acsl</i>	R↑AATTY		DIM
<i>AscI</i>		GG↑CGCGCC	?(5)	N
<i>AsuI</i>		G↑GNCC		
	<i>AspS9I</i>	G↑GNCC		I
	<i>BsiZI</i>	G↑GNCC		T
	<i>Cfr13I</i>	G↑GNCC	4(5)	DFKO
	<i>FmuI</i>	GGNC↑C		
	<i>Sau96I</i>	G↑GNCC	4(5)	AJLMNORS
<i>AsuII</i>		TT↑CGAA		
	<i>Bpu14I</i>	TT↑CGAA		I
	<i>BsiCI</i>	TT↑CGAA		T
	<i>Bsp119I</i>	TT↑CGAA		DF
	<i>BstBI</i>	TT↑CGAA		NS
	<i>CblI</i>	TT↑CGAA		J

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Table 1 cont.

Prototype	Isoschizomers	Recognition sequence ¹	Me site ²	Commercial source ³
	<i>Csp45I</i>	TT↑CGAA		OR
	<i>LspI</i>	TT↑CGAA		L
	<i>NspV</i>	TT↑CGAA		ABJKO
	<i>SfuI</i>	TT↑CGAA		M
<i>AvaI</i>		C↑YCGRG	?(4)	ABEGJKLMNPQRS
	<i>Ama87I</i>	C↑YCGRG		DI
	<i>BcoI</i>	C↑YCGRG		AT
	<i>BsoBI</i>	C↑YCGRG	?(4)	N
	<i>Eco88I</i>	C↑YCGRG		F
	<i>Nli387/7I</i>	CYCGR↑G		
<i>AvaII</i>		G↑GWCC	?(5)	ABEGJKLMNPQRS
	<i>Bme18I</i>	G↑GWCC		I
	<i>Eco47I</i>	G↑GWCC		FO
	<i>HgiEI</i>	G↑GWCC	?(5)	D
	<i>SinI</i>	G↑GWCC	4(5)	LR
	<i>VpaK11AI</i>	↑GGWCC		
<i>AvaIII</i>		ATGCA↑T		
	<i>BfrBI</i>	ATG↑CAT		
	<i>EcoT22I</i>	ATGCA↑T		AKO
	<i>Mph1103I</i>	ATGCA↑T		F
	<i>NsiI</i>	ATGCA↑T		BDEHJLMNRS
	<i>Ppu10I</i>	A↑TGCAT		DFN
	<i>Zsp2I</i>	ATGCA↑T		I
<i>AvrII</i>		C↑CTAGG		N
	<i>BlnI</i>	C↑CTAGG		AKM
	<i>BspA2I</i>	C↑CTAGG		I
<i>BaeI</i> ⁴		(10/15)ACNNNNGTAYC(12/7)	1(6), -4(6)	
<i>BalI</i>		TGG↑CCA	4(4)	AJKR
	<i>MluNI</i>	TGG↑CCA		M
	<i>MscI</i>	TGG↑CCA		BDNOS
<i>BamHI</i>		G↑GATCC	5(4)	ABCDEFGHJKLMNPQRST
<i>BbvI</i>		GCAGC(8/12)	2(5), -2(5)	N
	<i>Bst71I</i>	GCAGC(8/12)		R
<i>BbvII</i>		GAAGAC(2/6)		
	<i>BbsI</i>	GAAGAC(2/6)		N
	<i>BpiI</i>	GAAGAC(2/6)		DF
	<i>BpuAI</i>	GAAGAC(2/6)		M
<i>BccI</i>		CCATC		
<i>Bce83I</i>		CTTGAG(16/14)		
<i>BceI</i>		ACGGC(12/13)		
<i>BcgI5</i>		(10/12)CGANNNNNNTGC(12/10)	3(6), -3(6)	N
<i>BciVI</i>		GGATAC (6/5)		N
<i>BclI</i>		T↑GATCA		BCDEFGJLMNPRS
	<i>BsiQI</i>	T↑GATCA		T
	<i>FbaI</i>	T↑GATCA		AK
	<i>Ksp22I</i>	T↑GATCA		I
<i>BetI</i>		W↑CCGGW		
	<i>BsaWI</i>	W↑CCGGW		N
<i>BfiI</i>		ACTGGG(4/5)		F
<i>BglI</i>		GCCNNNN↑NGGC	?(4)	ABCDEFGHJKLMNPQRST

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Appendix F

Table 1 cont.

Prototype	Isoschizomers	Recognition sequence ¹	Me site ²	Commercial source ³
<i>Bgl</i> I		A↑GATCT	5(4)	ABCDEFGHIJKLMNQRST
<i>Bin</i> I		GGATC(4/5)		
	<i>Ac</i> /Wl	GGATC(4/5)		I
	<i>Al</i> wl	GGATC(4/5)		NS
<i>Bmg</i> I		GKGCCC		
<i>Bp</i> I		GAGNNNNNCTC		
<i>Bpu</i> 10I		CCTNAGC(-5/-2)		
<i>Bsa</i> AI		YAC↑GTR		N
	<i>Bs</i> +tBAI	YAC↑GTR		I
<i>Bsa</i> BI		GATNN↑NNATC		N
	<i>Bse</i> 8I	GATNN↑NNATC		I
	<i>Bsh</i> 1365I	GATNN↑NNATC		DF
	<i>Bsi</i> BI	GATNN↑NNATC		T
	<i>Bsr</i> BRI	GATNN↑NNATC		R
	<i>Mam</i> I	GATNN↑NNATC	?(6)	M
<i>Bsa</i> XI		ACNNNNNCTCC		
<i>Bsb</i> I		CAACAC		
<i>Bsc</i> GI		CCCGT		
<i>Bse</i> PI		G↑CGCGC		I
	<i>Bss</i> HII	G↑CGCGC	?(5)	ABDEJKLMNQR
	<i>Paul</i> I	G↑CGCGC		F
<i>Bse</i> RI		GAGGAG(10/8)		N
<i>Bsg</i> I		GTGCAG(16/14)		N
<i>Bsi</i> I		CACGAG(-5/-1)		
	<i>Bss</i> SI	CACGAG(-5/-1)		N
	<i>Bst</i> 2BI	CACGAG(-5/-1)		DI
<i>Bsi</i> YI		CCNNNNN↑NNGG		MT
	<i>Bsc</i> 4I	CCNNNNN↑NNGG		I
	<i>Bsl</i> I	CCNNNNN↑NNGG		N
<i>Bsm</i> I		GAATGC(1/-1)		ABDEJLMNOS
	<i>Bsa</i> MI	GAATGC(1/-1)		R
	<i>Bsc</i> CI	GAATGC(1/-1)		T
	<i>Mva</i> 1269I	GAATGC(1/-1)		F
<i>Bsm</i> AI		GTCTC(1/5)		N
	<i>Al</i> w26I	GTCTC(1/5)	3(5),-4(6)	DFNR
<i>Bsp</i> 24I5		(8/13)GACNNNNNNTGG(12/7)		
<i>Bsp</i> 1407I		T↑GTACA		DF
	<i>Bsr</i> GI	T↑GTACA		N
	<i>Ssp</i> BI	T↑GTACA		M
<i>Bsp</i> GI		CTGGAC		
<i>Bsp</i> HI		T↑CATGA		AN
	<i>Rca</i> I	T↑CATGA		BM
<i>Bsp</i> LU11I		A↑CATGT		M
<i>Bsp</i> MI		ACCTGC(4/8)		N
<i>Bsp</i> MII		T↑CCGGA		
	<i>Acc</i> III	T↑CCGGA		EJKQR
	<i>Bse</i> AI	T↑CCGGA		CM
	<i>Bsi</i> MI	T↑CCGGA		AT
	<i>Bsp</i> 13I	T↑CCGGA		I
	<i>Bsp</i> EI	T↑CCGGA		N

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Table 1 cont.

Prototype	Isoschizomers	Recognition sequence ¹	Me site ²	Commercial source ³
	<i>Kpn2I</i>	T↑CCGGA		BDF
	<i>MroI</i>	T↑CCGGA		MO
<i>BsrI</i>		ACTGG(1/-1)		N
	<i>Bse1I</i>	ACTGG(1/-1)		I
	<i>BseNI</i>	ACTGG(1/-1)		DF
	<i>BsrSI</i>	ACTGG(1/-1)		R
<i>BsrBI</i>		CCGCTC(-3/-3)		N
	<i>AccBSI</i>	CCGCTC(-3/-3)		DI
<i>BsrDI</i>		GCAATG(2/0)		N
	<i>Bse3DI</i>	GCAATG(2/0)		I
<i>BstEII</i>		G↑GTNACC		BDGHJLMNORS
	<i>BstPI</i>	G↑GTNACC		K
	<i>Eco91I</i>	G↑GTNACC		F
	<i>EcoO65I</i>	G↑GTNACC		AK
	<i>PspEI</i>	G↑GTNACC		I
<i>BstXI</i>		CCANNNNN↑NTGG		ABDEFGHJKLMNOQRS
<i>Cac8I</i>		GCN↑NGC		N
<i>CauII</i>		CC↑SGG		
	<i>AsuC2I</i>	CC↑SGG		I
	<i>BcnI</i>	CC↑SGG	2(4)	ADFK
	<i>EcoHI</i>	↑CCSGG		
	<i>NciI</i>	CC↑SGG		BEJLMNORST
<i>CfrI</i>		Y↑GGCCR	4(5)	F
	<i>EaeI</i>	Y↑GGCCR	4(5)	ADKLMNS
<i>Cfr10I</i>		R↑CCGGY	2(5)	ADFKMO
	<i>Bse118I</i>	R↑CCGGY		I
	<i>BsrFI</i>	R↑CCGGY		N
	<i>BssAI</i>	R↑CCGGY		C
<i>CjeI</i> ⁴		(8/14)CCANNNNNNGT(15/9)		
<i>CjePI</i> ⁴		(7/13)CCANNNNNNNNTC(14/8)		
<i>ClaI</i>		AT↑CGAT	5(6)	ABKMNPRST
	<i>BanIII</i>	AT↑CGAT		O
	<i>Bsa29I</i>	AT↑CGAT		I
	<i>BscI</i>	AT↑CGAT		L
	<i>BseCI</i>	AT↑CGAT	5(6)	C
	<i>BsiXI</i>	AT↑CGAT		T
	<i>Bsp106I</i>	AT↑CGAT	5(6)	E
	<i>BspDI</i>	AT↑CGAT		N
	<i>BspXI</i>	AT↑CGAT		G
	<i>Bsu15I</i>	AT↑CGAT	5(6)	DF
<i>CviJI</i>		RG↑CY	3(5)	Q
<i>CviRI</i>		TG↑CA	4(6)	
<i>DdeI</i>		C↑TNAG	1(5)	ABEGLMNOR
	<i>BstDEI</i>	C↑TNAG		DI
<i>DpnI</i> ⁵		GmATTC		ABLMNRS
<i>DraII</i>		RG↑GNCCY		AGMS
	<i>EcoO109I</i>	RG↑GNCCY		ABDEFJKLN
	<i>PssI</i>	RGGNC↑CY		
<i>DraIII</i>		CACNNN↑GTG		AEGMNS
<i>DrdI</i>		GACNNNN↑NNGTC		N

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Table 1 cont.

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	<i>Dse</i> DI	GACNNNN↑NNGTC		I
<i>Drd</i> II		GAACCA		
<i>Dsa</i> I		C↑CRYGG		M
	<i>Bst</i> DSI	C↑CRYGG		DI
<i>Eam</i> 1105I		GACNNN↑NNGTC		DFK
	<i>Ahd</i> I	GACNNN↑NNGTC		N
	<i>Asp</i> EI	GACNNN↑NNGTC		M
	<i>Ecl</i> HKI	GACNNN↑NNGTC		R
	<i>Nru</i> GI	GACNNN↑NNGTC		I
<i>Eci</i> I		TCCGCC		
<i>Eco</i> 31I		GGTCTC(1/5)	4(5),-4(6)	DF
	<i>Bsa</i> I	GGTCTC(1/5)		N
<i>Eco</i> 47III		AGC↑GCT		BDEFLMNOR
	<i>Afe</i> I	AGC↑GCT		I
	<i>Aor</i> 51HI	AGC↑GCT		AK
<i>Eco</i> 57I		CTGAAG(16/14)	5(6),-5(6)	DFNS
<i>Eco</i> NI		CCTNN↑NNNAGG		ANS
<i>Eco</i> RI		G↑AATTC	3(6)	ABCDEFGHIJKLMNOPQRST
<i>Eco</i> RII ⁶		↑CCWGG	2(5)	BEJMOS
+	<i>Bsi</i> LI	CC↑WGG		T
+	<i>Bst</i> NI	CC↑WGG	2(4)	CENS
+	<i>Bst</i> OI	CC↑WGG		R
+	<i>Bst</i> 2UI	CC↑WGG		I
+	<i>Mva</i> I	CC↑WGG	2(4)	ADFKMO
<i>Eco</i> RV		GAT↑ATC	2(6)	ABCDEFGHIJKLMNOPQRST
	<i>Eco</i> 32I	GAT↑ATC		F
<i>Esp</i> I		GC↑TNAGC		
	<i>Blp</i> I	GC↑TNAGC		N
	<i>Bpu</i> 1102I	GC↑TNAGC		BDEFK
	<i>Bsp</i> 1720I	GC↑TNAGC		I
	<i>Cel</i> II	GC↑TNAGC		ALM
<i>Esp</i> 3I		CGTCTC(1/5)	4(5),-4(6)	DF
	<i>Bsm</i> BI	CGTCTC(1/5)		N
<i>Fau</i> I		CCCGC(4/6)		I
<i>Fin</i> I		GGGAC		
	<i>Bsm</i> FI	GGGAC(10/14)		N
<i>Fnu</i> DII		CG↑CG		
	<i>Acc</i> II	CG↑CG		AJKQ
	<i>Bsh</i> 1236I	CG↑CG		DEF
	<i>Bst</i> UI	CG↑CG		NS
	<i>Mvn</i> I	CG↑CG		M
	<i>Se</i> I	↑CGCG		
	<i>Tha</i> I	CG↑CG		B
<i>Fnu</i> 4HI		GC↑NGC		N
	<i>Bso</i> FI	GC↑NGC		N
	<i>Fsp</i> 4HI	GC↑NGC		I
	<i>Ita</i> I	GC↑NGC		M
<i>Fok</i> I		GGATG(9/13)	3(6),-2(6)	ADGIJLMNRS
	<i>Bst</i> F5I	GGATG(2/0)		I
	<i>Sts</i> I	GGATG(10/14)	3(6),-2(6)	

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Table 1 cont.

Prototype	Isoschizomers	Recognition sequence ¹	Me site ²	Commercial source ³
<i>FseI</i>		GGCCGG↑CC	?(5)	AKN
<i>GdiII</i>		CGGCCR(-5/-1)		
<i>GsuI</i>		CTGGAG(16/14)		DFS
	<i>BpmI</i>	CTGGAG(16/14)		N
<i>HaeI</i>		WGG↑CCW		
<i>HaeII</i>		RGCGC↑Y		ABDEGJKLMNOPRS
	<i>Bsp143II</i>	RGCGC↑Y		F
	<i>BstH2I</i>	RGCGC↑Y		I
	<i>LpnI</i>	RGCTGCY		
<i>HaeIII</i>		GG↑CC	3(5)	ABCDGHIJKLMNOPQRST
	<i>BshI</i>	GG↑CC		T
	<i>BsuRI</i>	GG↑CC	3(5)	FI
	<i>PaiI</i>	GG↑CC		E
<i>HgaI</i>		GACGC(5/10)	3(5),-2(5)	AN
<i>HgiAI</i>		GWGCW↑C		
	<i>Alw21I</i>	GWGCW↑C		DF
	<i>AspHI</i>	GWGCW↑C		M
	<i>Bbv12I</i>	GWGCW↑C		I
	<i>BsiHKA</i>	GWGCW↑C		N
<i>HgiCI</i>		G↑GYRCC	5(5)	
	<i>AccB1I</i>	G↑GYRCC		I
	<i>BanI</i>	G↑GYRCC	?(5)	AEMNORS
	<i>BshNI</i>	G↑GYRCC		F
	<i>Eco64I</i>	G↑GYRCC		DF
<i>HgiEI</i>		ACNNNNNNNGGT		
<i>HgiJII</i>		GRGCT↑C		
	<i>BanII</i>	GRGCT↑C	4(5)	ABKLMNOQRS
	<i>Eco24I</i>	GRGCT↑C		DF
	<i>EcoT38I</i>	GRGCT↑C		J
	<i>FriOI</i>	GRGCT↑C		I
<i>HhaI</i>		GCG↑C	2(5)	ABGJKNOPRS
	<i>AspLEI</i>	GCG↑C		I
	<i>CfoI</i>	GCG↑C		BLMRS
	<i>Hin6I</i>	G↑CGC		DF
	<i>HinP1I</i>	G↑CGC		N
	<i>HspAI</i>	G↑CGC		I
<i>Hin4I</i>		GABNNNNNVTC		
<i>HindII</i>		GTY↑RAC	5(6)	EM
	<i>HincII</i>	GTY↑RAC	5(6)	ABCDEFGHJKLMNOPQRS
<i>HindIII</i>		A↑AGCTT	1(6)	ABCDEFGHJKLMNOPQRST
<i>HinfI</i>		G↑ANTC		ABDEFGHIJKLMNOPQRST
<i>HpaI</i>		GTT↑AAC	5(6)	ABDEFGJKLMNOPQRST
	<i>BstHPI</i>	GTT↑AAC		I
<i>HpaII</i>		C↑CGG	2(5)	BDEFGILMNOPQRS
	<i>BsiSI</i>	C↑CGG		C
	<i>HapII</i>	C↑CGG	2(5)	AK
	<i>Hin2I</i>	C↑CGG		F
	<i>MspI</i>	C↑CGG	1(5)	ABDEFGHIJKLMNOPQRS
<i>HphI</i>		GGTGA(8/7)	5(6),-2(5)	AFNS
	<i>AsuHPI</i>	GGTGA(8/7)		I

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Appendix F

Table 1 cont.

Prototype	Isoschizomers	Recognition sequence ¹	Me site ²	Commercial source ³
<i>KpnI</i>		GGTAC↑C	4(6)	ABCDEFGHIJKLMNQRST
	<i>Acc65I</i>	G↑GTACC		DFINR
	<i>Asp718I</i>	G↑GTACC		M
<i>Ksp632I</i>		CTCTTC(1/4)		M
	<i>Bsu6I</i>	CTCTTC(1/4)		I
	<i>Eam1104I</i>	CTCTTC(1/4)		DEF
	<i>EarI</i>	CTCTTC(1/4)		N
<i>MaeI</i>		C↑TAG		M
	<i>BfaI</i>	C↑TAG		N
<i>MaeII</i>		A↑CGT		M
	<i>TaiI</i>	ACGT↑		FN
	<i>TscI</i>	ACGT↑		L
<i>MaeIII</i>		↑GTNAC		MP
<i>MboI</i> ⁷		↑GATC	2(6)	ABCEFGKLNQRST
	<i>BscFI</i>	GATC		T
+	<i>Bsp143I</i>	↑GATC		DF
+	<i>BspKT6I</i>	GAT↑C	2(6)	
	<i>ChaI</i>	GATC↑		
	<i>DpnII</i>	↑GATC	2(6)	N
	<i>Kzo9I</i>	↑GATC		I
	<i>NdeII</i>	↑GATC		BDGJM
+	<i>Sau3AI</i>	↑GATC	4(5)	ABEGHJKLMNQRST
<i>MbolI</i>		GAAGA(8/7)	5(6)	ABDFGJKNOQRS
<i>McrI</i>		CGRY↑CG		
	<i>BsaOI</i>	CGRY↑CG		R
	<i>Bsh1285I</i>	CGRY↑CG		DF
	<i>BsiEI</i>	CGRY↑CG		N
	<i>BstMCI</i>	CGRY↑CG		I
<i>MfeI</i>		C↑AATTG		N
	<i>MunI</i>	C↑AATTG	3(6)	BDEFKM
<i>MluI</i>		A↑CGCGT		ABDEFGIJKLMNQRST
<i>MmeI</i>		TCCRAC(20/18)		
<i>MnlI</i>		CCTC(7/6)		AEFNQ
<i>MseI</i>		T↑TAA		BN
	<i>TruII</i>	T↑TAA		F
	<i>Tru9I</i>	T↑TAA		DILMRS
<i>MslI</i>		CAYNN↑NNRTG		N
<i>MstI</i>		TGC↑GCA		
	<i>Acc16I</i>	TGC↑GCA		DI
	<i>AviII</i>	TGC↑GCA		M
	<i>FspI</i>	TGC↑GCA		ABJKNOS
<i>MwoI</i>		GCNNNN↑NNGC	?(5)	N
<i>NaeI</i>		GCC↑GGC		ADEKLMNOR
	<i>MroNI</i>	G↑CCGGC		I
	<i>NgoAIV</i>	G↑CCGGC		B
	<i>NgoMI</i>	G↑CCGGC	2(5)	NR
<i>NarI</i>		GG↑CGCC		BEJMNOPRS
	<i>BbeI</i>	GGCGC↑C		AK
	<i>EgeI</i>	GGC↑GCC		I
	<i>EheI</i>	GGC↑GCC		ADFNO

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Table 1 cont.

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	<i>KasI</i>	G↑GCGCC		N
	<i>Mly113I</i>	GG↑CGCC		I
<i>NcoI</i>		C↑CATGG	?(4)	ABCDEFGHJKLMNOPQRST
	<i>Bsp19I</i>	C↑CATGG		I
<i>NdeI</i>		CA↑TATG		ABDEFGJKLMNPRS
	<i>FauNDI</i>	CA↑TATG		DI
<i>NheI</i>		G↑CTAGC		ABDEFGJKLMNPRS
	<i>AceII</i>	GCTAG↑C		
	<i>AsuNHI</i>	G↑CTAGC		I
<i>NlaIII</i>		CATG↑	2(6)	ANOS
	<i>CviAII</i>	C↑ATG	2(6)	
	<i>Hsp92II</i>	CATG↑		R
<i>NlaIV</i>		GGN↑NCC		NS
	<i>BscBI</i>	GGN↑NCC		T
	<i>BspLI</i>	GGN↑NCC		F
	<i>PspNAI</i>	GGN↑NCC		I
<i>NotI</i>		GC↑GGCCGC	?(4)	ABCDEFGHJKLMNOPQRST
	<i>CciNI</i>	GC↑GGCCGC		I
<i>NruI</i>		TCG↑CGA		ABCDEFGHIJKLMNOQRST
	<i>Bsp68I</i>	TCG↑CGA		F
<i>NspI</i>		RCATG↑Y		ABKM
	<i>BstNSI</i>	RCATG↑Y		DI
<i>NspBII</i>		CMG↑CKG		A
	<i>MspAII</i>	CMG↑CKG		INR
<i>PacI</i>		TTAAT↑TAA		NO
<i>Pfi1108I</i>		TCGTAG		
<i>PfiMI</i>		CCANNNN↑NTGG		ANS
	<i>AccB7I</i>	CCANNNN↑NTGG		IR
	<i>Esp1396I</i>	CCANNNN↑NTGG		F
	<i>Van91I</i>	CCANNNN↑NTGG		DFKM
<i>PleI</i>		GAGTC(4/5)		N
	<i>MlyI</i>	GACTC(5/5)		
<i>PmaCI</i>		CAC↑GTG		AK
	<i>BbrPI</i>	CAC↑GTG		MO
	<i>Eco72I</i>	CAC↑GTG		DEFR
	<i>PmlI</i>	CAC↑GTG		N
<i>PmeI</i>		GTTT↑AAAC		N
<i>PpuMI</i>		RG↑GWCCY		ANO
	<i>Psp5II</i>	RG↑GWCCY		BDF
	<i>PspPPI</i>	RG↑GWCCY		I
<i>PshAI</i>		GACNN↑NNGTC		AKN
<i>PstI</i>		CTGCA↑G	5(6)	ABCDEFGHJKLMNOPQRST
<i>PvuI</i>		CGAT↑CG		ABDEFGJKLMNQRS
	<i>BspCI</i>	CGAT↑CG		E
	<i>Ple19I</i>	CGAT↑CG		I
<i>PvuII</i>		CAG↑CTG	4(4)	ABCDEFGHJKLMNOPQRST
<i>RleAI</i>		CCCACA(12/9)		
<i>RsaI</i>		GT↑AC		ABCDEFGHIJLMNOPQRST
	<i>AfaI</i>	GT↑AC		K

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Appendix F

Table 1 cont.

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	<i>Csp6I</i>	G↑TAC		DFN
<i>RsrII</i>		CG↑GWCCG		BMNS
	<i>CpoI</i>	CG↑GWCCG		ADFK
	<i>CspI</i>	CG↑GWCCG		EOR
<i>SacI</i>		GAGCT↑C	5(?)	ACEFGHJKLMNOPQRST
	<i>Ecl136II</i>	GAG↑CTC		DFN
	<i>EcoICRI</i>	GAG↑CTC		R
	<i>Psp124BI</i>	GAGCT↑C		DI
	<i>SstI</i>	GAGCT↑C		BCS
<i>SacII</i>		CCGC↑GG		ACEGHJKLMNOPQRST
	<i>Cfr42I</i>	CCGC↑GG		DF
	<i>KspI</i>	CCGC↑GG		M
	<i>Mlu113I</i>	CC↑GCGG		
	<i>Sfr303I</i>	CCGC↑GG		I
	<i>SstII</i>	CCGC↑GG		BS
<i>SalI</i>		G↑TCGAC	5(6)	ABCDEFGHJKLMNOPQRST
<i>SanDI</i>		GG↑GWCCC		E
<i>SapI</i>		GCTCTTC(1/4)		N
<i>SauI</i>		CC↑TNAGG		
	<i>AocI</i>	CC↑TNAGG		M
	<i>AxyI</i>	CC↑TNAGG		J
	<i>Bse21I</i>	CC↑TNAGG		I
	<i>Bsu36I</i>	CC↑TNAGG		ENRS
	<i>CvnI</i>	CC↑TNAGG		B
	<i>Eco81I</i>	CC↑TNAGG		ADFKO
<i>Scal</i>		AGT↑ACT		ABCDEFGHJKLMNOPQRS
	<i>Acc113I</i>	AGT↑ACT		I
	<i>Eco255I</i>	AGT↑ACT		F
<i>ScrFI</i>		CC↑NGG	?(5)	DJMNOS
	<i>BssKI</i>	↑CCNGG		N
	<i>MspR9I</i>	CC↑NGG		I
<i>SduI</i>		GDGCH↑C		F
	<i>BmyI</i>	GDGCH↑C		M
	<i>Bsp1286I</i>	GDGCH↑C		ADJKNR
<i>SecI</i>		C↑CNNGG		
	<i>BsaJI</i>	C↑CNNGG		N
	<i>BseDI</i>	C↑CNNGG		DF
<i>SexAI</i>		A↑CCWGGT		M
<i>SfaNI</i>		GCATC(5/9)		IN
	<i>BscAI</i>	GCATC(4/6)		
<i>SfeI</i>		C↑TRYAG		
	<i>BfmI</i>	C↑TRYAG		F
	<i>BstSFI</i>	C↑TRYAG		DI
	<i>SfcI</i>	C↑TRYAG		N
<i>SfiI</i>		GGCCNNNN↑NGGCC	?(4)	ABCDEGJKLMNOPQRST
<i>SgfI</i>		GCGAT↑CGC		R
<i>SgrAI</i>		CR↑CCGGYG		M
<i>SimI</i>		GGGTC(-3/0)		
<i>SmaI</i>		CCC↑GGG	2(4)	ABCDEFGHJKLMNOPQRST

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Table 1 cont.

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	<i>Cfr9I</i>	C↑CCGGG	2(4)	FO
	<i>PspAI</i>	C↑CCGGG		E
	<i>XmaI</i>	C↑CCGGG	?(4)	ADINPRS
	<i>XmaCI</i>	C↑CCGGG		M
<i>SmlI</i>		C↑TYRAG		N
<i>SnaI</i>		GTATAC		
	<i>Bst1107I</i>	GTA↑TAC		DFKM
	<i>BstZ17I</i>	GTA↑TAC		N
<i>SnaBI</i>		TAC↑GTA		ACEKLMNRS
	<i>BstSNI</i>	TAC↑GTA		DI
	<i>Eco105I</i>	TAC↑GTA		FO
<i>SpeI</i>		A↑CTAGT		ABDEHJKLMNQRST
	<i>Ac/NI</i>	A↑CTAGT		I
<i>SphI</i>		GCATG↑C		ABCDEFGHIJKLMNQRST
	<i>BbuI</i>	GCATG↑C		R
	<i>PaeI</i>	GCATG↑C		F
<i>SplI</i>		C↑GTACG		AK
	<i>BsiWI</i>	C↑GTACG		MNO
	<i>Pfl23II</i>	C↑GTACG		DF
	<i>PspLI</i>	C↑GTACG		I
	<i>SunI</i>	C↑GTACG		B
<i>SrfI</i>		GCCC↑GGGC		EO
<i>Sse8387I</i>		CCTGCA↑GG		AK
	<i>SbfI</i>	CCTGCA↑GG		DI
	<i>SdaI</i>	CCTGCA↑GG		F
<i>Sse8647I</i>		AG↑GWCCT		
<i>SspI</i>		AAT↑ATT	?(6)	ABCDEFGHIJKLMNOQR
<i>StuI</i>		AGG↑CCT		ABEJKLMNPQRST
	<i>AatI</i>	AGG↑CCT		O
	<i>Eco147I</i>	AGG↑CCT		DF
	<i>Pme55I</i>	AGG↑CCT		I
	<i>SseBI</i>	AGG↑CCT		C
<i>StyI</i>		C↑CWWGG		BCEJMNRS
	<i>BssT1I</i>	C↑CWWGG		I
	<i>Eco130I</i>	C↑CWWGG		DF
	<i>EcoT14I</i>	C↑CWWGG		AK
	<i>ErhI</i>	C↑CWWGG		I
	<i>SwaI</i>	ATTT↑AAAT		M
	<i>SmiI</i>	ATTT↑AAAT		DI
<i>TaqI</i>		T↑CGA	4(6)	ABCDEFGJLMNOPQRST
	<i>TthHB8I</i>	T↑CGA	4(6)	AK
<i>TaqII⁸</i>		CACCCA(11/9)		
<i>TaqII⁸</i>		GACCGA(11/9)		
<i>TatI</i>		WGTA CW		
<i>TauI</i>		GCSGC		
<i>TfiI</i>		G↑AWTC		N
<i>TseI</i>		G↑CWGC		N
<i>Tsp45I</i>		↑GTSAC		N
<i>Tsp4CI</i>		ACN↑GT		

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Appendix F

Table 1 cont.

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<i>Tsp</i> El		↑AATT		LO
	<i>Sse</i> 9I	↑AATT	2(6)	DI
	<i>Tsp</i> 509I	↑AATT		N
<i>Tsp</i> RI		CAGTG(2/-7)		N
<i>Tth</i> 111I		GACN↑NNGTC		ADIKNQR
	<i>Asp</i> I	GACN↑NNGTC		M
	<i>Pfl</i> FI	GACNNNGTC		N
<i>Tth</i> 111II		CAARCA(11/9)		
<i>Uba</i> DI		GAACNNNNNTCC		
<i>Uba</i> EI		CACCTGC		
<i>Vsp</i> I		AT↑TAAT	5(6)	BDEFIR
	<i>Ase</i> I	AT↑TAAT		AJNO
	<i>Asn</i> I	AT↑TAAT		MS
	<i>Psh</i> BI	AT↑TAAT		K
<i>Xba</i> I		T↑CTAGA	6(6)	ABCDEFGHIJKLMNQRST
<i>Xcm</i> I		CCANNNNN↑NNNTGG		AN
<i>Xho</i> I		C↑TCGAG		ABCDEFGHIJKLMNQRST
	<i>Pae</i> R7I	C↑TCGAG	5(6)	N
	<i>Sci</i> I	CTC↑GAG		
	<i>Sfr</i> 274I	C↑TCGAG		I
<i>Xho</i> II		R↑GATCY		EMR
	<i>Bst</i> X2I	R↑GATCY		DI
	<i>Bst</i> YI	R↑GATCY		BN
	<i>Mfi</i> I	R↑GATCY		AK
<i>Xma</i> III		C↑GGCCG	4(5)	BE
	<i>Bst</i> ZI	C↑GGCCG		R
	<i>Eag</i> I	C↑GGCCG	?(5)	NPS
	<i>Ecl</i> XI	C↑GGCCG		M
	<i>Eco</i> 52I	C↑GGCCG		ADFKOR
<i>Xmn</i> I		GAANN↑NNTTC		DENR
	<i>Asp</i> 700I	GAANN↑NNTTC		M
	<i>Mro</i> XI	GAANN↑NNTTC		I

1. Recognition sequences use the standard abbreviations (*European Journal of Biochemistry* 150, 1-5 (1985)) to represent ambiguity.

R = A or G	B = not A (C or G or T)
Y = C or T	D = not C (A or G or T)
M = A or C	H = not G (A or C or T)
K = G or T	V = not T (A or C or G)
S = G or C	N = A or C or G or T
W = A or T	

Recognition sequences are written from 5' to 3', only one strand being given, and the point of cleavage is indicated by an arrow (↑). When no arrow appears, the precise cleavage site has not been determined. For enzymes such as *Bci*VI, which cleaves away from its recognition sequence, the sites of cleavage are indicated in parentheses. Thus GTATCC (6/5) indicates cleavage as shown below:

5'-GTATCCNNNNNN↑-3'
3'-CATAGNNNNN↑-5'

In all cases the recognition sequences are oriented so that the cleavage sites lie on their 3'-side.



2. The site of methylation by the cognate methylase when known is indicated as follows. The first number shows the base within the recognition sequence that is modified. A negative number indicates the complementary strand, numbered from the 5' base of that strand. The number in parentheses indicates the specific methylation involved. (6) = *N*⁶-methyladenosine; (5) = 5-methylcytosine; (4) = *N*⁴-methylcytosine. Thus for *Alw26I* the designation 3(5), -4(6) signifies the following pattern of methylation:

	1	2	3	4	5	
5'	G	T	5mC	T	C	3'
3'	C	6mA	G	A	G	5'
	-1	-2	-3	-4	-5	

3. Commercial sources of restriction enzymes are abbreviated as follows:

A, Amersham Life Sciences-USB (10/96); B, Life Technologies Inc., Gibco-BRL (10/96); C, Minotech Molecular Biology Products (10/96); D, Angewandte Gentechnologie Systeme (10/96); E, Stratagene (10/96); F, Fermentas MBI (10/96); G, Appligène Oncor (10/96); H, American Allied Biochemical, Inc. (10/96); I, Sib Enzyme Ltd (10/96); J, Nippon Gene Co., Ltd (10/96); K, Takara Shuzo Co. Ltd (10/96); L, NBL Gene Sciences Ltd (10/96); M, Boehringer-Mannheim (11/96); N, New England Biolabs (11/96); O, Toyobo Biochemicals (10/96); P, Pharmacia Biotech Inc. (10/96); Q, CHIMERx (5/96); R, Promega Corporation (10/96); S, Sigma (10/96); T, Advanced Biotechnologies Ltd (10/96). The date in parentheses indicates the most recent update.

4. Enzymes that cut on both sides of their recognition sequence, such as *BaeI*, *BcgI*, *Bsp24I*, *CjeI* and *CjePI*, have 4 cleavage sites each instead of 2. For example, *Bsp24I* cleaves as indicated:



This is described in Table 2 and some REBASE reports as *Bsp24I* (8/13)GACNNNNNTGG(12/7). Thus the recognition site is excised in a fragment, 32 base pairs long, with 5-base 3'-extensions at each end.

5. *DpnI* requires the presence of 6-methyladenosine within the recognition sequence GmATC.
6. *EcoRII* isoschizomers fall into two classes based upon their sensitivity to methylation. *EcoRII* will not cleave when the second cytosine in the recognition sequence is methylated to 5-methylcytosine whereas *MvaI* will cleave such a sequence. Isoschizomers of *EcoRII* that are like *MvaI* are indicated by +.
7. *MboI* isoschizomers fall into two classes based upon their sensitivity to methylation. *MboI* will not cleave when the recognition sequence contains 6-methyladenosine whereas *Sau3AI* will not cleave when its recognition sequence contains 5-methylcytosine. Isoschizomers of *MboI* that are like *Sau3AI* are indicated by +.
8. *TaqII* differs from other restriction enzymes in recognizing two distinct sequences: GACCG and CACCCA.

Table 2. Type I enzymes

Prototype enzyme	Recognition sequence ¹	Me site ²
<i>CfrAI</i>	GCANNNNNNNNGTGG	
<i>EcoAI</i>	GAGNNNNNNNGTCA	2(6), -3(6)
<i>EcoBI</i>	TGANNNNNNNNTGCT	3(6), -4(6)
<i>EcoDI</i>	TTANNNNNNNNGTCY	
<i>EcoDR2</i>	TCANNNNNNNGTGCG	
<i>EcoDR3</i>	TCANNNNNNNATCG	
<i>EcoDXXI</i>	TCANNNNNNNRRTTC	
<i>EcoEI</i>	GAGNNNNNNNATGC	
<i>EcoKI</i>	AACNNNNNNNGTGC	2(6), -3(6)
<i>EcoR124I</i>	GAANNNNNNNRTCG	3(6), -3(6)
<i>EcoR124II</i>	GAANNNNNNNRTCG	-3(6)
<i>EcoRD2</i>	GAANNNNNNNRRTC	
<i>EcoRD3</i>	GAANNNNNNNRRTC	
<i>EcoprfI</i>	CCANNNNNNNRRTGC	
<i>StyLTI</i>	GAGNNNNNNNRATYG	2(6), -4(6)
<i>StySJ</i>	GAGNNNNNNNGTRC	
<i>StySPI</i>	AACNNNNNNNGTRC	2(6), -3(6)
<i>StySQ</i>	AACNNNNNNNRATYG	

For notes and abbreviations see Table 1



Appendix F

Table 3. Type III enzymes

Prototype enzyme	Recognition sequence ¹	Me site ²
<i>EcoPI</i>	AGACC	3(6)
<i>EcoP15I</i>	CAGCAG(25/27)	5(6)
<i>HinfIII</i>	CGAAT	
<i>StyLTI</i>	CAGAG	4(6)

For notes and abbreviations see Table 1

Table 4. Homing endonucleases

Enzyme	Sequence around one cleavage site ¹	Commercial source ²
Endo. <i>SceI</i>	GATGCTGT_AGGC↑ATAGGCTTGGTT	
HO-endo	CTTCCGC_AACA↑GTATAATT	
I- <i>CeuI</i>	TCTAACTATAACGGTC_CTAA↑GGTAGCGAG	N
I- <i>ChuI</i>	GAAGGTTTGGCAC_CTCG↑ATGTCGGCTCATC	
I- <i>CpaI</i>	CGATCCTAAGGTAG_CGAA↑ATTCA	
I- <i>CpaII</i>	TCCGGCTA_ACTC↑TGTGCCAG	
I- <i>CreI</i>	CTGGGTTCAAAACGTC_GTGA↑GACAGTTTGG	
I- <i>CsmI</i>	GTACTAGCATGGGGTCAAATGTCTTTCTGG	
I- <i>DmoI</i>	ATGCCTTGCCGG_GTAA↑GTTCCGGCGCGCAT	
I- <i>PorI</i>	GCGAGCCCGTAA_GGGT↑GTGTACGGG	
I- <i>PpoI</i>	CTATGACTCTC_TTAA↑GGTAGCCAAAA	NR
I- <i>ScaI</i>	TGTCACATTGAGGTGCACTAGTTATTAC	
I- <i>SceI</i>	TTACGCTAGGG_ATAA↑CAGGGTAATA	M
I- <i>SceII</i>	TTCTGATTCTTTGGTC_ACCC↑TGAAGTATAT	
I- <i>SceIII</i>	CTTGGAGGTTTTGG_TAAC↑TATTTATTACCA	
I- <i>SceIV</i>	AATTTTCTCATG_ATTAT↑GCTCTAATCCATGG	
I- <i>TevI</i>	AGTGGTATCA_AC↑GCTCAGTAGATG	
I- <i>TevII</i>	GCTTATGAGTATGAAGTGAACAC_GT↑TATTC	
I- <i>TevIII</i>	↑TA_TGTATCTTTGCGTGTACCTTTAACTTC	
PI- <i>PspI</i>	AAAATCCTGGCAAACAGCTA_TTAT↑GGGTAT	N
PI- <i>SceI</i>	ATCTATGTCGG_GTGC↑GGAGAAAGAGGTAAT	N
PI- <i>TliI</i>	TCTTTATGCGG_ACAC↑TGACGGCTTTTAT	N
PI- <i>TliII</i>	AAATTGCTTGCAAACAGCTATTACGGCTAT	

1. The cleavage sites, where known, are indicated by ↑ on the strand presented and by _ on the complementary strand.

2. The commercial sources are M: Boehringer-Mannheim (11/96); N: New England BioLabs (11/96); R: Promega Corporation (10/96).

Appendix G

Sequence-rule priorities of some common ligands in molecular entities

The names of some common ligands are listed alphabetically below, the numbers preceding them indicating the relative priorities assigned to the ligands under the **sequence rule**. A higher number denotes greater preference; for example, a hydroxy group (numbered 57) takes precedence over an amino group (numbered 43). It should be noted that any change to the structure of a listed ligand may alter the order of preference. Where a ligand consists of or contains a heavy or radioactive isotope of hydrogen, the relative priorities are tritium > deuterium > protium. Racemates with a single chiral centre are labelled (*RS*); extensions of the sequence rule are used in the *E/Z*, *pro-R/pro-S*, *pro-E/pro-Z*, and *Re/Si* nomenclature systems.

64 acetoxyl	28 3,5-dinitrophenyl	70 methylthio (CH ₃ S-)
36 acetyl	59 ethoxy	11 neopentyl
48 acetylamino	40 ethoxycarbonyl	56 nitro
21 acetylenyl	3 ethyl	27 <i>m</i> -nitrophenyl
10 allyl	46 ethylamino	33 <i>o</i> -nitrophenyl
43 amino	68 fluoro	24 <i>p</i> -nitrophenyl
44 ammonio (⁺ H ₃ N-)	35 formyl	55 nitroso
37 benzoyl	63 formyloxy	6 pentyl
49 benzoylamino	62 glycosyloxy	61 phenoxy
65 benzoyloxy	7 hexyl	22 phenyl
13 benzyl	1 hydrogen	47 phenylamino
60 benzyloxy	57 hydroxy	54 phenylazo
41 benzyloxycarbonyl	76 iodo	18 propenyl
50 benzyloxycarbonylamino	9 isobutyl	4 propyl
75 bromo	8 isopentyl	29 1-propynyl
42 <i>tert</i> -butoxycarbonyl	20 isopropenyl	12 2-propynyl
5 butyl	14 isopropyl	73 sulfo (HO ₃ S-)
16 <i>sec</i> -butyl	69 mercapto (HS-)	25 <i>m</i> -tolyl
19 <i>tert</i> -butyl	58 methoxy	30 <i>o</i> -tolyl
38 carboxyl	39 methoxycarbonyl	23 <i>p</i> -tolyl
74 chloro	2 methyl	53 trimethylammonio
17 cyclohexyl	45 methylamino	32 trityl
52 diethylamino	71 methylsulfinyl	15 vinyl
51 dimethylamino	66 methylsulfinyloxy	31 2,6-xylyl
34 2,4-dinitrophenyl	72 methylsulfonyl	26 3,5-xylyl
	67 methylsulfonyloxy	

Appendix H

Species names

Below is a list of genera/species cited frequently in this Dictionary, together with their common names, where applicable. The asterisked names have their own entries in the Dictionary.

- **Agrobacterium*
- Amanita* a genus of toadstools; *A. muscaria* is fly agaric,
- A. phalloides* the 'death-cap' (deadly agaric)
- **Arabidopsis thaliana*
- **Aspergillus*
- **Bacillus*
- Bos taurus* cattle
- Brassica* a genus of cruciferous plants; *B. napus* is the swede, *B. oleracea* includes the cabbage, cauliflower, etc.
- Caenorhabditis elegans* a nematode
- Candida* a genus of yeasts
- Canis familiaris* the dog
- Chlamydomonas reinhardtii* a unicellular green alga
- **Clostridium*
- Corynebacterium* a genus of **coryneform** (def. 2) bacteria
- **Dictyostelium discoideum*
- Digitalis* the foxglove genus
- **Drosophila*
- **Escherichia coli*
- Euglena* a unicellular, flagellate alga belonging to the Euglenophyta
- Gallus gallus* the domestic fowl
- Haemophilus influenzae* an aerobic, nonmotile Gram-negative rod
- Halobacterium halobium* a halophilic archaeobacterium
- Klebsiella* a genus of Gram-negative rod non-motile bacteria
- Lactobacillus* a genus of **lactic-acid bacteria**
- Lycopersicon esculentum* the tomato
- Mus musculus* the mouse
- **Mycobacterium*
- **Neurospora*
- Nocardia* a genus of rod or filamentous Gram-positive nonmotile bacteria
- **Paramecium*
- Penicillium* mould
- Pisum sativum* the garden pea
- **Plasmodium*
- Pseudomonas* a genus of Gram-negative rod-shaped motile bacteria
- Rattus norvegicus* the rat
- **Rhizobium*
- **Saccharomyces*
- **Salmonella*
- **Schizosaccharomyces pombe*
- Shigella* a genus of nonmotile Gram-negative rod **coliform** bacteria
- **Staphylococcus*
- **Streptococcus*
- **Streptomyces*
- Sus scrofa* the pig
- **Tetrahymena*
- Thermus aquaticus* a thermophilic archaeobacterium
- Trypanosoma* a genus of parasitic protozoans; see **trypanosome**
- **Xenopus laevis*
- Zea mays* maize (corn)